II

VOLUME II · QUESTION BANK

Clinical SBA Question Bank

A complete Diploma examination practice companion · Edition 2

1,000 single-best-answer questions with five response options
Clinical scenarios integrating diagnosis, investigation and management
Topic-based Blocks A–M and fifty cross-unit integrated cases
Answers, justifications and explanations of distractors
Learning-outcome pointers, revision links and a progress mark sheet

Mapped to the Diploma in Prosthodontics – Learning Outcomes (August 2026); paired with Notes In Prosthodontics (Volume I) for focused revision.

For candidates preparing for the Part 1 written examination

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The clinical SBA question bank

One thousand five-option single-best-answer questions designed to be used alongside the integrated Diploma revision notebook. The questions follow the August 2026 learning-outcome domains; the answer cards explain not only what is right, but why alternatives are weaker.

I	**Topic blocks A–M** Q1–950 · Thirteen curriculum units	Anatomy, pathology, materials, clinical assessment, removable and fixed prosthodontics, occlusion, tooth wear, implants and governance.
II	**Integrated clinical cases** Block N · Q951–1000	Fifty multi-domain scenarios combining patient history, examination findings and safe management or referral decisions.
III	**Answer cards & mark sheet** All 1,000 answers explained	Each response includes a best answer, justification, reasons the distractors fail and pointers to the notebook and learning outcomes.

A simple study cycle

1 READ	2 TEST	3 REVIEW	4 REPAIR
Volume I notes	timed block	answer cards	weak LO

Work at examination pace, then read every justification, including the questions you answered correctly. Record the result by block. A score below 70% directs you back to the relevant Volume I unit and learning-outcome code.

Blueprint at a glance

A–M	13 topic-based blocks	950 questions
N	Integrated clinical cases	50 questions
A–N	Five options; one best answer	1,000 questions
Part 2	Answer explanations and revision links	1,000 responses
End	Progress mark sheet	Self-assessment

Scope note. Clinical decisions are framed for Level 2 / enhanced-skills practice, not specialist Fellowship training. This bank reflects its supplied source; examination structure, clinical doses and guidance should be checked against current official information.

How to use this bank

This bank holds 1,000 single-best-answer (SBA) questions written from *Notes in Prosthodontics — Integrated Revision Notebook* (Volume I). Each question maps to a notebook unit and to the August 2026 Diploma learning outcomes. Most stems are clinical scenarios that combine history, findings and a decision, as the real paper does.

Question styles. Every question has five options, A to E, and one best answer. Some options combine others ("Both A and B", "A, B and C", "All of the above", "None of the above"). The best answer may be a single option, a combination, all of them, or none. Read the lead-in carefully: *first*, *most appropriate*, *least invasive* and *most likely* each ask for something different.

Timing. The notebook describes Part 1 as 150 SBAs in 180 minutes, about 72 seconds a question; confirm the current format with RCSEd. Sit each block timed at that pace, then mark it.

Answer cards. Every answer in the second half of this doc is a card:

Part of the card	What it gives you
Answer	The letter and the option in words
Justification	Why it is the best answer, using the facts the stem gives you
Why not the others	Why the tempting distractors fail
Revise	The notebook unit and section to go back to, and the learning outcome tested

Scoring. Under 70% on a block: re-read the unit named on its cards, then re-sit the block a fortnight later. Log each sitting on the mark sheet at the end.

Scope. Questions test Level 2 (enhanced skills) judgement. Where a stem describes complex, specialist-level care, the best answer is often to stabilise, explain and refer. Doses, figures and guidance reflect the notebook; check them against current sources before the exam.

Blueprint

The 1,000 questions cover all six learning-outcome domains (1.1–1.11, 2a.1–2a.21, 2b.1–2b.12, 2c.1–2c.27, 3a.1–3a.12, 3b.1–3b.5). Blocks A–M follow the notebook's 13 units; Block N is a set of integrated cases that cross units, as long exam stems do.

Block	Notebook unit	Learning outcomes	Questions	Number
A	1 Applied anatomy	1.1, 1.5	1–70	70
B	2 Physiology and pathology	1.2–1.4, 1.8, 1.10–1.11, 2a.7–2a.8	71–130	60
C	3 Dental materials science	1.9	131–220	90

Block	Notebook unit	Learning outcomes	Questions	Number
D	4 Pharmacology, pain control, emergencies, the medically compromised patient	1.7, 2a.13, 2b.1–2b.2, 2b.11–2b.12, 2c.7–2c.8, 2c.15	221–290	70
E	5 Assessment, diagnosis, imaging, classifications and planning	2a.1–2a.21, 2b.3–2b.10	291–370	80
F	6 Removable partial dentures	2c.2–2c.6, 2c.13, 2c.20	371–470	100
G	7 Complete dentures	2c.4, 2c.9, 2c.13, 2c.19, 2c.21–2c.27	471–570	100
H	8 Fixed prosthodontics	2c.2, 2c.4, 2c.6, 2c.11, 2c.13–2c.14, 2c.20	571–670	100
I	9 Occlusion and TMD	2a.15, 2a.20, 2c.19	671–730	60
J	10 Tooth surface loss, failing dentition, perio–endo–pros interface, trauma	2c.1, 2c.10–2c.12, 2c.16–2c.18, 2c.20	731–800	70
K	11 Implants at Level 2 and digital workflows	2a.19, 2c.5	801–860	60
L	12 Complications, failures and maintenance	2c.21–2c.27	861–890	30
M	13 Governance, law and communication	3a.1–3a.12, 3b.1–3b.5	891–950	60
N	Integrated cases (all units)	Across 1, 2a–2c, 3a–3b	951–1000	50

Part 1 — Questions

Block A — Applied anatomy (Q1–70)

> *Notebook Unit 1 · LO 1.1, 1.5 (with 1.11, 2a.5–2a.6) · Allow 84 minutes*

1 A 58-year-old bruxist has firm, bilateral swellings at the angles of the mandible that bulge on clenching. You suspect masseteric hypertrophy.
Which nerve supplies the affected muscle?
A Buccal nerve
B Masseteric nerve, from the anterior division of V3, passing through the mandibular notch
C Facial nerve
D Auriculotemporal nerve
E Nerve to medial pterygoid

2 A 34-year-old with tension-type headaches and TMD has a muscle tender to palpation above the zygomatic arch, in the temple.
Which muscle is this most likely to be?
A Masseter
B Medial pterygoid
C Lateral pterygoid
D Temporalis
E Anterior belly of digastric

3 After a fall, a 22-year-old has a fracture of the left condylar neck. The radiograph shows the condylar head displaced anteromedially.
Which muscle has displaced it?
A Lateral pterygoid (inferior head)
B Masseter
C Temporalis
D Medial pterygoid
E Posterior belly of digastric

4 A patient has lost function of the right nerve to lateral pterygoid after surgery.
On wide opening, how will the mandible move?
A It deviates to the left
B It opens straight
C It deviates to the right
D It cannot open at all
E It protrudes without opening

5 You are explaining to a student which muscles close the mouth during a denture wearer's power stroke.
Which muscles elevate the mandible?
A Masseter
B Temporalis
C Medial pterygoid
D Both A and B only
E All of the above

6 A 41-year-old has 22 mm of pain-free opening, then a click and further opening to 45 mm.
Which movement occurs in the lower (condyle–disc) compartment during the first 20–25 mm?
A Translation
B Rotation (hinge movement)
C Lateral shift only
D Protrusion
E No movement

7 A patient with TMD describes earache with no ear disease on examination.
Which nerve provides the main sensory supply to the TMJ and explains the referral?
A Facial nerve
B Great auricular nerve

C Chorda tympani
D Masseteric nerve alone
E Auriculotemporal nerve

8 A patient with disc displacement without reduction has pain on loading the joint.
Which tissue is the main source of the pain?
A Retrodiscal (bilaminar) tissue
B The central part of the disc
C Fibrocartilage on the articular eminence
D The sphenomandibular ligament
E The condylar cartilage

9 You are reviewing TMJ anatomy before examining a patient with a dislocation history.
Which ligament is the true stabiliser of the joint, limiting posterior and inferior movement of the condyle?
A Sphenomandibular ligament
B Stylomandibular ligament
C Pterygomandibular raphe
D Lateral (temporomandibular) ligament
E Stylohyoid ligament

10 A dental student asks why the TMJ heals and remodels differently from other synovial joints.
What covers the articular surfaces of the TMJ?
A Hyaline cartilage
B Elastic cartilage
C Fibrocartilage
D Synovial membrane
E Periosteum only

11 A patient gags strongly during upper impressions.
Which nerves form the afferent and efferent limbs of the gag reflex?
A Trigeminal afferent, facial efferent
B Glossopharyngeal afferent, vagus efferent
C Vagus afferent, glossopharyngeal efferent
D Hypoglossal afferent, vagus efferent
E Facial afferent, hypoglossal efferent

12 A new lower complete denture lifts every time the patient protrudes the tongue. The anterior lingual flange is over-extended.
Which muscle is displacing it, and what is its nerve supply?
A Mylohyoid; nerve to mylohyoid
B Genioglossus; hypoglossal nerve
C Geniohyoid; C1 via the hypoglossal nerve
D Palatoglossus; vagus
E Buccinator; facial nerve

13 A lower denture rises when the patient swallows. The lingual flange in the molar region extends straight down below the mylohyoid ridge.
Which muscle is responsible?
A Mylohyoid
B Genioglossus
C Masseter
D Buccinator
E Superior constrictor

14 A patient reports a sore spot at the distobuccal corner of a lower denture whenever they clench.
Which muscle is pressing on the flange?
A Buccinator
B Temporalis
C Mylohyoid
D Medial pterygoid
E Masseter

15 An upper denture is dislodged when the patient opens wide or moves the jaw sideways. The distobuccal flange is very thick.
Which structure is interfering with it?
A Zygomatic buttress
B Hamular notch
C Masseter
D Coronoid process
E Maxillary tuberosity

16 You are marking the post-dam on a master cast.
Which landmark is a useful guide to its position?
A Incisive papilla
B Palatal rugae
C Fovea palatini, near the vibrating line
D Midline raphe
E Zygomatic buttress

17 On the master cast of an edentulous maxilla, you identify the hamular (pterygomaxillary) notches.
What do they mark?
A The posterior limit of the tuberosity and the lateral extent of the post-dam
B The anterior limit of the soft palate
C The parotid duct opening
D The position of the greater palatine foramen
E The buccal frenum

18 A 76-year-old with a severely resorbed mandible has tingling in the lower lip when chewing with a new denture.
What is the most likely cause?
A Pressure on the lingual nerve
B Pressure on the mental nerve where the foramen now lies near the ridge crest

C Pressure on the buccal nerve
D Pressure on the nerve to mylohyoid
E An allergy to the denture base

19 A patient has a burning pain behind the upper incisors under a new upper denture. The mucosa there is slightly red.
Which area needs relief?
A Rugae
B Fovea palatini
C Midline raphe posteriorly
D Zygomatic buttress
E Incisive papilla (nasopalatine nerve)

20 A patient reports swelling of the cheek on eating, with an over-extended upper buccal flange opposite the upper second molar.
Which structure could be compressed?
A Submandibular duct
B Sublingual ducts
C Parotid duct
D Buccal fat pad
E Facial artery

21 A 45-year-old has an abscess on a lower second molar, with swelling below the lower border of the mandible.
Which space is involved?
A Sublingual space
B Buccal space
C Canine space
D Submandibular space
E Submental space

22 A lower first premolar abscess has produced a swelling in the floor of the mouth, above the mylohyoid.
Which space is involved?
A Sublingual space
B Submandibular space
C Submasseteric space
D Pterygomandibular space
E Parapharyngeal space

23 A 50-year-old presents with bilateral swelling below the mandible and under the chin, a raised floor of the mouth, drooling and a muffled voice.
What is the most appropriate action?
A Antibiotics and review in 48 hours
B Incise and drain intraorally in the practice
C Extract the cause and send the patient home

D Warm saline mouthwashes
E Emergency referral for airway management

24 A patient has an upper canine abscess with swelling near the inner corner of the eye.
What is the main danger of this spread?
A Ludwig's angina
B Cavernous sinus thrombosis via the angular and ophthalmic veins
C Mediastinitis
D Parotitis
E Trismus

25 You explain to a colleague why infection in the infratemporal region is dangerous.
How does the pterygoid venous plexus communicate with the cavernous sinus?
A Through the facial artery
B Through the parotid gland
C Through emissary veins
D Through the external jugular vein
E There is no communication

26 An inferior alveolar nerve block is given.
Into which space is the solution deposited?
A Submandibular space
B Buccal space
C Sublingual space
D Pterygomandibular space
E Canine space

27 After an infected lower third molar, a patient has marked trismus and swelling over the angle of the mandible.
Which space is most likely involved?
A Submasseteric space
B Canine space
C Sublingual space
D Submental space
E Space of the body of the mandible

28 After a successful inferior alveolar block, the buccal gingiva of 46 is still sensitive during a crown preparation.
Which nerve needs anaesthetising?
A Lingual nerve
B Nerve to mylohyoid
C Mental nerve
D Auriculotemporal nerve
E Long buccal nerve, the sensory branch of the anterior division

29 You are describing the nerve to mylohyoid to a student.
Which muscles does it supply?

A Mylohyoid and posterior belly of digastric
B Mylohyoid and anterior belly of digastric
C Geniohyoid and mylohyoid
D Genioglossus and mylohyoid
E Mylohyoid alone

30 A patient has a lingual nerve injury after surgery, distal to where the chorda tympani joins it.
What does the chorda tympani carry within the lingual nerve?
A Taste from the anterior two-thirds of the tongue
B Secretomotor fibres to the submandibular and sublingual glands
C Motor fibres to the tongue
D Both A and B
E All of the above

31 After parotid surgery, a patient sweats and flushes over the cheek when eating.
Which nerve is involved?
A Great auricular nerve
B Facial nerve
C Auriculotemporal nerve
D Lingual nerve
E Buccal nerve

32 A patient with a skull-base injury is noted to have damaged the artery passing through the foramen spinosum.
Which branch of the maxillary artery is this?
A Middle meningeal artery
B Inferior alveolar artery
C Sphenopalatine artery
D Infraorbital artery
E Deep temporal artery

33 During palatal surgery, brisk bleeding occurs from the greater palatine foramen.
From which vessel does the greater palatine artery arise?
A Facial artery
B Lingual artery
C First part of the maxillary artery
D Descending palatine artery (third part of the maxillary artery)
E Ascending pharyngeal artery

34 You are explaining the blood supply to a hypertrophied masseter.
Which part of the maxillary artery gives the masseteric artery?
A First (mandibular) part
B Second (pterygoid) part
C Third (pterygopalatine) part
D The facial artery directly
E The superficial temporal artery alone

35 Which statement about the nerve to medial pterygoid is correct?
A It arises from the anterior division of V3
B It arises from the posterior division of V3
C It is a branch of the facial nerve
D It is a branch of the buccal nerve
E It arises from the main trunk of V3, before it divides

36 A patient with a fractured mandibular angle has displacement resisted on both surfaces of the ramus by two muscles.
Which pair forms the sling around the angle of the mandible?
A Masseter and medial pterygoid
B Temporalis and lateral pterygoid
C Masseter and temporalis
D Medial and lateral pterygoids
E Mylohyoid and digastric

37 A coronoid fracture fragment is pulled upwards towards the temporal fossa.
Which muscle inserts into the coronoid process?
A Masseter
B Lateral pterygoid
C Temporalis
D Medial pterygoid
E Buccinator

38 A patient protrudes during a jaw-relation record and the operator guides the mandible back.
Which muscle is the main retractor of the mandible?
A Lateral pterygoid
B Masseter (superficial)
C Medial pterygoid
D Anterior belly of digastric
E Temporalis (posterior, horizontal fibres)

39 You ask a patient to slide the mandible to the left during an excursive check.
Which muscles on which side mainly produce this movement?
A Left lateral and medial pterygoids
B Left masseter
C Right temporalis
D Right lateral and medial pterygoids
E Left digastric

40 A 28-year-old has a reciprocal click in the right TMJ.
Which muscle attaches to the articular disc and is most associated with disc displacement?
A Medial pterygoid
B Lateral pterygoid (superior head)
C Masseter (deep)
D Temporalis
E Posterior belly of digastric

41 A patient has weakness of the palatoglossal arch after a brainstem lesion, while tongue protrusion is normal.
Which nerve supplies palatoglossus?
A Hypoglossal nerve
B Glossopharyngeal nerve alone
C Facial nerve
D Mandibular nerve
E Vagus via the pharyngeal plexus

42 You are explaining soft-palate anatomy before a post-dam is cut.
Which soft palate muscle is not supplied by the pharyngeal plexus?
A Levator veli palatini
B Musculus uvulae
C Tensor veli palatini (nerve to medial pterygoid, V3)
D Palatopharyngeus
E Palatoglossus

43 After a Bell's palsy, a lower denture wearer finds food collecting in the buccal sulcus and the denture less stable.
Which weakened muscle explains this?
A Buccinator
B Masseter
C Mylohyoid
D Medial pterygoid
E Temporalis

44 You are recording the neutral zone for a patient with an unstable lower denture.
Which structures mainly determine lip support and the labial boundary of the neutral zone?
A Mylohyoid and genioglossus
B Masseter and temporalis
C Buccinator alone
D Orbicularis oris and the modiolus
E Lateral pterygoid

45 A student asks which nerve supplies geniohyoid.
What is the correct answer?
A Nerve to mylohyoid
B C1 fibres travelling with the hypoglossal nerve
C Facial nerve
D Lingual nerve
E Glossopharyngeal nerve

46 A patient with facial nerve palsy is examined for swallowing problems.
Which hyoid muscle is supplied by the facial nerve?
A Mylohyoid
B Anterior belly of digastric
C Geniohyoid

D Thyrohyoid
E Posterior belly of digastric (and stylohyoid)

47 You are checking the posterior extension of a new lower complete denture.
How much of the retromolar pad should the base cover?
A None of it
B A narrow strip at its front edge
C At least the anterior half to two-thirds
D All of it, onto the pterygomandibular raphe
E It depends only on the OVD

48 A 72-year-old has a moderately resorbed mandible.
Which area provides the primary support for the lower complete denture?
A The buccal shelf
B The genial tubercles
C The lingual sulcus
D The mylohyoid ridge
E The labial vestibule

49 A new upper complete denture is planned for a patient with a large torus palatinus.
Which areas normally need relief on the fitting surface?
A Incisive papilla
B Midline raphe
C Torus palatinus
D Both A and B only
E All of the above

50 A new lower denture lifts at the back each time the patient opens wide. The base extends over the whole retromolar pad.
Which structure is tightening against the over-extension?
A Mylohyoid
B Pterygomandibular raphe
C Masseter
D Buccinator
E Genioglossus

51 A 79-year-old has a severely resorbed, knife-edged mandibular ridge.
Which features may need relief in the fitting surface?
A Genial tubercles
B The mental foramen, now near the crest
C The knife-edged crest
D Both A and B only
E All of the above

52 A patient on anticholinergic medication complains of a dry mouth at rest.
Which gland produces most of the unstimulated saliva?
A Parotid

B Sublingual
C Submandibular (about 65–70%)
D Minor palatal glands
E Von Ebner's glands

53 A patient has had radiotherapy to both parotid glands.
Which saliva is reduced most?
A Stimulated saliva
B Unstimulated saliva only
C Mucous saliva only
D Minor gland saliva only
E None, because the submandibular glands compensate fully

54 You see a small salivary stone at the duct opening beside the lingual frenum.
Which duct opens here?
A Parotid duct
B Sublingual ducts of Rivinus
C Duct of Bartholin only
D Submandibular (Wharton's) duct, at the sublingual papilla
E Nasolacrimal duct

55 You explain parotid secretomotor supply to a student.
What is the correct parasympathetic pathway?
A Facial nerve, chorda tympani, lingual nerve
B Glossopharyngeal nerve, lesser petrosal nerve, otic ganglion, auriculotemporal nerve
C Vagus nerve, pharyngeal plexus
D Superior cervical ganglion, external carotid plexus
E Facial nerve, greater petrosal nerve, pterygopalatine ganglion

56 A patient has reduced submandibular flow after a middle-ear operation.
Which parasympathetic pathway supplies the submandibular gland?
A Glossopharyngeal nerve, otic ganglion
B Vagus nerve
C Facial nerve, chorda tympani, lingual nerve, submandibular ganglion
D Lesser petrosal nerve, auriculotemporal nerve
E Hypoglossal nerve

57 A patient has a ranula-like swelling in the floor of the mouth.
Which statement about the sublingual gland is correct?
A It is purely serous
B It drains through a single duct opposite the upper second molar
C It is supplied by the otic ganglion
D It produces most of the stimulated saliva
E It is mainly mucous and drains through multiple ducts of Rivinus on the sublingual fold

58 A patient has TMJ inflammation and the vascular supply is discussed.
Which arteries supply the TMJ?

A Superficial temporal and maxillary arteries
B Facial and lingual arteries
C Inferior alveolar artery only
D Occipital artery
E Ascending pharyngeal artery only

59 A student asks how to classify the TMJ.
Which description is correct?
A Fibrous joint
B Primary cartilaginous joint
C Synovial hinge joint only
D Synovial ginglymoarthrodial (hinge and gliding) joint
E Secondary cartilaginous joint

60 During an inferior alveolar block, the solution may be deflected by a ligament running from the spine of the sphenoid to the lingula.
Which ligament is this?
A Lateral ligament
B Stylomandibular ligament
C Sphenomandibular ligament
D Pterygomandibular raphe
E Stylohyoid ligament

61 A new lower denture is retentive at rest but is displaced whenever the patient speaks or smiles.
What should you do first?
A Recommend a denture adhesive
B Assess the border extension and polished-surface contours against muscle action
C Adjust the occlusion
D Remake the denture at once
E Increase the OVD

62 On a lower complete denture, the lingual flange in the molar region ends 4 mm short of the functional depth of the floor of the mouth.
What is the most likely consequence?
A Gagging
B Ulceration of the floor of the mouth
C Cheek biting
D A speech defect
E Loss of the peripheral seal and reduced retention

63 A 70-year-old complete denture wearer reports 6 weeks of numbness of the left lower lip. The denture fits well and does not press on the mental foramen.
What is the most appropriate action?
A Urgent referral to investigate the cause, including malignancy
B Relieve the denture and review in 6 months
C Fit a soft liner

D Reassure the patient
E Prescribe an antifungal

64 A patient with a lower molar infection has a temperature of 38.6°C, trismus of 15 mm and difficulty swallowing.
What is the most appropriate action?
A Prescribe amoxicillin and review in a week
B Extract the tooth under LA in the practice
C Advise hot mouthwashes
D Refer urgently to the maxillofacial team
E Prescribe metronidazole only

65 A 68-year-old with an atrophic mandible is assessed for an implant-retained overdenture.
Which structure most often limits safe implant placement in the premolar region?
A Lingual nerve
B The mental foramen and the anterior loop of the inferior alveolar nerve
C Facial artery
D Submandibular gland
E Hyoid bone

66 You are deciding the posterior extent of a new upper complete denture.
What should be assessed before planning the posterior palatal seal?
A Rugae pattern
B Tongue size
C Soft-palate form and movement at the vibrating line
D Lip length
E Shade of the mucosa

67 An edentulous patient has large undercuts on both maxillary tuberosities.
What is the usual approach?
A Choose a path or block out one side, reducing surgically only if both still prevent insertion
B Ignore them
C Fill both undercuts with acrylic
D Use a soft liner alone
E Extract any remaining molars

68 A partially dentate patient needs a lower Kennedy Class I RPD. Large bilateral mandibular tori leave no room for a lingual bar.
What is appropriate?
A A lingual bar over the tori
B A lingual plate over the tori
C No major connector
D A labial bar, or surgical removal of the tori if a sound design needs it
E A sublingual bar

69 A patient with an inferior alveolar block infection develops severe trismus and dysphagia.
Infection in the pterygomandibular space can spread directly to which space?

A Canine space
B Submental space
C Sublingual space
D Buccal space
E Lateral pharyngeal space

70 After a successful inferior alveolar block, the lower lip is numb but the first molar pulp is still sensitive.
Supplementary anaesthesia of which nerve is most likely to help?
A Buccal nerve
B Nerve to mylohyoid
C Mental nerve
D Lingual nerve
E Auriculotemporal nerve

Block B — Physiology and pathology (Q71–130)

> *Notebook Unit 2 · LO 1.2–1.4, 1.8, 1.10–1.11, 2a.7–2a.8 (with 2b.3–2b.5, 2c.1) · Allow 72 minutes*

71 A 58-year-old taking amitriptyline and oxybutynin complains that her dentures are loose. Her unstimulated whole saliva flow is 0.08 mL/min.
How should this result be interpreted?
A Normal flow
B Borderline; repeat in a year
C Hyposalivation
D Hypersalivation
E It cannot be interpreted without a stimulated sample

72 A student asks how saliva retains a complete denture.
Which factors contribute?
A Adhesion and cohesion
B Interfacial surface tension
C Atmospheric pressure through the peripheral seal
D Both A and B only
E All of the above

73 Two edentulous patients have similar ridges and well-made dentures. One has thin, watery saliva; the other has thick, ropy saliva.
Which saliva usually gives better retention?
A Thin, watery saliva
B Thick, ropy saliva
C Neither affects retention
D Thick saliva in the mandible only
E Only saliva mixed with adhesive

74 A dentate 64-year-old with Sjögren's syndrome asks what to use for her dry mouth. She has several root caries lesions.
What is most appropriate?
A A citric-acid-based saliva substitute
B Lemon and glycerine swabs
C Sugared lozenges to stimulate flow
D A neutral mucin- or carboxymethylcellulose-based substitute, with high-fluoride toothpaste
E An alcohol-based mouthwash

75 A patient with Sjögren's syndrome also has poorly controlled asthma.
Which treatment for her dry mouth is unsuitable?
A A saliva substitute
B Pilocarpine
C Sugar-free gum
D 5000 ppm fluoride toothpaste
E Frequent sips of water

76 A patient sips an acidic drink and stimulated saliva begins to neutralise it.
What is the main buffer in stimulated saliva?
A Bicarbonate
B Phosphate
C Proteins
D Urea
E Ammonia

77 You explain to a patient why saliva protects against infection.
Which are antimicrobial components of saliva?
A Lysozyme
B Lactoferrin
C Secretory IgA and histatins
D Both A and B only
E All of the above

78 A 70-year-old with new complete dentures says they cannot chew steak as they did with natural teeth.
About what proportion of dentate bite force and chewing efficiency do complete denture wearers achieve?
A The same
B About half
C About one-fifth to one-sixth
D About one-third
E About one-twentieth

79 You are explaining why complete denture wearers find fine biting control harder than dentate patients.
On which receptors does a complete denture wearer depend for feedback?
A Periodontal ligament mechanoreceptors

B Mucosal and TMJ receptors
C Pulpal receptors
D Osseoperception
E Muscle spindles alone

80 A patient with an implant crown asks whether the implant will "feel" like a tooth.
Compared with a natural tooth, feedback from an implant (osseoperception) is:
A Identical
B More sensitive
C Absent
D Less sensitive
E Thermal only

81 A healthy 72-year-old has lost all four first and second molars but has sound premolars and anterior teeth in both arches. She is comfortable and chews well.
What is the most appropriate advice?
A A removable partial denture to replace the molars
B Implants to replace the molars
C Cantilever bridges from the premolars
D Extraction of the remaining teeth for complete dentures
E No replacement; a shortened dental arch usually gives adequate function

82 A public-health lecture quotes the WHO goal for tooth retention.
How many functional natural teeth should be retained throughout life?
A 10
B 20
C 28
D 24
E 32

83 An 84-year-old with Parkinson's disease has worn the same well-liked complete dentures for 12 years. They are worn and loose.
Which approach best suits her ability to adapt?
A New dentures of a completely new design
B Implant-retained overdentures
C Immediate dentures
D Copy (replica) dentures
E No dentures

84 A patient edentulous for 30 years appears Class III, with the upper denture teeth set inside the lower ridge.
What explains this appearance?
A The maxilla resorbs upwards and inwards, and the mandible downwards and outwards
B The maxilla keeps growing
C The mandible resorbs inwards (centripetally)
D TMJ remodelling only
E Wear of the denture teeth

85 You are explaining long-term ridge resorption to a patient after extractions.
In the long term, how much more height does the anterior mandible lose than the anterior maxilla?
A The same
B About twice as much
C About four times as much
D About ten times as much
E Less

86 A patient asks when her ridges will shrink most after extractions.
When is resorption fastest?
A After 5 years
B In the first 6–12 months
C After 10 years
D At a constant rate for life
E It does not resorb

87 A 66-year-old with two sound lower canines wants to slow the loss of her lower ridge.
Which measures help preserve the ridge?
A Retaining roots as overdenture abutments
B Implants
C Removing dentures at night
D Both A and B only
E All of the above

88 A patient has worn the same ill-fitting dentures day and night for 15 years.
Which habit is most likely to have accelerated ridge resorption?
A Continuous 24-hour wear of ill-fitting dentures
B Removing the dentures at night
C Regular relines
D Well-fitting dentures
E Implants

89 A 56-year-old lost 36 ten years ago and wants it replaced. 26 has over-erupted by 3 mm.
What is the main consequence for planning?
A None
B Increased OVD
C Posterior crossbite
D Reduced interocclusal space, so the over-erupted tooth must be managed first
E TMJ ankylosis

90 An 80-year-old drinks sweet tea through the day and has several root caries lesions.
At about what pH does root dentine begin to demineralise?
A 4.5
B 5.5
C 6.2–6.7
D 7.4
E 3.0

91 Patient X has sugar in ten cups of tea a day. Patient Y has the same total sugar in one dessert.
What is the most important caries advice for patient X?
A Only total sugar intake matters
B Brush after every cup
C Use a mouthwash after every cup
D Reduce the frequency of sugar intake
E Rinse with water after each cup

92 A partially dentate patient with recession has multiple root caries lesions.
Which organism is particularly associated with root caries?
A Actinomyces
B Lactobacillus only
C Candida
D Streptococcus sanguinis
E Porphyromonas gingivalis

93 A histology slide shows an early enamel caries lesion.
Which zone shows the greatest demineralisation?
A Surface zone
B Body of the lesion
C Dark zone
D Translucent zone
E Sound enamel

94 A deep carious lesion on 36 is close to the pulp. The tooth is vital with no symptoms of irreversible pulpitis.
What is the most appropriate caries removal approach?
A Remove all softened dentine to hard dentine over the pulp
B Extract the tooth
C Root canal treatment
D Leave infected dentine at the periphery
E Remove infected dentine, leave affected dentine near the pulp, and seal

95 A 45-year-old has high caries risk and active root caries.
Which toothpaste does Delivering Better Oral Health recommend?
A 1000 ppm fluoride
B 1450 ppm fluoride
C 5000 ppm fluoride
D 22,600 ppm fluoride
E No fluoride

96 An adult with partial dentures is at high caries risk.
Which professional fluoride application is recommended?
A Fluoride varnish 22,600 ppm at least twice a year
B 5000 ppm varnish monthly
C 1450 ppm varnish once a year

D 2800 ppm gel weekly
E 226 ppm rinse daily in the surgery

97 A 76-year-old with a Co-Cr RPD, poor dexterity and gingival recession is reviewed.
Which factors increase her root caries risk?
A Recession
B Hyposalivation
C Partial dentures
D All of the above
E None of the above

98 A 21-year-old has palatal erosion of the upper anterior teeth, bilateral parotid enlargement and calluses on her knuckles.
What is the most likely cause?
A Dietary erosion from fruit juice
B Bulimia nervosa
C Toothbrush abrasion
D Bruxism
E Abfraction

99 A 38-year-old has smooth, silky enamel on the upper incisors, cupped cusps, and amalgam restorations standing proud of the surrounding tooth.
What is the most likely diagnosis?
A Attrition
B Abrasion
C Abfraction
D Caries
E Erosion

100 A 50-year-old has flat, shiny wear facets that match on opposing teeth, fractured cusps and hypertrophic masseters.
What is the most likely diagnosis?
A Erosion
B Abrasion
C Attrition
D Abfraction
E Caries

101 A 45-year-old has sharp-edged, V-shaped cervical notches on the upper canines and premolars, worse on the left. She is right-handed and brushes hard.
What is the most likely diagnosis?
A Erosion
B Abrasion
C Attrition
D Caries
E Perimylolysis

102 A bruxist has wedge-shaped cervical lesions with sharp internal angles, some extending below the gingival margin, on teeth with heavy lateral contacts.
Which process is proposed to explain these lesions?
A Erosion
B Abrasion
C Root caries
D Abfraction (flexural stress at the cervical margin)
E Attrition

103 A 55-year-old has slow, localised wear of the upper anterior teeth over many years, yet the posterior teeth still meet and there is no space to restore the worn teeth.
Why is there no interocclusal space?
A Compensatory alveolar growth has kept the worn teeth in contact
B The OVD has been lost
C TMD has developed
D The bruxism has stopped
E The posterior teeth have intruded

104 You screen a new patient's tooth wear with the BEWE.
How is the BEWE scored?
A 0–4 for each tooth surface
B As a DMFT count
C 0–5 for each tooth
D Using the Smith and Knight index
E 0–3 for the worst surface in each sextant, with the sum guiding management

105 You need to record wear on each surface for a research audit.
Which index scores 0–4 on every tooth surface?
A BEWE
B BPE
C Smith and Knight Tooth Wear Index
D ICDAS
E DMFS

106 You suspect an eating disorder in a 19-year-old with palatal erosion, who denies vomiting.
What is the most appropriate approach?
A Confront her with the diagnosis
B Approach with empathy, liaise with her GMP with consent, and focus on prevention and monitoring before restoration
C Place crowns immediately
D Contact the police
E Take no action

107 An upper denture wearer has nodular, papillary tissue in the centre of the palate under the denture.
Which Newton type of denture stomatitis is this?
A Type III (papillary hyperplasia)
B Type I (pinpoint hyperaemia)

C Type II (diffuse erythema)
D Type IV
E It is not denture stomatitis

108 A 74-year-old has diffuse erythema of the palate under an ill-fitting upper acrylic denture that she wears day and night. She has no symptoms.
What is the most appropriate first management?
A Oral fluconazole for 14 days
B Biopsy of the palate
C A new denture immediately
D Denture hygiene, removal at night, soaking, and correcting the fit
E Topical steroid

109 A patient on warfarin has denture stomatitis that has not settled with hygiene and night removal.
Which antifungal is most appropriate?
A Miconazole oral gel
B Fluconazole
C Itraconazole
D Miconazole cream
E Nystatin suspension

110 A 70-year-old denture wearer has painful fissures at both corners of the mouth.
Which investigations are indicated?
A Biopsy of each corner
B A swab and no other tests
C Full blood count, haematinics and blood glucose
D A radiograph of the mandible
E No investigations

111 A patient with angular cheilitis grows both Candida and Staphylococcus aureus from swabs. He is not on any interacting drugs.
Which topical treatment covers both organisms?
A Miconazole cream
B Nystatin cream alone
C Hydrocortisone cream alone
D Aciclovir cream
E Chlorhexidine gel

112 A leaf of firm tissue lies in the lower labial sulcus under an over-extended flange of a 10-year-old denture.
What is the most appropriate management?
A Excise it immediately without adjusting the denture
B Ignore it
C Prescribe an antifungal
D Trim the flange, use a tissue conditioner, and excise with histology if it does not resolve
E Prescribe a topical steroid

113 An ulcer under a lower flange is still present 3 weeks after the flange was adjusted.
What is the most appropriate next step?
A Adjust the flange again and review in 3 months
B Urgent referral on the suspected cancer pathway
C Prescribe chlorhexidine
D Fit a soft liner
E Reassure the patient

114 An edentulous upper anterior ridge is mobile and displaceable. The patient has natural lower anterior teeth.
Which statement is correct?
A It should always be excised
B It is bone hypertrophy
C It is caused by Candida
D It should be compressed in a putty impression
E It is fibrous replacement of bone, best recorded with a mucostatic or selective-pressure impression

115 A 62-year-old smoker has a well-demarcated red patch on the floor of the mouth that does not wipe off.
Which lesion carries the highest risk of malignant transformation?
A Homogeneous leukoplakia
B Frictional keratosis
C Erythroplakia
D Linea alba
E Geographic tongue

116 You are performing a soft-tissue examination on a high-risk patient.
Which sites carry the highest risk of oral squamous cell carcinoma?
A Hard palate and gingiva
B Dorsum of the tongue and lips
C Buccal mucosa and palate
D Lateral border of the tongue and floor of the mouth
E Retromolar pad only

117 A new patient smokes 20 a day, drinks 30 units a week and chews areca nut.
Which of these are risk factors for oral cancer?
A Tobacco
B Alcohol
C Areca nut
D Both A and B only
E All of the above

118 At a denture review you find a red and white patch on the lateral tongue that the patient had not noticed.
What does NICE NG12 advise?
A Urgent referral on the suspected cancer pathway
B Review in 6 months

C Antifungal treatment and review
D Routine referral
E Photograph only

119 A 79-year-old's teeth respond weakly to cold testing, and root canal access on a molar proves difficult.
What explains both findings?
A Pulp necrosis
B Secondary dentine deposition, giving smaller pulp chambers and lower sensibility responses
C Allergy
D Root resorption
E Enamel hypoplasia

120 A 52-year-old has glossitis, recurrent ulcers, angular cheilitis and a burning tongue.
What is the most likely underlying cause?
A Lichen planus
B Herpes simplex
C Haematinic deficiency (iron, B12 or folate)
D Smoking
E Toothbrush trauma

121 A 58-year-old with poorly controlled type 2 diabetes has periodontitis.
How are the two conditions related?
A They are unrelated
B Diabetes affects only caries
C Periodontitis improves diabetes, but not the reverse
D Diabetes affects only implants
E The relationship is bidirectional

122 A 66-year-old with osteoporosis asks about implants. She has not been prescribed drugs for it.
Which statement is correct?
A Osteoporosis is generally not a contraindication; antiresorptive drug treatment is the bigger issue
B Osteoporosis is an absolute contraindication
C Implants always fail in osteoporosis
D Osteoporosis prevents denture wear
E A bone density scan is required before any denture

123 A patient asks how to clean her cobalt–chromium partial denture.
Which advice is correct?
A Soak it overnight in household bleach
B Do not soak it in hypochlorite, because it damages Co-Cr; use a suitable non-abrasive cleaner
C Boil it weekly
D Scrub it with abrasive toothpaste
E Never remove it

124 An 86-year-old care-home resident with poor dexterity needs replacement of missing teeth. Attendance and cleaning are unreliable.

Which approach is most appropriate?
A A long-span fixed bridge
B Multiple implants
C Precision-attachment RPD
D A simple, hygienic, maintainable design
E Full-mouth crowns

125 A 50-year-old with generalised periodontitis wants a bridge.
What is the usual sequence of periodontal care before definitive prosthodontics?
A Surgery, then hygiene instruction
B Bridge first, then periodontal care
C Hygiene and risk-factor control, subgingival instrumentation, surgery where needed, then supportive care
D Antibiotics alone
E Extraction of all mobile teeth first

126 A 60-year-old has lost 47 only. There is no drifting or over-eruption and he has no complaints.
What is the most appropriate advice?
A No replacement is a valid option
B An implant is essential
C A bridge is essential
D An RPD is essential
E Extract 17

127 A 62-year-old has a dry mouth from her antidepressant and has new root caries.
Which measures are appropriate?
A Ask her GMP whether the medication can be reviewed
B Sugar-free salivary stimulants
C High-fluoride toothpaste
D Both A and B only
E All of the above

128 After head and neck radiotherapy, a patient's saliva is scant, thick and ropy, and his upper denture keeps dropping.
What explains the poor retention?
A An over-extended flange
B Thick, scant saliva gives a poor adhesive and cohesive film
C Excess OVD
D Candida
E A thick palate

129 You review a patient's medication list for causes of dry mouth.
Which drug group is least associated with dry mouth?
A Antidepressants
B Antihistamines
C Antimuscarinics

D Penicillin antibiotics
E Diuretics

130 A patient with frequent sugar intake and low saliva develops rapidly progressive caries.
Which concept explains the shift towards acid-producing species?
A The specific plaque hypothesis alone
B The non-specific plaque hypothesis
C The ecological plaque hypothesis
D The Stephan curve
E The focal infection theory

Block C — Dental materials science (Q131–220)

> *Notebook Unit 3 · LO 1.9 (with 2c.6, 2c.13–2c.14) · Allow 108 minutes*

131 A technician designs a cobalt–chromium clasp to engage a 0.25 mm undercut, whereas a cast gold clasp on the same tooth would engage 0.5 mm.
Which property of Co-Cr explains the difference?
A Lower yield strength
B Greater ductility
C Lower density alone
D Higher elastic modulus (about 220 GPa, roughly twice that of Type IV gold)
E Higher thermal conductivity

132 A clasp arm must flex over the height of contour every time the denture is inserted and still return to its original shape.
Which property must the strain not exceed?
A Proportional (elastic) limit
B Ultimate tensile strength
C Hardness
D Fracture toughness
E Coefficient of thermal expansion

133 A patient repeatedly bent her Co-Cr clasp to tighten it, and it snapped at the shoulder.
Which property of Co-Cr made this likely?
A High elastic modulus
B High hardness
C Low density
D High yield strength
E Low ductility (low percentage elongation)

134 A patient with Parkinson's disease drops her upper denture in the sink every few weeks.
Which property should the new denture base be chosen for?
A Hardness
B Toughness (a high-impact acrylic)
C Resilience

D Elastic modulus
E Low thermal expansion

135 An upper complete denture opposing natural lower teeth cracks down the midline after 3 years, without ever being dropped.
What is the failure mechanism?
A Impact
B Creep
C Fatigue
D Corrosion
E Hydrolysis

136 A sports mouthguard and a soft denture liner both need to absorb energy without permanent deformation.
Which property describes this?
A Toughness
B Resilience
C Hardness
D Ductility
E Creep

137 A clinician chooses zirconia rather than a glass ceramic for a posterior bridge in a heavy biter.
Which property gives zirconia its advantage in resisting crack growth?
A Elastic modulus
B Translucency
C Hardness
D Thermal conductivity
E Fracture toughness (KIc)

138 An elastomeric impression is removed slowly over several seconds and the casts do not fit.
Why should it be removed with a snap?
A Viscoelastic materials recover better when deformed quickly
B It reduces the setting time
C It prevents hydrogen release
D It increases the material's hardness
E It stops imbibition

139 A wax pattern for a crown is left on the bench overnight before investing. The casting does not seat.
What has happened to the wax?
A It has absorbed water
B It has hardened
C It has polymerised
D Stress relaxation has distorted it
E It has corroded

140 A researcher measures the hardness of a thin glaze layer on porcelain.
Which hardness test suits brittle materials and thin coatings?

A Brinell
B Rockwell
C Knoop
D Shore A
E Charpy

141 A large acrylic restoration repeatedly leaks at its margins with hot and cold drinks.
How does the thermal expansion of acrylic compare with tooth?
A Acrylic (about 80 ppm/°C) expands far more than tooth (about 11 ppm/°C)
B They are identical
C Acrylic expands less than tooth
D Acrylic does not expand
E Tooth expands about 80 ppm/°C

142 A technician explains why the porcelain on a metal–ceramic crown resists cracking on cooling.
How should the coefficient of thermal expansion of the porcelain relate to the metal?
A Much higher than the metal
B Exactly equal
C Unrelated
D Slightly lower than the metal, so the porcelain cools into compression
E Zero

143 A clinician wants a post whose stiffness is close to dentine, to spread stress more evenly in a root-filled premolar.
Which post has a modulus closest to dentine (15–20 GPa)?
A Zirconia
B Titanium
C Glass-fibre
D Cobalt–chromium
E Stainless steel

144 A sales representative says a material with a high elastic modulus must be the best choice for a posterior crown.
What does a high elastic modulus necessarily mean?
A The material is tough
B The material resists wear
C The material bonds strongly
D The material has high flexural strength
E None of these; modulus describes stiffness only

145 You need a mucostatic impression of a flabby ridge and do not need to record undercuts.
Which material records tissue at rest but cannot record undercuts?
A Alginate
B Impression plaster
C Addition silicone putty
D Polyether
E Agar

146 A primary impression in compound is distorted, with soft surface and hard centre.
Why must compound be softened thoroughly?
A It sets by chemical reaction
B It is hydrophilic
C It has low thermal conductivity, so heat penetrates slowly
D It contains eugenol
E It shrinks on heating

147 A patient complains of a burning feeling after a wash impression with zinc oxide–eugenol paste.
Which is a disadvantage of ZOE impression paste?
A Eugenol irritation, and it cannot record undercuts
B It is too elastic
C It releases hydrogen
D It needs a water bath
E It is hydrophobic

148 An alginate impression for study casts is left in a sealed bag over the weekend before pouring.
What is the main problem?
A It becomes too elastic
B It releases hydrogen
C It polymerises further
D It bonds to the tray
E Syneresis and imbibition distort it; alginate should be poured within about 10–15 minutes

149 An alginate impression tears in the interproximal areas.
Which design feature helps prevent tearing?
A A thinner layer
B A metal stock tray with no adhesive
C Pouring in plaster
D At least 3–4 mm bulk of material
E Removing it slowly

150 A condensation silicone impression was poured 2 days after it was taken. The crown made on it is tight.
Why?
A It absorbed water
B It shrank as ethanol evaporated
C Hydrogen was released
D It expanded on standing
E Latex inhibited its set

151 An addition silicone wash stays sticky around a preparation where the nurse handled the retraction cord with gloved fingers.
What is the most likely cause?
A Too much catalyst
B High room temperature
C Hydrogen release

D Surfactant in the material
E Sulphur from latex gloves inhibiting the set

152 Casts poured immediately from an addition silicone impression have bubbly, porous die surfaces.
What explains this?
A Syneresis
B Ethanol release
C Hydrogen release from the set material; wait about an hour or use a material with a hydrogen absorber
D Imbibition
E Latex contamination

153 A polyether impression is taken for a lower implant case with large embrasure spaces. The stone dies break on removal.
Which property of polyether caused the problem?
A Hydrophobicity
B High stiffness
C Low tear strength
D Hydrogen release
E Low dimensional stability

154 A clinician wants an elastomer that wets a moist preparation well without added surfactant.
Which material is inherently hydrophilic?
A Polyether
B Addition silicone
C Condensation silicone
D Impression compound
E Polysulphide

155 A patient has a flabby anterior maxilla that must be recorded undisplaced.
Which technique records the tissue at rest?
A Compound in a close-fitting tray
B Heavy-bodied silicone in a close-fitting tray
C Putty with no spacer
D A low-viscosity material in a spaced (or windowed) tray
E Finger pressure during setting

156 A special tray is made for an alginate secondary impression.
What spacer thickness is typical?
A About 3 mm
B No spacer
C 0.1 mm
D 10 mm
E 0.5 mm

157 Several casts from one practice are distorted. The practice uses thin plastic stock trays without adhesive for elastomers.

What is the main cause?
A The material brand
B The disinfectant
C The stone
D Tray flexure and lack of adhesive
E Room temperature

158 A one-stage (simultaneous) putty–wash impression shows putty at the preparation margin.
What is the recognised risk of this technique?
A Hydrogen release
B The putty displaces the wash from the margins
C Excessive wash thickness
D Imbibition
E Tray flexure

159 A two-stage putty–wash impression gives a tight crown.
What is the recognised risk of this technique?
A Putty displacing the wash
B Hydrogen release
C Too little working time
D Latex inhibition
E Distortion if the set putty is reseated without adequate relief

160 A nurse asks how to disinfect an alginate impression before sending it to the laboratory.
What is correct?
A Autoclave it
B Soak it overnight in hypochlorite
C Rinse it, then use a material-specific disinfectant for the time on the label, avoiding prolonged immersion
D Do nothing
E Dry it with hot air

161 A technician adds extra water to stone so it pours easily.
What is the effect on the cast?
A Stronger, with greater expansion
B Weaker, with less expansion and a slower set
C No change
D Faster set
E Harder surface

162 A technician spatulates stone for twice the recommended time.
What is the effect?
A Slower set and less expansion
B Weaker cast
C No change
D The stone will not set
E A faster set and more expansion

163 A cast poured from an impression with blood on it has a soft, chalky surface.
Why?
A Blood and saliva retard the setting of gypsum, so impressions should be rinsed
B Blood accelerates the set
C Blood increases expansion
D Blood reacts with the tray adhesive
E Blood has no effect

164 A technician wants a quicker-setting stone for a chairside cast.
Which additive accelerates gypsum setting?
A Borax
B Citrate
C Potassium sulphate
D Acetate
E Blood

165 A setting stone cast is left under water in a soaking bath and comes out oversized.
What explains this?
A Thermal expansion
B Syneresis
C Imbibition of the impression
D Hygroscopic expansion
E Polymerisation shrinkage

166 A crown on a 3 mm high, over-tapered molar preparation has come off twice, despite resin cement.
What is most appropriate?
A Use more cement
B Change to zinc phosphate
C Roughen the fitting surface only
D Improve resistance form (grooves or boxes, or crown lengthening) before a new crown
E Accept repeated recementation

167 Dies for a metal–ceramic bridge are being poured.
Which gypsum is appropriate?
A Plaster (Type II)
B Die stone (Type IV or V)
C Dental stone (Type III)
D Impression plaster (Type I)
E Investment

168 Base-metal castings consistently fit tightly on Type IV dies.
Which gypsum could compensate?
A Type I plaster
B Type II plaster
C Type V high-strength, high-expansion die stone
D Type III stone
E Orthodontic stone

169 A technician uses gypsum-bonded investment for a cobalt–chromium framework. The casting is rough and brittle.
Why is gypsum-bonded investment unsuitable?
A Calcium sulphate breaks down above about 700°C, releasing sulphur dioxide
B It expands too much
C It cannot be poured
D It sets too slowly
E It is too porous

170 A metal–ceramic crown framework is to be cast in a high-fusing alloy.
Which investment is appropriate?
A Gypsum-bonded
B Plaster
C Silicone
D Type IV stone
E Phosphate-bonded

171 A gold crown casting shrinks about 1.4% as it cools.
How is this shrinkage compensated?
A By grinding the die
B By using a thicker cement
C By setting, hygroscopic and thermal expansion of the investment
D By casting at a lower temperature
E It cannot be compensated

172 A wax pattern is finished at 5 p.m. and the technician plans to invest it the next morning.
What is the best advice?
A Invest promptly, because wax has high thermal expansion and releases stresses on standing
B Store it in the fridge overnight
C Leave it in hot water
D Re-wax the margins in the morning
E It makes no difference

173 A student asks how heat-cured PMMA polymerisation starts.
What is the initiator?
A Hydroquinone
B Camphorquinone
C Tertiary amine
D Benzoyl peroxide
E Ethylene glycol dimethacrylate

174 Liquid monomer in an old bottle has stayed liquid for two years.
Which component prevents premature polymerisation?
A Benzoyl peroxide
B Cross-linking agent
C Pigment

D Tertiary amine
E Hydroquinone

175 Acrylic denture teeth from one manufacturer resist crazing around the necks better than another brand.
Which component improves craze resistance?
A Hydroquinone
B A cross-linking agent (ethylene glycol dimethacrylate)
C Pigments
D Plasticiser
E Water

176 A technician mixes heat-cure acrylic and waits for it to pass through its stages.
At which stage should it be packed into the flask?
A Sandy
B Stringy (sticky)
C Rubbery
D Stiff
E Dough

177 A processed complete denture has bubbles confined to the thick palatal and tuberosity areas.
What is the most likely cause?
A Too little dough in the flask
B Too little packing pressure
C Heating too fast, so monomer boiled (gaseous porosity)
D Dough left uncovered on the bench
E Too much cross-linking agent

178 A processed denture shows irregular voids throughout, including at the surface.
What is the most likely cause?
A Heating too fast
B Monomer evaporation from uncovered dough
C Too much hydroquinone
D Too little dough or packing pressure (contraction porosity)
E A long curing cycle

179 A patient develops mucosal irritation under a chairside self-cure reline but not under her heat-cured denture.
What explains the difference?
A Self-cure acrylic contains no initiator
B Self-cure acrylic has much higher residual monomer (about 2–5% against 0.2–0.5%)
C Heat-cured acrylic is radiopaque
D Self-cure acrylic has no plasticiser
E Heat-cured acrylic contains hydroquinone

180 A patient sterilises her acrylic dentures by soaking them in boiling water. They no longer fit.
Why?

A Stress relief and warpage occur well below the glass transition temperature (about 105°C)
B The monomer evaporated
C The teeth expanded
D Water absorption shrank them
E The pigments degraded

181 An upper denture has fractured twice at a deep labial frenal notch. The patient has a full lower natural dentition.
What is the most effective remedy?
A A metal palate or fibre reinforcement, and reducing stress concentration at the notch
B A thicker layer of self-cure acrylic
C Softer acrylic teeth only
D A soft liner
E Repair with cyanoacrylate

182 A practice switches to digitally milled, pre-polymerised PMMA denture bases.
What is the main material advantage?
A Higher thermal conductivity
B Very low residual monomer and porosity, with no polymerisation shrinkage in the clinic
C Radiopacity
D Self-repairing surface
E Chemical bond to Co-Cr

183 A patient may have swallowed a fragment of a fractured acrylic partial denture, but it is not visible on the chest radiograph.
Why?
A It has dissolved
B The exposure was too high
C Acrylic absorbs X-rays strongly
D The fragment was too small to matter
E PMMA is radiolucent

184 A complete denture with porcelain posterior teeth opposes natural lower premolars and molars.
What is the main risk?
A Fracture of the base
B Discolouration
C Debonding of the porcelain teeth
D Abrasion of the opposing natural teeth
E Candida

185 A tissue conditioner is used on inflamed mucosa before new impressions.
What is it made from?
A Heat-cured PMMA
B Silicone rubber
C PEMA powder with an aromatic ester plasticiser in ethanol
D Zinc oxide and eugenol
E Polyether

186 A tissue conditioner placed 4 weeks ago has become hard and rough.
Why?
A It has absorbed saliva
B Ethanol and plasticiser have leached out; it should be changed every few days to weeks
C Candida has hardened it
D The base has reacted with it
E It has polymerised further from heat

187 A silicone long-term soft liner has partly separated from the base and is colonised by Candida.
Which statement about silicone soft liners is correct?
A They bond less well to the base and support Candida growth
B They harden within days
C They are best trimmed with hot burs
D They release ethanol
E They never need replacing

188 A patient with a Co-Cr RPD and an acrylic complete denture asks how to soak them overnight.
Which advice is correct?
A Hypochlorite for both
B Boiling water for both
C Hypochlorite only for the all-acrylic denture; an alkaline peroxide cleaner for the Co-Cr
D Vinegar for both
E No soaking for either

189 A technician asks why cobalt–chromium frameworks resist corrosion in saliva.
What provides corrosion resistance?
A Molybdenum carbides
B A gold coating
C Nickel
D Carbon
E A passive chromium oxide film (chromium content about 25–30%)

190 A Co-Cr clasp fractured on first insertion. The laboratory reports a high carbon content in the alloy.
What does excess carbon do?
A Improves castability
B Increases ductility
C Reduces stiffness
D Forms excess carbides, making the alloy brittle
E Increases corrosion resistance

191 A 50-year-old woman with a nickel allergy needs a resin-bonded bridge.
Which statement about nickel–chromium alloys is correct?
A Nickel allergy is rare in women
B Ni-Cr is safe for anyone with a nickel allergy
C Ni-Cr is a noble alloy
D Beryllium in some Ni-Cr alloys is harmless
E Nickel allergy is more common in women; avoid Ni-Cr and consider Co-Cr

192 A clinician compares implant abutment metals.
What is the elastic modulus of titanium, and how is it protected from corrosion?
A About 220 GPa; chromium oxide
B About 20 GPa; tin oxide
C About 2 GPa; no oxide
D About 110 GPa; a titanium dioxide film
E About 90 GPa; nickel oxide

193 A clasp on a lower premolar needs to engage a 0.5–0.75 mm undercut and be adjustable.
Which material is most suitable?
A Wrought wire with a cold-worked, fibrous grain structure
B Cast cobalt–chromium
C Cast titanium
D Cast nickel–chromium
E Acrylic

194 A cast gold clasp is planned for a premolar abutment.
Which undercut should it engage?
A 0.1 mm
B 0.25 mm
C 0.5 mm
D 1.0 mm
E 2.0 mm

195 A patient feels a sharp twinge when foil touches a new gold crown that contacts an amalgam in the opposing tooth.
What explains this?
A Crevice corrosion
B Galvanic current between dissimilar metals in saliva
C Stress corrosion
D Tarnish
E Allergy to gold

196 A technician explains how porcelain bonds to a metal coping.
Which bond mechanism is most important?
A Mechanical retention from air abrasion
B Van der Waals forces
C Chemical bonding through the metal oxide layer
D Compressive fit
E Cement

197 Porcelain repeatedly delaminates from a base-metal coping fired with a thick, dark oxide layer.
What explains the failure?
A Too little oxide
B Porcelain CTE too low
C Gold contamination

D Insufficient firing
E An oxide layer that is too thick weakens the bond

198 The incisal edge of a metal–ceramic upper central crown has chipped. The porcelain at the incisal edge was 3 mm thick with no metal support.
What is the design rule?
A Porcelain should be at least 3 mm thick everywhere
B Unsupported porcelain should not exceed about 2 mm
C Metal should be at least 2 mm thick
D Porcelain does not need metal support
E Opaque porcelain should be 1 mm thick

199 A metal–ceramic crown on 11 looks opaque and bulky.
What labial reduction is usually needed for a metal–ceramic crown?
A 0.3 mm
B 0.5 mm
C 0.7 mm
D 1.2–1.5 mm
E 3 mm

200 A lithium disilicate onlay is to be bonded to a premolar.
What is the correct surface treatment of the restoration?
A Hydrofluoric acid (about 5–10%), silane, then resin cement
B Air abrasion and MDP primer only
C Phosphoric acid only
D Tin plating
E No treatment

201 A monolithic zirconia crown with limited retention is to be cemented.
What surface treatment is recommended?
A Hydrofluoric acid and silane
B Phosphoric acid and silane
C No treatment and zinc oxide eugenol
D Low-pressure air abrasion (about 50 µm alumina) with an MDP-containing primer or cement
E Coarse diamond roughening and silane

202 A clinician applies silane alone to the fitting surface of a zirconia crown before resin cementation. It debonds.
Why did silane fail?
A Silane was out of date
B Zirconia is too rough
C Silane needs a silica-based surface; zirconia contains no glass phase
D Resin cement cannot bond to tooth
E The crown was too thick

203 A student asks why zirconia resists crack propagation.
What is transformation toughening?

A Cubic to tetragonal change on heating
B Stress at a crack tip converts tetragonal to monoclinic zirconia, with about 3–5% volume expansion that closes the crack
C Glass infiltration of cracks
D Heat treatment after milling
E Ion exchange at the surface

204 A zirconia hip implant failed in the 1990s after roughening in body fluids.
What is this phenomenon?
A Low-temperature degradation (ageing): spontaneous tetragonal to monoclinic change in moisture
B Galvanic corrosion
C Creep
D Fatigue only
E Hydrolysis of silane

205 A laboratory mills a zirconia bridge noticeably larger than the final size.
Why?
A To allow for cementation
B Because the scanner is inaccurate
C To allow polishing
D For heat expansion during glazing
E It is milled pre-sintered and shrinks about 20–25% linearly on sintering

206 A chairside lithium disilicate crown is blue after milling.
Why?
A It is milled in the lithium metasilicate state and then crystallised
B It is contaminated with cobalt
C It has been over-sintered
D The block was the wrong shade
E It needs a glaze

207 A patient's lower incisors are wearing rapidly against new upper ceramic crowns that were adjusted chairside and not repolished.
Which surface is most abrasive to opposing enamel?
A Polished monolithic zirconia
B Unglazed or roughened feldspathic porcelain
C Gold
D Glazed lithium disilicate
E Composite

208 A patient feels sensitivity after a gold crown is cemented with zinc phosphate.
Which property of zinc phosphate explains this, and how is it mixed?
A High pH; mixed quickly in bulk
B Fluoride release; mixed on a warm slab
C Eugenol content; mixed on paper
D Moisture sensitivity; mixed on a pad
E Low initial pH; mixed incrementally on a cool glass slab because the reaction is exothermic

209 A leucite-reinforced glass-ceramic crown cracked a month after it was cemented with resin-modified glass ionomer.
What is the likely cause?
A Too thin a cement film
B Eugenol contamination
C Hygroscopic expansion of the RMGIC fracturing a weak glass ceramic
D Galvanic corrosion
E Silane failure

210 A glass ionomer cement is chosen for a metal–ceramic crown on a patient with high caries risk.
What are its main properties?
A No fluoride release; insoluble immediately
B High bond to zirconia via MDP
C Eugenol content and sedation
D Fluoride release, CTE close to tooth, and sensitivity to moisture early on
E The highest bond strength of any cement

211 Porcelain veneers are to be fitted to four upper incisors.
Which cement is essential?
A Zinc phosphate
B Resin cement
C Glass ionomer
D Zinc oxide–eugenol
E Polycarboxylate

212 Provisional crowns are cemented with a eugenol-containing temporary cement before definitive resin-bonded crowns.
What is the problem?
A Eugenol inhibits polymerisation of the later resin cement
B Eugenol etches enamel
C Eugenol stains zirconia
D Eugenol is radiolucent
E There is no problem

213 A crown does not seat fully and the cement layer is thick.
What is the ISO maximum film thickness for luting cements?
A 5 µm
B 10 µm
C 25 µm
D 100 µm
E 250 µm

214 You are bonding a resin-bonded bridge to enamel.
What is the standard enamel etch?
A 10% polyacrylic acid for 10 s
B 5% hydrofluoric acid for 60 s
C 2% chlorhexidine for 30 s

D 30–37% phosphoric acid for 15–30 s
E Air abrasion only

215 After etching dentine with phosphoric acid, an assistant dries the cavity with prolonged air until it looks chalky.
What is the consequence?
A Better resin penetration
B A stronger bond
C Reduced sensitivity
D A thicker hybrid layer
E Collagen collapse and a weaker bond

216 A composite is placed in a sclerotic cervical wear lesion in a 70-year-old.
How can bonding be improved?
A Longer drying
B Using a eugenol cement first
C Hydrofluoric acid
D Silane
E Roughening the sclerotic dentine and selectively etching the enamel

217 A resin-bonded bridge is made with a cobalt–chromium wing.
What is the current standard for bonding the wing?
A Perforated wings retained by resin (Rochette)
B Electrolytically etched metal (Maryland)
C Lost-salt crystals (Virginia)
D Air-abraded base metal bonded with an MDP-containing resin cement
E Zinc phosphate cement

218 A resin-bonded bridge is cast in a high-gold alloy.
What extra step is needed for a chemical bond?
A Phosphoric acid etching of the metal
B Hydrofluoric acid etching of the metal
C Tin plating or a noble-metal primer
D Nothing; gold bonds directly
E A eugenol cement

219 A root-filled premolar is to be crowned with a post-retained core.
Which factor is most important for the long-term survival of the tooth?
A A ferrule of 1.5–2 mm of sound dentine all round
B A wide, long post
C A metal post
D Leaving no root filling apically
E A threaded post

220 A clinician plans a full-arch zirconia bridge from an intraoral scan of an edentulous maxilla.
Which statement about intraoral scanner accuracy is correct?
A It is greatest over full arches

B It is greatest over short spans and lower over full arches and mobile, featureless edentulous mucosa
C It is unaffected by saliva
D It equals conventional impressions in every situation
E It is best for edentulous ridges

Block D — Pharmacology, pain control, emergencies and the medically compromised patient (Q221–290)

> *Notebook Unit 4 · LO 1.7, 2a.13, 2b.1–2b.2, 2b.11–2b.12, 2c.7–2c.8, 2c.15 · Allow 84 minutes · Doses are adult UK doses unless stated; check the current BNF, SDCEP and Resuscitation Council UK guidance*

221 A healthy 60 kg adult needs extensive crown preparations under local anaesthetic. You use the dental-text maximum of 4.4 mg/kg for lidocaine with adrenaline.
About how many 2.2 mL cartridges of 2% lidocaine with 1:80,000 adrenaline is that?
A 3
B 6
C 10
D 13
E 2

222 Buccal infiltration in the lower first molar region gives good anaesthesia with one agent but not another.
Why is articaine effective for mandibular buccal infiltration?
A It has a very long half-life
B It is a pure ester
C It contains no vasoconstrictor
D It is neurotoxic to bone
E It diffuses well through bone, and its ester side chain allows rapid plasma metabolism

223 A patient who is 36 weeks pregnant needs an urgent extraction for a painful tooth.
Which local anaesthetic is best avoided?
A Prilocaine with felypressin
B Lidocaine with adrenaline
C Articaine with adrenaline
D Mepivacaine plain
E None; all are equally suitable

224 Minutes after several cartridges, a patient reports tingling around the lips and ringing in the ears, then becomes drowsy.
What is happening?
A A vasovagal faint
B Anaphylaxis
C Hypoglycaemia
D Early local anaesthetic toxicity, which affects the CNS first
E An adrenaline reaction only

225 A patient taking propranolol receives several cartridges of lidocaine with adrenaline.
What interaction may follow?
A Hypotension with tachycardia
B Methaemoglobinaemia
C Hypertension with reflex bradycardia
D Bronchospasm only
E No interaction

226 An inferior alveolar block repeatedly fails on a lower molar with an acute periapical abscess.
What is the main pharmacological reason?
A Low tissue pH reduces the non-ionised fraction of the anaesthetic
B The anaesthetic has expired
C Adrenaline is inactivated
D Accessory supply from the facial nerve
E The patient is resistant to amides

227 A patient with trismus of 15 mm needs anaesthesia for a lower molar.
Which technique is designed for limited opening?
A Standard inferior alveolar block
B Mental nerve block
C Akinosi (closed-mouth) block
D Palatal infiltration
E Long buccal block only

228 An inhalation sedation machine is being checked before use.
What is the minimum oxygen concentration it must deliver?
A 10%
B 15%
C 21%
D 25%
E 30% (a maximum of 70% nitrous oxide)

229 An anxious patient asks for inhalation sedation for impressions.
Which of these is a relative contraindication?
A Mild hypertension
B First-trimester pregnancy
C Controlled asthma
D Age over 60
E Dental anxiety

230 During IV sedation with midazolam, a patient's respiratory rate falls and they become unresponsive.
Which drug reverses midazolam?
A Naloxone
B Glucagon
C Atropine
D Flumazenil
E Adrenaline

231 A patient is to have IV midazolam sedation for a lengthy procedure.
What post-operative instruction is correct?
A They may drive home after an hour
B No escort is needed
C They may return to work that afternoon
D They may drink alcohol that evening
E A responsible adult escort, and no driving or operating machinery until the next day

232 A patient with poorly controlled heart failure (ASA III) asks for IV sedation in your practice.
Which patients are generally suitable for sedation in primary care?
A Any ASA grade
B ASA I–II
C ASA III–IV
D ASA IV only
E Only patients over 70

233 A patient has a dental abscess with facial swelling, a temperature of 38.5°C and malaise. You have drained it.
Which antibiotic regimen does SDCEP recommend?
A Amoxicillin 3 g single dose
B Clindamycin 150 mg four times daily for 14 days
C Metronidazole 200 mg once daily
D Phenoxymethylpenicillin 500 mg four times daily for 5 days (or amoxicillin 500 mg three times daily)
E Doxycycline 100 mg daily

234 The patient in the previous scenario is allergic to penicillin.
What is an appropriate alternative?
A Clarithromycin 500 mg twice daily (or metronidazole 400 mg three times daily for anaerobes)
B Amoxicillin at a lower dose
C Co-amoxiclav
D Phenoxymethylpenicillin with antihistamine cover
E Flucloxacillin

235 A smoker has painful, bleeding, punched-out interdental papillae and halitosis.
What antimicrobial, if indicated, is recommended?
A Amoxicillin 500 mg for 14 days
B Doxycycline 20 mg for 3 months
C Metronidazole 400 mg three times daily for 3 days
D Fluconazole 50 mg for 7 days
E Aciclovir 200 mg five times daily

236 A patient has a localised abscess on a lower premolar. There is no spreading swelling and no systemic upset.
What is the first-line management?
A Amoxicillin for 5 days and review
B Metronidazole alone
C Clindamycin

D Drainage and removal of the cause; antibiotics are not needed
E Analgesics only and review in a month

237 A patient taking simvastatin and warfarin has denture stomatitis that has not settled with hygiene and night removal.
Which antifungal must be avoided?
A Fluconazole (and miconazole)
B Nystatin
C Chlorhexidine soaks
D Hypochlorite soaks for the acrylic denture
E Improved denture fit

238 A patient with a prosthetic aortic valve needs crown preparation with retraction cord.
What is the current guidance on antibiotic prophylaxis?
A Amoxicillin 3 g before every appointment
B Clindamycin for all invasive procedures
C Not routinely recommended; for patients at special risk, consider it in consultation with the cardiologist
D Never consider it
E Give it only for impressions

239 A patient uses 0.2% chlorhexidine mouthwash straight after brushing and finds it ineffective.
Why?
A Chlorhexidine needs to be swallowed
B It is inactivated by fluoride
C It is too dilute
D It only works at night
E Sodium lauryl sulphate in toothpaste inactivates it; wait about 30 minutes

240 You plan analgesia after a difficult extraction for several patients.
Which patient should not be given ibuprofen?
A A 30-year-old with no medical history
B A patient with NSAID-sensitive asthma
C A patient taking a statin
D A patient with controlled hypothyroidism
E A patient with a penicillin allergy

241 A 48 kg, 82-year-old woman with mild hepatic impairment needs paracetamol.
What is correct?
A Give 2 g four times daily
B No limit applies
C The usual maximum is 4 g in 24 hours, reduced for low body weight and hepatic impairment
D Paracetamol is contraindicated
E Give 1 g hourly as needed

242 A healthy adult has moderate pain after crown preparation.
Which analgesic approach is most effective?

A Codeine alone
B Aspirin alone
C Paracetamol alone
D Ibuprofen and paracetamol combined
E No analgesia

243 A patient on warfarin needs an extraction before a partial denture. Her INR this morning is 3.2.
What is appropriate?
A Stop warfarin for 3 days
B Proceed in primary care with local haemostatic measures
C Refer to hospital
D Give vitamin K
E Defer until the INR is below 2

244 A patient on warfarin has an INR of 4.6 on the day of a planned extraction.
What should you do?
A Proceed with extra sutures
B Proceed with tranexamic acid only
C Proceed under IV sedation
D Halve the next warfarin dose and proceed
E Defer treatment and seek advice from the anticoagulation service

245 A patient taking apixaban twice daily needs several extractions (a higher bleeding-risk procedure).
What does SDCEP advise?
A Miss the morning dose
B Stop apixaban for a week
C Switch to warfarin
D Take a double dose
E No change at all

246 A patient taking rivaroxaban once each morning needs a higher-risk extraction.
What does SDCEP advise?
A Stop for 5 days
B Delay the morning dose until about 4 hours after haemostasis
C Miss two days
D Give vitamin K
E Bridge with heparin

247 A patient on dual antiplatelet therapy (aspirin and clopidogrel) after a coronary stent needs a tooth extracted before a denture.
What is correct?
A Stop clopidogrel for 7 days
B Stop aspirin only
C Stop both
D Refer for hospital admission
E Do not stop either; use local haemostatic measures

248 A patient with moderate haemophilia A needs a lower molar extracted.
What is the safe approach?
A Liaise with the haemophilia centre; avoid an inferior alveolar block without factor cover
B Proceed with an inferior alveolar block and sutures
C Give aspirin for pain
D Prescribe tranexamic acid and proceed with no liaison
E Give vitamin K

249 A patient on apixaban needs crown preparation with retraction cord and impressions.
How should the apixaban be managed?
A Miss the morning dose
B Stop for 48 hours
C No interruption; these are low bleeding-risk procedures
D Switch to aspirin
E Refer to hospital

250 A patient has exposed bone in the lower lingual region after a sore spot from a denture. She has taken alendronic acid for 7 years and has never had radiotherapy.
When does this meet the definition of MRONJ?
A Immediately
B After 2 weeks
C After 4 weeks
D When the bone has been exposed for more than 8 weeks
E Only if it is painful

251 A patient stopped denosumab 18 months ago; another stopped IV zoledronic acid 18 months ago.
Which statement is correct?
A Denosumab's effect wears off within months; bisphosphonates persist in bone for years
B Both effects stop at once
C Bisphosphonate effects wear off in weeks
D Denosumab persists for decades
E Neither drug is associated with MRONJ

252 You are assessing MRONJ risk in several patients.
Who is at highest risk?
A Osteoporosis patient on oral alendronate for 2 years
B Osteoporosis patient with no drug treatment
C Cancer patient on high-dose IV zoledronic acid
D Patient on vitamin D
E Patient who stopped alendronate 10 years ago after 1 year

253 A cancer patient on IV zoledronic acid has a lower canine with a heavily broken-down crown but a sound root. A denture is planned.
What is the preferred management of the canine?
A Extract it before making the denture
B Extract it with antibiotic cover
C Leave it untreated

D Crown it
E Retain the root (root treatment and decoronation as an overdenture abutment)

254 A patient on oral bisphosphonates for 6 years needs a simple extraction.
What does SDCEP advise about a drug holiday?
A Stop for 3 months beforehand
B Drug holidays are not recommended for routine dental care
C Stop for 6 months afterwards
D Switch to denosumab
E Stop permanently

255 An oncologist plans IV zoledronic acid for a patient with bone metastases. The patient has an ill-fitting partial denture and a broken-down tooth.
What should happen first?
A Start the drug and see the dentist later
B Extract every tooth
C Give antibiotics only
D A dental assessment, getting the mouth healthy before treatment starts if possible
E Implants to support the denture

256 A 58-year-old will start radiotherapy for an oropharyngeal cancer, with both parotids in the field. A lower molar has a poor prognosis.
When should it ideally be extracted?
A During radiotherapy
B Six months after radiotherapy
C Only if it becomes painful
D The day before radiotherapy starts
E At least 10–14 days before radiotherapy, to allow healing

257 A dentate patient finished head and neck radiotherapy 6 months ago and now has rampant cervical caries, a dry mouth and limited opening.
Which measures are appropriate?
A Routine 1450 ppm toothpaste only
B Extraction of all carious teeth
C Antibiotics for the caries
D 5000 ppm toothpaste, fluoride gel in custom trays, saliva substitutes and trismus exercises
E Complete dentures at once

258 A patient who had mandibular radiotherapy 8 years ago asks whether the risk of osteoradionecrosis has gone.
Which statement is correct?
A The risk ends after 5 years
B The risk is lifelong, dose-dependent and much higher in the mandible than the maxilla
C The maxilla is at higher risk
D Only implants carry a risk
E The risk applies only during radiotherapy

259 A patient who had a maxillectomy and radiotherapy needs an obturator and asks about implants in the irradiated bone.
At which NHS complexity level is this care?
A Level 3
B Level 1
C Level 2
D Not provided by the NHS
E Hygienist level

260 A patient's GP letter gives an HbA1c result.
At or above which value is diabetes diagnosed?
A 30 mmol/mol
B 42 mmol/mol
C 48 mmol/mol (6.5%)
D 55 mmol/mol
E 70 mmol/mol

261 An undiagnosed 54-year-old with periodontitis and recurrent candidosis has an HbA1c of 45 mmol/mol.
What does this indicate?
A Normal glycaemia
B Type 1 diabetes
C Well-controlled diabetes
D Prediabetes (42–47 mmol/mol)
E Hypoglycaemia

262 A patient with insulin-treated diabetes is booked for a long denture appointment.
Which arrangement is best?
A A morning appointment after a normal meal and medication, with oral glucose available
B A late-afternoon appointment, fasting
C Ask the patient to omit insulin
D Any time; no precautions
E Admit to hospital

263 Minutes after an impression, a patient develops wheeze, facial swelling, urticaria and a falling blood pressure.
What is the first drug and dose?
A Chlorphenamine 10 mg IV
B Hydrocortisone 200 mg IM
C Adrenaline 1:1000, 500 micrograms IM into the anterolateral thigh, repeated after 5 minutes if needed
D Salbutamol 100 micrograms
E Adrenaline 1:10,000, 1 mg IV

264 An 8-year-old develops anaphylaxis in the surgery.
What is the IM adrenaline dose?
A 150 micrograms
B 300 micrograms

C 500 micrograms
D 1 mg
E 50 micrograms

265 A 65-year-old with known angina develops central chest tightness during a crown preparation.
What is the first drug to give?
A Aspirin 75 mg
B Oxygen only
C Adrenaline 500 micrograms
D Salbutamol
E GTN spray 400 micrograms sublingually

266 The chest pain in the previous patient is unrelieved after GTN and lasts over 20 minutes. He is pale and sweaty.
What should you do?
A Give more GTN and send him home
B Give glucose
C Call 999, give aspirin 300 mg chewed, and give oxygen if he is hypoxic
D Give adrenaline
E Lie him flat with legs raised and wait

267 A patient with asthma becomes wheezy and breathless during an appointment but can still speak.
What is the first treatment?
A Adrenaline IM
B Salbutamol 100 micrograms per puff, 4–6 puffs via a spacer, repeated up to 10 puffs
C Glucagon
D GTN
E Hydrocortisone orally

268 A patient with diabetes becomes sweaty, confused and irritable but can still swallow.
What should you give?
A Glucagon 1 mg IM
B Insulin
C Adrenaline
D Oral glucose, 15–20 g
E Nothing; wait for recovery

269 The same patient becomes unconscious and cannot swallow.
What should you give?
A Oral glucose gel between the cheeks
B Insulin
C Adrenaline
D Midazolam
E Glucagon 1 mg IM

270 A patient with epilepsy has a tonic–clonic seizure in the chair that is still continuing after 5 minutes.
What should you give?

A Buccal midazolam 10 mg
B Glucagon 1 mg
C Adrenaline 500 micrograms
D Diazepam tablets
E Nothing; wait for it to stop

271 A young patient becomes pale, sweaty and dizzy, then briefly loses consciousness during an injection. The pulse is slow but present.
What is the immediate management?
A Adrenaline IM
B CPR
C Glucagon
D Sit the patient upright
E Lie the patient flat, raise the legs and give oxygen

272 A patient on long-term corticosteroids collapses with profound hypotension after a stressful procedure.
What is appropriate?
A Glucagon
B GTN
C Salbutamol
D Call 999 and give hydrocortisone if prescribed and you are trained
E Adrenaline 1 mg IV

273 A patient collapses, is unresponsive and is not breathing normally.
What is correct adult CPR?
A 30 compressions to 2 breaths, at 100–120 a minute and 5–6 cm deep, with an AED as soon as available
B 15:2 at 60 a minute
C 5 breaths, then 15 compressions
D Compressions 2–3 cm deep
E Wait for the ambulance before compressions

274 A patient inhales part of a crown preparation's debris, coughs ineffectively and cannot speak. She is conscious.
What is the first action?
A Start CPR
B Give adrenaline
C Up to 5 back blows, then up to 5 abdominal thrusts
D Encourage her to lie down
E Give water

275 A patient becomes acutely unwell in the surgery.
How should oxygen be given?
A 2 L/min via nasal cannula
B 15 L/min via a non-rebreathe mask
C 5 L/min via a venturi mask

D Only if the patient asks
E Never before the ambulance arrives

276 A 30-year-old with poorly controlled epilepsy has lost 11 and 21 and takes phenytoin.
Which prosthodontic approach is most appropriate?
A Prefer a fixed option, avoiding small removable prostheses that could be inhaled or fractured
B A small acrylic partial denture worn at night
C Porcelain teeth on an acrylic base
D No replacement because of the epilepsy
E A spoon denture

277 A patient is 10 weeks pregnant and asks when to have elective bridge work.
When is the safest time for necessary dental treatment?
A First trimester
B Immediately, regardless of stage
C Third trimester
D During labour
E Second trimester; elective prosthodontics can usually wait until after delivery

278 A patient on haemodialysis three times a week needs crown work.
Which arrangement is appropriate?
A Treat on a dialysis day
B Treat on a non-dialysis day and avoid NSAIDs and nephrotoxic drugs
C Give ibuprofen for pain
D Refuse treatment
E Give double-dose antibiotics

279 A patient with a confirmed latex allergy needs addition silicone impressions.
Which precaution is needed?
A Latex gloves are fine with a barrier cream
B Use latex rubber dam only
C Premedicate with an antihistamine and use latex
D Use non-latex gloves and dam; latex also inhibits the setting of addition silicone
E Avoid all impressions

280 A denture wearer has a confirmed methyl methacrylate allergy on patch testing.
Which base material may be considered?
A Self-cured acrylic
B A tissue conditioner
C Fully cured heat-cured PMMA, polyamide or polycarbonate
D Chairside reline resin
E Light-cured composite base

281 A patient with an implanted cardiac defibrillator needs gingival management before an impression.
Which instrument needs caution?
A Hand curettes
B Electrosurgery (and some ultrasonic devices)

C Retraction cord
D A slow-speed handpiece
E Air–water syringe

282 A colleague has told a patient to stop rivaroxaban for 3 days before impressions for a new denture.
What is the best response?
A Continue the rivaroxaban; denture impressions are low bleeding risk and need no interruption
B Agree, and stop it for a week instead
C Switch to warfarin
D Give vitamin K
E Refer to hospital for the impressions

283 A 52-year-old wants a 3-unit bridge. She has active, untreated periodontitis with bleeding and 6 mm pockets around the proposed abutments.
What is the most appropriate plan?
A Prepare the bridge now and treat the periodontitis later
B Use resin-bonded bridges to avoid preparation
C Extract the abutments
D Stabilise the periodontal disease first, then reassess before any definitive bridge
E Prescribe antibiotics and proceed

284 A patient on IV zoledronic acid is considering an RPD or implants.
What should the consent discussion include?
A The expected benefit
B The alternatives, including no treatment
C The medical risks (such as MRONJ) and the maintenance needs
D Both A and B only
E All of the above

285 An 85-year-old care-home resident with Parkinson's disease needs replacement complete dentures.
Which approach is most appropriate?
A An entirely new design to improve appearance
B Implant-retained fixed bridges
C Copy dentures with minimal changes, and labelling for identification
D Immediate dentures
E No dentures

286 A patient on long-term oral prednisolone for arthritis needs a new partial denture.
Which consequence is most relevant to the prosthodontic care?
A Steroid cover is needed for impressions
B Steroids prevent denture stomatitis
C Increased risk of candidosis and delayed healing
D Steroids cause gingival overgrowth
E No relevance at all

287 An edentulous patient with Sjögren's syndrome has poor retention despite well-made complete dentures.

Which option should be considered?
A A thicker denture base
B An implant-retained overdenture, after assessment
C Porcelain teeth
D A higher OVD
E Removing the post-dam

288 A 35-year-old has erosion from gastro-oesophageal reflux that is not yet controlled. She wants her worn teeth crowned.
What is the most appropriate approach?
A Crown all affected teeth now
B Extract the worst teeth
C Refuse all treatment
D Prescribe antibiotics
E Manage the cause with her GMP first; delay definitive restorations until the condition is stable

289 A patient with recently diagnosed, unstable angina asks for a complex bridge.
What is the most appropriate approach?
A Stabilise oral disease, liaise with her physician, and defer elective definitive prosthodontics until her condition is stable
B Proceed with all treatment in one long session
C Refuse all care permanently
D Use IV sedation and proceed
E Prescribe GTN and proceed

290 An ASA III patient with severe COPD requests IV sedation for implant-supported overdenture work.
What is most appropriate?
A Sedate in the practice
B Use inhalation sedation at 80% nitrous oxide
C Treat without sedation regardless of anxiety
D Refer to a secondary care setting
E Give oral diazepam and proceed

Block E — Assessment, diagnosis, imaging, classifications and planning (Q291–370)

> *Notebook Unit 5 · LO 2a.1–2a.21, 2b.3–2b.10 · Allow 96 minutes · FDI notation*

291 A 68-year-old attends saying, "I want new teeth." She has an upper complete denture and a lower RPD.
What is the most important first step in the history?
A Record her smoking history
B Take radiographs
C Record her complaint in her own words and what she expects from treatment
D Check her BPE
E Take impressions

292 You are assessing edentulous patients' attitudes using House's classification.
Which category carries the worst prognosis?
A Hysterical
B Philosophical
C Exacting
D Indifferent
E All carry the same prognosis

293 A 32-year-old wants a bridge to replace 21. When she smiles broadly, more than 3 mm of gingiva shows.
Why is this finding important?
A It contraindicates any bridge
B It predicts periodontitis
C It shows a reduced OVD
D It is a sign of bruxism
E A high smile line increases the aesthetic risk of the margins and pontic

294 A new patient has a BPE code 4 in one sextant and code 2 elsewhere.
What does the BSP recommend?
A Oral hygiene instruction only
B A full six-point periodontal chart of the whole dentition
C A chart of that sextant only
D A repeat BPE in a year
E Immediate referral for surgery

295 You examine an 80-year-old's existing complete dentures before planning replacements.
What should the assessment include?
A Retention and stability
B Extension and support
C OVD, occlusion, appearance and wear
D All of the above
E None; the old dentures are irrelevant

296 An upper arch is missing 16, 17, 26 and 27. The third molars are absent and will not be replaced.
What is the Kennedy classification?
A Class II
B Class I
C Class III
D Class IV
E Class II modification 1

297 A lower arch is missing 46, 47 and 35. 48 is absent and will not be replaced.
What is the Kennedy classification?
A Class I
B Class III
C Class II

D Class II modification 1
E Class III modification 1

298 An upper arch is missing 14 and 15 only.
What is the Kennedy classification?
A Class III
B Class I
C Class II
D Class IV
E Class III modification 1

299 An upper arch is missing 12, 11, 21 and 22 only.
What is the Kennedy classification?
A Class III
B Class III modification 1
C Class IV
D Class I
E Class IV modification 1

300 A lower arch is missing 36 and 37. 38 is present and will be used as an abutment.
What is the Kennedy classification?
A Class I
B Class II
C Class IV
D Class II modification 1
E Class III

301 A lower arch is missing 44, 45 and 46. 47 has a hopeless prognosis and will be extracted before the denture is made.
How should the arch be classified?
A After the planned extraction (Class II)
B Before the extraction (Class III)
C It cannot be classified
D As Class IV
E By Craddock only

302 A student classifies an arch as "Kennedy Class IV modification 1".
Why is this wrong?
A Class IV is always modification 2
B Modifications apply only to Class I
C Class IV has no modifications; any space behind it would make it a different class
D Class IV does not exist
E Modifications are counted by size

303 An upper arch is missing 17 (not to be replaced), 24 and 25. All other teeth are present.
What is the Kennedy classification?
A Class II

B Class II modification 1
C Class I
D Class III
E Class IV

304 A lower RPD will be supported by rests on the abutments and by the free-end saddle on the ridge.
What is its Craddock class?
A Class 1, tooth-borne
B Class 2, mucosa-borne
C Class 4, implant-borne
D It has no Craddock class
E Class 3, tooth- and mucosa-borne

305 A mandibular ridge has adequate height but a narrow, sharp crest.
What is its Cawood and Howell class?
A Class III
B Class IV (knife-edge)
C Class V
D Class VI
E Class II

306 An edentulous mandible has a flat ridge of inadequate height and width.
What is its Cawood and Howell class?
A Class II
B Class III
C Class IV
D Class VI
E Class V

307 An edentulous mandible is depressed, with loss of basal bone.
What is its Cawood and Howell class?
A Class V
B Class VI
C Class IV
D Class III
E Class I

308 You are predicting retention for a new upper complete denture.
Which palatal vault form gives the best retention and resistance to lateral displacement?
A V-shaped
B Flat
C U-shaped
D High and narrow
E Asymmetric

309 The soft palate of an edentulous patient drops steeply at about 70° from the hard palate.
What is its House class and implication?

A Class III: a narrow post-dam area and poorer posterior seal
B Class I: the widest post-dam area
C Class II: an average post-dam
D Class III: the best posterior seal
E It has no effect on the post-dam

310 A 50-year-old's worst interproximal bone loss reaches the middle third of the root.
What is the BSP stage?
A Stage I
B Stage II
C Stage IV
D Stage III
E It cannot be staged

311 A 45-year-old has 40% interproximal bone loss at the worst site.
What is the BSP grade?
A Grade A
B Grade C
C It cannot be graded
D Grade B (40 ÷ 45 ≈ 0.9)
E Grade D

312 A 30-year-old has 60% interproximal bone loss at the worst site.
What is the BSP grade?
A Grade A
B Grade B
C Stage only
D It cannot be graded
E Grade C (60 ÷ 30 = 2.0)

313 A patient's periodontitis affects 6 of 26 teeth.
Is the extent localised or generalised?
A Generalised, because more than 5 teeth are affected
B Localised, because fewer than 30% of teeth are affected
C Generalised, because more than 25% are affected
D Molar–incisor pattern
E Extent is not recorded

314 After treatment, a patient has bleeding on probing at 6% of sites, all pockets 4 mm or less, and no bleeding at 4 mm sites.
What is the periodontal status?
A Currently stable
B In remission
C Unstable
D Stage IV
E Grade C

315 A treated patient has bleeding on probing at 15% of sites, all pockets 4 mm or less, and no bleeding at 4 mm sites.
What is the periodontal status?
A Currently stable
B Unstable
C Currently in remission
D Untreated
E Stage IV

316 A treated patient still has several 5 mm pockets that bleed.
What is the periodontal status?
A Currently stable
B In remission
C Currently unstable
D Healthy
E Stage I

317 A 58-year-old has lost 6 teeth to periodontitis, has fewer than 20 teeth, and has drifting with masticatory dysfunction.
Which stage is this likely to be?
A Stage IV, which usually needs complex rehabilitation
B Stage I
C Stage II
D Stage III Grade A
E It cannot be staged

318 After steps 1 and 2 of periodontal therapy in primary care, a patient has Stage III Grade A periodontitis.
At what NHS complexity level does further care sit?
A Level 1
B Level 3
C Not provided
D Hygienist only
E Level 2

319 A 35-year-old has Stage III Grade C periodontitis.
At what NHS complexity level does care sit?
A Level 1
B Level 2
C Hygienist only
D Level 3
E Not provided

320 During a BPE, the black band of the probe is partly visible in a sextant.
What code is this?
A Code 2
B Code 3 (pocket 3.5–5.5 mm)

C Code 4
D Code *
E Code 1

321 A lower incisor moves more than 1 mm horizontally and can be depressed.
What is its Miller mobility grade?
A Grade 1
B Grade 3
C Grade 2
D Grade 0
E Grade 4

322 A lower first molar has a furcation lesion that a probe passes right through from buccal to lingual.
What is the Hamp furcation grade?
A Grade I
B Grade II
C Grade 0
D Grade III
E Grade IV

323 On BEWE screening, the worst surface in a sextant shows a distinct defect with less than 50% of the surface lost.
What is the score?
A 2
B 0
C 1
D 3
E 4

324 A patient's BEWE scores across the six sextants sum to 15.
What does this indicate?
A No risk
B Low risk
C High risk: oral hygiene and dietary assessment, find the cause, monitor, and consider restoration
D Medium risk only
E Immediate full-mouth crowns

325 A patient's upper central incisors are retroclined with a deep overbite, and the lower incisal edges lie behind the cingulum plateau.
What is the incisor relationship?
A Class I
B Class II division 1
C Class III
D Class I with crowding
E Class II division 2

326 A patient's lower incisal edges lie anterior to the cingulum plateau of the upper incisors.
What is the incisor relationship?
A Class III
B Class I
C Class II division 1
D Class II division 2
E Bimaxillary proclination

327 A lateral cephalogram shows an ANB angle of 3°.
What is the skeletal pattern?
A Class II
B Class III
C Severe Class II
D It cannot be interpreted
E Class I (normal is about 2–4°)

328 You assess a lower premolar as a bridge abutment.
What are the ideal and minimum acceptable crown–root ratios?
A Ideal 1:1, minimum 2:1
B Ideal 1:2, minimum 3:1
C Ideal 3:1, minimum 1:1
D Ideal 2:3, minimum 1:1
E Any ratio is acceptable

329 You need to assess bone level, root form and periapical status of a proposed bridge abutment.
Which radiograph is most appropriate?
A Panoramic radiograph
B Periapical radiograph using the paralleling technique
C Occlusal radiograph
D Lateral cephalogram
E CBCT

330 A patient with moderate periodontitis needs bone levels recorded in the posterior segments.
Which radiograph is best?
A Horizontal bitewings
B Lateral cephalogram
C Vertical bitewings
D Occlusal radiographs
E CBCT

331 A panoramic radiograph shows a hazy radiopacity over the right ramus. The patient has an earring in the left ear.
Which statement explains the artefact?
A It is a calcified lymph node
B It appears on the same side, smaller and lower
C It is a sialolith

D It indicates a retained root
E It is a ghost image, projected on the opposite side, higher and larger

332 A colleague asks for CBCT on every new denture patient "as a baseline".
What is the correct principle?
A CBCT is suitable for routine screening
B CBCT replaces all intraoral radiographs
C CBCT is justified only for a three-dimensional question that lower-dose imaging cannot answer
D CBCT has a lower dose than a bitewing
E CBCT needs no clinical evaluation

333 Under IR(ME)R 2017, a dental nurse takes a radiograph requested by the dentist.
Who justifies the exposure?
A The practitioner (here, the dentist)
B The operator
C The employer
D The radiation protection adviser
E The patient

334 An audit finds radiographs in the notes with no written findings.
What does IR(ME)R require of every exposure?
A Justification only
B Optimisation only
C A second opinion
D Justification, optimisation (ALARP) and a recorded clinical evaluation
E Patient signature

335 A patient worried about radiation asks which dental radiograph gives the highest dose.
What is the order from lowest to highest effective dose?
A CBCT, panoramic, intraoral
B Intraoral, panoramic, CBCT
C Panoramic, intraoral, CBCT
D All are the same
E Intraoral, CBCT, panoramic

336 You need a reliable sensibility test on a lower premolar abutment.
Which thermal test is more reliable?
A Hot gutta-percha
B Warm water bath
C Ice from the freezer
D Cold, using tetrafluoroethane or ethyl chloride on a pledget
E Hot air

337 An upper central incisor showed no response to the electric pulp test 2 weeks after a sporting injury. There is no discolouration or tenderness.
What is the correct interpretation?
A The pulp is necrotic; start root canal treatment now

B The EPT measures nerve response, not blood flow; recently traumatised teeth may not respond, so monitor
C The tooth must be extracted
D The EPT is always accurate after trauma
E The tooth is ankylosed

338 A researcher wants to measure true pulpal vitality (blood flow) in traumatised teeth.
Which test assesses blood flow?
A Electric pulp testing
B Cold testing
C Heat testing
D Percussion
E Laser Doppler flowmetry or pulse oximetry

339 A patient has sharp pain on biting with a lower molar that has a large amalgam; the pain stops on release.
Which test helps confirm a cracked cusp?
A Electric pulp test
B Periapical radiograph alone
C A bite test on individual cusps (for example a Tooth Slooth)
D Percussion of the opposite tooth
E Mobility testing

340 You are planning a lower Co-Cr RPD and a crown on an abutment.
What is the main value of mounted study casts on a semi-adjustable articulator with a facebow and RCP record?
A To assess interocclusal space, plan a wax-up and survey the RPD design
B To measure caries risk
C To assess pulp vitality
D To replace radiographs
E To check periodontal pockets

341 A patient wants composite build-ups for worn incisors but is unsure about the look.
What is the best way to test aesthetics and the proposed OVD before treatment?
A A diagnostic wax-up and intraoral mock-up
B A panoramic radiograph
C A photograph only
D A verbal description
E Proceeding directly with composite

342 A denture wearer with angular cheilitis and glossitis has a full blood count showing a low MCV.
What deficiency is most likely?
A Vitamin B12
B Iron
C Folate
D Vitamin D
E Zinc

343 A patient with a sore tongue and recurrent ulcers has a high MCV.
What does this suggest?
A Iron deficiency
B Thalassaemia
C Vitamin B12 or folate deficiency
D Diabetes
E Normal blood

344 A heavily restored lower premolar is being assessed as a key abutment.
Which features support restorability?
A A possible ferrule of 1.5–2 mm without harmful crown lengthening
B No vertical root fracture
C An acceptable crown–root ratio and strategic value
D Both A and B only
E All of the above

345 A patient asks for root canal treatment of a lower molar under the NHS before a crown.
What does the NHS standard require for the tooth to enter endodontic care?
A Any tooth may be treated
B The tooth must already be crowned
C The patient must be over 50
D Sound coronal tissue above the crest (about 2 mm high and 1 mm wide, or over 30% of the original crown)
E A periapical lesion must be present

346 A post-crowned upper premolar has a deep, narrow, isolated periodontal pocket and a J-shaped radiolucency.
What is the most likely diagnosis and prognosis?
A Periodontitis; good prognosis
B Caries; restorable
C Normal anatomy
D Endodontic lesion; good prognosis
E Vertical root fracture; hopeless prognosis

347 A colleague orders a full radiographic survey for a patient attending only for denture adjustment, with no clinical indication.
Which principle does this break?
A Imaging should always be routine
B Radiographs must be taken every visit
C Investigations should be chosen only when their results may change the diagnosis, prognosis or plan
D Imaging is optional if the patient pays
E None

348 A denture wearer has a persistent, indurated ulcer under the lower flange. A panoramic radiograph looks normal.
What should you do?
A Reassure, because the radiograph is normal

B Refer urgently; a normal radiograph does not overrule a suspicious clinical finding
C Adjust the flange and review in 3 months
D Take a CBCT
E Prescribe antibiotics

349 A 45-year-old presents in pain from a periapical abscess. She also has active caries, untreated periodontitis and missing teeth she wants replaced.
What is the correct sequence of care?
A Definitive bridges, then caries control
B Implants first
C Maintenance, then urgent care
D Urgent care, stabilisation, re-evaluation, definitive treatment, supportive maintenance
E Re-evaluation, then urgent care

350 A 40-year-old has lost 25 and asks what can be done.
Which options should be discussed?
A Leaving the space, a resin-bonded bridge, a conventional bridge, an RPD or an implant, with their pros and cons
B An implant only
C A conventional bridge only
D An RPD only
E Only the cheapest option

351 A patient needs a single crown on 36. She has a stable, comfortable intercuspal position with no symptoms.
Which occlusal approach is appropriate?
A Reorganise to RCP
B Increase the OVD
C Full-mouth equilibration first
D A conformative approach, matching the existing ICP
E Splint therapy first

352 A patient with generalised severe wear needs an increase in OVD and restoration of most teeth.
What is the appropriate level of care?
A Complex reorganisation with careful diagnostic testing, usually with specialist involvement (Level 3)
B Level 1 care
C Single-visit composite in general practice without planning
D No treatment is possible
E Hygienist care

353 You are consenting a patient for a conventional bridge.
Which principle does Montgomery establish?
A The dentist decides what risks to mention
B Material risks and reasonable alternatives, including no treatment, must be discussed
C Only risks over 10% need mentioning
D Written consent is enough
E The Bolam test alone applies

354 You refer a patient with a complex failing dentition to a restorative consultant.
What should the referral contain?
A The patient's name only
B A request to "see and treat"
C The diagnosis, key-tooth prognosis, proposed phases, responsibilities and relevant records
D Only radiographs
E The patient's payment details

355 A patient with active periodontitis requests four new crowns on the upper anteriors.
What is the main flaw in starting with the crowns?
A Crowns are not available on the NHS
B The patient is too young
C Crowns need no periodontal support
D Crowns cure periodontitis
E Definitive restorations before disease control are unsound

356 A complete denture wearer has angular cheilitis, deep nasolabial folds and a prominent chin when the dentures are in occlusion.
What does this suggest, and how would you confirm it?
A Excessive OVD; measure the overjet
B Allergy; patch test
C Reduced OVD; compare the rest vertical dimension with the occlusal vertical dimension (freeway space)
D Candida only; swab
E TMD; MRI

357 You measure the freeway space of a complete denture wearer.
What is the normal range?
A 0 mm
B 0.5 mm
C 10 mm
D About 2–4 mm
E 8–12 mm

358 Before preparing several crowns, you find a 2 mm slide from RCP to ICP.
Why should this be recorded?
A It may affect whether a conformative or reorganised approach is needed
B It indicates caries
C It shows periodontitis
D It has no relevance
E It means the patient has TMD

359 You are examining an edentulous mouth before complete dentures.
Which features should be recorded?
A Ridge height, width and shape
B Flabby tissue and undercuts
C Tori and the denture-bearing mucosa

D Both A and B only
E All of the above

360 A complete denture wearer complains of poor retention and a dry mouth.
How is unstimulated whole saliva flow measured?
A A blood test
B Timed collection by spitting or drooling for 5–15 minutes
C A panoramic radiograph
D Chewing paraffin wax only
E A swab

361 You are grading the difficulty of edentulous and partially dentate cases.
What does the ACP Prosthodontic Diagnostic Index grade?
A Caries risk
B Complete and partial edentulism by ridge height, ridge relationship, muscle attachments and complicating factors
C Periodontal stage
D Tooth wear
E Implant success

362 A 19-year-old has bone loss confined to the incisors and first molars.
How is the extent described in the 2017 classification?
A Generalised
B Localised
C Stage IV
D Necrotising
E Molar–incisor pattern

363 A 40-year-old has localised Stage III Grade B periodontitis after initial therapy.
At what NHS complexity level is further care?
A Level 1
B Level 2
C Level 3
D Not provided
E Hygienist only

364 You need to check for proximal caries on the abutments of an RPD.
Which radiograph is best?
A Horizontal bitewings
B Panoramic radiograph
C Occlusal radiograph
D Lateral cephalogram
E CBCT

365 A colleague uses panoramic radiographs alone to screen for early caries.
Why is this inadequate?
A Panoramic images have no dose

B They are three-dimensional
C They show enamel detail perfectly
D Focal trough distortion and overlap limit fine detail; they are not suitable for detecting early caries
E They cannot show teeth

366 A patient asks how much radiation a CBCT gives compared with a periapical.
Which approximate figures are correct?
A Intraoral about 0.001–0.008 mSv; CBCT about 0.02–1 mSv depending on field of view
B Both about 1 Sv
C Intraoral higher than CBCT
D CBCT about 10 mSv always
E Both negligible and equal

367 An anxious patient reports a response to the electric pulp test on a tooth with a periapical radiolucency.
What could explain a false-positive result?
A Recent trauma
B An immature apex
C Anxiety, or liquefactive necrosis conducting the current
D A calcified pulp
E Old age

368 An 82-year-old's lower canine, with a sound restoration and no radiographic lesion, does not respond to the EPT.
What is a likely explanation?
A Liquefactive necrosis
B Anxiety
C Too much current
D Calcification and reduced pulp size with age, giving a false negative
E Pulpitis

369 An arch has one large free-end saddle and two small bounded spaces.
How are the bounded spaces handled in Kennedy classification?
A They change the class
B They are counted as modification spaces by number, whatever their size
C Only the largest counts
D They are ignored
E Each is a separate class

370 You are deciding whether to keep a compromised lower molar.
Which factors determine its prognosis?
A Periodontal support
B Restorability and endodontic status
C Strategic value
D The patient's risk factors
E All of the above

Block F — Removable partial dentures (Q371–470)

> *Notebook Unit 6 · LO 2c.2–2c.6, 2c.13, 2c.20 · Allow 120 minutes · FDI notation*

371 You have surveyed a study cast for a lower RPD and are about to draw the design.
What is decided first?
A The clasps
B The major connector
C The indirect retainers
D The outline of the saddles (which spaces will be restored)
E The tooth shade

372 You have outlined the saddles on a design for an upper RPD.
Which design element comes next?
A Clasps
B Support
C The major connector
D Review
E Retention

373 A Co-Cr RPD rocks sideways when the patient chews.
Which components resist lateral forces?
A Rests
B Retentive clasp tips
C Mesh
D Acrylic teeth
E Bracing and reciprocal elements, and guide planes

374 A healthy 74-year-old has premolars and anterior teeth in both arches and chews comfortably. Her molars were lost years ago.
What should you consider first when outlining saddles?
A Whether the posterior spaces need restoring at all (shortened dental arch)
B A full palatal plate
C Implants
D Precision attachments
E Clasps on the canines

375 You are measuring undercuts on 35 for a cast Co-Cr clasp.
Which undercut gauge do you use?
A 0.75 mm
B 0.5 mm
C 0.25 mm
D 1.0 mm
E 0.1 mm

376 A student marks the survey line on a premolar.
What does the survey line represent?

A The gingival margin
B The occlusal plane
C The greatest convexity of the tooth relative to the chosen path of insertion
D The cementoenamel junction
E The long axis

377 You are choosing the path of insertion on the surveyor.
Which factors determine it?
A Guide planes
B Retention (suitable undercuts) and interference
C Aesthetics
D Both A and B only
E All of the above

378 You tilt a lower cast on the surveyor table by lowering its front (heels up).
What happens to the undercuts?
A Distal undercuts increase
B Mesial undercuts increase
C Buccal undercuts disappear
D Nothing changes
E All undercuts are eliminated

379 You have chosen a tilt for the cast and need the laboratory to reproduce it.
What should you do?
A Describe it in words
B Photograph it
C Send the surveyor
D Tripod the cast
E Nothing is needed

380 Guide planes are being prepared on abutments for an upper Co-Cr RPD.
How high should they typically be?
A 2–4 mm, parallel to the path of insertion
B 0.5 mm
C 8 mm
D The full crown height
E Angled 20° to the path

381 The survey line on the lingual of 34 is so high that a reciprocal arm would sit above the occlusal third.
How can this be corrected?
A Use a thicker arm
B Use acrylic instead
C Remove the reciprocal arm
D Use a lingual bar
E Recontour the enamel to lower the height of contour

382 15 has no retentive undercut on the buccal surface for the chosen path.
How can an undercut be created conservatively?
A Add composite to create a contour with a suitable undercut
B Crown the tooth
C Grind the occlusal surface
D Extract it
E Tilt the cast until any undercut appears

383 A student lists the functions of an occlusal rest.
Which is NOT a function of a rest?
A Transmitting load along the long axis of the tooth
B Preventing gingival impingement
C Providing retention by engaging an undercut
D Maintaining the vertical position of the denture
E Preventing over-eruption of the abutment

384 You are preparing an occlusal rest seat on 36 for a Co-Cr framework.
Which description is correct?
A Flat floor, 0.5 mm deep
B A box with a floor angled away from the tooth centre
C Half the buccolingual width, 3 mm deep
D Spoon-shaped, about 1–1.5 mm deep and one-third of the buccolingual width, with the floor angled under 90° to the minor connector
E Placed on the buccal cusp tip

385 You plan guide planes and rest seats on several abutments.
When are rest seats prepared?
A Before surveying
B After the guide planes, before the master impression
C After the framework is made
D At the fit appointment
E After the guide planes and after the master impression

386 A rest is needed on a lower canine for an RPD.
Which rest is preferred?
A An incisal rest
B A cingulum rest (on a natural or composite-built cingulum)
C A rest on the labial surface
D A rest on the cusp tip
E No rest

387 45 is the terminal abutment of a lower free-end saddle.
Where should its rest be placed?
A Distally
B Buccally
C On the cusp tip

D Mesially
E Lingually

388 An embrasure clasp is planned between 34 and 35.
What rest arrangement is needed?
A No rest
B An incisal rest
C A single distal rest on 35
D A lingual rest
E Embrasure (double) rests on both teeth

389 A student lists the requirements of a clasp unit.
How much of the tooth's circumference should the clasp encircle?
A More than 180°
B 90°
C Less than 120°
D Exactly 360°
E Encirclement does not matter

390 47 is tipped mesially into the space of 46 and has a mesiolingual undercut.
Which clasp suits 47?
A An I-bar
B An RPA
C A ring clasp with a supporting strut
D An embrasure clasp
E A gingivally approaching T-bar

391 A Kennedy Class II lower arch has no modification space on the dentate side.
How is retention usually gained on that side?
A An embrasure clasp, with the embrasure prepared for space
B An I-bar on the canine
C A ring clasp on the canine
D No retention
E A labial bar

392 An RPI clasp assembly is planned for 44.
Which finding contraindicates the I-bar?
A A 5 mm buccal sulcus
B A mesiobuccal undercut
C A sulcus under about 4 mm, a soft-tissue undercut or a prominent frenum
D A vital tooth
E A mesial rest seat

393 A free-end abutment needs a flexible clasp engaging a 0.5–0.75 mm undercut.
Which clasp is suitable?
A Cast Co-Cr
B Wrought wire

C Cast gold
D Acrylic
E Cast titanium

394 A technician makes a clasp arm twice as long as planned.
How does its flexibility change?
A It halves
B It doubles
C No change
D It increases about 4 times
E It increases about 8 times

395 You review a clasp design on a premolar.
Where should the reciprocal arm sit?
A Below the survey line
B In the undercut
C On the gingiva
D On or above the survey line, contacting the tooth before or with the retentive arm
E It is not needed

396 The retentive arm of a new clasp is pressing on the tooth when the denture is fully seated.
What is the correct position of the retentive tip?
A In the occlusal third, under tension
B On the survey line
C Above the survey line
D In the gingival third undercut, passive when fully seated
E On the gingiva

397 A patient asks why her lower free-end RPD moves while her other partial denture does not.
What explains the problem of free-end saddles?
A Teeth move about 0.02–0.1 mm under load, while mucosa compresses about 0.4–2 mm
B Mucosa is less compressible than teeth
C Teeth move more than mucosa
D Clasps are always loose
E Saliva is thinner posteriorly

398 You are planning the extension of a lower free-end saddle.
What should it cover?
A Only the ridge crest
B Half the length of the ridge
C The retromolar pad and buccal shelf
D The floor of the mouth
E The external oblique ridge only

399 A lower Kennedy Class I RPD is planned for a patient with a resorbed, displaceable ridge.
Which impression technique records the saddle area functionally?
A Alginate in a stock tray

B Putty only
C A mucostatic plaster impression
D A one-stage impression of the teeth and ridge in compound
E The altered cast technique (Applegate)

400 An RPI clasp is designed for 34, the abutment of a free-end saddle.
What are its three components?
A Retentive arm, plate, I-bar
B Mesial rest, distal proximal plate and buccal I-bar
C Distal rest, mesial plate and lingual arm
D Ring, plate, I-bar
E Rest, plate and wrought wire

401 A patient bites on the free-end saddle of a lower RPD with an RPI assembly on 34.
What happens to the I-bar and proximal plate?
A They tighten against the tooth
B They move gingivally and mesially, disengaging from the tooth and reducing torque
C They move occlusally
D They lock the denture in place
E Nothing

402 44 is the terminal abutment of a free-end saddle, but the buccal sulcus is 3 mm deep with a soft-tissue undercut.
Which clasp assembly is appropriate?
A RPI
B Ring clasp
C Gingivally approaching T-bar
D No clasp
E RPA, replacing the I-bar with an Akers-type arm

403 You are designing indirect retention for a lower Kennedy Class I RPD.
Where does the fulcrum line run?
A Along the lingual bar
B Across the incisors
C Through the clasp tips
D Through the rests nearest to the free-end saddles
E Along the ridge crest

404 A patient's lower bilateral free-end RPD lifts at the back when she eats toffee.
Where should an indirect retainer be placed?
A On the fulcrum line
B Behind the fulcrum line
C As far anterior to the fulcrum line as possible
D On the saddle
E On the retromolar pad

405 A student asks what else indirect retainers achieve besides resisting rotation.
What is correct?
A They give support and stop the major connector settling
B They provide the main retention
C They replace rests
D They improve the shade
E They have no other role

406 A long lower free-end saddle on a resorbed ridge is being set up.
How should the saddle teeth be arranged?
A Wide, cusped teeth
B Extra teeth to the retromolar pad
C Porcelain teeth
D High occlusal contacts
E Narrow, reduced occlusal tables in harmonious occlusion

407 At review, pressing on the lower free-end saddle lifts the canine rests from their seats.
What is needed?
A Tighter clasps
B New rests
C Reline of the saddles
D Removal of the indirect retainers
E A new framework

408 A 70-year-old has only 13–23 remaining. The palate is flat and resorbed.
Which maxillary connector is appropriate?
A Full palatal plate
B Narrow palatal bar
C Horseshoe
D Labial bar
E Mid-palatal strap

409 An upper arch has short bounded saddles on both sides (16 and 26 missing).
Which maxillary connector is appropriate?
A Full palatal plate
B Mid-palatal strap, at least 8 mm wide
C Horseshoe
D Anterior palatal bar
E Labial bar

410 An upper arch has anterior and posterior saddles and a large torus that the patient will not have removed.
Which connector is appropriate?
A Full palatal plate over the torus
B Narrow mid-palatal strap over the torus
C Horseshoe

D Anteroposterior strap avoiding the torus
E Lingual bar

411 A student asks which maxillary connector is least rigid.
Which is it, and when is it used?
A Full plate; for free-end saddles
B Mid-palatal strap; for bounded saddles
C Horseshoe; for anterior saddles only or an inoperable torus
D Anteroposterior strap; for any case
E All are equally rigid

412 A lower RPD is planned. From the gingival margin to the functional floor of the mouth there are 9 mm.
Which connector and dimensions are correct?
A Lingual plate, because the sulcus is deep
B Labial bar
C Sublingual bar
D Lingual bar: half-pear, about 4 mm high, at least 3 mm clear of the gingival margin
E No connector

413 A 66-year-old has mobile lower incisors, a 6 mm lingual sulcus, and may lose more teeth.
Which lower connector is appropriate?
A Lingual plate
B Lingual bar
C Labial bar
D Sublingual bar
E Dental bar

414 A shallow lingual sulcus would place a lingual bar on the gingivae, but the incisors are healthy.
Which connector lies deeper in the sulcus and needs a functional impression?
A Lingual plate
B Sublingual bar
C Labial bar
D Horseshoe
E Continuous clasp

415 A patient has markedly lingually inclined lower premolars and large tori.
Which connector is indicated?
A Lingual bar
B Lingual plate
C Sublingual bar
D Dental bar
E Labial bar

416 A dental bar (continuous clasp) on the cingula is suggested for a patient with recession.
What is its main drawback?
A It traps food

B It gives no indirect retention
C It must cover the gingiva
D It is invisible
E It is very flexible

417 You review the edges of a Co-Cr major connector.
How should they relate to the gingival margins?
A They should end on the gingival margin
B At least 3 mm (mandible) to 6 mm (maxilla) away, or covering the margin fully with relief
C 1 mm away
D Within the gingival crevice
E It does not matter

418 A minor connector runs from a lingual bar to a rest on 34.
How should it cross the gingival margin?
A Diagonally and wide
B Along the gingival margin
C Over the labial surface
D At right angles, as narrow as possible, lying in the embrasure
E Embedded in acrylic

419 A technician proposes a thin, flexible lingual bar to improve comfort.
Why is this wrong?
A Thin bars are more expensive to cast
B Rigidity does not matter
C Major connectors must be rigid so that forces spread across the arch
D Flexible bars improve retention
E Lingual bars must be acrylic

420 A 60-year-old has worn a mucosa-borne acrylic RPD covering the gingival margins for 10 years.
What is the typical consequence?
A Better periodontal health
B Less caries
C No effect
D Bone gain
E Gingival stripping, with more periodontal disease and caries

421 Several teeth of uncertain prognosis remain, and further extractions are likely within a year.
Which RPD is most appropriate?
A A Co-Cr framework with precision attachments
B A swing-lock
C A flexible nylon denture
D Implants
E An acrylic interim (transitional) denture

422 An upper acrylic RPD is designed with point contacts on the teeth, open embrasures, free gingival margins and distal stabilising wires.

What design is this?
A A spoon denture
B An Every denture
C A swing-lock
D A horseshoe Co-Cr
E An immediate denture

423 A patient needs a small, interim replacement for 11 with no gingival contact.
Which design is this?
A A spoon denture
B An Every denture
C A swing-lock
D A full plate
E A precision-attachment RPD

424 A patient has missing anterior teeth and periodontally compromised remaining teeth with labial undercuts, where conventional clasps work poorly.
Which design is indicated?
A An Every denture
B A spoon denture
C A flexible nylon denture
D A swing-lock denture
E A horseshoe

425 A patient asks for a flexible nylon denture because she dislikes metal clasps.
What is its main disadvantage?
A It is radiopaque
B It is heavy
C It is hard to adjust, reline or add to
D It is always tooth-borne
E It corrodes

426 A patient asks about precision attachments for a lower RPD.
What is correct?
A They need no crowns
B They are simple Level 1 care
C They need crowned abutments and usually lie beyond Level 2
D They are cheaper than clasps
E They cannot be used with Co-Cr

427 26 has over-erupted into a lower edentulous space where an RPD is planned.
What is appropriate before designing?
A Correct the over-eruption and occlusal plane (pre-prosthetic optimisation, a Level 2 task)
B Ignore it
C Extract 26
D Increase the OVD
E Use a soft liner

428 The metal framework of a Co-Cr RPD is tried in before the teeth are set.
What should be checked?
A Fit with disclosing medium
B That rests seat fully
C That clasps are passive and the occlusion is not disturbed
D Both A and B only
E All of the above

429 A patient removes her new RPD by pulling on the clasp tips.
What instruction should she be given?
A Pull harder on the clasps
B Remove it only at the dentist
C Use a tool on the clasps
D Insert and remove it along the path of insertion, using the saddles, not the clasps
E Never remove it

430 A new Co-Cr RPD is fitted.
When should it first be reviewed?
A At 6 months
B About 1 week, then at regular recalls
C Only if there are problems
D At 2 years
E The same day

431 A patient has worn a Co-Cr RPD for 5 years.
What is the most important long-term maintenance?
A Tightening clasps every year
B Polishing the framework
C Replacing the teeth
D Monitoring caries and periodontal health at the abutments, and relining saddles as needed
E Remaking every 2 years

432 A patient is worried that an RPD will damage her remaining teeth.
What does the evidence (Bergman) show?
A Any RPD destroys the abutments
B Acrylic RPDs are harmless
C A well-designed, tooth-supported Co-Cr RPD with good hygiene and recall does little harm
D RPDs prevent caries
E Only implants are safe

433 A laboratory returns a Co-Cr framework with clasps on different teeth from those planned.
Who is responsible for the design?
A The technician
B The patient
C The practice manager
D The dental nurse
E The dentist, whose prescription must specify every component

434 A student designs a retentive clasp on 45 with no reciprocal element.
Why is the design incomplete?
A Retention is too high
B Without reciprocation, the retentive arm pushes the tooth as it flexes
C It is too rigid
D It looks bad
E Reciprocation is optional

435 A patient has pain on biting on 36 a week after fitting a Co-Cr RPD. A rest on 36 marks heavily with articulating paper.
What is the likely cause and remedy?
A A high rest; adjust the rest or deepen the seat
B A loose clasp; tighten it
C Candida; antifungal
D Allergy; remake in acrylic
E Caries; restore

436 An upper arch is missing 24, 25 and 26 only, with sound 23 and 27.
What support and indirect retention does this design need?
A Mucosal support with indirect retainers
B Tooth support from rests at both ends; indirect retention is generally not needed
C Free-end design with an RPI
D A full palatal plate
E No rests

437 A Kennedy Class IV RPD replaces 12 to 22.
Which design principles apply?
A Rests on the teeth next to the anterior saddle
B Indirect retainers placed posteriorly
C Attention to aesthetics and lip support
D Both A and B only
E All of the above

438 A lower Kennedy Class I RPD is planned with 33 to 44 remaining and a deep lingual sulcus.
Which design is most appropriate?
A Mesial rests and RPI clasps on 34 and 44, cingulum rests on the canines, a lingual bar and maximum saddle extension
B Distal rests and Akers clasps on the premolars
C An acrylic mucosa-borne plate
D A labial bar with no rests
E Ring clasps on the canines

439 An upper arch has 13, 12, 22 and 23 remaining, with bilateral free-end saddles and a high, well-formed vault.
Which connector gives the best support and retention?
A Horseshoe
B Narrow palatal bar

C Full palatal plate
D Mid-palatal strap
E Labial bar

440 A student asks when guide planes and rest seats should be prepared for a Co-Cr RPD.
What is the correct sequence?
A Prepare freehand at the first visit
B Prepare after the framework arrives
C Prepare after the master impression
D Survey the study cast and plan, prepare the teeth, then take the master impression
E Prepare at the fit appointment

441 An RPI assembly is planned on 34.
How long should the distal guide plane be?
A Short, about 2–3 mm, so the plate can disengage under load
B The full crown height
C 6 mm
D There should be none
E Angled away from the path

442 35 has a 0.75 mm undercut on its buccal surface.
Which clasp suits this depth?
A Cast Co-Cr
B Cast gold
C Cast titanium
D Acrylic
E Wrought wire

443 You tilt a lower cast on the surveyor so that its heels are lowered.
What happens to the undercuts?
A Mesial undercuts increase
B Distal undercuts increase
C Lingual undercuts vanish
D Nothing changes
E All undercuts double

444 A technician makes a Co-Cr clasp arm twice as thick as planned.
How does its flexibility change?
A It doubles
B It halves
C It falls to about one-eighth
D It increases eightfold
E No change

445 A patient dislikes the buccal circumferential clasp showing on 13.
Which is a more aesthetic option if the sulcus allows?
A A longer circumferential arm

B A ring clasp
C An embrasure clasp
D A gingivally approaching I-bar into a mesiobuccal undercut, or moving retention to a premolar
E No retention at all

446 A lower lingual plate is planned for a patient with mobile incisors.
What is required for it to succeed?
A A deep sulcus
B No rests
C Gingival contact without relief
D Avoiding the cingula
E Excellent oral hygiene, as it covers the gingival margins

447 A patient with xerostomia and high caries risk needs an RPD.
Which design principle is most important?
A Maximum tooth coverage
B Acrylic, mucosa-borne design
C Plates covering all gingival margins
D A minimal, open design with little tooth and gingival coverage
E Precision attachments

448 A patient asks how to soak her Co-Cr RPD overnight.
What is correct?
A Dilute hypochlorite
B An alkaline peroxide cleanser
C Boiling water
D Vinegar
E Bleach

449 A patient asks whether to wear her new RPD at night.
What is the standard advice?
A Remove it at night
B Wear it day and night
C Wear it only at night
D Wear it only for meals
E It makes no difference

450 A lower Kennedy Class II RPD fitted 6 years ago now rocks, and the indirect retainer lifts when the saddle is pressed.
What is the correct management?
A Tighten the clasps
B Add another clasp
C Reline the free-end saddle
D Remove the indirect retainer
E Extract the abutment

451 An abutment tooth has over-erupted slightly since the patient stopped wearing her RPD.
Which function of rests helps prevent this?
A Retention
B Preventing over-eruption and maintaining tooth position
C Bracing only
D Aesthetics
E Reciprocation

452 A bounded saddle abutment is tilted towards the space, so the denture cannot be inserted without a gap.
What helps?
A Preparing guide planes parallel to the path of insertion
B Ignoring the tilt
C A flexible clasp only
D Removing the rest
E Using acrylic only

453 Food collects between a saddle and the abutment of a Co-Cr RPD.
What design feature is likely missing?
A A rest
B A reciprocal arm
C A lingual bar
D A proximal plate on a guide plane
E A labial flange

454 A Co-Cr RPD is being set up for a patient with a stable natural occlusion.
How should the denture occlusion be arranged?
A Balanced occlusion like a complete denture
B At an increased OVD
C With heavy contacts on the saddle teeth
D Without contact on the saddle teeth
E In harmony with the existing ICP

455 A tilted lower molar and over-erupted upper molar complicate an RPD design.
How is correcting them before design classed?
A Level 1, always simple
B Level 3 only
C Pre-prosthetic optimisation, within Level 2
D Unnecessary
E Specialist orthodontics only

456 A 70-year-old has grade 2 mobile lower incisors with stable periodontal disease and needs a free-end RPD.
Which lower connector helps?
A Lingual bar
B Labial bar
C Sublingual bar

D Lingual plate, splinting the incisors
E No connector

457 A lower lingual bar has sunk onto the lingual gingiva of the incisors.
What is the most likely cause?
A The bar is too thick
B The bar is too rigid
C Inadequate rest support, so the denture settles
D Allergy
E Saliva

458 A lower Kennedy Class I RPD needs indirect retention.
Which components can act as indirect retainers?
A Canine cingulum rests, a lingual plate or rests on premolars, placed anterior to the fulcrum line
B Clasp tips behind the fulcrum line
C Mesh on the saddle
D Acrylic teeth
E The retromolar pad

459 A student designs an RPD by first choosing an attractive clasp for each tooth.
What is wrong with this approach?
A Nothing
B Clasps should be chosen last of all
C Clasps are optional
D Rests are not needed
E It reverses the design logic; support and the path of insertion come before clasp selection

460 A patient's RPD lifts at the back when she eats sticky food. The only rests are on the premolars next to the saddles.
Where should an extra rest be added?
A On the premolar distal surface
B As far forward of the fulcrum line as possible, such as a cingulum rest on a canine
C On the saddle
D On the retromolar pad
E Nowhere

461 A distal-extension RPD is being planned.
How is it supported?
A Teeth only
B Mucosa only
C Teeth and mucosa together
D Implants only
E Clasps only

462 A colleague plans Akers clasps engaging the distobuccal undercuts of the terminal abutments of a lower free-end RPD.
What is the problem?

A Too little retention
B Poor aesthetics only
C Clasps are too flexible
D The clasps will torque the abutments distally as the saddle moves; mesial rests with RPI or RPA are better
E There is no problem

463 A patient has lost several anterior teeth and the remaining teeth have reduced support with labial undercuts.
Which design gives retention without conventional clasps?
A An Every denture
B A swing-lock denture
C A spoon denture
D A lingual plate
E A flexible denture

464 An RPD is needed while a patient's periodontal treatment outcome is uncertain.
Which type is most appropriate?
A An acrylic interim RPD
B A Co-Cr RPD with precision attachments
C Implants
D A swing-lock
E A conventional bridge

465 A patient requests an RPD with precision attachments on crowned canines.
At what level is this care usually provided?
A Level 1
B Hygienist level
C Not provided
D Level 2
E Beyond Level 2 (specialist)

466 A Co-Cr framework will not seat. Disclosing medium shows contact on the inner surface of a minor connector.
What should you do?
A Adjust only the marked area and re-check
B Remake immediately
C Bend the clasps
D Force it into place
E Reline it

467 A week after fitting, a patient reports a sore abutment. The rest on it marks heavily in occlusion.
What should you do?
A Tighten the clasp
B Adjust the high rest
C Extract the tooth

D Remake the framework
E Prescribe analgesics only

468 You send a Co-Cr RPD case to the laboratory.
What must the prescription include?
A The patient's address only
B A request to "design as you see fit"
C A drawn design specifying every component, on a surveyed cast with tripod marks
D The shade only
E The cost

469 A patient lisps on "s" with a new upper Co-Cr RPD whose anterior strap is thick over the rugae.
What should be adjusted?
A The clasps
B The rests
C The saddle teeth
D The thickness and position of the anterior strap
E The OVD

470 A patient asks why a well-made Co-Cr RPD is kinder to the remaining teeth than an acrylic one.
Which features reduce harm?
A Tooth support through rests
B Minimal gingival coverage
C Regular recall and hygiene
D Both A and B only
E All of the above

Block G — Complete dentures (Q471–570)

> *Notebook Unit 7 · LO 2c.4, 2c.9, 2c.13, 2c.19, 2c.21–2c.27 · Allow 120 minutes*

471 A new upper complete denture stays up at rest but slides sideways when the patient chews.
Which property is deficient?
A Retention
B Support
C Stability
D Aesthetics
E Vertical dimension

472 A student asks what holds a complete denture in place.
Which factors contribute to retention?
A Physical factors such as adhesion, cohesion and the peripheral seal
B Muscular factors such as polished-surface shape and the neutral zone
C Mechanical factors such as undercuts, implants or adhesive
D Both A and B only
E All of the above

473 A lower denture sinks into a soft, flat ridge under load, causing pain.
Which property describes resistance to movement towards the tissues, and how is it improved?
A Support; by maximum extension over the stress-bearing areas
B Retention; by adhesive
C Stability; by steeper cusps
D Support; by shortening the flanges
E Retention; by a post-dam

474 A well-fitting lower denture is displaced by the cheeks and tongue during speech.
Which group of retentive factors is failing?
A Physical factors
B Mechanical factors
C Adhesive factors
D Muscular factors: polished-surface shape and the neutral zone
E None

475 You prescribe special trays for complete dentures from primary impressions.
How far should the tray borders be from the sulcus depth?
A At full sulcus depth
B About 2 mm short, to allow border moulding
C 10 mm short
D 2 mm over-extended
E It does not matter

476 You take primary impressions of an edentulous patient in stock trays.
What is the main aim?
A A mucostatic record of the ridge only
B Recording the full sulcus depth so that special trays can be made
C Recording the occlusion
D Recording the post-dam
E Choosing the shade

477 You are border moulding a special tray before the secondary impression.
Which material is traditionally used?
A Alginate
B Putty
C Plaster
D Greenstick impression compound
E Wax

478 The laboratory pours the secondary impressions for complete dentures.
Which stone is used for the master casts?
A Type I impression plaster
B Type II plaster
C Type V die stone
D Investment
E Type III stone

479 You are taking secondary (master) impressions in close-fitting special trays.
Which materials are suitable?
A ZOE paste, light-bodied silicone or polyether
B Alginate in a stock tray
C Compound only
D Putty only
E Wax

480 You are at the jaw registration visit with wax rims.
What is recorded at this visit?
A Only the shade
B Only OVD
C OVD, RCP, occlusal plane, centre and canine lines, lip line, tooth mould and shade
D The post-dam only
E Nothing; the laboratory decides

481 At the wax try-in of new complete dentures, everything looks satisfactory.
What else should be done before processing?
A Nothing
B Take new impressions
C Fit the dentures
D Change the shade without asking
E Record the patient's approval and mark the post-dam

482 You are fitting new complete dentures.
How do you identify pressure points on the fitting surface?
A Pressure-indicating paste
B Articulating paper
C Visual inspection only
D A radiograph
E The patient's report only

483 New complete dentures have been fitted.
When is the first review usually arranged?
A The same day
B About 1 week later
C 6 months later
D Only if there is pain
E 2 years later

484 You are contouring an upper wax rim for lip support.
How far anterior to the centre of the incisive papilla do the labial surfaces of the central incisors usually lie?
A 2–3 mm
B 4–5 mm
C 8–10 mm

D 15 mm
E Directly over it

485 You set the incisal level of an upper wax rim for a 45-year-old.
How much rim usually shows below the relaxed upper lip?
A None
B 4–5 mm
C 6 mm
D About 1–2 mm (less with age)
E 8 mm

486 You check the upper occlusal rim with a Fox's plane guide.
How should the occlusal plane be oriented?
A Parallel to the Frankfort plane
B Tilted down anteriorly by 10°
C Parallel to the lower ridge only
D Parallel to the interpupillary line anteriorly and to the ala–tragal line (Camper's plane) posteriorly
E Perpendicular to the facial midline

487 You set the lower posterior occlusal plane.
What is the usual landmark?
A The lower lip
B The mylohyoid ridge
C About two-thirds up the retromolar pad, level with the lateral border of the tongue
D The top of the retromolar pad
E The external oblique ridge

488 You mark guidelines on the upper rim.
What do the centre line and canine lines follow?
A The facial midline (philtrum) and the corners of the mouth at rest
B The lower midline and the commissures in a smile
C The nose tip and the pupils
D The incisive papilla only
E The labial frenum and the buccal frena

489 An edentulous patient's rest vertical dimension measures 70 mm.
What OVD would you aim for?
A 70 mm
B 74 mm
C 60 mm
D 72 mm
E About 67 mm (a freeway space of about 3 mm)

490 You use a Willis gauge during jaw registration.
What does it measure?
A Pupil to lip
B Base of the nose to the underside of the chin

C Interpupillary distance
D Inter-alar width
E Ridge height

491 You test the OVD of trial dentures phonetically.
What is the normal closest speaking space on "s"?
A About 1–2 mm
B 0 mm
C 5 mm
D 8 mm
E 10 mm

492 Rest vertical dimension readings vary between visits for the same patient.
Why, and how should you respond?
A The gauge is faulty
B Rest position is fixed for life
C Measurement is pointless
D Rest position varies with posture, fatigue and emotion; measure several times with the patient upright and relaxed
E Use the old denture's OVD only

493 A week after fitting, a patient complains of generalised soreness of the lower ridge. Her teeth click when she speaks and her lips look strained.
What is the most likely cause?
A OVD too low
B Candida
C OVD too high (freeway space too small)
D An over-extended upper flange
E Porcelain teeth only

494 A patient's old dentures give him a collapsed appearance, angular cheilitis and a prominent chin.
What do these signs indicate?
A OVD too high
B Over-extension
C Candida only
D Allergy
E OVD too low (freeway space too large)

495 A colleague builds the OVD of new dentures well above the rest position.
Which long-term consequences may follow?
A Improved chewing
B Muscle fatigue, soreness and faster ridge resorption
C Better retention
D Angular cheilitis
E Reduced clicking

496 A student asks why complete dentures are made in the retruded contact position.
What is the reason?
A It is the most comfortable
B It gives the most protrusion
C It increases OVD
D It is the position patients habitually use
E It is reproducible

497 You are recording the horizontal jaw relation.
Which technique is appropriate?
A Ask the patient to bite hard in any position
B Guide the mandible gently to the retruded arc, with the patient relaxed and the tongue curled back
C Push the chin forcibly backwards
D Ask the patient to protrude
E Record it with the patient lying flat and asleep

498 You use a Gothic arch (needle-point) tracer because registration is unreliable.
Where is RCP recorded?
A At the end of the right lateral path
B In the centre of the tracing
C At the most protruded point
D At the apex of the arrowhead
E Anywhere on the left path

499 At try-in, the posterior teeth on one side do not meet and there is a slide on closure.
What is the correct action?
A Grind the teeth until they meet
B Process the dentures and adjust at fit
C Take a new RCP registration and re-articulate
D Add wax to the teeth
E Accept it

500 A technician uses an average-value articulator for complete dentures.
What are its fixed settings?
A Condylar guidance about 30° and a Bonwill triangle with sides of about 10 cm
B No lateral movement at all
C Settings taken from a protrusive record
D A fully adjustable Bennett angle
E Condylar guidance of 0°

501 A semi-adjustable articulator is used with a facebow.
How is its condylar guidance set?
A From average values
B From a protrusive interocclusal record
C From the occlusal plane
D From the incisive papilla
E It cannot be set

502 A technician needs the Bennett angle but has no lateral records.
Which formula estimates it from condylar guidance (H)?
A L = H × 2
B L = 30 − H
C L = H/8 + 12 (Hanau)
D L = H/2
E L = H + 15

503 A student is comparing articulators.
What defines an arcon articulator?
A It has no condylar elements
B It is a simple hinge
C It has fixed average values
D Its condylar element is on the upper member
E Its condylar element is on the lower member, as in the anatomy (for example Whip Mix, Denar)

504 A facebow record is taken for complete dentures on a semi-adjustable articulator.
What does it record?
A The relationship of the maxillary arch to the hinge axis
B The OVD
C RCP
D The lower occlusal plane
E The shade

505 In protrusion, the posterior parts of the record rims separate.
What is this phenomenon, and what does balanced occlusion need?
A Bennett movement; steeper anterior teeth
B Hanau's phenomenon; flat teeth
C Curve of Spee only
D Christensen's phenomenon; a compensating curve
E Freeway space; reduced OVD

506 A technician refers to Hanau's quint when setting balanced occlusion.
What are its five factors?
A OVD, RCP, shade, mould, lip line
B Freeway space, OVD, overjet, overbite, post-dam
C Condylar guidance, incisal guidance, cusp angle, plane of orientation, compensating curve
D Ridge height, width, shape, saliva, tongue
E Retention, stability, support, aesthetics, speech

507 A student asks why bilateral balanced occlusion is used in complete dentures.
What is its purpose?
A Simultaneous contacts on working and non-working sides keep the dentures stable in excursions
B It improves appearance
C It increases bite force
D It reduces OVD
E It prevents Candida

508 A technician sets a lingualised occlusion.
How is it arranged?
A All cusps contact
B Only the upper palatal cusps contact the lower central fossae
C Only the lower buccal cusps contact
D Anterior teeth only
E Flat teeth with no contact

509 A patient has a severely resorbed mandible, a Class III jaw relationship and poor neuromuscular control.
Which posterior teeth are most suitable?
A Steep-cusped teeth
B Porcelain anatomic teeth
C 33° cusped teeth
D Monoplane (zero-degree) teeth
E No posterior teeth

510 You estimate the width of an upper central incisor for a patient without pre-extraction records.
Which guide (Berry) relates it to facial width?
A One-quarter of the inter-alar width
B Equal to the canine lines
C One-tenth of the face height
D One-half of the interpupillary distance
E About one-sixteenth of the bizygomatic width

511 A colleague insists that tooth shape must match the face form (Williams) and personality (dentogenics).
What is the evidence position?
A They are mandatory
B They are proven by trials
C They are required by the GDC
D They guide shape but are weakly evidence-based
E They are discredited and must never be used

512 An 80-year-old wants new dentures and asks for "bright white, perfect teeth".
Which approach to shade is best?
A Always the lightest shade
B Involve her in the choice, usually favouring a darker, characterised shade that suits her age
C Choose without discussion
D Grey teeth
E Match her old dentures regardless

513 A technician sets the lower second molar on the ascending slope of the ramus.
What is the consequence?
A The lower denture is pushed forward in function
B Better support
C Less cheek biting

D Better aesthetics
E No effect

514 A patient has a severely resorbed maxilla that is much narrower than the mandible.
How are the posterior teeth usually set?
A Lingual to the upper ridge
B In normal overjet regardless
C Omitted
D On the palate
E In crossbite, with the upper teeth possibly buccal to the resorbed ridge

515 A patient's lower denture has been unstable for years on a resorbed ridge.
How is the neutral zone recorded?
A With alginate in a stock tray
B From the old denture's teeth
C With a mouldable material on a lower base, shaped during swallowing and speaking
D With a facebow
E From a radiograph

516 A colleague asks when a neutral-zone technique is most useful.
Which case suits it best?
A A resorbed lower ridge with a history of unstable lower dentures
B A first-time denture wearer with excellent ridges
C An immediate denture
D A partial denture
E An upper denture only

517 A patient keeps biting his cheeks with new complete dentures.
What setting feature prevents this?
A Higher OVD
B Steeper cusps
C A thicker base
D Lingual setting of all teeth
E Adequate posterior horizontal overlap (buccal overjet)

518 You are carving the post-dam on an upper master cast.
Where is it deepest?
A Over the median raphe
B At the hamular notch only
C About 1–1.5 mm deep either side of the midline, shallower over the raphe
D Uniformly 3 mm
E On the anterior palate

519 A student asks why a post-dam is needed.
Which is one of its functions?
A It adds retention by engaging an undercut
B It compensates for polymerisation contraction lifting the posterior border

C It increases OVD
D It improves aesthetics
E It reduces the weight of the denture

520 You mark the vibrating line for an upper complete denture.
How are the posterior and anterior vibrating lines identified?
A By palpation only
B With a radiograph
C At the incisive papilla
D Posterior: by asking the patient to say "ah"; anterior: with the Valsalva manoeuvre
E By the fovea alone

521 At try-in, a patient's "f" sounds like "v".
What does the "f/v" test check?
A Posterior palatal thickness
B The OVD alone
C Lower incisor position
D The post-dam
E That the upper incisal edges touch the wet–dry line of the lower lip (upper incisor length and labial position)

522 A patient whistles on "s" with new dentures.
What is a likely cause?
A A narrow anterior palate or anterior teeth set too far palatally
B OVD too low
C Over-extended lower flange
D Porcelain teeth
E Candida

523 A patient develops a lisp with a new upper denture.
Which feature is most likely to blame?
A An over-extended lower lingual flange
B A short post-dam
C Lower incisors set too far labially
D Excessive palatal thickness in the rugae area
E Porcelain posterior teeth

524 A new upper complete denture drops when the patient talks or opens wide.
What is the most likely cause?
A OVD too high
B An over-extended distobuccal flange (coronoid process)
C Lingualised occlusion
D Thin saliva
E An under-extended lower flange

525 A new upper denture is loose even at rest, with no displacement on opening.
What is the most likely cause?

A Excessive OVD
B Porcelain teeth
C An inadequate post-dam or peripheral seal
D Cheek biting
E An over-extended distobuccal flange

526 A new lower complete denture lifts when the patient swallows.
What is the most likely cause?
A Excessive OVD
B An over-extended lingual flange (mylohyoid) or retromolar region
C A short post-dam
D Porcelain teeth
E Candida

527 A patient's denture-bearing areas are sore all over after a week. The occlusion and extension look satisfactory, and she has anaemia.
Which causes of generalised soreness should be considered?
A An over-extended flange only
B Cheek biting only
C Excessive OVD, an RCP error, parafunction, inadequate support, or systemic factors such as anaemia or diabetes
D Monomer allergy only
E Thin saliva only

528 A patient has a single ulcer in the depth of the lower labial sulcus.
What is the most likely cause?
A Excessive OVD
B An occlusal interference
C Candida
D An RCP error
E An over-extended flange or sharp flange edge

529 A patient has an ulcer on the crest of the lower ridge under a new denture.
What is the most likely cause?
A An occlusal interference, a pressure spot or a sharp underlying bony crest
B An over-extended flange
C A short post-dam
D Lingualised occlusion
E A deep labial notch

530 A patient with a severely resorbed mandible has pain and numbness of the lower lip with a new denture.
What is the cause?
A Candida
B Excessive OVD
C Allergy

D The denture pressing on the mental nerve
E Anaemia

531 A patient bites his cheeks with new complete dentures. The posterior teeth meet edge to edge.
What is the cause?
A Lack of posterior buccal overjet
B Excessive OVD
C A short post-dam
D An over-extended lingual flange
E Candida

532 A patient's teeth click together when he talks.
What are the likely causes?
A Low OVD only
B Candida
C Lingualised occlusion
D A deep post-dam
E Excessive OVD, porcelain teeth or loose dentures

533 A patient gags with a new upper denture. The posterior border is thick and slightly over-extended.
What is the management?
A Add adhesive
B Increase the OVD
C Correct the posterior border, and use a training base, desensitisation or palateless (implant) designs if needed
D Remove the post-dam entirely
E Prescribe antiemetics only

534 A patient complains that food collects under her upper denture.
What is the most likely cause?
A Excessive OVD
B Porcelain teeth
C Cheek biting
D A poor peripheral seal or under-extension
E Lingualised occlusion

535 A denture wearer has a burning sensation under the upper denture.
Which causes should be considered?
A Excessive OVD only
B Denture stomatitis, xerostomia, haematinic deficiency or diabetes, and very rarely monomer allergy
C Occlusal error only
D Over-extension only
E It is always psychological

536 An 80-year-old denture wearer has angular cheilitis.
Which factors are most likely to contribute?
A Excessive OVD

B Porcelain teeth
C Over-extension
D Low OVD, Candida and haematinic deficiency
E A short post-dam

537 An upper complete denture has a midline fracture. The patient has a large midline torus and natural lower teeth.
What is the failure mechanism?
A Impact
B Fatigue from flexure over a midline fulcrum, worsened by opposing natural teeth
C Creep
D Corrosion
E Thermal shock

538 The occlusion of newly processed complete dentures is uneven.
What is the most accurate way to correct processing errors?
A A clinical remount: a new RCP record on the processed dentures, remounted and adjusted on the articulator
B Grinding in the mouth by eye
C A soft liner
D Ignoring it
E Adding self-cure acrylic to the teeth

539 An 84-year-old who has worn the same dentures for 15 years needs replacements because they are loose and worn.
Which technique best preserves her adaptation?
A Immediate dentures
B Implant-retained fixed bridges
C A conventional new design
D A neutral-zone denture
E Copy (replica) dentures, changing only what is necessary

540 A patient asks whether an immediate upper denture can be made without a labial flange.
What is the disadvantage of an open-face (socketed) immediate denture?
A It looks worse
B It is heavier
C Poor retention, and it cannot be relined labially
D It causes more swelling
E It needs more appointments to fit

541 An immediate upper denture is fitted after extractions.
What is the correct aftercare?
A Remove it at night from the first day
B Review in 6 months
C Leave it in for the first 24 hours, review the next day, and reline at about 3–6 months
D Never reline it
E Replace it after 1 month

542 Two lower canines with good support remain under a planned complete lower denture.
How are they prepared as overdenture abutments?
A Left full height with crowns
B Extracted
C Reduced flush with the bone
D Root-filled and reduced to 2–3 mm above the gingiva with domed surfaces
E Splinted with a bridge

543 A patient with overdenture abutments asks how to protect them.
What is the key preventive measure?
A Antibiotics
B Fluoride gel placed in the fitting surface of the denture, with meticulous hygiene
C Wearing the denture continuously
D Avoiding brushing the roots
E Chlorhexidine only

544 A 70-year-old with an atrophic mandible has struggled with lower complete dentures.
What do the McGill (2002) and York (2009) consensus statements recommend?
A A two-implant mandibular overdenture as the minimum standard of care
B A conventional denture as the standard
C Four implants as the minimum
D A fixed implant bridge
E Denture adhesive

545 An upper complete denture opposes lower natural anterior teeth and a lower free-end RPD that is rarely worn.
Which are features of combination syndrome (Kelly)?
A Anterior maxillary resorption with a flabby ridge
B Tuberosity enlargement and papillary hyperplasia
C Over-eruption of the lower anteriors and resorption under the lower free-end saddles
D Both A and B only
E All of the above

546 You are planning for the patient in the previous question.
How is combination syndrome best prevented?
A Higher OVD
B Porcelain teeth
C Removing the lower RPD
D Heavy anterior contact
E Restoring posterior support (a well-fitting lower RPD) and reducing anterior loading

547 An upper complete denture is needed over a flabby anterior ridge.
Which impression technique is appropriate?
A A windowed special tray: a compressive material for the firm ridge, then low-viscosity material or plaster over the flabby tissue
B Compound throughout
C Putty in a stock tray

D Alginate with finger pressure
E Heavy-bodied polyether, compressing the tissue

548 A lab note says a denture was "rebased" rather than "relined".
What is the difference?
A They are identical
B A rebase adds a thin layer to the fitting surface
C A reline replaces the teeth
D A rebase replaces the whole base; a reline adds a new fitting surface
E A reline changes the occlusion

549 After a closed-mouth laboratory reline, the patient's dentures feel "too tall".
What has likely happened?
A The teeth have worn
B The palate is thinner
C The patient bit too firmly during the reline impression, increasing the OVD
D The post-dam was lost
E Candida

550 A patient's dentures are unstable because of enlarged tuberosities and a shallow sulcus. Surgery is being considered.
At which level is pre-prosthetic surgery usually provided?
A Level 1
B Level 3
C Level 2
D By the dental technician
E It is never done

551 A chairside hard reline is planned for a loose lower denture.
What is its main drawback compared with a laboratory reline?
A It needs a second visit
B It is more expensive
C It increases OVD automatically
D The material is more porous
E It cannot be polished at all

552 A new lower denture keeps lifting because its lingual flange is over-extended. A colleague suggests adhesive and a higher OVD.
Why is this wrong?
A Adhesive is never acceptable
B A higher OVD always helps
C Adhesive and OVD changes rarely cure an over-extended flange; correct the border
D The patient needs implants
E It is the correct approach

553 A patient with failing complete dentures asks about implants at a Level 2 practice.
What is appropriate?

A Place implants at once
B Refuse to discuss implants
C Tell the patient implants are never suitable
D Offer adhesive only
E Discuss implant-assisted dentures as an option and refer for implant surgery

554 At the try-in, only the left posterior teeth contact and the mandible slides on closure.
What does this indicate?
A Excessive OVD
B An RCP registration error
C A short post-dam
D Porcelain teeth
E Thin saliva

555 An upper denture was retentive in wax at the try-in but is loose after processing.
Which step is likely to have been missed?
A Cutting the post-dam on the master cast along the marked vibrating line
B Relining
C Changing the shade
D Adding porcelain teeth
E Increasing OVD

556 You use a Fox's plane guide at the jaw registration visit.
What does it check?
A The occlusal plane against the interpupillary and ala–tragal lines
B The OVD
C The shade
D The centre line
E The post-dam depth

557 A patient bites his tongue and the lower denture lifts. The lower posterior teeth are set lingual to the ridge crest.
What is the correction?
A Increase the OVD
B Add a post-dam
C Use porcelain teeth
D Reset the teeth over the ridge crest, in the neutral zone
E Shorten the labial flange

558 An 82-year-old with early dementia has loose but familiar dentures she manages well.
Which approach is most appropriate?
A Implant-retained dentures
B A conventional new design
C Copy dentures with minimal changes
D No dentures
E Immediate dentures

559 A patient needs an upper complete denture against a full natural lower dentition.
Which design feature reduces the risk of fracture?
A A thin acrylic palate
B A metal (Co-Cr) or reinforced palate, with balanced contacts
C Porcelain teeth
D No post-dam
E A soft liner

560 A student asks what mechanical retention means for a complete denture.
Which are mechanical retentive factors?
A Adhesion and cohesion
B Atmospheric pressure
C The neutral zone
D Surface tension
E Undercuts, implants and adhesives

561 With new dentures in, a patient's lips are apart at rest and look strained.
What is the most likely cause?
A OVD too low
B OVD too high
C A short post-dam
D Candida
E Over-extension of the lower flange

562 A patient has generalised soreness, clicking teeth and no freeway space with new dentures.
What is the definitive management?
A Adhesive
B A soft liner
C Analgesics
D Leaving them out for a week
E Re-establish the OVD with a new jaw record and reset or remake the dentures

563 A patient who had an immediate denture fitted removed it the same evening and now cannot reinsert it.
Why should an immediate denture be left in for 24 hours?
A To improve the shade
B To stop it fracturing
C To allow relining
D It acts as a splint and limits swelling
E To reduce caries

564 A 75-year-old, edentulous for 20 years, has a severely resorbed mandible and has failed with several well-made lower dentures.
What is the most appropriate next step?
A Another conventional denture
B Denture adhesive only
C Assessment and referral for a two-implant overdenture

D A soft liner
E Accept the situation

565 An edentulous patient asks whether she still needs dental check-ups.
What is the correct advice?
A Yes: regular review of the oral mucosa and denture fit
B No; edentulous patients need no recall
C Only if in pain
D Every 10 years
E Only for denture repairs

566 Three weeks after new dentures, a patient says they are secure except with very fibrous food. Examination finds no fault.
What is the best response?
A Remake the dentures
B Increase the OVD
C Explain realistic expectations and the adaptation period
D Prescribe adhesive at once
E Refer for implants immediately

567 You are selecting the mould of the upper anterior teeth.
Which anatomical guides help choose the width of the six anterior teeth?
A The incisive papilla only
B The canine lines and inter-alar width
C The lower lip line
D The palatal vault
E The post-dam

568 A laboratory asks which articulator to use for routine complete dentures.
What is usually adequate?
A An average-value articulator
B A fully adjustable articulator with a kinematic facebow
C A simple hinge
D No articulator
E A fully programmed digital articulator

569 A student asks which articulator uses a Bonwill triangle with sides of about 10 cm.
What is the answer?
A A simple hinge
B A fully adjustable articulator
C A semi-adjustable arcon
D An average-value articulator
E A virtual articulator only

570 You write the laboratory prescription for new complete dentures.
What should it include?
A The tooth mould and shade

B The occlusal scheme
C The post-dam and any relief areas
D Both A and B only
E All of the above

Block H — Fixed prosthodontics (Q571–670)

> *Notebook Unit 8 · LO 2c.2, 2c.4, 2c.6, 2c.11, 2c.13–2c.14, 2c.20 (with 2a.16, 2b.6, 2b.10) · Allow 120 minutes · FDI notation*

571 A colleague prepares a vital upper central incisor of a 22-year-old for an all-ceramic crown with 2.5 mm of axial reduction.
Which Shillingburg principle has been neglected?
A Retention and resistance form
B Preservation of tooth structure and the pulp
C Structural durability
D Marginal integrity
E Preservation of the periodontium

572 You assess the convergence of a molar crown preparation.
What total occlusal convergence is typical and acceptable clinically?
A 0°
B 2–4°
C 30–40°
D 10–20° (theoretical ideal about 6°)
E 45°

573 A lower molar preparation is only 2.5 mm high after reduction.
What minimum axial wall heights are usually recommended?
A About 3 mm for incisors and premolars, 4 mm for molars
B 1 mm for all teeth
C 6 mm for molars
D 2 mm for molars
E Height does not matter

574 A wide molar preparation is short.
Which ratio improves resistance form?
A Height-to-width of 0.1
B Width-to-height of 2
C Taper-to-height of 1
D Height-to-taper of 0.4
E Height-to-width of at least 0.4

575 A crown on a short, tapered molar preparation keeps coming off.
What can improve retention and resistance?
A A thicker cement layer

B More taper
C Grooves or boxes, an adhesive cement, or crown lengthening
D A temporary cement
E A deeper margin alone

576 You are preparing a vital molar for a crown.
Which measures protect the pulp?
A Sharp burs, copious water spray, light intermittent pressure, avoiding desiccation and well-fitting temporaries
B Dry cutting for visibility
C Heavy continuous pressure
D Blunt burs
E Leaving the tooth without a temporary

577 You prepare a lower molar for a full gold crown.
Where is the functional cusp bevel placed?
A Lingual cusps of lower molars
B Buccal cusps of upper molars
C Buccal cusps of lower molars (and palatal cusps of upper molars)
D All cusps equally
E No bevel is needed

578 You plan a full-coverage cast metal crown on 36.
What reduction and margin are appropriate?
A 3 mm occlusal reduction and a shoulder
B 0.5 mm occlusal reduction and a knife edge
C 2 mm occlusal reduction and a heavy chamfer
D No occlusal reduction
E 1.5 mm on functional and 1 mm on non-functional cusps, with a light chamfer margin

579 You prepare 21 for a metal–ceramic crown.
What margin design is typical?
A Knife edge all round
B Labial shoulder (with or without a bevel) or heavy chamfer; lingual chamfer
C Lingual shoulder only
D Feather edge labially
E No defined margin

580 You prepare 11 for a lithium disilicate crown.
What margin is typical?
A Knife edge
B Feather edge
C Bevelled shoulder with sharp internal angles
D Rounded shoulder or deep chamfer of about 1 mm, with rounded internal angles
E No margin

581 You prepare 12 for a porcelain veneer.
How deep is the labial reduction?
A 1.5–2 mm
B No reduction ever
C 1.2 mm into dentine
D 0.3–0.7 mm, staying within enamel
E 3 mm

582 You are deciding where to place the margin of a molar crown with no aesthetic concern.
Where is it best placed?
A Supragingivally
B Deeply subgingivally
C On the bone crest
D At the furcation
E 3 mm subgingivally

583 You plan a subgingival margin on 21 for aesthetics.
How far should it stay from the bone crest?
A 0.5 mm
B 1 mm
C On the crest
D 5 mm
E About 3 mm (a supracrestal attachment of about 2 mm plus the sulcus)

584 A crown margin placed 1 mm from the bone crest has caused chronic gingival inflammation.
What has happened?
A Allergy to the crown
B Caries
C Invasion of the supracrestal tissue attachment, causing chronic inflammation, recession or bone loss
D Normal healing
E Occlusal trauma only

585 A crown on 46 is bulbous at its buccal furcation area, and the gingiva bleeds.
What is the cause?
A Undercontour
B Overcontour trapping plaque; emergence profiles should be flat
C Wrong shade
D Occlusal overload
E Cement type

586 You need gingival retraction for 24 in a patient with poorly controlled hypertension.
Which retraction medicament should be avoided?
A Aluminium chloride
B Ferric sulphate
C Plain cord
D Retraction paste
E Adrenaline-impregnated cord

587 A student asks why provisional crowns are needed.
What are their purposes?
A Only aesthetics
B To protect the pulp, keep tooth position, protect the gingiva, restore function and appearance, and test the planned shape or OVD
C Only to keep the patient happy
D To retain the definitive cement
E None

588 You make chairside provisional crowns.
Which material is commonly used, and why?
A PMMA, for its low exotherm
B Zinc oxide–eugenol
C Metal crowns only
D Bis-acryl composite, for its low exotherm and good strength
E Wax

589 A lithium disilicate crown will be resin-bonded next week.
Which temporary cement should be used now?
A A eugenol-free temporary cement
B Zinc oxide–eugenol
C Zinc phosphate
D Glass ionomer
E Resin cement

590 You are restoring a single crown on 36 in a patient with a stable, comfortable ICP.
Which occlusal approach is appropriate?
A Reorganise to RCP
B Increase the OVD
C Conformative: restore to the existing ICP
D Full-mouth equilibration
E Canine guidance on all teeth

591 Extensive restoration is planned with a change of OVD, and the ICP is unstable.
Which approach and level apply?
A Conformative, Level 1
B No approach needed
C Reorganised (RCP), usually Level 3 when major
D Conformative, Level 2
E Hygienist level

592 You need a shade for a crown on 11.
When should the shade be taken?
A After preparation
B At the fit appointment
C After the impression

D Under the operating light
E Before preparation, because teeth dehydrate and lighten during treatment

593 You are choosing a shade for an anterior crown.
Which dimension of colour is most important?
A Hue
B Value
C Chroma
D Translucency
E Fluorescence

594 A colleague takes a dual-arch (triple) tray impression for a 4-unit bridge.
Why is this inappropriate?
A Dual-arch trays suit single units with stable intercuspation only
B They are too accurate
C They cannot record margins
D They are only for dentures
E They need alginate

595 The laboratory applies die spacer to the die.
What thickness is typical, and where does it stop?
A 200 µm, over the margin
B 1 mm, everywhere
C None is needed
D About 20–40 µm, stopping short of the margin
E 5 µm, on the margin only

596 A lower first molar has had root canal treatment and has an MOD cavity.
What restoration improves survival?
A An amalgam without cusp coverage
B Cuspal coverage (an onlay or crown)
C A composite without coverage
D A veneer
E No restoration

597 An upper central incisor has had root canal treatment through a small access cavity, with little other tissue loss.
What restoration is appropriate?
A A composite restoration of the access cavity
B A post and crown
C A full metal–ceramic crown
D Extraction
E A veneer and post

598 A root-filled lower molar has a large pulp chamber and adequate remaining walls.
How can the core be retained without a post?
A It cannot

B With a threaded post only
C A Nayyar core (amalgam or composite retained in the pulp chamber)
D With pins only
E With temporary cement

599 A molar has just had root canal treatment and will be crowned in 3 months.
What is important meanwhile?
A Leave the access open
B A cotton pellet only
C Nothing
D A good coronal seal placed promptly, because coronal leakage contributes to failure
E Antibiotics

600 A root-filled upper premolar has no ferrule above the bone.
Which option can create a ferrule?
A A longer post
B A wider post
C A metal core alone
D Resin cement
E Surgical crown lengthening or orthodontic extrusion, with appropriate specialist input

601 A student mixes up bridge terms.
What is a retainer?
A The replacement tooth
B The joint between units
C The supporting tooth
D The edentulous space
E The restoration on the abutment tooth that holds the bridge

602 You assess possible bridge abutments.
Which root form makes the best abutment?
A Conical, fused roots
B Short single roots
C Long roots, and divergent multi-rooted forms
D Roots with resorption
E Dilacerated roots

603 A student quotes Ante's law.
What does it state?
A The pontic should be wider than the abutment
B Cantilevers are always contraindicated
C Bridges should be no longer than three units
D The total root surface area of the abutments should equal or exceed that of the teeth being replaced
E Abutments must be vital

604 An examiner asks about the status of Ante's law.
What is correct?

A It is supported by strong trial evidence
B It is historical and not well supported by evidence, though still examined
C It is a legal requirement
D It applies only to implants
E It has never been taught

605 You are choosing a single-rooted upper abutment for a cantilever.
Which single-rooted upper tooth has the largest root surface area?
A Canine
B Central incisor
C Lateral incisor
D First premolar
E Second premolar

606 A bridge span is doubled in length with the same connectors.
How does its deflection change?
A It doubles
B It halves
C It quadruples
D It increases about eightfold
E It is unchanged

607 A technician doubles the occlusogingival height of a bridge connector.
How does deflection change?
A It doubles
B It halves
C It quadruples
D It is unchanged
E It falls to about one-eighth

608 A bridge replaces all four upper incisors between the canines.
What extra mechanical problem does arch curvature create?
A The pontics lie outside the line joining the abutments and act as a lever arm
B The pontics are too small
C The connectors become thicker
D The canines are too long
E None

609 A fixed-fixed conventional bridge is planned between two heavily restored abutments.
What is a recognised drawback of this design?
A The least tooth destruction
B No need for parallel preparations
C The most tooth destruction, the need for parallel preparations, and a loose retainer may go unnoticed
D No retainers
E It cannot be used for bounded spans

610 A young patient has lost 12; 13 is sound with a long root.
Which conventional design is suitable?
A A fixed-fixed bridge from 11 to 13
B A cantilever from 13 replacing 12
C A distal cantilever from 11
D A spring cantilever
E A fixed-movable bridge from 14

611 You are designing a cantilever bridge.
What rules apply?
A The pontic should be no larger than the abutment; avoid long or posterior distal cantilevers
B The pontic should be larger than the abutment
C Long distal cantilevers are preferred
D Use two pontics
E There are no rules

612 47 is tilted mesially and will be the distal abutment of a bridge replacing 46, with 45 as the mesial abutment.
Which design avoids the need for parallel preparations?
A A fixed-fixed bridge
B A fixed-movable bridge
C A cantilever from 47
D A spring cantilever
E A conventional fixed-fixed with deeper preparations

613 You design a fixed-movable bridge.
Where is the keyway placed?
A In the mesial of the distal retainer
B In the pontic
C In the distal of the mesial (minor) retainer, so mesial drift seats the key
D In the occlusal surface
E There is no keyway

614 A student describes an old bridge with a palatal bar from a premolar to a distant incisor pontic.
What is it, and why is it rarely used?
A A Maryland bridge; too strong
B A fixed-movable bridge; too flexible
C A resin-bonded cantilever; poor aesthetics
D A spring cantilever; the palatal bar is unhygienic
E A hybrid bridge; too costly

615 A 24-year-old has lost 22. 23 is unrestored with plenty of enamel, and the occlusion is favourable.
What is usually the first choice?
A A conventional fixed-fixed bridge
B An implant immediately
C A cantilever metal–ceramic bridge

D An acrylic RPD
E A single-wing resin-bonded cantilever bridge

616 A two-winged fixed-fixed resin-bonded bridge has been in place for 4 years. The patient reports a bad taste.
What is the likely problem?
A Wing fracture
B Pontic fracture
C Differential movement has debonded one wing unnoticed, leading to caries
D Allergy
E Normal ageing

617 A bridge has a conventional retainer at one end and a resin-bonded retainer at the other.
What is this design called, and what joint is usually used?
A Fixed-fixed; rigid
B Spring cantilever; none
C Maryland; rigid
D Hybrid bridge; usually a movable joint
E Cantilever; none

618 A five-unit bridge from 13 to 17 has a pier abutment at 15. One terminal retainer keeps debonding.
Why, and what is the solution?
A The span is too short; add a pontic
B The pier acts as a fulcrum; use a non-rigid connector at the pier or separate bridges
C A stiffer, thicker bridge
D Temporary cement
E Remove the pier abutment

619 A patient asks why a resin-bonded bridge is recommended over a conventional bridge.
What are its advantages?
A Minimal or no preparation and no anaesthesia
B Reversible and quicker, with the pulp not at risk
C Failures are usually rebondable
D Both A and B only
E All of the above

620 A resin-bonded wing is planned on 23 but the occlusion is tight.
How much clearance does a Co-Cr wing need, and how can it be gained without preparation?
A About 0.7 mm; using a Dahl approach
B About 2 mm; by extracting the opposing tooth
C None
D About 3 mm; by orthodontics
E About 0.1 mm; by grinding the wing thin

621 You are designing a resin-bonded cantilever bridge from 13 replacing 12.
Which design features improve its success?
A A thin wing covering the minimum enamel

B No rest
C Two wings for safety
D Maximum enamel coverage within aesthetic limits, about 180° wraparound, a cingulum rest and minimal preparation for a path
E A labial wing

622 You are cementing a Co-Cr resin-bonded bridge wing.
What is the correct bonding protocol?
A Zinc phosphate on dry enamel
B Glass ionomer without isolation
C Temporary cement first
D Hydrofluoric acid on the metal
E Air-abrade the metal (50 µm alumina), etch the enamel, and use an MDP-containing resin cement under rubber dam

623 A 45-year-old bruxist with heavy anterior contacts asks for a resin-bonded bridge replacing 21.
What is the main concern?
A Too much enamel
B Bruxism and heavy occlusion make debonding more likely
C Age
D The lack of anaesthesia
E The cost

624 A single-wing resin-bonded bridge debonds after 3 years. The wing and pontic are intact and the abutment is sound.
What is the usual management?
A Clean and rebond it
B Extract the abutment
C Make a conventional bridge immediately
D Place an implant
E Leave the space

625 A pontic is needed for 46 in a patient with no aesthetic concern.
Which pontic design is most hygienic?
A Saddle
B Ovate
C Sanitary (hygienic), with no ridge contact and at least a 3 mm gap
D Ridge lap
E Modified ridge lap

626 A pontic replaces 14 in a patient who smiles broadly.
Which design balances appearance and hygiene?
A Sanitary
B Saddle
C Conical
D Ovate in a thin ridge
E Modified ridge lap, contacting the labial side of the ridge only

627 A patient with a thick ridge after socket preservation wants the most natural emergence for a 21 pontic.
Which pontic is indicated?
A Sanitary
B Saddle
C Ovate, sitting in a concavity in the ridge
D Conical
E Hygienic

628 A bridge has a concave pontic covering the whole ridge, and the mucosa beneath is inflamed.
What is this pontic, and why has it failed?
A A saddle (ridge lap) pontic; it is unhygienic
B An ovate pontic; it is too small
C A sanitary pontic; it traps food
D A modified ridge lap; it is too narrow
E A conical pontic; it is too short

629 You check a new bridge pontic at try-in.
How should its tissue surface relate to the mucosa?
A It should press hard enough to blanch the mucosa
B Convex, smooth and glazed or highly polished, without pressing on the mucosa
C Concave and unglazed
D Rough to aid adhesion
E Separated by 1 mm in the aesthetic zone

630 A posterior bridge is planned for a patient with heavy occlusal forces.
How can load on the pontic be reduced?
A Widen the pontic
B Use steep cusps
C Increase the OVD
D Narrow the buccolingual width of the pontic
E Use porcelain occlusal surfaces

631 A three-unit all-ceramic bridge fractures.
Where is the failure most likely?
A The pontic centre
B The connector
C The occlusal surface
D The margin
E The abutment

632 A lithium disilicate bridge is planned from 13 to 15 replacing 14.
What is the minimum connector cross-section, and how far back can such a bridge extend?
A About 16 mm^2 in the premolar region; three units, no further back than the second premolar
B 4 mm^2; any length
C 9 mm^2; to the second molar

D 25 mm²; five units
E No minimum

633 A three-unit posterior zirconia bridge is planned.
What approximate minimum connector cross-section is needed?
A 2 mm²
B 4 mm²
C 30 mm²
D About 9 mm²
E 50 mm²

634 A technician can enlarge a connector either in height or in width.
Which matters more, and why?
A Width, because strength rises with the cube of width
B Neither
C Width, for aesthetics
D Both equally
E Height, because strength rises with the square of height and deflection falls with its cube

635 A 30-year-old has tetracycline staining that has not responded to bleaching and wants better-looking teeth.
Which indication supports porcelain veneers?
A Heavy bruxism
B Class III incisors
C Discolouration unresponsive to bleaching
D Little enamel
E Poor oral hygiene

636 A patient asks for veneers on her upper incisors.
Which finding contraindicates them?
A Bruxism with an edge-to-edge incisor relationship
B Mild discolouration
C Small diastemas
D Good enamel thickness
E A request for minor shape change

637 You are choosing a preparation design for upper incisor veneers.
Which design is most common, for stress distribution and seating?
A Window
B Feathered incisal
C Butt joint
D Full crown
E Incisal overlap with a palatal finish line

638 You are fitting feldspathic porcelain veneers.
What is the bonding protocol?
A Zinc phosphate

B Glass ionomer
C Air abrasion and MDP
D HF-etched, silanated ceramic and light-cured resin cement on etched enamel
E Temporary cement

639 A lower first premolar has an MOD restoration, and its buccal cusp is less than half as wide as it is high.
What restoration is indicated?
A An intracoronal inlay
B A larger amalgam
C Cusp coverage with an onlay
D A veneer
E No restoration

640 You prepare a cavity for a gold inlay.
How should the walls be shaped?
A Undercut for retention
B Diverging occlusally by about 6–10°, with no undercuts
C Parallel
D Converging occlusally
E Rough and irregular

641 You write a laboratory prescription for a metal–ceramic bridge.
What must it contain?
A The patient's name only
B Only the shade
C A request to "do your best"
D The fee
E Patient and clinician details, materials, design, shade, margin, pontic form, connector type and the delivery date

642 A crown arrives from the laboratory.
What regulatory documentation should accompany it?
A None
B A receipt only
C A statement of manufacture, from a manufacturer registered with the MHRA
D A CE mark only
E A GDC certificate

643 A dental technician asks whether she must register with a professional regulator.
Who regulates dental technicians and clinical dental technicians?
A The GDC
B The MHRA
C The CQC
D The BDA
E No one

644 A patient asks for a "Hollywood smile".
In what order should the smile be analysed?
A Teeth first, then face
B From outside inward: face and midline, lips and smile line, incisal display and plane, then tooth proportions, shade and gingival contour
C Shade only
D Gingiva only
E Random order

645 A colleague insists every anterior restoration must follow the golden proportion.
What is correct?
A It is mandatory
B It always applies to natural smiles
C It is a legal requirement
D It is not a mandatory rule for a natural smile
E It applies only to veneers

646 A 25-year-old has a single darkened, root-filled upper incisor and asks for a crown.
What should be considered first?
A A crown
B Investigating the cause, then non-vital (internal) bleaching
C A veneer
D Extraction and an implant
E An RBB

647 A 19-year-old has white-spot lesions on the labial surfaces of the upper incisors after orthodontics.
What is the least invasive option?
A Crowns
B Veneers
C Composite build-ups
D Resin infiltration or microabrasion
E No treatment possible

648 A 17-year-old has peg-shaped upper lateral incisors after orthodontic space redistribution.
What is the most conservative restorative option?
A Crowns
B Porcelain veneers
C Additive direct composite
D Extraction and implants
E An RBB

649 A 30-year-old has crowded, rotated upper incisors and wants them "straightened" with crowns.
What should be offered first?
A Crowns
B Veneers
C Extraction

D Composite only
E Orthodontics, before any elective tooth preparation

650 A patient insists on porcelain veneers for her upper incisors.
What should be done before any preparation?
A A diagnostic wax-up and mock-up, assessment of enamel, parafunction and overbite, and a discussion of irreversibility and replacement
B Immediate preparation
C A shade guide only
D Whitening the day before bonding
E Nothing

651 A 28-year-old with sound, healthy upper incisors wants crowns because one is slightly darker.
Why are crowns inappropriate?
A They are too weak
B They are too cheap
C They are less conservative than whitening, orthodontics or composite, which can meet the aim
D They cannot match shade
E Crowns are illegal for aesthetic reasons

652 At a 5-year review, a bridge is still in place but has recurrent caries and recession.
How should its outcome be described?
A A success
B It has survived but is not a success
C A failure, because it is present
D A complete failure
E Not assessable

653 A heavy bruxist needs a posterior crown.
How does parafunction influence the plan?
A It does not
B Use layered porcelain
C Use temporary cement
D Choose feldspathic veneers
E It favours monolithic materials and protective measures such as a night guard

654 A patient needs crowns or bridges on most teeth with an OVD increase.
At what level is this care?
A Level 1
B Level 2
C Hygienist
D Level 3 (referral)
E Not provided

655 A patient with uncontrolled hypertension needs gingival retraction before an impression.
Which retraction method should be avoided?
A Adrenaline-impregnated cord

B Aluminium chloride cord
C Ferric sulphate
D Retraction paste
E Plain cord

656 You are taking a shade for 11 in a busy surgery.
Which conditions improve accuracy?
A Under the operating light, after preparation
B With the patient wearing lipstick
C After a 20-minute look
D In colour-corrected daylight, quickly, with photographs of shade tabs sent to the laboratory
E After rubber dam isolation

657 A colleague plans to scan a full arch digitally for a 10-unit bridge.
Which statement about intraoral scans is correct?
A They suit short-span work best
B They are most accurate for full arches
C They need no calibration
D They replace occlusal records
E They avoid all errors

658 A patient with a 3-unit fixed-fixed bridge notices a bad taste. One retainer has come loose but the bridge has not fallen out.
What is the main risk?
A Pontic fracture
B Allergy
C Gingival recession only
D Loss of the opposing tooth
E Caries under the loose retainer

659 A lower first molar has had root canal treatment and has an MOD cavity with thin cusps.
What is the appropriate restoration?
A An MOD composite without coverage
B Cuspal coverage with an onlay or crown
C A post and amalgam
D A veneer
E No restoration

660 You are choosing between an inlay, an onlay and a crown for a premolar with a large cavity.
Which statement is correct?
A A crown always preserves most tooth
B An inlay protects weak cusps
C An adhesive onlay preserves more tooth than a crown while covering the cusps
D Onlays cannot be bonded
E Crowns need no preparation

661 You are assessing the lower anterior teeth as abutments.
Which lower tooth has the smallest root surface area?
A Central incisor
B Lateral incisor
C Canine
D First premolar
E Second premolar

662 46 is missing. 45 is upright, but 47 has tipped mesially by 25°.
Which approach is appropriate?
A A fixed-fixed bridge with parallel preparations through 47
B A distal cantilever from 45
C Extract 47
D A fixed-movable bridge, or uprighting 47 orthodontically
E A spring cantilever

663 A patient wants 46 replaced by a cantilever pontic from 45 alone. He has a heavy bite.
What is the best advice?
A This is ideal
B Use a larger pontic
C Avoid it; long or posterior distal cantilevers torque the abutment
D Add a second pontic
E Use temporary cement

664 A rigid bridge from 13 to 17 (14 and 16 missing) has repeatedly debonded at 13.
What is the design problem and solution?
A Too few pontics; add another
B The bridge is too short
C The shade is wrong
D Temporary cement is needed
E 15 acts as a pier fulcrum; use a non-rigid connector at the pier or separate bridges

665 A patient with a deep overbite and bruxism requests upper incisor veneers.
What is the best response?
A Proceed with thin veneers
B Explain that heavy occlusion contraindicates veneers, and discuss alternatives
C Use temporary cement
D Prepare deeply into dentine
E Proceed without a mock-up

666 A 16-year-old finishes orthodontics with small lateral incisors and spaces, and growth is ongoing.
What is the preferred restoration?
A Porcelain veneers
B Crowns
C Direct composite build-ups
D Implants
E Conventional bridges

667 A patient has just finished whitening and wants composite bonding the next day.
Why should you wait?
A Whitening strengthens enamel
B The shade may still change and residual oxygen reduces the resin bond for a time
C Composite cannot be used after whitening
D Whitening etches enamel
E There is no reason to wait

668 You use a double-cord technique before an impression for 21.
What is its purpose?
A The first cord controls fluid and the second displaces the tissue until just before the impression
B To stop bleeding only
C To push the margin deeper
D To shrink the gingiva permanently
E To replace the impression material

669 A crown on 36 is overcontoured buccally over the furcation, and the gingiva there is chronically inflamed.
What is the definitive solution?
A Chlorhexidine
B Antibiotics
C Gingival surgery only
D Recontour or remake the crown with a flat emergence profile
E Ignore it

670 You review a 5-year-old bridge.
What should be checked?
A Biological health and margins
B Contacts, caries and cracks
C Occlusion, appearance and patient satisfaction
D Both A and B only
E All of the above

Block I — Occlusion and temporomandibular disorders (Q671–730)

> *Notebook Unit 9 · LO 2a.15, 2a.20, 2c.19 · Allow 72 minutes*

671 A student is asked to define ICP.
Which definition is correct?
A The first contact on the retruded arc
B The position of the mandible at rest
C The condyles in their most superior position
D Maximum intercuspation of the teeth, independent of condylar position
E The position used for complete dentures

672 You guide a patient's mandible along the terminal hinge arc until the first tooth contact.
What position is this?
A The retruded contact position (RCP)
B ICP
C Rest position
D Protrusive position
E Habitual position

673 A colleague talks about recording centric relation.
What is centric relation?
A Maximum intercuspation
B A tooth-guided position
C A jaw-to-jaw relation with the condyles in their most superior-anterior position against the eminence, discs correctly interposed, independent of the teeth
D The rest position
E The edge-to-edge position

674 On guiding a patient into RCP, you find a slide into ICP.
What is normal?
A A slide is always pathological
B A slide occurs in fewer than 10% of people
C A normal slide is 5 mm
D A slide must always be eliminated
E A slide of about 1 mm forwards occurs in about 90% of people

675 You measure a dentate patient's rest position.
What is the normal freeway space?
A 0 mm
B About 2–4 mm
C 6–8 mm
D 10 mm
E 0.5 mm

676 In right lateral excursion, only the right canines contact.
What is this guidance called?
A Group function
B Balanced occlusion
C A non-working interference
D Mutually protected occlusion only
E Canine guidance

677 In left lateral excursion, the left canine, premolars and mesiobuccal cusp of the first molar all contact.
What is this guidance called?
A Canine guidance
B Group function
C A non-working interference

D Balanced occlusion
E Deflective contact

678 A textbook describes an occlusion in which the anterior teeth protect the posteriors in excursions, and the posteriors protect the anteriors in ICP.
What is this?
A Balanced occlusion
B Group function
C Lingualised occlusion
D Mutually protected occlusion
E Monoplane occlusion

679 In right lateral excursion, the left second molars contact heavily.
What is this and when is it desirable?
A A non-working (balancing) interference; it is damaging in natural dentitions and desirable only in complete dentures
B A working-side contact; always desirable
C Canine guidance
D Group function; ideal
E Anterior guidance

680 In right lateral excursion, the right first molars contact before the right canine.
What is this?
A Group function
B Canine guidance
C A working-side interference
D A non-working interference
E Balanced occlusion

681 On closing, a contact on 17 diverts the mandible forward and sideways.
What is this?
A A deflective contact
B Canine guidance
C Group function
D Freeway space
E The Bennett angle

682 A technician needs the angle of the non-working condyle path to the sagittal plane.
What is this, and its typical value?
A The condylar guidance angle; 45°
B The Frankfort angle; 30°
C The Bennett angle; about 7–15°
D The ANB angle; 3°
E The incisal guidance angle; 60°

683 You inspect the mediolateral curvature of the posterior occlusal plane.
What is it called?

A Curve of Spee
B Compensating curve
C Bonwill triangle
D Balkwill angle
E Curve of Wilson

684 A patient's maximum mouth opening is 34 mm.
How should this be interpreted?
A Normal
B Restricted (normal is about 40–55 mm)
C Hypermobile
D Normal for women
E Excessive

685 You record mandibular movements.
What are typical lateral excursion and protrusion ranges?
A 1–2 mm
B 20 mm each
C 4 mm lateral only
D About 8–10 mm each
E 30 mm

686 A patient opens with pure hinge rotation of the condyles before translation.
About how much incisal opening does pure rotation give?
A 5 mm
B 20–25 mm
C 40 mm
D 55 mm
E 70 mm

687 You review the features of an "ideal" functional occlusion before restoration.
Which features apply?
A A stable ICP with even, simultaneous contacts
B Axial loading of posterior teeth
C Anterior guidance discluding the posteriors, with no non-working interferences
D Both A and B only
E All of the above

688 A 30-year-old has aching cheeks, worse in the morning, and familiar pain when the masseters are palpated. Joint movements are painless.
What is the DC/TMD diagnosis?
A Myalgia
B Arthralgia
C Disc displacement with reduction
D Degenerative joint disease
E Subluxation

689 A patient has pain localised to the left TMJ on palpation and when moving the jaw.
What is the DC/TMD diagnosis?
A Myalgia
B Disc displacement with reduction
C Arthralgia
D Headache attributed to TMD
E Subluxation

690 A patient has a click on opening and another on closing (reciprocal clicking), without pain or locking.
What is the DC/TMD diagnosis?
A Disc displacement without reduction
B Degenerative joint disease
C Subluxation
D Disc displacement with reduction
E Myalgia

691 A patient has reciprocal clicking and occasional brief episodes when the jaw catches and will not open fully.
What is the DC/TMD diagnosis?
A Disc displacement without reduction, with limited opening
B Myalgia
C Disc displacement with reduction, with intermittent locking
D Subluxation
E Degenerative joint disease

692 A patient's click stopped suddenly a week ago. She now opens 28 mm, deviating to the right, with restricted left lateral movement.
What is the diagnosis?
A Disc displacement with reduction
B Subluxation
C Myalgia
D Disc displacement without reduction, with limited opening (right closed lock)
E Degenerative joint disease

693 A patient had a closed lock 2 years ago. She now opens 44 mm without clicking.
What is the diagnosis?
A Disc displacement with reduction
B Disc displacement without reduction, without limited opening
C Subluxation
D Myalgia
E Normal joint

694 A 68-year-old has crepitus in both TMJs, and a radiograph shows flattening and osteophytes on the condyles.
What is the diagnosis?
A Degenerative joint disease
B Disc displacement with reduction

C Myalgia
D Subluxation
E Arthralgia only

695 After a wide yawn, a patient cannot close her mouth. The condyles are palpable in front of the eminences.
What is the diagnosis?
A Closed lock
B Disc displacement with reduction
C Myalgia
D Degenerative joint disease
E Dislocation (open lock)

696 A patient has temporal headaches that worsen with chewing and are reproduced by palpating the temporalis.
What is the DC/TMD diagnosis?
A Headache attributed to TMD
B Migraine
C Giant cell arteritis
D Trigeminal neuralgia
E Arthralgia

697 A patient asks whether her "bad bite" caused her TMD.
What is the current understanding of TMD aetiology?
A Malocclusion is the main cause
B It is purely genetic
C It is multifactorial and biopsychosocial; occlusal factors have only a weak association
D It is always caused by trauma
E It is infective

698 You are taking the history of a patient with TMD.
In which group is TMD commonest?
A Men over 60
B Children under 10
C Men aged 20–40
D People over 80
E Women aged about 20–40

699 A 25-year-old has a painless click on opening, with no locking or limitation.
What is the appropriate management?
A An anterior repositioning splint full-time
B Occlusal equilibration
C An MRI scan
D Reassurance only
E Referral for surgery

700 A patient with masticatory myalgia asks what she can do herself.
Which self-care advice is appropriate?
A Chew gum to exercise the muscles
B A soft diet, avoiding wide opening and gum, warm compresses, jaw rest, and keeping "lips together, teeth apart"
C Open as wide as possible several times a day
D Avoid all analgesics
E Sleep on the painful side

701 A stabilisation splint is planned for a patient with myalgia.
Which design is correct?
A Soft vacuum-formed, worn full-time
B Anterior repositioning, worn full-time
C Partial-coverage anterior bite plane worn continuously
D A posterior-only splint
E Full-coverage hard acrylic (often maxillary), with even contacts in RCP and canine guidance

702 A patient wants a soft splint for jaw pain because it is cheaper.
What is the concern?
A Soft splints are illegal
B They are for short-term use and may increase clenching in some patients
C They cause caries
D They are always more effective
E They need monthly replacement

703 A patient has worn an anterior repositioning splint day and night for 9 months for a clicking joint.
What complication is likely?
A Anterior open bite
B Increased overbite
C Canine wear
D Posterior open bite
E Gingival recession

704 A patient has worn an anterior bite plane continuously for a year.
What complication may follow?
A Anterior open bite
B Increased freeway space
C Over-eruption of the uncovered posterior teeth
D Recession of the incisors
E Root resorption of the canines

705 A patient's left TMJ locked 2 days ago, with sudden loss of clicking and restricted opening.
What might be tried first?
A Manipulation to try to recapture the disc, with reassurance, a splint and exercises
B Immediate surgery
C Occlusal grinding

D Extraction of the third molars
E Nothing

706 A patient's jaw dislocates during an extraction. The condyles are anterior to the eminences.
How is it reduced?
A Pull the chin forwards
B Push the chin upwards
C Wait for spontaneous reduction
D Press the mandible downwards and backwards, with the thumbs on the lower molars or external oblique ridges
E Ask the patient to bite hard

707 A 72-year-old has new jaw pain on chewing that eases with rest, temporal headaches and a tender scalp.
What is the most appropriate action?
A Same-day medical referral for suspected giant cell arteritis (urgent ESR or CRP)
B A stabilisation splint
C Ibuprofen and review in a month
D Occlusal adjustment
E TMJ MRI

708 A 58-year-old has progressive trismus, numbness of the lower lip, unexplained weight loss and earache with a normal ear examination.
What must be excluded?
A Myalgia
B Malignancy (for example nasopharyngeal, parotid or infratemporal)
C Disc displacement with reduction
D Bruxism
E Dental caries

709 A 64-year-old has jaw ache that comes on when climbing stairs and settles with rest.
What must be considered?
A Myalgia
B Pulpitis
C Disc displacement
D Sinusitis
E Cardiac pain referred to the jaw

710 A patient has brief, electric-shock pains in the right cheek triggered by light touch when shaving.
What is the likely diagnosis and treatment?
A Myalgia; splint
B Arthralgia; NSAIDs
C Trigeminal neuralgia; carbamazepine via the GMP or a specialist
D Pulpitis; RCT
E Sinusitis; antibiotics

711 A colleague offers selective occlusal grinding to "cure" a patient's TMD.
What is the evidence position?
A It is first-line treatment
B It is irreversible and not an evidence-based TMD treatment
C It is required before splints
D It cures disc displacement
E It is reversible

712 A patient's myofascial pain persists after 6 months of self-care, splint therapy and physiotherapy.
Which secondary-care options exist?
A Low-dose tricyclics
B Botulinum toxin
C Arthrocentesis or arthroscopy for joint disorders
D Both A and B only
E All of the above

713 A patient's chronic TMD pain persists despite conservative care.
Who would usually start low-dose tricyclic medication?
A The GMP or specialist
B The dental nurse
C The technician
D The patient alone
E The pharmacist without prescription

714 You examine a patient who says she grinds her teeth.
Which signs suggest bruxism?
A Erosion cupping only
B Abrasion notches
C Root caries
D Attrition facets, fractured restorations and cusps, masseteric hypertrophy, a scalloped tongue and linea alba
E Gingival recession only

715 A known bruxist needs new posterior crowns.
How should the new work be protected?
A Layered feldspathic porcelain
B No protection is needed
C A hard night-time splint, with metal or monolithic zirconia occlusal surfaces
D A soft daytime splint
E Temporary crowns only

716 Multiple posterior crowns are planned.
What should precede them?
A Occlusal grinding
B A soft splint
C Examination on mounted casts to find interferences, the slide and the guidance

D Extraction of the third molars
E Nothing

717 A patient needs restorations on most teeth with a planned OVD increase, and has an unstable ICP.
Which approach is appropriate?
A Conform to ICP
B Restore one tooth at a time without a plan
C Ignore the occlusion
D Reorganise to RCP
E Splint therapy only

718 A colleague plans occlusal equilibration.
When is it justified?
A Only to remove specific interferences for restorative reasons
B As a routine TMD cure
C In all new patients
D To prevent caries
E Before any splint

719 A 28-year-old has typical masticatory myalgia with no red flags.
Is routine imaging needed?
A Yes, MRI in all cases
B No; imaging is targeted to specific questions such as bone pathology or an alternative diagnosis
C Yes, a panoramic radiograph always
D Yes, CBCT always
E Yes, a skull radiograph

720 A closed lock of 4 months has not improved with self-care, a splint and exercises.
What is appropriate?
A Continue indefinitely
B Occlusal grinding
C Extraction of molars
D Full-mouth crowns
E Referral

721 A heavy bruxist with sound canines is having new posterior crowns.
Why is canine guidance favoured?
A Canine guidance discludes the posterior teeth, reducing lateral load on the new crowns
B It increases posterior contact
C It improves appearance
D It reduces OVD
E It prevents caries

722 A patient with myofascial pain and high stress asks what else may help.
Which adjuncts are appropriate?
A Physiotherapy and jaw exercises
B Cognitive behavioural therapy

C Stress management
D Both A and B only
E All of the above

723 A patient with acute TMJ arthralgia asks about pain relief.
What is appropriate?
A Long-term opioids
B Antibiotics
C Simple analgesia, such as short-term NSAIDs
D Steroid injections at the first visit
E No medication

724 A patient has an acute left disc displacement without reduction.
Which way does the mandible deviate on opening?
A Right
B Left (towards the affected side)
C No deviation
D Forwards only
E It alternates

725 A student draws the outline of all possible incisal-point movements in the sagittal plane.
What is it called?
A The Bonwill triangle
B The curve of Spee
C The Gothic arch
D Posselt's envelope of motion
E The Bennett angle

726 A colleague prescribes a "Michigan splint".
What is it?
A A soft lower splint
B A full-coverage hard maxillary stabilisation splint
C An anterior repositioning splint
D A partial anterior bite plane
E A night guard for snoring

727 You are examining a new TMD patient.
What should be documented?
A Only the bite
B Only radiographs
C Only the shade of the teeth
D Pain location, movement limitation, joint noises, locking and functional impact
E Nothing until imaging

728 A patient's upper canine has reduced periodontal support, so it cannot take all lateral load alone.
Which guidance is preferable when restoring?
A Canine guidance on that canine

B A non-working contact
C Group function, sharing the working-side load
D Balanced occlusion
E No guidance

729 A patient has a stable, comfortable ICP and needs a single onlay.
Which approach is correct?
A Reorganise to RCP
B Full-mouth equilibration
C Increase the OVD
D Splint first
E Conform to the existing ICP

730 A patient denies grinding but has a marked ridge on the buccal mucosa and indentations along the tongue.
What do these signs suggest?
A Clenching or bruxism (linea alba and a scalloped tongue)
B Candida
C Lichen planus
D Iron deficiency
E Normal anatomy only

Block J — Tooth wear, the failing dentition, interfaces and trauma (Q731–800)

> *Notebook Unit 10 · LO 2c.1, 2c.10–2c.12, 2c.16–2c.18, 2c.20 (with 1.4, 2a.18) · Allow 84 minutes · FDI notation*

731 A patient referred for NHS crowns has active caries, poor plaque control and bleeding gums.
What do the NHS prosthodontic entry criteria require first?
A Radiographs only
B A stable oral environment: good hygiene, managed caries risk, treated caries and stable periodontal disease
C Payment in advance
D Implants
E Nothing

732 You plan care for a patient with pain, caries, periodontitis and missing teeth.
What is the correct order of phases?
A Definitive, urgent, maintenance
B Stabilisation, definitive, urgent
C Maintenance, reassessment, urgent
D Definitive first
E Urgent, stabilisation, reassessment, definitive, maintenance

733 A patient is entering the stabilisation phase.
Which care belongs here?

A Hygiene and diet advice, caries removal with temporisation, periodontal steps 1–2, extraction of hopeless teeth and interim prostheses
B Definitive crowns
C Implant placement
D Full-mouth rehabilitation
E Veneers

734 At reassessment, a patient's plaque and bleeding scores have halved since stabilisation began.
What does this suggest?
A No change in motivation
B Treatment should stop
C The patient needs extractions
D An engaging patient (about 50% improvement)
E Poor prognosis

735 A patient has completed stabilisation and reassessment.
What does the definitive phase include?
A Only hygiene
B Only extractions
C Endodontics, periodontal surgery if needed, then fixed and removable prosthodontics
D Only splints
E Only maintenance

736 Treatment is complete for a patient with a history of periodontitis and new bridgework.
How is the maintenance recall set?
A Every 2 years for all
B Never
C Matched to risk, with supportive periodontal care and prosthesis review
D Only if in pain
E Every week

737 A 32-year-old has rapidly progressing palatal erosion of the upper anteriors.
What is the first step?
A Identify and remove the cause (diet diary, history of reflux or eating disorders), and refer to the GMP for intrinsic causes
B Crown the teeth
C Place veneers
D Extract the worn teeth
E Prescribe antibiotics

738 A patient with dietary erosion asks for advice.
Which advice is appropriate?
A Sip acidic drinks through the day
B Brush immediately after acidic drinks
C Use a whitening toothpaste
D Avoid fluoride

E Confine acids to mealtimes, use a straw, avoid brushing straight after acid, and use high-fluoride toothpaste

739 You decide to monitor a patient's tooth wear.
Which tools help?
A Pulp tests only
B Photographs, study casts, silicone indices and BEWE or Smith and Knight scores
C Bitewings only
D CBCT
E Patient memory

740 A patient with moderate erosion is being monitored.
When should restoration be considered?
A Never
B At the first sign of wear
C Only for cosmetic reasons
D For pain or sensitivity, aesthetic or functional concern, structural compromise or pulpal risk, or documented progression despite prevention
E When the patient turns 50

741 A 45-year-old with localised anterior wear needs restoration.
What is the first-line restorative approach?
A Conventional crowns
B Extraction and implants
C Porcelain veneers
D Direct resin composite: additive, adhesive and repairable
E Post crowns

742 A colleague proposes conventional crowns as the first option for anterior wear.
Where do crowns sit in the restorative ladder?
A First
B Last, because they remove the most tooth
C Second
D Never used
E Equal to composite

743 A student asks how the Dahl concept began.
What did Dahl use in 1975?
A Composite build-ups
B An orthodontic bracket
C A removable cobalt–chromium anterior bite platform
D Implants
E Crown lengthening

744 Composite is placed on worn upper anteriors at an increased OVD.
How do the posterior contacts re-establish?
A Intrusion of the anterior teeth and over-eruption of the posterior teeth, with some mandibular

repositioning
B Wear of the composite only
C Extrusion of the anterior teeth only
D Posterior intrusion
E They never re-establish

745 A patient asks how long the back teeth will be apart after Dahl composites.
What is typical?
A A day
B A week
C 3 years
D They never meet
E About 4–6 months, occasionally up to 18

746 A 50-year-old has wear localised to the upper anteriors with no space to restore them, but the posterior teeth are unworn.
Which approach suits this?
A The Dahl approach
B Full-mouth crowns at an increased OVD
C Extraction
D Crown lengthening of all teeth
E Elective root canal treatment

747 A patient with localised anterior wear has implant crowns replacing 36 and 46.
Why may the Dahl approach fail?
A The implants are too strong
B Composite cannot be used near implants
C The anterior teeth cannot intrude
D Implants cannot erupt to re-establish posterior contact
E It will succeed faster

748 You consider the Dahl approach for a patient with localised anterior wear.
Which is a relative contraindication?
A Good compliance
B Stable periodontal health
C Long-term bisphosphonate therapy
D Unworn posterior teeth
E A deep overbite

749 You place Dahl composites today.
What should the patient be warned about?
A Nothing
B Immediate perfect chewing
C Permanent speech change
D Pulp death
E Temporary speech change, tenderness and eating difficulty, with posterior contact taking months

750 A patient has generalised severe wear and needs an OVD increase on most teeth.
How is this planned?
A Direct composite on all teeth in one visit without testing
B A diagnostic wax-up and a provisional trial at the new OVD; extensive reorganised rehabilitation is generally Level 3
C Extraction of all teeth
D Crowns at the existing OVD
E Dahl on posterior teeth only

751 You list ways of creating space for restorations.
Which method should be avoided?
A The Dahl approach
B Increasing the OVD
C Orthodontic intrusion or extrusion
D Minor adjustment of opposing teeth
E Elective root canal treatment to gain space

752 Surgical crown lengthening is proposed to gain height on worn upper incisors.
What is correct?
A It is Level 1 care
B It has no risks
C It is Level 3, and risks root exposure and an unfavourable crown–root ratio
D It always improves aesthetics
E It preserves the crown–root ratio

753 A colleague assumes a patient's OVD is reduced because the teeth are worn.
Why is this unsafe?
A Worn teeth always mean reduced OVD
B Dentoalveolar compensation often maintains OVD, so it should not be inferred from wear alone
C OVD cannot be measured
D OVD is irrelevant
E Wear always increases OVD

754 A patient with active, untreated reflux has severe palatal erosion.
Why should crowns not be planned yet?
A Crowns are too expensive
B Erosion stops naturally
C Reflux does not affect crowns
D The cause must be stabilised and the remaining tissue reviewed first
E Crowns cure reflux

755 A patient has erosion, a bruxism habit and toothbrush abrasion.
How should the wear be approached?
A Recognise that mixed mechanisms are common and address all relevant risk factors
B Treat only the erosion
C Treat only the bruxism

D Assume one cause
E Ignore the brushing

756 You are deciding which teeth to keep in a failing dentition.
What gives a tooth strategic value?
A Its colour
B Its abutment potential, occlusal stops and role in canine guidance
C Its size
D Being the last molar
E Previous treatment cost

757 A 70-year-old has sound premolars and anteriors, and compromised lower molars needing complex treatment.
What approach is reasonable?
A Heroic molar treatment
B Full clearance
C Implants on all molars
D Crowns on all teeth
E A shortened dental arch, avoiding heroic treatment of the molars when function is adequate

758 A 60-year-old will gradually lose several teeth of poor prognosis over the next few years.
How should the transition to edentulism be managed?
A Transitional acrylic RPDs to which teeth can be added, with overdenture abutments where possible
B Immediate total clearance
C No prosthesis
D Implants in every socket
E Co-Cr frameworks immediately

759 A root is to be retained under a complete denture.
What must be true of it?
A It can be carious
B It must be crowned
C It must be root-filled or asymptomatic and free of caries
D It must be mobile
E It must be above the ridge by 5 mm

760 A bridge is planned between 14 and 16. 16 has a guarded prognosis.
What is the principle?
A Use 16 as an abutment anyway
B Splint 16 with the bridge to strengthen it
C Extract 14
D Teeth of doubtful prognosis should not be bridge abutments
E Use a longer bridge

761 A patient finishing periodontal therapy wants a bridge.
When is the periodontium stable enough?
A Any pocket depth

B 6 mm pockets that bleed
C PPD 4 mm or less with no bleeding at those sites
D BOP at 50%
E Mobility grade 3

762 You check readiness for definitive prosthodontics in primary care.
What are indicative full-mouth targets?
A Plaque under 20% and bleeding under 30%
B Plaque under 80%
C Bleeding under 60%
D No targets
E Plaque 0% only

763 You restore a patient with a history of periodontitis.
Which features protect the periodontium?
A Subgingival margins everywhere
B Overcontoured crowns
C Closed embrasures
D Saddle pontics
E Supragingival margins, correct contour, and cleansable embrasures and pontics

764 A partial denture is needed for periodontally involved abutments.
Which design is appropriate?
A Acrylic mucosa-borne with gingival coverage
B Tooth-supported Co-Cr with rests, an open design away from the gingivae, guide planes and minimal clasp torque
C Rigid clasps in deep undercuts
D A horseshoe with no rests
E A flexible nylon denture

765 A patient's lower incisors are mobile, and the mobility now makes biting uncomfortable.
When is splinting indicated, and which type is preferred?
A Always; with wire
B Never
C Only for aesthetics
D When increasing mobility interferes with function or comfort; adhesive splints
E When mobility is grade 0

766 A lower first molar with a Grade III furcation is proposed as a bridge abutment.
What is correct?
A It is an ideal abutment
B Root resection is routine Level 2 care
C Furcation does not affect prognosis
D Splint it with composite only
E Furcation-involved molars are poor abutments; root resection belongs in specialist care

767 Two patients have problems around implants: one has bleeding without bone loss; the other has bone loss.
Which is correct?
A Both are Level 1
B Both are Level 3
C Peri-implant mucositis (reversible, no bone loss) is Level 2; peri-implantitis (bone loss) is Level 3
D Both need explantation
E Peri-implantitis is reversible

768 A heavily broken-down lower premolar is to be crowned and needs root canal treatment.
What should be decided before the root canal treatment?
A The crown shade
B Nothing
C The post length
D Restorability: sound coronal tissue, ferrule potential and periodontal support
E The cost of an implant

769 A patient asks about the long-term risk to vital teeth prepared as conventional bridge abutments.
About how many become non-vital?
A About 10–15%
B Less than 1%
C 50%
D 90%
E None

770 A strategic upper canine has a combined endodontic–periodontal lesion and is the key abutment of a planned bridge.
At which level is this managed?
A Level 1
B Level 3
C Level 2
D Hygienist
E Not treatable

771 A 60-year-old has lost 46. There is no drifting and she is not bothered by the space.
What are the disadvantages of leaving it?
A Appearance, drifting and over-eruption, which are often minimal
B Inevitable TMD
C Bone loss throughout the jaw
D Certain loss of 47
E None at all

772 A patient asks about an implant to replace 21.
Which disadvantages of implants should be discussed?
A None
B Only cost
C Implants damage adjacent teeth

D Surgery, cost, time, bone requirements, peri-implant disease and limited NHS access
E Implants never fail

773 A patient needs routine conforming fixed and removable restorations with healthy tissues.
At what NHS complexity level is this?
A Level 3
B Level 1
C Level 2
D Not provided
E Hygienist only

774 You classify care at Level 2.
Which falls within Level 2?
A Cleft rehabilitation
B Major occlusal reorganisation
C Implant placement
D Severe gagging preventing any denture
E Replacing and temporising multiple fixed units

775 A 20-year-old with hypodontia needs multiple replacements.
At which level is this care?
A Level 1
B Level 2
C Level 3, along with cleft, amelogenesis imperfecta and head and neck cancer rehabilitation
D Hygienist
E Not provided

776 A patient's severe gag reflex has prevented every attempt at a denture.
At which level does care sit?
A Level 1
B Level 2
C Hygienist
D Level 3
E Not provided

777 A patient has undiagnosed facial pain and wants new crowns to cure it.
At which level is this managed?
A Level 1
B Level 3
C Level 2
D Not provided
E Hygienist

778 A 15-year-old's permanent upper central incisor was avulsed 10 minutes ago and is clean.
What is the immediate action?
A Replant it promptly if appropriate, and arrange urgent care
B Store it dry

C Discard it
D Plan an implant
E Wait for a week

779 A 4-year-old's primary upper central incisor has been avulsed.
What is correct?
A Replant it immediately
B Store it in milk and replant later
C Splint for 6 months
D Root-fill it extraorally and replant
E Never replant an avulsed primary tooth

780 An avulsed permanent tooth cannot be replanted at the scene.
What should be done?
A Wrap it in tissue
B Scrub it clean
C Store it in an appropriate medium and arrange urgent expert care
D Leave it dry
E Discard it

781 A replanted tooth is under review.
Which long-term complications are monitored?
A Caries only
B Pulpal necrosis, inflammatory or replacement resorption, ankylosis, colour change and root fracture healing
C Gingival recession only
D Allergy
E None

782 A 13-year-old lost 21 in an accident and asks for an implant.
What is correct?
A Place it immediately
B Implants grow with the jaw
C Age does not matter
D An implant is the only option
E A fixed implant is generally deferred until growth is complete

783 The 13-year-old in the previous question needs a replacement until growth finishes.
Which interim options preserve future choices?
A Implants
B Conventional bridges
C Crowns on adjacent teeth
D A resin-bonded bridge, a removable partial denture or orthodontic space management
E No replacement

784 A replanted incisor in a 12-year-old has become progressively infra-occluded.
What is the likely cause?

A Caries
B Pulpitis
C Ankylosis (replacement resorption)
D Bruxism
E Normal eruption

785 An upper central incisor has a crown–root fracture extending below the gingiva.
What determines restorability?
A The fracture position, root length, periodontal support and ferrule potential
B Colour only
C The patient's age only
D The shade guide
E Nothing; all are restorable

786 A 10-year-old has complex trauma to several incisors.
How should care be planned?
A With multidisciplinary advice
B Immediate implants
C Crowns on all teeth
D Extraction of all affected teeth
E No follow-up

787 A 24-year-old with bulimia has palatal erosion and wants veneers.
What is the most appropriate approach?
A Immediate veneers
B Full-mouth crowns
C Liaise with the GMP, prevent and monitor, and delay definitive restoration until the condition is stable
D Extraction
E Ignore the cause

788 A bruxist has flat, shiny facets and recently fractured cusps.
Which preventive measure is appropriate?
A An acid-free diet only
B A night-time hard splint
C Whitening
D Orthodontics
E Extraction of canines

789 An adult with active erosion and dentine sensitivity asks which toothpaste to use.
What is appropriate?
A A whitening toothpaste
B A charcoal toothpaste
C Non-fluoride toothpaste
D 5000 ppm fluoride toothpaste
E Baking soda only

790 Direct composite has failed repeatedly on worn upper anteriors.
What are the next options before conventional crowns?
A Extraction
B Implants
C Full-coverage crowns immediately
D Dentures
E Indirect composite, or gold or ceramic onlays and palatal veneers

791 A patient with mild, stable erosion and no symptoms was told by another dentist she needs crowns on all teeth.
What is the most appropriate advice?
A Crown all teeth
B Veneer all teeth
C Extract and replace
D Full-mouth composite now
E Prevention and monitoring

792 Dahl composites have just been placed on 13 to 23.
What is the immediate occlusal situation?
A The posterior teeth are out of contact
B All teeth contact evenly
C Only molars contact
D The OVD is reduced
E The canines are out of contact

793 Conventional bridge preparation is planned on vital abutments.
What should be done before preparation?
A Nothing
B Elective root canal treatment of both
C Assess pulp status pre-operatively and record it
D Extract one abutment
E Prescribe antibiotics

794 You plan the transition to complete dentures in a 70-year-old with a failing dentition.
Which teeth are most worth keeping as overdenture abutments?
A Mobile incisors
B Canines, with their long roots and strategic position
C Third molars
D Teeth with Grade III furcation
E Any carious tooth

795 A patient with several failing teeth asks for them all to be removed at once.
What is generally preferred?
A Immediate total clearance always
B No prosthesis
C Dental implants on the same day

D A planned, staged transition, avoiding sudden total clearance where possible
E Leaving all teeth untreated

796 An upper lateral incisor has had root canal treatment through a small access with little other tissue loss.
What restoration is appropriate?
A A post crown
B A full ceramic crown
C A conservative composite restoration
D A veneer with post
E Extraction

797 A lingual plate is proposed to splint mobile lower incisors.
When is this appropriate?
A Always
B Never
C Only with an acrylic base
D When hygiene is poor
E Only if the patient's oral hygiene allows

798 A patient's full-mouth plaque score is 45% and bleeding score 50%. She wants a bridge now.
What is appropriate?
A Proceed with the bridge
B Use a resin-bonded bridge instead
C Prescribe antibiotics and proceed
D Improve hygiene and disease control first; she is not yet ready for definitive work
E Extract the abutments

799 A patient with treated periodontitis has completed prosthodontic care.
What does the maintenance phase include?
A Supportive periodontal care, with prosthesis review at a risk-based recall
B No recall
C Annual radiographs only
D Antibiotics every 6 months
E A new bridge every 5 years

800 A 30-year-old missing 12 is offered orthodontic space closure.
What is the main disadvantage of this option?
A It needs an implant
B Time, appliances and Level 3 planning
C It damages adjacent teeth
D It is irreversible surgery
E It is never possible

Block K — Implants and digital prosthodontics (Q801–860)

> *Notebook Unit 11 · LO 2a.19, 2c.5 (with 2c.2, 2c.4, 2c.21) · Allow 72 minutes · FDI notation*

801 A general dentist working at Level 2 is asked to place an implant for 36.
What is within the Level 2 scope?
A Implant assessment, planning and placement
B Sinus grafting
C Providing and maintaining simple implant-retained prostheses and diagnosing peri-implant disease; assessment, planning and placement are Level 3
D Full-arch fixed implant bridges
E Ridge augmentation

802 A student defines osseointegration.
Which definition (Brånemark) is correct?
A A fibrous attachment between bone and implant
B Mechanical retention in a socket
C A periodontal ligament around the implant
D Cementation of the implant into bone
E A direct structural and functional connection between living bone and the surface of a load-bearing implant, with no intervening soft tissue

803 An implant feels firm at placement.
How do primary and secondary stability differ?
A Primary is mechanical (at placement); secondary is biological (bone remodelling over weeks)
B Primary is biological; secondary is mechanical
C Both are mechanical
D Both are biological
E Secondary stability is achieved at placement

804 A colleague notices implant stability values dropped a few weeks after placement.
When does the "stability dip" occur?
A Immediately at placement
B About 2–4 weeks after placement
C At 6 months
D At 1 year
E It never occurs

805 An implant is planned next to a natural tooth.
What minimum distance from implant to tooth is advised?
A 0.5 mm
B 3 mm
C 5 mm
D 1.5 mm
E None

806 Two adjacent implants are planned for 35 and 36.
What minimum distance between them is advised?
A 3 mm
B 1 mm
C 1.5 mm
D 6 mm
E 10 mm

807 An implant is planned in the posterior mandible.
What safety margin above the inferior alveolar canal is advised?
A 0 mm
B 0.5 mm
C 5 mm
D 2 mm
E 8 mm

808 A CBCT shows a thin buccal plate at a planned implant site.
How much buccal bone is usually wanted around an implant?
A 0.2 mm
B None
C 5 mm
D 10 mm
E At least 1.5–2 mm

809 A temporary crown is placed on an implant 3 days after surgery.
Which ITI loading protocol is this?
A Conventional
B Early
C Immediate (within 1 week)
D Delayed
E Submerged

810 An implant is restored 6 weeks after placement.
Which ITI loading protocol is this?
A Immediate
B Early (1 week to 2 months)
C Conventional
D Delayed
E Late

811 A patient's older implant crown keeps loosening.
Which implant–abutment connection is associated with more screw loosening?
A Conical (Morse taper)
B External hex
C Internal octagon
D Internal conical with platform switching
E One-piece zirconia

812 A colleague prefers a particular implant connection for its seal and stability.
Which connection offers these?
A External hex
B No connection
C Butt joint
D Conical (Morse taper)
E Slip fit

813 An abutment narrower than the implant platform is used.
What is this called, and what may it achieve?
A Platform matching; more bone loss
B Bone grafting
C Platform switching; it may reduce crestal bone loss
D Immediate loading
E External hex

814 A patient asks whether an implant is suitable for her.
Which history most increases the risk of peri-implantitis?
A A history of periodontitis, especially with smoking
B Good oral hygiene
C Controlled hypertension
D A penicillin allergy
E Previous orthodontics

815 A 75-year-old edentulous patient struggles with a loose lower denture.
What is the minimum standard of care for the edentulous mandible?
A A conventional denture
B Denture adhesive
C A single implant
D Four implants with a fixed bridge
E A two-implant overdenture (McGill 2002, York 2009)

816 Unsplinted ball or locator attachments are chosen for a two-implant mandibular overdenture.
What are their characteristics?
A Easier hygiene and maintenance, less restorative space needed; inserts wear and need replacing
B They need the most vertical space
C They splint the implants rigidly
D They never wear
E They are hardest to clean

817 Implants for an overdenture are markedly divergent.
Which attachment system suits them?
A A bar, which splints the implants and gives more retention but needs more vertical space and hygiene effort
B Single unsplinted balls
C Magnets only

D No attachment
E Denture adhesive

818 An edentulous patient wants an implant overdenture in the maxilla.
What is usual?
A One implant
B Usually 4 or more implants; a palateless design is possible
C Two implants
D No implants
E Ten implants

819 A single implant crown on 36 is planned and the patient wants it to be retrievable.
Which retention type is preferred?
A Cement-retained with resin cement
B Cement-retained with glass ionomer
C Friction-retained
D Screw-retained: retrievable, with no subgingival cement
E Adhesive bridge

820 A cement-retained implant crown has bleeding and suppuration around it a year after fitting.
What is a common cause?
A Occlusal overload only
B Shade mismatch
C Residual subgingival cement
D Platform switching
E A bar attachment

821 A full-arch fixed implant bridge is planned.
Which statement is correct?
A It is Level 1
B Long distal cantilevers are preferred
C Passive fit is unimportant
D It is Level 3; passive fit is essential and distal cantilevers are kept short
E It needs no maintenance

822 A patient with a two-implant mandibular overdenture complains it is loose after 18 months.
What is the commonest cause?
A Wear of the attachment inserts, needing replacement
B Implant fracture
C Peri-implantitis
D Mandibular fracture
E Allergy

823 A single implant crown has been in place for 3 years.
What are its commonest complications?
A Implant fracture
B Nerve injury

C Sinus perforation
D Bone graft failure
E Abutment screw loosening and ceramic chipping

824 You tighten the abutment screw of an implant crown.
How should the screw be torqued?
A By hand until tight
B With pliers
C With a calibrated torque driver to the manufacturer's value
D As tight as possible
E It does not matter

825 A colleague plans a bridge joining a natural premolar to an implant.
Why should this be avoided?
A Implants are too weak
B Implants have no periodontal ligament, so differential mobility causes problems
C Teeth cannot be crowned
D It is illegal
E It is always successful

826 You review a patient with an implant crown.
What defines peri-implant health?
A Bleeding on probing
B Progressive bone loss
C No erythema, bleeding or suppuration, and no bone loss beyond initial remodelling
D A 6 mm pocket
E Suppuration

827 An implant shows bleeding on gentle probing, with no bone loss beyond initial remodelling.
What is the diagnosis and management level?
A Peri-implantitis; Level 3
B Peri-implant mucositis: reversible; Level 2 debridement, OHI and a prosthesis design check
C Peri-implant health
D Implant failure; remove
E Occlusal trauma

828 No baseline radiograph exists for an implant with bleeding.
What findings indicate peri-implantitis?
A Any bleeding
B A 4 mm pocket alone
C Mucosal redness alone
D Bone 3 mm or more apical to the implant platform, with PPD 6 mm or more and bleeding
E No bone loss

829 You fit a new implant-retained crown.
What should be recorded at fitting?
A A baseline radiograph and probing depths

B The shade only
C Nothing
D A CBCT
E A photograph only

830 A patient with an implant bar overdenture asks how to clean it.
Which maintenance is appropriate?
A No special cleaning
B Abrasive toothpaste on the abutments
C Metal scalers on the implants
D Chlorhexidine only
E Interdental brushes and floss under the bar, professional debridement with instruments that will not scratch titanium, and annual review

831 A hygienist is about to debride around a titanium implant abutment.
Which instruments should be used?
A Instruments that will not scratch titanium (such as plastic, titanium or carbon-fibre tips)
B Stainless-steel scalers
C Diamond burs
D Coarse prophylaxis paste
E None; implants need no cleaning

832 A patient asks whether an implant crown will "feel" like a natural tooth.
What is correct?
A It has a periodontal ligament
B It has normal proprioception
C It has no periodontal ligament or physiological movement; feedback comes from osseoperception only
D It moves more than a tooth
E It has pulpal sensation

833 An implant is still in place after 10 years but has 5 mm of bone loss and recurrent suppuration.
How should its outcome be described?
A A success
B It has survived, but is not a success
C A failure because it is present
D Fully healthy
E Not assessable

834 A patient asks for an implant to replace 25.
What should the assessment start with?
A Implant brand
B Bone graft options
C The CBCT
D The fee
E The prosthesis: the missing-tooth problem, desired restoration, smile line, restorative space, occlusion, cleansability and the patient's priorities

835 A Level 2 dentist is asked to perform a sinus lift.
What is appropriate?
A Proceed
B Proceed under sedation
C Proceed with a guide
D Refer; sinus grafting and ridge augmentation are not routine Level 2 care
E Proceed if the patient pays privately

836 A colleague plans to scan a full arch digitally for a cross-arch bridge.
Which statement about intraoral scanners is correct?
A They are most accurate over full arches
B They are excellent for short spans; accuracy falls over full arches through stitching errors
C They are unaffected by saliva
D They record functional borders
E They record mucodisplaced edentulous tissue well

837 You consider an intraoral scan for an edentulous upper arch.
What are the limitations of intraoral scanning?
A Subgingival margins
B Blood and saliva
C Mobile edentulous mucosa, which cannot be mucodisplaced or border-moulded
D Both A and B only
E All of the above

838 A laboratory asks for a scan in STL format.
What does an STL file contain?
A Volumetric radiographic data
B Colour and texture
C Surface geometry as triangles, in an open format without colour
D Jaw movement data
E A patient's medical history

839 A clinician wants a digital model that shows gingival colour for shade communication.
Which file formats can carry colour and texture?
A STL
B DICOM
C PDF
D PLY or OBJ
E JPG only

840 A surgical guide is planned for an implant at 36.
Which data are merged?
A CBCT (DICOM) with surface scans (STL)
B Two STL files only
C Photographs only
D A panoramic radiograph and an alginate cast
E A facebow record only

841 A manufacturer quotes the trueness and precision of its scanner.
What do these terms mean?
A Trueness is speed; precision is colour
B Both mean resolution
C Trueness is repeatability; precision is closeness to the reference
D Both mean file size
E Trueness is closeness to the reference; precision is repeatability

842 A laboratory mills a crown from a pre-sintered zirconia block.
What are the features of subtractive manufacture?
A Porous, layered material
B Dense, homogeneous material, but fine detail is limited by bur size
C No material waste
D Unlimited internal detail
E It is used only for wax

843 A laboratory 3D-prints a model and a Co-Cr framework.
Which processes are additive?
A Milling
B Casting
C Grinding
D Stereolithography (SLA), digital light processing (DLP) and selective laser melting (for metals)
E Turning

844 A practice buys a 3D printer.
What are its typical prosthodontic uses?
A Models, surgical guides, trays, splints and temporary crowns
B Definitive implant bodies
C Titanium abutments only
D Zirconia blocks
E Nothing clinical

845 A patient asks about digital complete dentures.
What are their advantages and limitations?
A They cannot be duplicated
B They need more visits
C Fewer visits, stored records and low-monomer milled bases with good fit; limited recording of functional borders and complex jaw relations
D They always fit better than conventional dentures in all cases
E They need no clinical records

846 A patient's smile has been designed digitally.
What is still useful before treatment?
A Nothing
B Immediate preparation
C Only the digital image

D A physical mock-up from a wax-up for the patient's approval
E A CBCT

847 A clinician wants to record a patient's dynamic occlusion digitally.
Which technology does this?
A An STL scan
B A panoramic radiograph
C Jaw motion tracking
D A shade photograph
E An alginate impression

848 A practice designs crowns digitally.
How should the records be managed?
A Save source files securely and document revisions
B Delete them after milling
C Keep only the final file
D Share them publicly
E No records are needed

849 A printed provisional crown is weaker than expected.
What affects the properties of printed restorations?
A Nothing
B Only the colour
C Only the printer brand
D The patient's age
E Post-curing and using a material indicated for the purpose

850 A multi-implant full-arch prosthesis needs a passive fit.
What is appropriate?
A An intraoral scan alone without verification
B A validated technique, with a verification jig or conventional splinted impression where indicated
C Alginate
D Any technique
E Guessing

851 An implant crown on 46 has loosened three times.
What should be checked?
A Shade
B Nothing; just retighten
C Screw torque, occlusal loading (including bruxism) and the connection
D The opposing tooth's caries
E The patient's diet only

852 A patient has a two-implant mandibular overdenture opposing an upper complete denture.
What complication of the upper denture is common?
A Fracture of the upper denture
B Implant failure in the maxilla

C Upper denture loosening is impossible
D Lower implant fracture
E None

853 A bruxist has a new implant crown.
What is appropriate?
A No precautions
B Remove the crown at night
C A soft daytime splint
D Layered porcelain
E A protective night guard, as bruxism raises the risk of complications

854 A patient had radiotherapy to the mandible and wants implants there.
What is correct?
A Implants in irradiated bone have the same success
B They are routine Level 2
C They are contraindicated in all cases
D Higher failure rates; Level 3 care
E Only in the maxilla

855 A patient with bone metastases on high-dose IV zoledronic acid asks for implants.
What is appropriate?
A Implants are routine
B Implants only after careful risk assessment and consent; they are generally avoided because of MRONJ risk
C Stop the drug and place implants
D No precautions
E Implants reduce MRONJ risk

856 A patient with an HbA1c of 85 mmol/mol wants implants.
What is appropriate?
A Proceed immediately
B Ignore diabetes
C Prescribe antibiotics and proceed
D Refuse permanently
E Recognise the raised risk and optimise glycaemic control first

857 A Level 2 dentist manages peri-implant mucositis on a bar overdenture.
What does management include?
A Implant removal
B Bone grafting
C Surgical access
D Mechanical debridement, OHI and a check of the prosthesis design for cleansability
E Systemic antibiotics only

858 An implant has suppuration, a 7 mm pocket and radiographic bone loss.
What is appropriate?

A Debride and review in a year
B Refer as peri-implantitis (Level 3)
C Ignore it
D Retighten the screw
E Replace the crown only

859 A patient's locator-retained overdenture has lost its retention.
What is the usual first step?
A Replace the worn matrix (insert)
B Remove the implants
C Remake the denture
D Add denture adhesive permanently
E Reline with a soft liner

860 A clinician plans aesthetic rehabilitation using a facial scan and a virtual articulator.
What do these add?
A They replace consent
B They guarantee results
C They integrate facial references into the digital plan
D They replace clinical examination
E They remove the need for records

Block L — Complications, failures and maintenance (Q861–890)

> *Notebook Unit 12 · LO 2c.21–2c.27 (with 3a.2–3a.8) · Allow 36 minutes · FDI notation*

861 A crown on 46 has come off for the third time. The preparation looks short and tapered.
What is the correct approach?
A Recement with a stronger cement and review
B Make an identical new crown
C Check for caries and assess the preparation; recement only if sound and retentive, otherwise improve retention or redesign
D Extract the tooth
E Leave it out

862 A patient with a 4-year-old fixed-fixed bridge reports a bad taste. One retainer is loose, but the bridge is still held by the other.
What is appropriate?
A Remove the bridge, treat the caries and redesign
B Recement through a hole in the retainer
C Monitor for a year
D Prescribe chlorhexidine
E Polish the margins

863 A small chip has come off the porcelain of a metal–ceramic crown, without exposing metal.
What is appropriate?

A Remake the crown immediately
B Extract the tooth
C Leave it rough
D Use zinc phosphate to fill it
E Polish it, or repair with composite (HF and silane on the ceramic)

864 Porcelain has repeatedly fractured from the same metal–ceramic crown. It was more than 2 mm thick and unsupported.
What is the definitive solution?
A Composite repair again
B Replace and redesign with proper metal support, or a metal or zirconia occlusal surface
C Thicker porcelain
D A soft night guard only
E Accept repeated repairs

865 A patient has sensitivity to cold a week after a crown was cemented on a vital tooth.
What is the first management?
A Immediate root canal treatment
B Extraction
C Remove the crown
D Check the occlusion, desensitise and monitor sensibility
E Antibiotics

866 A crowned upper premolar has become non-vital with a periapical lesion; the crown is otherwise sound.
What is appropriate?
A Extract the tooth
B Replace the crown first
C Antibiotics only
D Monitor
E Root canal treatment through the crown, then seal the access (or replace the crown)

867 A post-crowned premolar has a vertical root fracture with no ferrule.
What is the usual outcome?
A Recementation
B Extraction; prevention depends on ferrule and post design
C A longer post
D Composite repair
E Splinting

868 Bleeding and swelling develop around one newly fitted crown.
What should be suspected first?
A Allergy
B Caries
C Root fracture
D Residual excess cement, until proven otherwise
E Normal healing

869 A patient complains of food packing between a new crown on 36 and 37.
What is the likely cause and remedy?
A An open or weak contact; adjust or replace the restoration
B Caries; restore 37
C Gingivitis; chlorhexidine
D Occlusal overload; splint
E Normal; reassure

870 A single-wing resin-bonded bridge has debonded. The wing is intact and there is no caries.
What is appropriate?
A Remake as a conventional bridge
B Extract the abutment
C Clean, re-sandblast and rebond under rubber dam with MDP cement, check the occlusion, and redesign if it debonds repeatedly
D Rebond with temporary cement
E Replace with an implant immediately

871 You audit failures of conventional bridges in your practice.
What are the leading causes of failure in the literature?
A Caries and loss of retention
B Pontic fracture and allergy
C Shade mismatch
D Gingival recession only
E Opposing tooth wear

872 You consent a patient for crowns on vital teeth.
Which common biological complication should be mentioned?
A Allergy to gold
B Root resorption
C Tooth discolouration only
D The later need for endodontic treatment
E TMD

873 A Co-Cr framework will not seat at try-in. Disclosing medium shows widespread contacts, with no single interference.
What is the most likely cause and action?
A Normal; force it
B Grind every contact
C A distorted impression or cast; re-impression and remake
D Too much retention; open the clasps
E Reline it

874 A patient's RPD clasp has become loose.
What is a common cause?
A Too much reciprocation
B Excess undercut
C Porcelain teeth

D Night removal
E Distortion from the patient removing the denture by the clasp

875 A Co-Cr clasp fractures at the shoulder after several adjustments.
What is the cause?
A Corrosion
B Fatigue, aggravated by repeated adjustment of low-ductility Co-Cr or casting porosity
C Too little undercut
D Acid erosion
E A loose rest

876 An RPD abutment has become painful and mobile.
Which causes should be considered?
A Shade
B Saliva
C Denture adhesive
D Occlusal overload, an unseated rest, free-end saddle rotation or a clasp active at rest
E Night removal

877 A lower free-end saddle sinks and the indirect retainer lifts when pressed.
What is the remedy?
A Tighten the clasps
B Remove the indirect retainer
C Add a rest
D Extract the abutment
E Reline the saddle

878 A patient with an RPD has new caries on the abutments.
What is the likely cause?
A Plaque retention at clasps and minor connectors, and high sugar intake
B The metal
C Fluoride
D Over-tight clasps
E Night removal

879 An edentulous patient asks how often to have her dentures checked.
What is the recommendation?
A Every 10 years
B Only when broken
C At least annually, for mucosal examination, fit and occlusal wear
D Never
E Weekly

880 A patient's dentures are loose, but the appearance, OVD and occlusion are good.
What is the appropriate treatment?
A Remake
B Reline or rebase

C Adhesive only
D Implants
E No treatment

881 A denture has fractured in two.
What principles apply to the repair?
A Reassemble the pieces roughly and add acrylic
B Reassemble the fracture accurately before pouring a cast; note that self-cure repairs are weaker, and investigate the cause of repeated fractures
C Use superglue
D Make a new denture each time
E Heat the pieces

882 A patient asks how to clean her acrylic complete dentures.
What is the correct advice?
A Abrasive toothpaste
B Boiling water
C Brush with a soft brush and soap or non-abrasive cleanser over a basin of water, soak overnight, and leave them out at night
D Bleach for metal dentures
E Wear them day and night

883 Newly processed complete dentures are too tall at fitting, though the wax try-in was correct.
Which laboratory error is likely?
A Over-polished fitting surface
B Abraded dies
C A wrong shade
D Raised OVD from flash during processing
E Distorted impression

884 A practice wants fewer laboratory errors.
What are the best preventive measures?
A Clear written prescriptions, inspection of impressions and casts at each stage, photographs and shade information, and agreed delivery times
B Changing laboratory every month
C Sending no prescription
D Phoning the technician only
E Accepting all work

885 A paper reports 95% survival of a prosthesis at 10 years.
What does "survival" mean, and how does it differ from success?
A They are identical
B Survival means free of complications
C Success means still in place
D Survival includes patient satisfaction
E Survival means still in place; success means in place and free of complications

886 A patient asks how long a conventional bridge lasts.
What is the approximate benchmark survival?
A About 50% at 10 years
B About 99% at 20 years
C About 89% at 10 years
D About 20% at 5 years
E It cannot be estimated

887 A patient asks about the longevity of a single metal–ceramic crown.
What is the approximate benchmark?
A About 95% at 5 years
B About 50% at 5 years
C About 70% at 1 year
D 100% for life
E About 30% at 10 years

888 A patient asks how long a modern resin-bonded bridge lasts.
What is the approximate benchmark?
A About 20% at 5 years
B About 80–90% at 5–10 years, with cantilevers performing well
C About 99% at 20 years
D About 50% at 1 year
E They always fail

889 A practice plans maintenance appointments for its patients.
Which prosthesis needs frequent maintenance?
A A single gold crown
B A composite filling
C A metal–ceramic bridge
D A veneer
E Implant overdenture attachments

890 A lower Co-Cr RPD has fractured for the third time at the same place.
What is the most appropriate approach?
A Repair it again
B Use self-cure acrylic
C Tell the patient to eat softer food
D Diagnose the cause (thickness, support, occlusal forces) and change the design
E Make an identical new one

Block M — Governance, law and communication (Q891–950)

> *Notebook Unit 13 · LO 3a.1–3a.12, 3b.1–3b.5 · Allow 72 minutes · Rules are for England unless stated; check current guidance*

891 A patient signed a consent form for a bridge, but the risks were never discussed and she felt pressured.
What makes consent valid?
A A signature alone
B It is informed, voluntary, given by a person with capacity, specific to the treatment, documented, and can be withdrawn
C Verbal agreement only
D The dentist's judgement
E Payment

892 You are consenting a professional violinist for a lower molar extraction near the lingual nerve.
Which case set the standard for disclosure of risks?
A Bolam (1957)
B Bolitho (1997)
C Gillick (1985)
D Caparo (1990)
E Montgomery v Lanarkshire (2015): material risks to a reasonable patient, or to this patient, and reasonable alternatives

893 A clinician's crown preparation technique is challenged in a negligence claim.
Which test applies to the standard of treatment?
A Bolam, qualified by Bolitho (opinion must withstand logical analysis)
B Montgomery
C Gillick
D The Mental Capacity Act
E The Equality Act

894 A patient alleges negligence after a crown.
What must be proved?
A Only that harm occurred
B Only breach of duty
C Duty of care, breach of that duty, causation, and resulting harm
D Only lack of consent
E Only that the patient is unhappy

895 An associate's error injures a patient. The practice owner is also sued.
What principle allows this?
A Duty of candour
B Montgomery
C Caldicott
D Vicarious liability: an employer is liable for employees' acts in the course of employment
E Gillick

896 A patient with capacity declines a recommended bridge and chooses no replacement.
Which Mental Capacity Act principle applies?
A Assume incapacity in older patients
B Override the decision if unwise

C Decide in the clinician's best interests
D An unwise decision does not mean the person lacks capacity
E Refer to court

897 You need to assess an 80-year-old's capacity to consent to extractions.
What is the two-stage test?
A An impairment of mind or brain; then whether the person can understand, retain, use or weigh, and communicate the decision
B Age and diagnosis
C A memory test only
D A family member's opinion
E A GP letter only

898 A patient with fluctuating dementia consented last week but seems confused today.
Which principle about capacity applies?
A Capacity once given lasts
B Capacity is global
C Consent from last week stands
D The carer decides
E Capacity is decision-specific and time-specific

899 A 17-year-old wants an RBB and her parent disagrees.
What is the legal position?
A Parents must consent
B 16–17-year-olds are presumed to have capacity to consent (Family Law Reform Act 1969)
C Consent is impossible under 18
D A court must decide
E The dentist decides

900 A 14-year-old requests treatment without her parents knowing.
When may she consent?
A Never
B Only with parental consent
C If she is Gillick competent
D At age 16 only
E If she pays

901 A Scottish dentist treats an adult lacking capacity.
Which legislation applies?
A Mental Capacity Act 2005
B Children Act 1989
C Equality Act 2010
D Data Protection Act 2018
E Adults with Incapacity (Scotland) Act 2000

902 You use a decision aid with a patient choosing between a bridge and an implant.
What does NICE NG197 (shared decision-making) recommend discussing?

A Only the recommended option
B Only cost
C Benefits, risks, alternatives and doing nothing (BRAN)
D Only risks over 10%
E Nothing in writing

903 A student lists the GDC's nine principles.
Which is NOT one of them?
A Put patients' interests first
B Obtain valid consent
C Raise concerns if patients are at risk
D Maximise the practice's income
E Have a clear and effective complaints procedure

904 A dentist is planning CPD for the next cycle.
What are the enhanced CPD requirements?
A At least 100 verifiable hours per 5-year cycle, at least 10 hours in every 2-year period, with a personal development plan
B 50 hours per year
C No minimum
D 250 hours per 5 years
E 10 hours per 5 years

905 A newly qualified dentist asks about indemnity.
What is correct?
A It is optional
B Appropriate indemnity is mandatory
C Only needed for implants
D The NHS always covers it
E Only for private work

906 A dentist extracts a tooth and discovers it was the wrong one.
Who has a professional duty of candour?
A All registrants
B Only the practice owner
C Only the CQC
D No one
E Only the nurse

907 After an error, a dentist apologises to the patient.
What is the legal status of an apology?
A It admits negligence
B It must be avoided
C It voids indemnity
D It must be in writing only
E An apology is not an admission of liability

908 A CQC-registered practice has a notifiable safety incident.
Which duty applies to the provider?
A None
B A voluntary apology
C The statutory duty of candour
D Only internal discussion
E Reporting to the police

909 A patient develops a serious adverse reaction to a prescribed antibiotic.
How should it be reported?
A To the GDC
B To RIDDOR
C To the CQC only
D Through the MHRA Yellow Card scheme
E Not at all

910 A batch of defective impression trays causes repeated distortions.
Where are medical device incidents reported?
A The GDC
B The MHRA
C The HSE
D The police
E The BDA

911 A nurse suffers a serious injury from a workplace accident.
Under which regulations, and to whom, is it reported?
A IR(ME)R, to the CQC
B GDPR, to the ICO
C RIDDOR, to the HSE
D The Equality Act, to the court
E Montgomery, to the GDC

912 A healthy lower molar is extracted instead of its carious neighbour.
How is this classified?
A A never event (wrong-site surgery)
B A minor incident
C A complaint only
D Acceptable error
E A near miss

913 An NHS patient complains about her dentures.
What is the acknowledgement timescale and time limit for complaints?
A Acknowledge within 1 month; no time limit
B Acknowledge within 3 working days; complaints normally within 12 months of the event or of becoming aware
C Within 6 months; 3 years

D No acknowledgement needed
E Within a week; 6 months

914 A complaint is not resolved locally.
Where does it escalate?
A NHS: the GDC; private: the CQC
B Always to court
C To the BDA
D Both to the police
E NHS: the Parliamentary and Health Service Ombudsman; private: the Dental Complaints Service

915 A nurse raises concerns about unsafe practice by her employer.
Which law protects her?
A The Equality Act 2010
B The Mental Capacity Act 2005
C Data Protection Act 2018
D The Public Interest Disclosure Act 1998
E The Children Act 1989

916 A dentist realises an entry in yesterday's notes is wrong.
How should it be corrected?
A Delete and rewrite it
B Add a dated, signed note of the correction, leaving the original visible
C Use correction fluid
D Ignore it
E Ask the nurse to change it

917 A practice is clearing old records.
How long must adult and child records be kept (NHS England guidance)?
A Adults 3 years; children 5 years
B Adults 7 years; children 18 years
C Indefinitely
D Adults at least 11 years after last treatment; children until age 25 or 11 years after last treatment, whichever is longer
E Until the patient dies

918 A patient requests a copy of her records.
What does UK GDPR require?
A A response within one month, normally free of charge
B A response within 6 months with a fee
C Refusal
D Only summaries
E A response within a year

919 Police ask for a patient's records without a court order or consent, in a minor matter.
When may confidentiality be breached?
A Whenever police ask

B Never
C With consent, when required by law (such as a court order), or in the public interest to prevent serious harm, sharing the minimum necessary
D If the patient owes money
E At the dentist's discretion

920 A practice reviews how it shares patient information.
Which principles govern confidential information sharing?
A Bolam principles
B Montgomery principles
C Gillick principles
D Hanau principles
E Caldicott principles

921 A new practice installs intraoral X-ray sets.
What does IRR17 require of the employer?
A Nothing beyond buying the equipment
B Registering with the GDC
C Patient consent forms only
D Registering with the HSE, appointing a radiation protection adviser and supervisor, and setting local rules and a controlled area
E A CBCT unit

922 You set up the controlled area for an intraoral X-ray set.
What is the typical extent?
A The whole building
B Within about 1.5 m of the tube and patient, and within the primary beam until it is attenuated
C 10 m
D Only the patient's head
E No area is needed

923 You review who does what under IR(ME)R 2017.
Who are the duty holders?
A Patient and nurse
B HSE and CQC
C GDC and BDA
D Only the dentist
E Employer, referrer, practitioner and operator

924 An audit shows radiographs taken with round collimation and no film holders.
Which optimisation measures should be introduced?
A A higher exposure
B More repeat films
C Rectangular collimation, film holders, the fastest suitable receptor, appropriate exposure settings, and image-quality audit
D Panoramic views instead
E No change

925 You check the steam steriliser's display.
What is the standard sterilisation cycle?
A 134–137°C for at least 3 minutes
B 121°C for 1 minute
C 100°C for 10 minutes
D 160°C dry heat for 2 hours only
E 80°C for 30 minutes

926 A practice will sterilise wrapped handpieces and hollow instruments.
Which steriliser is needed?
A A type N (non-vacuum) steriliser
B A boiling water bath
C An ultrasonic bath
D A hot air oven
E A type B (vacuum) steriliser

927 You review the practice's instrument cleaning.
What is best practice under HTM 01-05?
A Manual scrubbing only
B No cleaning before sterilising
C A washer-disinfector
D Soaking in bleach
E Rinsing under the tap

928 A nurse asks which steriliser tests are done daily.
What is correct?
A None
B An automatic control test, plus a steam penetration test for vacuum sterilisers
C Annual validation only
D A spore test every cycle
E A weekly visual check

929 Impressions and try-ins move between the surgery and the laboratory.
What does HTM 01-05 require?
A Rinse and disinfect them in both directions, and label them as disinfected
B Disinfect only outgoing items
C Disinfect only incoming items
D Autoclave all impressions
E Nothing

930 You review dental unit waterline management.
What flushing is recommended?
A None
B Once a week
C Only at the end of the day
D About 2 minutes at the start of the day and 20–30 seconds between patients, with a biocide
E 10 minutes between patients

931 A nurse sustains a needlestick from a used syringe.
What is the immediate management?
A Encourage bleeding, wash with soap and water without scrubbing, cover, report, and contact occupational health urgently (HIV PEP ideally within 1 hour)
B Suck the wound
C Scrub with alcohol
D Ignore it if the patient seems healthy
E Wait until the end of the day

932 A newly employed dentist will perform exposure-prone procedures.
What is required?
A Nothing
B An annual flu vaccine only
C A chest radiograph
D Hepatitis B vaccination with a documented response
E A blood test for anaemia

933 A 6-year-old has unexplained facial bruising and multiple untreated carious teeth.
What is the dentist's role?
A Investigate the family
B Confront the parents
C Recognise, record and refer; do not investigate yourself
D Ignore it
E Treat and say nothing

934 A 15-year-old discloses female genital mutilation.
What must a regulated professional in England and Wales do?
A Nothing
B Report the known case to the police
C Tell only her parents
D Record it only
E Refer to a dermatologist

935 A wheelchair user cannot reach your upstairs surgery.
What does the Equality Act 2010 require?
A Nothing
B Refusal of care
C A higher fee
D Referral only to hospital
E Reasonable adjustments for a person with a protected characteristic (disability)

936 A practice manager lists the pillars of clinical governance.
Which list is correct?
A Profit, marketing and sales
B Only audit
C Clinical effectiveness and audit, risk management, education and training, patient involvement, information management, staff management and research

D Only complaints
E Only CPD

937 A dentist checks her RPD records against a published standard.
What is this activity?
A Research
B Audit, measuring practice against a standard
C Service evaluation
D A clinical trial
E A case report

938 An audit found poor recording of RPD designs, and changes have been made.
What closes the audit loop?
A Publishing the audit
B Discarding the data
C Writing a guideline
D Re-audit
E Disciplining staff

939 A practice uses small, repeated cycles to test a new recall system.
What is this quality-improvement method?
A Root cause analysis
B A randomised trial
C Audit only
D Significant event analysis
E Plan–Do–Study–Act (PDSA) cycles

940 You are appraising evidence for a new bonding protocol.
Which evidence is at the top of the hierarchy?
A A systematic review and meta-analysis of RCTs
B A single RCT
C A cohort study
D A case series
E Expert opinion

941 A guideline rates the evidence for a recommendation as "low certainty".
Which system grades certainty of evidence?
A Bolam
B Montgomery
C Caldicott
D SPIKES
E GRADE

942 A new caries test correctly identifies 90% of carious surfaces.
What does this figure describe?
A Specificity
B Positive predictive value

C Sensitivity (true positive rate)
D Accuracy
E Prevalence

943 The same test is used in a population with very low caries prevalence.
What happens to its positive predictive value?
A It falls, because PPV depends on prevalence
B It rises
C It is unchanged
D It equals sensitivity
E It equals specificity

944 A treatment reduces failure from 30% to 20%.
What is the number needed to treat?
A 1
B 3
C 5
D 10
E 20

945 A trial reports a risk ratio of 0.85 with a 95% confidence interval of 0.70 to 1.04.
What does this mean?
A A significant reduction
B The result is not statistically significant, because the interval crosses 1
C A significant increase
D The study was too large
E The result proves no effect

946 A dentist moves from England to Wales.
Which bodies regulate services in each of the four UK nations?
A CQC (England), Healthcare Improvement Scotland, Healthcare Inspectorate Wales, RQIA (Northern Ireland)
B The GDC everywhere
C The BDA everywhere
D The NHS everywhere
E The MHRA everywhere

947 A dentist applies to provide Level 2 prosthodontics.
What is required?
A Nothing beyond registration
B A formal link with a consultant-led managed clinical network, a minimum caseload, outcome recording (PROMs and PREMs), audit and complication reporting
C A specialist title
D A PhD
E Hospital employment

948 You must tell a patient that her failing teeth cannot be saved.
Which framework structures breaking bad news?
A BRAN
B PDSA
C SPIKES: setting, perception, invitation, knowledge, emotions, strategy and summary
D GRADE
E ABCDE

949 A patient speaks little English and her 12-year-old son offers to interpret.
What is appropriate?
A Use the son
B Proceed with gestures
C Use another patient
D Postpone indefinitely
E Use a professional interpreter

950 An NHS patient in England starts a course involving a bridge.
What supports informed consent and reduces complaints?
A A verbal estimate only
B No documentation
C A leaflet without discussion
D A written treatment plan and estimate (FP17DC)
E Payment in advance

Block N — Integrated cases (Q951–1000)

> *All units · Longer stems that combine medical, clinical and professional issues, as in the real paper · Allow 60 minutes · FDI notation*

951 A 72-year-old has an upper complete denture and a lower Kennedy Class I RPD. 33 is broken down to the gingival margin but its root is sound. She takes apixaban for atrial fibrillation and has taken alendronic acid for 7 years.
What is the most appropriate management of 33?
A Stop apixaban for 3 days and extract it
B Extract it under antibiotic cover after a 3-month drug holiday
C Root-treat and retain the root as an overdenture abutment, if feasible, rather than extracting
D Leave it untreated under the denture
E Crown it to act as a clasped abutment

952 A 28-year-old has palatal erosion of the upper incisors and reflux symptoms. There is no space to restore the teeth, and she asks for porcelain veneers.
What is the most appropriate first step?
A Liaise with her GMP about the reflux, give preventive advice, and monitor before any restoration
B Prepare the teeth for veneers now
C Place Dahl composites today

D Crown lengthening
E Extract and replace with implants

953 An 83-year-old care-home resident with moderate dementia has lost both dentures. Staff report that she cannot eat solid food.
What is the most appropriate approach?
A Implant overdentures
B Do nothing, as dementia contraindicates dentures
C Copy dentures from the lost originals
D An immediate denture
E Assess capacity, act in her best interests with carer involvement, and provide simple conventional dentures with labelling

954 A patient with a known latex allergy develops urticaria, wheeze and hypotension minutes after an impression was taken by a nurse wearing latex gloves.
What is the immediate management?
A Chlorphenamine orally and observation
B Remove the allergen, call for help, and give adrenaline 1:1000, 500 micrograms IM
C Salbutamol and discharge
D Hydrocortisone only
E Glucose gel

955 A cement-retained implant crown on 21 was fitted 6 months ago. There is bleeding on probing at a 4 mm pocket, and the radiograph shows a small radiopaque fleck at the margin with no bone loss.
What is the diagnosis and management?
A Peri-implantitis; refer for surgery
B Implant failure; explant
C Normal remodelling; no action
D Peri-implant mucositis from residual cement: remove the cement, debride, give OHI and review (Level 2)
E Occlusal overload; adjust the opposing tooth

956 A 64-year-old had radiotherapy to the mandible 3 years ago. A lower molar under his RPD has become non-restorable.
What is the most appropriate action?
A Extract it in primary care with antibiotics
B Refer for specialist assessment because of the lifelong osteoradionecrosis risk; consider endodontic retention if possible
C Leave it and reline the RPD
D Extract with a drug holiday
E Place an implant at the same visit

957 A 19-year-old has palatal erosion, parotid enlargement and calluses on her knuckles. She asks for crowns and does not want her parents to know.
What is the most appropriate approach?
A Tell her parents
B Refuse treatment

C Place crowns immediately
D Report to social services
E Approach with empathy, seek her consent to liaise with her GMP, and focus on prevention and monitoring before restoration

958 A 77-year-old with an acrylic upper denture has diffuse palatal erythema. She takes warfarin and simvastatin and wears the denture day and night.
What is the most appropriate management?
A Denture hygiene, night removal, soaking, correcting the fit, and nystatin if it persists
B Miconazole oral gel
C Fluconazole 50 mg daily
D Biopsy of the palate
E Stop warfarin

959 A resin-bonded bridge replacing 21 has debonded three times. The patient is a bruxist with a deep overbite and heavy contact on the wing.
What is the most appropriate next step?
A Rebond again with the same design
B Use temporary cement
C Add a second wing to 22
D Review the occlusion and design, and discuss alternatives such as a conventional bridge, an implant or an RPD
E Extract 11

960 A 55-year-old has generalised severe wear, dentine exposure on most teeth and a planned OVD increase of 4 mm.
What is the most appropriate plan?
A Crowns on all teeth at the current OVD
B Dahl composites on the posterior teeth only
C A diagnostic wax-up and provisional trial at the new OVD, with referral for Level 3 planning
D Extraction and complete dentures
E Monitor only, despite symptoms

961 A 58-year-old with treated, stable periodontitis and high caries risk needs a lower Kennedy Class II RPD. The lingual sulcus is 9 mm deep.
Which design is most appropriate?
A An acrylic mucosa-borne plate covering the gingivae
B A flexible nylon denture
C Co-Cr with a lingual plate and distal rests
D Co-Cr, tooth-supported, with a mesial rest and RPI on the terminal abutment, a lingual bar, an open design and maximum saddle extension
E A swing-lock

962 A new upper complete denture drops on wide opening, and the patient gags. The posterior border is thick, and the distobuccal flange is long.
What is the most appropriate correction?
A Add adhesive and increase the OVD

B Remake with porcelain teeth
C Trim the distobuccal flange and thin the posterior border, preserving the post-dam seal
D Remove the post-dam
E Reline with a soft liner

963 A 14-year-old's replanted upper central incisor is now infra-occluded and ankylosed. He is still growing.
What is the most appropriate approach?
A Monitor and seek multidisciplinary specialist advice; defer any implant until growth is complete
B Place an implant now
C Crown the tooth to level it
D Extract it and leave the space
E Orthodontic extrusion of the ankylosed tooth

964 A 70-year-old on IV zoledronic acid for myeloma wants implants to stabilise her lower complete denture.
What is the most appropriate advice?
A Implants are routine
B Stop the zoledronic acid and place implants
C Place implants with antibiotic cover
D Refuse any denture
E A well-fitting conventional denture with relief over bony prominences; implants are generally avoided because of high MRONJ risk

965 A 52-year-old with an HbA1c of 80 mmol/mol, generalised Stage III periodontitis and missing 46 wants a bridge.
What is the most appropriate sequence?
A Prepare the bridge now
B Periodontal therapy and referral to the GMP for diabetic control first; reassess before any definitive bridge
C Extract all molars
D Place an implant
E Prescribe antibiotics and proceed

966 A heavy bruxist had root canal treatment on 46 two weeks ago. The tooth has an MOD cavity with thin cusps and a temporary restoration.
What is the most appropriate restoration?
A Prompt cuspal coverage (onlay or crown) in metal or monolithic zirconia, plus a night guard
B A large composite without coverage
C Layered feldspathic porcelain onlay
D Leave the temporary for a year
E A post and amalgam core only

967 You realise after the patient has left that you extracted 15 instead of 14.
What is the most appropriate action?
A Say nothing unless she notices
B Alter the notes to read 15

C Wait for a complaint
D Contact her, explain, apologise, offer a remedy, report it as a never event and document everything
E Ask a colleague to see her next time

968 An NHS patient writes to complain about the cost and fit of her new denture.
What is the first required step?
A Ignore it until she phones
B Refund immediately without discussion
C Acknowledge it within 3 working days and investigate locally
D Refer it to the GDC
E Refer it to the Ombudsman at once

969 A patient with an implanted cardiac defibrillator and a high smile line needs a crown on 11 with a slightly subgingival margin.
What retraction method and margin position are appropriate?
A Electrosurgery and a deep margin at the bone crest
B Adrenaline cord and a 3 mm subgingival margin
C Laser troughing to the crest
D No retraction and a supragingival margin regardless of aesthetics
E Plain or aluminium chloride cord, with the margin just subgingival and about 3 mm from the bone crest

970 An upper RPD is needed for anterior and posterior saddles. The patient has a large torus and refuses surgery.
Which major connector is most appropriate?
A Full palatal plate
B Anteroposterior strap avoiding the torus
C Horseshoe
D Narrow mid-palatal bar over the torus
E Lingual bar

971 An 85-year-old attending for a denture reline has fingertip bruising on both arms. Her carer answers all questions for her.
What is the most appropriate action?
A Ignore it
B Ask the carer to explain
C Photograph the bruising and investigate
D Report directly to the police without discussion
E Record your findings and refer through the practice safeguarding lead to adult social care

972 A 20-year-old with hypodontia (missing 12, 22, 35 and 45) wants fixed replacements.
What is the most appropriate pathway?
A Implants in general practice immediately
B Interim RBBs or an RPD, with referral for multidisciplinary Level 3 care
C Conventional bridges on all sites
D No treatment
E Crowns on all adjacent teeth

973 During try-in, a crown slips from your fingers to the back of the patient's mouth. He is coughing forcefully and can speak.
What is the most appropriate immediate action?
A Start back blows at once
B Abdominal thrusts
C Lay him flat
D Encourage coughing; if the crown is not retrieved, arrange urgent chest imaging to exclude inhalation
E Ignore it

974 An implant supporting a crown on 36 shows suppuration, a 7 mm pocket and 4 mm of radiographic bone loss since the baseline radiograph.
What is the most appropriate action?
A Debride and review in a year
B Retighten the abutment screw
C Refer as peri-implantitis (Level 3)
D Remove the crown permanently
E Prescribe antibiotics only

975 A patient with masticatory myalgia insists on "having her bite adjusted" before her crowns are made.
What is the most appropriate approach?
A Explain that reversible care comes first (self-care and a stabilisation splint), and do not equilibrate to treat TMD
B Equilibrate the whole dentition
C Prepare the crowns at a new OVD
D Prescribe long-term opioids
E Refer for TMJ surgery

976 A 79-year-old with a severely resorbed mandible has burning and numbness of the lower lip whenever she wears her new lower denture. The symptoms resolve when it is out.
What is the most likely cause and action?
A Candida; antifungal
B Allergy; patch test
C Pressure on the mental nerve; relieve the fitting surface over the foramen
D Excessive OVD; remake
E Malignancy; urgent referral

977 A patient with lower canine overdenture abutments has new root caries on both.
What is the most appropriate preventive measure?
A Fluoride gel in the fitting surface of the denture, with meticulous hygiene and a diet review
B Extract both roots
C Wear the denture day and night
D Crown both roots
E Antibiotics

978 A patient has a two-implant lower overdenture opposing an upper complete denture. The upper anterior ridge has become flabby.

What is the most appropriate approach to a new upper denture?
A A compressive putty impression and heavy anterior contacts
B A selective-pressure (windowed) impression, with balanced occlusion avoiding heavy anterior contact
C Extraction of the lower implants
D Excision of the flabby ridge first, routinely
E A soft liner only

979 A 60-year-old with Sjögren's syndrome has a Co-Cr RPD and multiple new root caries lesions.
What is the most appropriate integrated plan?
A Remove the RPD permanently
B Use acidic saliva substitutes
C Extract all carious teeth
D High-fluoride toothpaste, a neutral saliva substitute, an open RPD design and a short recall
E Antibiotics

980 A trial reports that a new denture adhesive reduces denture failure from 40% to 30%.
What is the number needed to treat?
A 1
B 3
C 4
D 30
E 10

981 A colleague requests CBCT at every annual implant review "to be safe".
What is correct?
A Each exposure must be justified individually; a periapical radiograph against baseline usually answers the clinical question
B CBCT is the routine standard
C No radiographs are ever needed
D A panoramic radiograph every 3 months
E Patient preference decides

982 A patient asks the practice to delete all her records because she is moving abroad.
What is the correct response?
A Delete them immediately
B Delete only the radiographs
C Give her the original records
D Explain that records must be retained (adults at least 11 years after last treatment), and offer copies
E Shred them after a month

983 A 17-year-old with capacity wants a resin-bonded bridge. Her mother refuses.
What is the correct position?
A The mother's refusal stands
B The 17-year-old is presumed to have capacity and can consent
C A court order is needed
D Wait until she is 18
E The dentist decides

984 An 81-year-old with dementia whose capacity fluctuates needs a painful, infected tooth removed. On the day, she cannot weigh the information.
What is the most appropriate approach?
A Proceed because she agreed last month
B Ask a neighbour to sign
C Act in her best interests, consulting those close to her and choosing the least restrictive option, unless she regains capacity
D Never treat her
E Wait for a court order in every case

985 A crown arrives from a new laboratory with no statement of manufacture.
What should you do?
A Fit it anyway
B Throw it away
C Accept it if it fits well
D Ask the patient to sign a waiver
E Request the statement and confirm the laboratory is MHRA-registered before fitting

986 You take an impression for a patient with known hepatitis B.
How should the impression be handled?
A Autoclave it
B Treat it as high risk and refuse to send it
C Send it untreated with a warning label
D Bag it without disinfection
E Standard precautions: rinse, disinfect per protocol, label and send, as for every patient

987 A rigid five-unit bridge from 13 to 17, with a pier at 15, has debonded at 13. There is caries under the 13 retainer.
What is the most appropriate management?
A Recement it
B Cut the bridge and leave 13 untreated
C Remove it, treat the caries, and redesign with a non-rigid connector at the pier or as separate bridges
D Add a sixth unit
E Use temporary cement

988 A crown on 11 looks too light. The shade was taken after preparation under the operating light.
What explains the mismatch?
A Wrong cement
B Wrong material
C Laboratory error only
D Dehydration and poor lighting; the shade should be taken before preparation in colour-corrected daylight
E The patient's diet

989 A patient with moderate haemophilia A needs several extractions before an immediate upper denture.
What is the most appropriate management?

A Proceed in practice with local measures only
B Liaise with the haemophilia centre, avoiding regional blocks without factor cover
C Give aspirin for pain
D Avoid dentures
E Extract one tooth per year

990 A 68-year-old has taken oral alendronate for 2 years for osteoporosis, without steroids. A tooth under her RPD needs extraction.
What is the most appropriate approach?
A Proceed with standard care and no drug holiday; review healing
B Stop alendronate for 6 months first
C Refuse extraction
D Antibiotics for a month first
E Refer every case to hospital

991 A patient with a severe gag reflex has never tolerated an upper complete denture.
Which option may help?
A A thicker palate
B An implant-retained palateless prosthesis (after referral), with desensitisation
C A longer post-dam
D A soft liner
E Adhesive only

992 A patient asks whether her resin-bonded bridge has been a "success" because it is still present after 6 years, though it has debonded twice.
What is the most accurate explanation?
A It is a complete success
B It is a failure
C Debonding does not count
D Survival and success are identical
E It has survived, but success means in place and free of complications

993 A 30-year-old has lost 12. 13 is sound, unrestored and enamel-rich, and the occlusion is light.
What is the most appropriate first-choice replacement?
A A conventional fixed-fixed bridge from 11 to 13
B An immediate implant without assessment
C A single-wing resin-bonded cantilever from 13
D An acrylic RPD
E A spring cantilever

994 A patient with an upper complete denture and only lower anterior teeth has a flabby premaxilla and enlarged tuberosities.
What is the most appropriate management?
A Provide posterior support with a lower RPD, and use a selective-pressure impression for the flabby ridge
B Extract the lower anteriors immediately
C Increase anterior loading

D Excise all flabby tissue routinely
E No change

995 A 74-year-old attends for a denture adjustment and mentions new pain in the jaw when chewing, a headache and scalp tenderness.
What is the most appropriate action?
A Adjust the denture and review in a month
B A stabilisation splint
C Ibuprofen
D Same-day medical referral for suspected giant cell arteritis
E TMJ radiograph

996 A patient had a myocardial infarction 3 weeks ago and asks to start elective bridgework now.
What is the most appropriate approach?
A Proceed today with sedation
B Refuse care permanently
C Give GTN and proceed
D Liaise with the physician and defer elective treatment until he is medically stable
E Proceed without local anaesthetic

997 A patient on dual antiplatelet therapy needs a crown preparation with retraction cord.
What is correct?
A Stop clopidogrel for 7 days
B Continue both antiplatelets; it is a low bleeding-risk procedure
C Stop aspirin only
D Refer to hospital
E Give vitamin K

998 A 66-year-old smoker has an ulcer under her lower denture flange that persists 4 weeks after the flange was adjusted.
What is the most appropriate action?
A Adjust further and review in 3 months
B A soft liner
C Chlorhexidine
D Antifungal
E Urgent referral on the suspected cancer pathway

999 An audit showed incomplete RPD prescriptions. The practice introduced a template.
What is the next step?
A Re-audit to close the loop
B Publish the results
C Stop auditing
D Change laboratory
E Discipline the staff

1000 A patient has completed treatment with an implant overdenture, a Co-Cr RPD and several crowns.
What is the most appropriate long-term plan?

A No further review
B Review only when there is pain
C Risk-based recall with supportive periodontal and peri-implant care, prosthesis maintenance, and recording of patient-reported outcomes
D Annual CBCT
E Replace all prostheses every 3 years

Part 2 — Answers and justifications

Block A answers — Applied anatomy

1 · Answer B — Masseteric nerve.
Why: Masseter is supplied by the masseteric nerve, a motor branch of the anterior division of V3 that passes through the mandibular (sigmoid) notch. Masseteric hypertrophy is a recognised sign of bruxism.
Not the others: The buccal nerve is the only *sensory* branch of the anterior division; the facial nerve supplies muscles of facial expression, not mastication.
Revise: Unit 1 §1.1 · LO 1.1

2 · Answer D — Temporalis.
Why: Temporalis lies in the temporal fossa and is the muscle most often palpably tender in tension-type headache linked to TMD.
Not the others: Masseter is felt over the ramus; the pterygoids cannot be palpated reliably from the temple.
Revise: Unit 1 §1.1 (exam traps), Unit 9 · LO 1.1

3 · Answer A — Lateral pterygoid (inferior head).
Why: The inferior head inserts into the pterygoid fovea on the condylar neck, so after a neck fracture it pulls the head forwards and medially.
Not the others: Masseter, temporalis and medial pterygoid act on the ramus and coronoid, below the fracture.
Revise: Unit 1 §1.1 · LO 1.1

4 · Answer C — It deviates to the right.
Why: Each lateral pterygoid pulls its own condyle forward. If the right one fails, the left condyle translates alone and swings the chin towards the paralysed (right) side.
Not the others: Opening still occurs through the left muscle and the depressors, so D is wrong.
Revise: Unit 1 §1.1 · LO 1.1

5 · Answer E — All of the above.
Why: Masseter, temporalis and medial pterygoid are all elevators. Only the lateral pterygoid opens the mouth.
Not the others: D omits medial pterygoid, which also elevates and protrudes.
Revise: Unit 1, In one minute · LO 1.1

6 · Answer B — Rotation.
Why: The lower compartment (condyle against disc) allows hinge rotation for the first 20–25 mm; translation then occurs in the upper compartment.
Not the others: Translation is an upper-compartment movement.
Revise: Unit 1 §1.5 · LO 1.5

7 · Answer E — Auriculotemporal nerve.
Why: The auriculotemporal nerve is the main sensory supply of the TMJ (with the masseteric and deep temporal nerves, Hilton's law). It also supplies the ear, which explains referred earache.
Not the others: The chorda tympani carries taste; the great auricular supplies skin over the parotid.
Revise: Unit 1 §1.5 · LO 1.5

8 · Answer A — Retrodiscal (bilaminar) tissue.
Why: The disc is avascular and aneural centrally. When it displaces anteriorly, the condyle loads the vascular, innervated bilaminar zone, which is the source of pain.
Revise: Unit 1 §1.5 · LO 1.5

9 · Answer D — Lateral (temporomandibular) ligament.
Why: It is the true stabiliser, limiting posterior and inferior condylar movement.
Not the others: The sphenomandibular and stylomandibular ligaments are accessory and contribute little.
Revise: Unit 1 §1.5 · LO 1.5

10 · Answer C — Fibrocartilage.
Why: The TMJ surfaces are fibrocartilage, reflecting their intramembranous origin. This tissue can remodel under load.
Not the others: Hyaline cartilage covers most other synovial joints.
Revise: Unit 1 §1.5 · LO 1.5

11 · Answer B — Glossopharyngeal afferent, vagus efferent.
Why: IX carries the sensation from the posterior tongue and oropharynx; X drives the pharyngeal response.
Revise: Unit 1 §1.4 · LO 1.1

12 · Answer B — Genioglossus; hypoglossal nerve.
Why: Genioglossus and the lingual frenum limit the anterior lingual flange; protrusion tightens them and lifts an over-extended flange.
Not the others: Mylohyoid limits the flange in the molar region, not anteriorly.
Revise: Unit 1 §1.4, §1.6 · LO 1.1, 2a.5

13 · Answer A — Mylohyoid.
Why: Mylohyoid rises on swallowing. In the molar region the lingual flange should slope towards the tongue rather than run straight down below the ridge.
Revise: Unit 1 §1.6 · LO 1.1

14 · Answer E — Masseter.
Why: Masseter contracts on clenching and pushes on the distobuccal flange, forming the masseteric notch.

Not the others: Buccinator lies over the buccal shelf further forward.
Revise: Unit 1 §1.6 · LO 1.1

15 · Answer D — Coronoid process.
Why: On opening and lateral movement the coronoid moves forward into the space beside the tuberosity, limiting the width of the upper distobuccal flange.
Revise: Unit 1 §1.6 · LO 1.1

16 · Answer C — Fovea palatini, near the vibrating line.
Why: The post-dam lies between the anterior and posterior vibrating lines; the foveae sit close to this area and help locate it.
Revise: Unit 1 §1.6 · LO 1.1

17 · Answer A — Posterior limit of the tuberosity and lateral extent of the post-dam.
Why: The post-dam runs from one hamular notch to the other, and the denture must reach into the notches for a complete seal.
Revise: Unit 1 §1.6 · LO 1.1

18 · Answer B — Mental nerve at a foramen near the crest.
Why: In severe resorption the mental foramen comes to lie on or near the ridge crest, so the denture compresses the nerve. Relieve the fitting surface over it.
Not the others: Allergy causes generalised mucosal changes, not lip paraesthesia.
Revise: Unit 1 §1.6 · LO 1.1, 2a.5

19 · Answer E — Incisive papilla.
Why: The papilla overlies the incisive foramen and the nasopalatine nerve, a standard relief area. Pressure causes burning anterior pain.
Revise: Unit 1 §1.6 · LO 1.1

20 · Answer C — Parotid duct.
Why: It opens opposite the upper second molar and can be compressed by an over-extended or thick buccal flange.
Revise: Unit 1 §1.6, §1.8 · LO 1.1

21 · Answer D — Submandibular space.
Why: By the mylohyoid line rule, lower second and third molar apices lie below the mylohyoid attachment, so infection spreads into the submandibular space.
Revise: Unit 1 §1.7 · LO 1.1

22 · Answer A — Sublingual space.
Why: Teeth from the incisors to the first molar have apices above mylohyoid; infection spreads into the sublingual space.
Revise: Unit 1 §1.7 · LO 1.1

23 · Answer E — Emergency referral for airway management.
Why: Bilateral submandibular, sublingual and submental involvement is Ludwig's angina, an airway emergency.

Not the others: Treating in practice, or delaying with antibiotics, risks airway obstruction.
Revise: Unit 1 §1.7, companion note 03 (risk and referral) · LO 1.1, 2a.6

24 · Answer B — Cavernous sinus thrombosis.
Why: Upper canine infection above levator anguli oris enters the canine space. It can spread through the angular and ophthalmic veins to the cavernous sinus.
Revise: Unit 1 §1.7 · LO 1.1

25 · Answer C — Emissary veins.
Why: The pterygoid plexus connects to the cavernous sinus through emissary veins, a route for intracranial spread of infection.
Revise: Unit 1 §1.3 · LO 1.1

26 · Answer D — Pterygomandibular space.
Why: It contains the inferior alveolar and lingual nerves and vessels and is the target of the block.
Revise: Unit 1 §1.7 · LO 1.1

27 · Answer A — Submasseteric space.
Why: Submasseteric infection, often from a lower third molar, presents with marked trismus and swelling over the angle.
Revise: Unit 1 §1.7 · LO 1.1

28 · Answer E — Long buccal nerve.
Why: The buccal nerve is the only sensory branch of the anterior division of V3; the inferior alveolar block does not reach it, so the buccal gingiva of lower molars needs a separate injection.
Revise: Unit 1 §1.2 · LO 1.1

29 · Answer B — Mylohyoid and anterior belly of digastric.
Why: The nerve to mylohyoid leaves the inferior alveolar nerve and supplies both.
Not the others: The posterior belly of digastric is facial nerve; geniohyoid is C1.
Revise: Unit 1 §1.2, §1.4 · LO 1.1

30 · Answer D — Both A and B.
Why: The chorda tympani (VII) carries taste from the anterior two-thirds of the tongue and preganglionic secretomotor fibres to the submandibular ganglion.
Not the others: Motor supply to the tongue is the hypoglossal nerve.
Revise: Unit 1 §1.2, §1.8 · LO 1.1

31 · Answer C — Auriculotemporal nerve.
Why: It carries secretomotor fibres to the parotid. After surgery these fibres can regrow into sweat glands (Frey's syndrome).
Revise: Unit 1 §1.2 · LO 1.1

32 · Answer A — Middle meningeal artery.
Why: A branch of the first part of the maxillary artery, it enters the skull through the foramen spinosum. The auriculotemporal nerve embraces it.
Revise: Unit 1 §1.3 · LO 1.1

33 · Answer D — Descending palatine artery.
Why: The third (pterygopalatine) part of the maxillary artery gives the descending palatine artery, which divides into the greater and lesser palatine arteries.
Revise: Unit 1 §1.3 · LO 1.1

34 · Answer B — Second (pterygoid) part.
Why: The second part, related to the lower head of lateral pterygoid, gives the deep temporal, pterygoid, masseteric and buccal branches.
Revise: Unit 1 §1.3 · LO 1.1

35 · Answer E — It arises from the main trunk of V3.
Why: The main trunk gives the nervus spinosus and the nerve to medial pterygoid before dividing. That nerve also passes through the otic ganglion to tensor veli palatini and tensor tympani.
Revise: Unit 1 §1.1 (exam traps), §1.2 · LO 1.1

36 · Answer A — Masseter and medial pterygoid.
Why: Masseter on the lateral surface and medial pterygoid on the medial surface together form the pterygomasseteric sling around the angle.
Revise: Unit 1 §1.1 (exam traps) · LO 1.1

37 · Answer C — Temporalis.
Why: Temporalis inserts into the apex and medial surface of the coronoid and the anterior border of the ramus, so it pulls a coronoid fragment upwards.
Not the others: Lateral pterygoid inserts into the pterygoid fovea, disc and capsule, never the coronoid.
Revise: Unit 1 §1.1 · LO 1.1

38 · Answer E — Temporalis (posterior fibres).
Why: The near-horizontal posterior fibres are the main retractors; the deep masseter assists.
Not the others: Lateral pterygoid protrudes, the opposite movement.
Revise: Unit 1, In one minute · LO 1.1

39 · Answer D — Right lateral and medial pterygoids.
Why: To move the chin left, the right condyle must move forward and medially; the right pterygoids do this.
Revise: Unit 1 §1.1 · LO 1.1

40 · Answer B — Lateral pterygoid (superior head).
Why: The superior head attaches to the disc and capsule, so it is the muscle most associated with disc displacement, clicking and locking.
Revise: Unit 1 §1.1 (exam traps), Unit 9 · LO 1.1, 2a.15

41 · Answer E — Vagus via the pharyngeal plexus.
Why: Palatoglossus is the only tongue muscle not supplied by the hypoglossal nerve. Normal protrusion shows XII is intact.
Revise: Unit 1 §1.4 · LO 1.1

42 · Answer C — Tensor veli palatini.
Why: It is supplied by the nerve to medial pterygoid (V3). All other soft palate muscles are supplied by

the pharyngeal plexus.
Revise: Unit 1 §1.4 · LO 1.1

43 · Answer A — Buccinator.
Why: Buccinator (facial nerve) keeps food on the occlusal table and grips the buccal polished surface. Weakness lets food pocket and reduces muscular control of the denture.
Revise: Unit 1 §1.4 · LO 1.1, 2a.5

44 · Answer D — Orbicularis oris and the modiolus.
Why: These determine lip support, the neutral zone and the shape of the labial flange.
Revise: Unit 1 §1.4, Unit 7 (neutral zone) · LO 1.1

45 · Answer B — C1 via the hypoglossal nerve.
Why: Geniohyoid is supplied by C1 fibres that travel with XII; it depresses the mandible and pulls the hyoid forward.
Revise: Unit 1 §1.4 · LO 1.1

46 · Answer E — Posterior belly of digastric (and stylohyoid).
Why: Both are second-arch muscles supplied by the facial nerve. The anterior belly is first-arch, supplied by the nerve to mylohyoid.
Revise: Unit 1 §1.4 (exam traps) · LO 1.1

47 · Answer C — At least the anterior half to two-thirds.
Why: This gives support and a posterior seal without impinging on the pterygomandibular raphe.
Revise: Unit 1 §1.6 · LO 1.1

48 · Answer A — The buccal shelf.
Why: Between the external oblique ridge and the residual ridge, its cortical bone lies at right angles to occlusal load, making it the primary support area.
Revise: Unit 1 §1.6 · LO 1.1

49 · Answer E — All of the above.
Why: The incisive papilla, midline raphe and torus palatinus are thin, unyielding or innervated areas, listed as relief areas in the maxilla.
Revise: Unit 1 §1.6 · LO 1.1

50 · Answer B — Pterygomandibular raphe.
Why: The raphe runs from the hamulus to the retromolar region and tightens on wide opening, lifting a base that extends too far distally.
Revise: Unit 1 §1.6 · LO 1.1

51 · Answer E — All of the above.
Why: Genial tubercles, a crestal mental foramen, a sharp mylohyoid ridge and a knife-edged crest are all mandibular relief areas in severe resorption.
Revise: Unit 1 §1.6 · LO 1.1

52 · Answer C — Submandibular (about 65–70%).
Why: The submandibular gland produces most unstimulated (resting) saliva.

Not the others: The parotid dominates stimulated flow.
Revise: Unit 1 §1.8 · LO 1.1

53 · Answer A — Stimulated saliva.
Why: The parotid contributes most of the stimulated saliva, so its loss reduces flow during eating most.
Revise: Unit 1 §1.8, Unit 4 (radiotherapy) · LO 1.1

54 · Answer D — Submandibular (Wharton's) duct.
Why: It opens at the sublingual papilla beside the lingual frenum, the common site for salivary stones.
Revise: Unit 1 §1.8 · LO 1.1

55 · Answer B — IX, lesser petrosal, otic ganglion, auriculotemporal.
Why: This is the parasympathetic route to the parotid.
Not the others: A is the submandibular route; E supplies the lacrimal and nasal glands.
Revise: Unit 1 §1.8 · LO 1.1

56 · Answer C — VII, chorda tympani, lingual nerve, submandibular ganglion.
Why: The chorda tympani passes through the middle ear, so ear surgery can reduce submandibular and sublingual secretion.
Revise: Unit 1 §1.8 · LO 1.1

57 · Answer E — Mainly mucous, with multiple ducts of Rivinus.
Why: The sublingual gland drains through many small ducts on the sublingual fold and shares the submandibular parasympathetic supply.
Revise: Unit 1 §1.8 · LO 1.1

58 · Answer A — Superficial temporal and maxillary arteries.
Why: The TMJ is supplied by branches of these two vessels.
Revise: Unit 1 §1.5 · LO 1.5

59 · Answer D — Synovial ginglymoarthrodial joint.
Why: It hinges (rotation, lower compartment) and glides (translation, upper compartment).
Revise: Unit 1 §1.5 · LO 1.5

60 · Answer C — Sphenomandibular ligament.
Why: It runs from the spine of the sphenoid to the lingula, medial to the nerve at the mandibular foramen, and is an accessory TMJ ligament.
Revise: Unit 1 §1.5 · LO 1.5

61 · Answer B — Assess borders and polished surfaces against muscle action.
Why: A denture retentive at rest but displaced in function suggests over-extension or polished-surface contours that conflict with muscles. Diagnose the functional border first.
Not the others: Adhesive, occlusal changes or a remake do not address the cause.
Revise: Unit 1 companion note 01, 04 · LO 1.1, 2a.5

62 · Answer E — Loss of the peripheral seal.
Why: A short lingual flange fails to contact the floor of the mouth, breaking the seal and reducing retention.
Revise: Unit 1 companion note 04 · LO 1.1

63 · Answer A — Urgent referral.
Why: Unexplained, persistent lip numbness with no denture cause may signal malignancy or neuropathy and must be investigated.
Not the others: Relieving and reviewing in 6 months delays diagnosis.
Revise: Unit 1 companion note 03 · LO 1.1, 2a.6

64 · Answer D — Urgent maxillofacial referral.
Why: Fever, trismus and dysphagia show a spreading infection that may threaten the airway.
Revise: Unit 1 §1.7, companion note 03 · LO 1.1, 2a.6

65 · Answer B — Mental foramen and anterior loop of the inferior alveolar nerve.
Why: In atrophy the foramen lies high on the ridge and the nerve may loop forward, limiting premolar implant sites. Recognise it and refer for appropriate imaging and planning.
Revise: Unit 1 companion note 02, Unit 11 · LO 1.1, 2a.19

66 · Answer C — Soft-palate form and movement.
Why: The posterior seal depends on the vibrating line, and its position varies with soft-palate form; examine it before deciding the seal.
Revise: Unit 1 companion note 02 · LO 1.1, 2a.5

67 · Answer A — Choose a path or block out one side.
Why: A suitable path of insertion or relief on one side usually allows insertion; surgery is reserved for undercuts that still prevent it.
Revise: Unit 1 §1.6, Unit 7 · LO 1.1

68 · Answer D — Labial bar, or removal of the tori.
Why: Tori block a lingual connector; a labial bar avoids them, or the tori are removed if a sound design needs a lingual connector.
Revise: Unit 1 §1.6, Unit 6 (connectors) · LO 1.1, 2c.3

69 · Answer E — Lateral pharyngeal space.
Why: The pterygomandibular space communicates with the lateral pharyngeal space, which can lead to airway compromise.
Revise: Unit 1 §1.7 · LO 1.1

70 · Answer B — Nerve to mylohyoid.
Why: It can give accessory supply to lower molars; supplementary lingual infiltration or intraligamentary anaesthesia may help.
Revise: Unit 1 §1.2, Unit 4 (local anaesthesia) · LO 1.1

Under 70% on Block A? Re-read Unit 1 and Crash notebook Schematics 1–4, then re-sit in a fortnight.

Block B answers — Physiology and pathology

71 · Answer C — Hyposalivation.
Why: Unstimulated flow below 0.1 mL/min is hyposalivation (normal is about 0.3–0.4 mL/min). Amitriptyline and oxybutynin are both anticholinergic.
Not the others: A stimulated sample adds information, but the resting value alone already meets the

definition.
Revise: Unit 2 §2.1 · LO 1.2

72 · Answer E — All of the above.
Why: Retention comes from a thin saliva film (adhesion to mucosa and base, cohesion within the film), interfacial surface tension, viscosity, and atmospheric pressure maintained by the border seal.
Revise: Unit 2 §2.1 · LO 1.2

73 · Answer A — Thin, watery saliva.
Why: A thin film gives better adhesion and surface tension. Thick, ropy saliva forms a thicker film that reduces retention and can push the denture down.
Revise: Unit 2 §2.1 · LO 1.2

74 · Answer D — Neutral substitute with high-fluoride toothpaste.
Why: Substitutes for dentate patients must be pH-neutral; acidic products demineralise teeth. High-fluoride toothpaste protects against the high caries risk.
Not the others: Lemon, citric acid and sugared lozenges all increase caries or erosion.
Revise: Unit 2 §2.1 · LO 1.2, 2a.7

75 · Answer B — Pilocarpine.
Why: Pilocarpine is a muscarinic agonist that can provoke bronchospasm; it is contraindicated in uncontrolled asthma and glaucoma.
Revise: Unit 2 §2.1 · LO 1.2, 1.7

76 · Answer A — Bicarbonate.
Why: Bicarbonate concentration rises with flow rate and is the main buffer of stimulated saliva.
Revise: Unit 2 §2.1 · LO 1.2

77 · Answer E — All of the above.
Why: Lysozyme, lactoferrin, secretory IgA and histatins (antifungal) are all antimicrobial.
Revise: Unit 2 §2.1 · LO 1.2

78 · Answer C — About one-fifth to one-sixth.
Why: Complete denture wearers achieve about one-fifth to one-sixth of dentate bite force and chewing efficiency. Explaining this sets realistic expectations.
Revise: Unit 2 §2.2 · LO 1.2

79 · Answer B — Mucosal and TMJ receptors.
Why: Periodontal mechanoreceptors are lost with the teeth, so edentulous patients rely on mucosal and TMJ receptors for feedback.
Revise: Unit 2 §2.2 · LO 1.2

80 · Answer D — Less sensitive.
Why: Implants lack a periodontal ligament; their osseoperception is less sensitive than a natural tooth's feedback.
Revise: Unit 2 §2.2 · LO 1.2

81 · Answer E — No replacement; shortened dental arch.
Why: Käyser showed that premolar-to-premolar arches (about 3–5 occlusal units) usually give adequate

function in older adults. A free-end RPD often adds little and risks plaque and caries.
Revise: Unit 2 §2.2 · LO 1.2, 2c.1

82 · Answer B — 20.
Why: The WHO goal is retention of at least 20 functional teeth throughout life.
Revise: Unit 2 §2.2 · LO 1.2

83 · Answer D — Copy (replica) dentures.
Why: Ability to adapt falls with age and neuromuscular disease. Copying keeps the familiar shapes she has learned to control while correcting wear and fit.
Revise: Unit 2 §2.2, Unit 7 · LO 1.2, 2c.9

84 · Answer A — Centripetal maxilla, centrifugal mandible.
Why: The upper arch becomes smaller and the lower wider, giving an apparent Class III relationship and posterior crossbite.
Revise: Unit 2 §2.3 · LO 1.3

85 · Answer C — About four times as much.
Why: Tallgren showed the anterior mandible loses about four times the height of the anterior maxilla in the long term.
Revise: Unit 2, In one minute, §2.3 · LO 1.3

86 · Answer B — First 6–12 months.
Why: Resorption is fastest early after extraction and continues more slowly for life. This is why immediate dentures need relining.
Revise: Unit 2 §2.3 · LO 1.3

87 · Answer E — All of the above.
Why: Retained roots and implants load the bone and preserve it; night removal reduces continuous loading of the ridge.
Revise: Unit 2 §2.3 · LO 1.3

88 · Answer A — Continuous wear of ill-fitting dentures.
Why: Prosthetic factors (poor fit, continuous wear, inadequate support) accelerate resorption.
Revise: Unit 2 §2.3 · LO 1.3

89 · Answer D — Reduced interocclusal space.
Why: Over-eruption after tooth loss reduces the space for a replacement; the opposing tooth may need adjustment, a restoration or orthodontic intrusion first.
Revise: Unit 2 §2.3 · LO 1.3, 2c.1

90 · Answer C — 6.2–6.7.
Why: Root dentine and cementum demineralise at a higher pH than enamel (5.5), so root caries starts with milder acid challenges.
Revise: Unit 2 §2.4 (exam traps) · LO 1.4

91 · Answer D — Reduce frequency.
Why: The Stephan curve shows each intake drops plaque pH for 30–60 minutes. Frequency, not total

quantity, drives caries.
Revise: Unit 2 §2.4 · LO 1.4

92 · Answer A — Actinomyces.
Why: Actinomyces is particularly associated with root caries; Streptococcus mutans initiates coronal caries and Lactobacillus drives progression.
Revise: Unit 2 §2.4 · LO 1.4

93 · Answer B — Body of the lesion.
Why: Deep to surface, the zones are translucent, dark, body and surface; the body has the greatest mineral loss.
Not the others: The surface zone stays relatively intact.
Revise: Unit 2 §2.4 · LO 1.4

94 · Answer E — Selective removal.
Why: Remove soft, infected dentine; leave affected, remineralisable dentine over the pulp and seal well, avoiding pulp exposure.
Revise: Unit 2 §2.4, companion note 05 · LO 1.4

95 · Answer C — 5000 ppm.
Why: Delivering Better Oral Health (2021) recommends 5000 ppm for high-risk patients aged 16 and over.
Revise: Unit 2 §2.4 · LO 1.4, 2b.3

96 · Answer A — 22,600 ppm varnish at least twice yearly.
Why: This is the DBOH recommendation for adults at higher caries risk.
Revise: Unit 2 §2.4 · LO 1.4

97 · Answer D — All of the above.
Why: Recession exposes roots, hyposalivation removes protection, and partial dentures retain plaque around clasps and saddles. Poor dexterity adds to the risk.
Revise: Unit 2 §2.4 · LO 1.4

98 · Answer B — Bulimia nervosa.
Why: Palatal erosion (perimylolysis), parotid enlargement and Russell's sign (knuckle calluses) together point to bulimia.
Revise: Unit 2 §2.5 · LO 1.4, 2a.8

99 · Answer E — Erosion.
Why: Acid dissolves tooth around the restorations, leaving them standing proud, with smooth, silky enamel and cupped cusps.
Not the others: In attrition, restorations wear flush with the tooth.
Revise: Unit 2 §2.5 (exam traps) · LO 1.4

100 · Answer C — Attrition.
Why: Matching facets on opposing teeth, fractured cusps and masseteric hypertrophy show tooth-to-tooth wear from bruxism.
Revise: Unit 2 §2.5 · LO 1.4

101 · Answer B — Abrasion.
Why: Sharp, V-shaped cervical notches from a foreign object (the toothbrush) are worst on the side opposite the brushing hand — the left for a right-handed patient.
Revise: Unit 2 §2.5 · LO 1.4

102 · Answer D — Abfraction.
Why: Abfraction is the proposed (debated) flexural-stress mechanism for wedge-shaped cervical lesions with sharp angles, sometimes subgingival, on teeth under heavy lateral load.
Revise: Unit 2 §2.5 · LO 1.4

103 · Answer A — Compensatory alveolar growth.
Why: In slow wear, dentoalveolar compensation keeps worn teeth in contact and maintains the OVD, leaving no space. This is the rationale for the Dahl approach.
Not the others: OVD is usually maintained, not lost.
Revise: Unit 2 §2.5, Unit 10 · LO 1.4, 2c.10

104 · Answer E — 0–3 per sextant; the sum guides management.
Why: BEWE scores the most affected surface in each sextant 0–3; the cumulative score directs prevention, monitoring or treatment.
Not the others: A describes Smith and Knight.
Revise: Unit 2 §2.5, Unit 5 · LO 1.4, 2a.8

105 · Answer C — Smith and Knight Tooth Wear Index.
Why: It scores every surface 0–4 and suits detailed research or monitoring rather than screening.
Revise: Unit 2 §2.5 · LO 1.4

106 · Answer B — Empathy, liaison with the GMP, prevention first.
Why: Patients may be secretive. A non-judgemental approach, consent-based liaison with the GMP, and protective, monitoring-led care come before any restoration.
Revise: Unit 2 §2.5 · LO 1.4, 3b.1

107 · Answer A — Newton type III.
Why: Type III is papillary hyperplasia of the central palate. Type I is pinpoint hyperaemia; type II is diffuse erythema.
Revise: Unit 2 §2.6 (exam traps) · LO 1.8

108 · Answer D — Hygiene, night removal, soaking and correcting fit.
Why: Denture stomatitis is driven by denture biofilm and continuous wear; treat the denture first. Antifungals are added only if it persists.
Revise: Unit 2 §2.6, companion note 03 · LO 1.8

109 · Answer E — Nystatin.
Why: Miconazole (even as gel or cream) and systemic azoles interact with warfarin and raise the INR. Nystatin is not absorbed.
Revise: Unit 2 §2.6 (exam traps), Unit 4 · LO 1.8, 1.7

110 · Answer C — FBC, haematinics and glucose.
Why: Angular cheilitis is predisposed by haematinic deficiency, diabetes and immunosuppression, as

well as low OVD and denture infection.
Revise: Unit 2 §2.6 · LO 1.8, 2a.7

111 · Answer A — Miconazole cream.
Why: Miconazole is active against both Candida and Staphylococcus aureus. Treat the denture as the source as well.
Revise: Unit 2 §2.6 · LO 1.8

112 · Answer D — Trim, condition, excise with histology if it persists.
Why: Removing the irritant and using a tissue conditioner often lets the hyperplasia shrink. Persistent tissue is excised and examined histologically.
Revise: Unit 2 §2.6 · LO 1.8

113 · Answer B — Urgent suspected cancer referral.
Why: An ulcer persisting 3 weeks after removing the cause must be treated as possible malignancy.
Revise: Unit 2 §2.6, §2.7 · LO 1.8, 1.11

114 · Answer E — Fibrous replacement of bone; mucostatic or selective-pressure impression.
Why: A flabby ridge, often opposed by lower natural anteriors (combination syndrome), is recorded undisplaced with a windowed tray so it does not rebound under the denture.
Not the others: Excision loses ridge height and is rarely needed.
Revise: Unit 2 §2.6, Unit 7 · LO 1.8, 2c.9

115 · Answer C — Erythroplakia.
Why: Erythroplakia carries the highest malignant transformation risk of the potentially malignant disorders.
Revise: Unit 2 §2.7 · LO 1.11

116 · Answer D — Lateral tongue and floor of mouth.
Why: These are the highest-risk sites for oral squamous cell carcinoma.
Revise: Unit 2 §2.7 · LO 1.11

117 · Answer E — All of the above.
Why: Tobacco and alcohol act synergistically; areca nut is an independent carcinogen. HPV, immunosuppression and previous cancer also raise risk.
Revise: Unit 2 §2.7 · LO 1.11

118 · Answer A — Urgent suspected cancer referral.
Why: NICE NG12 lists a red or red-and-white patch, persistent ulceration over 3 weeks, or a persistent lump as triggers for urgent referral.
Revise: Unit 2 §2.7 · LO 1.11, 2a.6

119 · Answer B — Secondary dentine.
Why: Ageing teeth lay down secondary dentine, shrinking the pulp, lowering sensibility responses and making access harder.
Not the others: A weak response does not mean necrosis.
Revise: Unit 2 §2.8 · LO 1.10

120 · Answer C — Haematinic deficiency.
Why: Deficient iron, B12 or folate causes glossitis, angular cheilitis, recurrent ulcers and a burning sensation.
Revise: Unit 2 §2.8 · LO 1.10

121 · Answer E — Bidirectional.
Why: Poor glycaemic control worsens periodontitis, and periodontal inflammation worsens glycaemic control.
Revise: Unit 2 §2.8 · LO 1.10

122 · Answer A — Not a contraindication; antiresorptives matter more.
Why: Osteoporosis itself does not preclude implants. Antiresorptive drugs carry the MRONJ risk.
Revise: Unit 2 §2.8, Unit 4 · LO 1.10

123 · Answer B — No hypochlorite for Co-Cr.
Why: Sodium hypochlorite corrodes cobalt–chromium frameworks. Use non-abrasive cleaners suitable for metal.
Revise: Unit 2 companion note 03 · LO 1.4, 2b.4

124 · Answer D — Simple, hygienic, maintainable design.
Why: When cleaning and attendance are unreliable, favour designs the patient and carers can maintain over elaborate fixed or implant work.
Revise: Unit 2 companion note 04 · LO 2b.5, 2c.1

125 · Answer C — OHI and risk control, instrumentation, surgery, supportive care.
Why: This is the stepwise periodontal sequence; definitive prosthodontics follows stability.
Revise: Unit 2 companion note 02 · LO 2b.3

126 · Answer A — No replacement is a valid option.
Why: Not every missing tooth needs replacing. A single missing second molar with a stable occlusion and no complaint can be left, after the options are discussed.
Revise: Unit 2 companion note 05 · LO 2c.1

127 · Answer E — All of the above.
Why: Treat the cause where possible (a medication review with the GMP), stimulate saliva without sugar, and protect the teeth with high fluoride.
Revise: Unit 2 §2.1 · LO 1.2, 2a.7

128 · Answer B — Thick, scant saliva gives a poor film.
Why: Retention depends on a thin continuous saliva film. After radiotherapy, saliva is reduced and viscous, so adhesion and cohesion fail.
Revise: Unit 2 §2.1 · LO 1.2

129 · Answer D — Penicillin antibiotics.
Why: Antidepressants, antihistamines, antimuscarinics and diuretics commonly cause dry mouth; penicillins do not.
Revise: Unit 2 §2.1 · LO 1.2

130 · Answer C — Ecological plaque hypothesis.
Why: Environmental change, such as frequent sugar and low saliva, selects for acidogenic, aciduric species, shifting the plaque towards caries.
Revise: Unit 2 §2.4 · LO 1.4

Under 70% on Block B? Re-read Unit 2 and Crash notebook Schematic 5, then re-sit in a fortnight.

Block C answers — Dental materials science

131 · Answer D — Higher elastic modulus.
Why: Co-Cr (about 220 GPa) is about twice as stiff as Type IV gold (90–100 GPa). A stiffer clasp generates more force per unit of flexure, so it must engage a shallower undercut (0.25 mm against 0.5 mm).
Not the others: Density affects weight, not clasp design.
Revise: Unit 3 §3.1, §3.7 · LO 1.9

132 · Answer A — Proportional (elastic) limit.
Why: Below the proportional limit a clasp recovers fully; above it the clasp deforms permanently and loosens.
Not the others: UTS is the stress at fracture, far beyond the working range.
Revise: Unit 3 §3.1 · LO 1.9

133 · Answer E — Low ductility.
Why: Co-Cr has little permanent strain before fracture, so repeated bending work-hardens and cracks it. Wrought wire tolerates adjustment better.
Revise: Unit 3 §3.1 · LO 1.9

134 · Answer B — Toughness.
Why: Toughness is energy absorbed before fracture. High-impact acrylic, with a rubber phase, resists breakage when dropped.
Not the others: Resilience is energy absorbed elastically, relevant to liners and mouthguards.
Revise: Unit 3 §3.1, §3.5 · LO 1.9

135 · Answer C — Fatigue.
Why: Repeated sub-critical flexing, worsened by natural opposing teeth, a deep frenal notch or a torus, causes the classic midline fracture.
Not the others: Impact fracture follows a single event such as dropping.
Revise: Unit 3 §3.1, §3.5 · LO 1.9

136 · Answer B — Resilience.
Why: Resilience is the energy absorbed elastically (the area under the elastic part of the stress–strain curve).
Revise: Unit 3 §3.1 · LO 1.9

137 · Answer E — Fracture toughness.
Why: KIc measures resistance to crack propagation; zirconia's is much higher than glass ceramics'.
Not the others: High modulus means stiffness, not crack resistance.
Revise: Unit 3 §3.1, §3.9 · LO 1.9

138 · Answer A — Viscoelastic recovery is better with fast deformation.
Why: Elastomers behave more elastically under a rapid load; slow removal allows permanent deformation.
Revise: Unit 3 §3.1 · LO 1.9

139 · Answer D — Stress relaxation.
Why: Wax has high thermal expansion and stores stresses that release over time, distorting the pattern. Invest promptly.
Revise: Unit 3 §3.1, §3.4 · LO 1.9

140 · Answer C — Knoop.
Why: Knoop's elongated, shallow indenter suits brittle materials and thin coatings.
Revise: Unit 3 §3.1 · LO 1.9

141 · Answer A — Acrylic expands far more.
Why: Acrylic is about 80 ppm/°C against about 11 ppm/°C for tooth, so temperature change opens the margins.
Revise: Unit 3 §3.1 · LO 1.9

142 · Answer D — Slightly lower than the metal.
Why: The metal contracts slightly more on cooling and leaves the porcelain in compression, which brittle materials resist well.
Revise: Unit 3 §3.1, §3.8 · LO 1.9

143 · Answer C — Glass-fibre.
Why: Glass-fibre posts (about 20–40 GPa) are closest to dentine (15–20 GPa). Their failures tend to be debonds rather than root fractures.
Revise: Unit 3 §3.1, §3.12 · LO 1.9

144 · Answer E — None; modulus describes stiffness only.
Why: Modulus tells you stiffness. It says nothing about toughness, wear, bond strength or flexural strength — a classic exam trap.
Revise: Unit 3 companion note (SBA traps) · LO 1.9

145 · Answer B — Impression plaster.
Why: Plaster is mucostatic and accurate but brittle, so it fractures over undercuts.
Revise: Unit 3 §3.2 · LO 1.9

146 · Answer C — Low thermal conductivity.
Why: Compound must be heated through; a soft surface over a hard core distorts the impression.
Revise: Unit 3 §3.2 · LO 1.9

147 · Answer A — Eugenol irritation; no undercuts.
Why: ZOE paste is rigid when set, so it cannot record undercuts, and eugenol can irritate.
Revise: Unit 3 §3.2 · LO 1.9

148 · Answer E — Syneresis and imbibition.
Why: Alginate loses water (syneresis) or absorbs it (imbibition) and distorts; pour within about 10–15

minutes unless it is an extended-pour product.
Revise: Unit 3 §3.2 · LO 1.9

149 · Answer D — At least 3–4 mm bulk.
Why: Alginate has low tear strength; adequate bulk, a perforated or adhesive tray and snap removal reduce tearing.
Revise: Unit 3 §3.2 · LO 1.9

150 · Answer B — Ethanol evaporation shrinkage.
Why: Condensation silicone gives off ethanol as a by-product and keeps shrinking; pour promptly.
Revise: Unit 3 §3.2 · LO 1.9

151 · Answer E — Sulphur from latex gloves.
Why: Sulphur compounds in latex poison the platinum catalyst of addition silicone. Use vinyl or nitrile gloves when handling cord near the preparation.
Revise: Unit 3 §3.2 · LO 1.9

152 · Answer C — Hydrogen release.
Why: Set addition silicone can release hydrogen, causing porous die surfaces if poured immediately. Wait about an hour unless the product contains a hydrogen absorber.
Revise: Unit 3 §3.2 · LO 1.9

153 · Answer B — High stiffness.
Why: Set polyether is very rigid, so it is hard to remove from undercuts and can fracture stone dies.
Revise: Unit 3 §3.2 · LO 1.9

154 · Answer A — Polyether.
Why: Polyether is inherently hydrophilic; addition silicone needs added surfactant.
Revise: Unit 3 (exam traps) · LO 1.9

155 · Answer D — Low-viscosity material in a spaced or windowed tray.
Why: Mucostatic recording avoids displacing the flabby tissue, which would otherwise rebound under the denture.
Revise: Unit 3 §3.2 · LO 1.9

156 · Answer A — About 3 mm.
Why: Alginate needs about 3 mm bulk for strength; elastomers use 1–3 mm depending on viscosity; ZOE is close-fitting.
Revise: Unit 3 §3.2 · LO 1.9

157 · Answer D — Tray flexure and lack of adhesive.
Why: A flexible tray distorts during removal and springs back afterwards; elastomers also need the matching adhesive.
Revise: Unit 3 §3.2 · LO 1.9

158 · Answer B — Putty displaces the wash.
Why: In a simultaneous technique the stiff putty can push the wash away from the margins.
Revise: Unit 3 §3.2 · LO 1.9

159 · Answer E — Distortion when the set putty is reseated without relief.
Why: Reseating set putty over the wash compresses it; it rebounds after removal, giving undersized dies and tight crowns.
Revise: Unit 3 §3.2 · LO 1.9

160 · Answer C — Rinse, then a material-specific disinfectant.
Why: Follow the label time; prolonged immersion distorts alginate and polyether.
Revise: Unit 3 §3.2, Unit 13 · LO 1.9, 3a.4

161 · Answer B — Weaker, less expansion, slower set.
Why: Extra water leaves more porosity in the set gypsum.
Revise: Unit 3 §3.3 · LO 1.9

162 · Answer E — Faster set and more expansion.
Why: Spatulation breaks up crystals and creates more nuclei for crystallisation.
Revise: Unit 3 §3.3 · LO 1.9

163 · Answer A — Blood and saliva retard the set.
Why: They leave a soft, chalky cast surface; rinse impressions before disinfecting and pouring.
Revise: Unit 3 §3.3 · LO 1.9

164 · Answer C — Potassium sulphate.
Why: Potassium sulphate, gypsum seed and low-concentration sodium chloride accelerate; borax, citrates and acetates retard.
Revise: Unit 3 §3.3 · LO 1.9

165 · Answer D — Hygroscopic expansion.
Why: Gypsum immersed in water while setting expands more than when it sets in air.
Revise: Unit 3 §3.3 · LO 1.9

166 · Answer D — Improve resistance form.
Why: A short, over-tapered preparation cannot be rescued by cement. Add grooves or boxes, or gain height by crown lengthening.
Revise: Unit 3 companion note 02, Unit 8 · LO 1.9, 2c.13

167 · Answer B — Die stone (Type IV or V).
Why: Dies need high strength and abrasion resistance; plaster and Type III stone are too weak.
Revise: Unit 3 §3.3 · LO 1.9

168 · Answer C — Type V high-expansion die stone.
Why: Its greater setting expansion compensates for the higher casting shrinkage of base-metal alloys.
Revise: Unit 3 §3.3 · LO 1.9

169 · Answer A — Calcium sulphate breaks down above about 700°C.
Why: It releases sulphur dioxide and contaminates the casting, so gypsum-bonded investment is for gold alloys only.
Revise: Unit 3 §3.4 · LO 1.9

170 · Answer E — Phosphate-bonded.
Why: It withstands the higher temperatures of metal–ceramic and base-metal alloys.
Revise: Unit 3 §3.4 · LO 1.9

171 · Answer C — Setting, hygroscopic and thermal expansion of the investment.
Why: The mould is expanded to match the alloy's casting shrinkage (gold about 1.4%; Co-Cr more).
Revise: Unit 3 §3.4 · LO 1.9

172 · Answer A — Invest promptly.
Why: Wax relaxes and distorts as stored stresses release; delays reduce casting accuracy.
Revise: Unit 3 §3.4 · LO 1.9

173 · Answer D — Benzoyl peroxide.
Why: Benzoyl peroxide in the powder decomposes on heating to free radicals. In self-cure acrylic a tertiary amine activates it.
Not the others: Camphorquinone is the light-cure initiator; hydroquinone is the inhibitor.
Revise: Unit 3 §3.5 · LO 1.9

174 · Answer E — Hydroquinone.
Why: It is the inhibitor in the monomer, giving shelf life.
Revise: Unit 3 §3.5 · LO 1.9

175 · Answer B — Cross-linking agent.
Why: Ethylene glycol dimethacrylate cross-links the polymer and improves resistance to crazing.
Revise: Unit 3 §3.5 · LO 1.9

176 · Answer E — Dough.
Why: The stages are sandy, stringy, dough, rubbery, stiff. Pack at dough, when the mix is no longer sticky and flows under pressure.
Revise: Unit 3 §3.5 · LO 1.9

177 · Answer C — Gaseous porosity from overheating.
Why: Monomer boils at about 100.3°C. If heated too fast, the exothermic thick sections overheat and form bubbles.
Revise: Unit 3 §3.5 · LO 1.9

178 · Answer D — Contraction porosity.
Why: Insufficient dough or pressure leaves voids throughout and at the surface.
Not the others: Monomer evaporation from uncovered dough gives granular porosity.
Revise: Unit 3 §3.5 · LO 1.9

179 · Answer B — Higher residual monomer.
Why: Self-cure acrylic leaves about 2–5% residual monomer against 0.2–0.5% for heat-cure; monomer is an irritant and allergen.
Revise: Unit 3 §3.5 · LO 1.9

180 · Answer A — Warpage below the glass transition temperature.
Why: Stresses relieve and the base warps well below about 105°C. Advise against hot or boiling water.
Revise: Unit 3 §3.5 · LO 1.9

181 · Answer A — Metal palate or reinforcement, and reduce the notch stress.
Why: Fatigue fracture beside a stress-raising notch against natural teeth needs a stronger base (metal or high-impact acrylic) and a gentler notch.
Revise: Unit 3 §3.5, Unit 7 · LO 1.9

182 · Answer B — Very low residual monomer and porosity.
Why: Industrially polymerised blocks are milled, so there is no processing shrinkage and the fit is better.
Revise: Unit 3 §3.5, Unit 11 · LO 1.9

183 · Answer E — PMMA is radiolucent.
Why: Standard acrylic does not show on radiographs, which matters when fragments may be swallowed or inhaled.
Revise: Unit 3 §3.5 · LO 1.9

184 · Answer D — Abrasion of opposing natural teeth.
Why: Porcelain teeth resist wear but abrade enamel; they also click, are brittle and need more space. Acrylic teeth are preferred opposite natural teeth.
Revise: Unit 3 §3.6 · LO 1.9

185 · Answer C — PEMA with plasticiser in ethanol.
Why: This gel stays viscoelastic for days to weeks, conditioning traumatised mucosa and allowing functional impressions.
Revise: Unit 3 §3.6 · LO 1.9

186 · Answer B — Ethanol and plasticiser leach out.
Why: The material hardens and roughens, harbouring Candida; replace it every few days to weeks.
Revise: Unit 3 §3.6 · LO 1.9

187 · Answer A — Weaker bond and Candida growth.
Why: Silicone liners stay resilient longer but bond poorly to PMMA, are hard to trim and polish, and support Candida.
Revise: Unit 3 §3.6 · LO 1.9

188 · Answer C — Hypochlorite for acrylic only; alkaline peroxide for Co-Cr.
Why: Hypochlorite is antifungal but bleaches and corrodes metals.
Revise: Unit 3 §3.6 · LO 1.9

189 · Answer E — Passive chromium oxide film.
Why: At least 12–13% chromium is needed for passivation; Co-Cr has about 25–30%.
Revise: Unit 3 §3.7 · LO 1.9

190 · Answer D — Excess carbides; brittleness.
Why: Carbon forms carbides that harden the alloy, but too much makes it brittle.
Revise: Unit 3 §3.7 · LO 1.9

191 · Answer E — Nickel allergy is commoner in women; avoid Ni-Cr.
Why: Co-Cr is the usual alternative for a base-metal framework. Some Ni-Cr alloys also contain toxic beryllium.
Revise: Unit 3 §3.7 · LO 1.9

192 · Answer D — About 110 GPa; titanium dioxide film.
Why: Titanium is about half as stiff as Co-Cr and forms a passive TiO2 layer.
Revise: Unit 3 §3.7 · LO 1.9

193 · Answer A — Wrought wire.
Why: The cold-worked, fibrous grain gives flexibility and adjustability, suiting 0.5–0.75 mm undercuts.
Revise: Unit 3 §3.7 · LO 1.9

194 · Answer C — 0.5 mm.
Why: Cast gold engages 0.5 mm; Co-Cr 0.25 mm; wrought wire 0.5–0.75 mm.
Revise: Unit 3 §3.7 · LO 1.9

195 · Answer B — Galvanic current.
Why: Dissimilar metals in saliva form a cell; contact creates a current felt as a shock.
Revise: Unit 3 §3.7 · LO 1.9

196 · Answer C — Chemical bonding via the oxide layer.
Why: Tin and indium oxides on noble alloys, or chromium oxide on base metals, bond chemically to porcelain. Mechanical, compressive and van der Waals forces contribute less.
Revise: Unit 3 §3.8 · LO 1.9

197 · Answer E — Oxide layer too thick.
Why: Base metals form a thick oxide that fails cohesively, weakening the bond.
Revise: Unit 3 §3.8 · LO 1.9

198 · Answer B — Unsupported porcelain should not exceed about 2 mm.
Why: Porcelain needs uniform metal support; excess unsupported bulk chips.
Revise: Unit 3 §3.8 · LO 1.9

199 · Answer D — 1.2–1.5 mm.
Why: Space is needed for metal (0.3–0.5 mm), opaque (about 0.2 mm) and body porcelain; less reduction gives an opaque, bulky crown.
Revise: Unit 3 §3.8, Unit 8 · LO 1.9, 2c.14

200 · Answer A — HF, silane, resin cement.
Why: HF etches the glass phase of the silica-based ceramic and silane couples it to resin.
Revise: Unit 3 §3.9, companion note 01 · LO 1.9

201 · Answer D — Low-pressure air abrasion with MDP.
Why: Zirconia has no glass phase to etch; MDP bonds chemically to its oxide. Conventional GIC or RMGIC is acceptable when the preparation is retentive.
Revise: Unit 3 §3.9 · LO 1.9

202 · Answer C — Silane needs a silica surface.
Why: Zirconia is polycrystalline with no silica. Use MDP, or tribochemical silica coating before silane.
Revise: Unit 3 §3.9 (exam traps) · LO 1.9

203 · Answer B — Tetragonal to monoclinic with volume expansion.
Why: The 3–5% expansion at a crack tip compresses and arrests the crack.
Revise: Unit 3 §3.9 · LO 1.9

204 · Answer A — Low-temperature degradation.
Why: In moisture, zirconia can transform spontaneously, roughening and weakening its surface.
Revise: Unit 3 §3.9 · LO 1.9

205 · Answer E — About 20–25% sintering shrinkage.
Why: Zirconia is milled pre-sintered (green or white state) and enlarged to compensate.
Revise: Unit 3 §3.9 · LO 1.9

206 · Answer A — Milled as lithium metasilicate, then crystallised.
Why: The blue, softer state is easy to mill; firing produces lithium disilicate with its final shade and strength.
Revise: Unit 3 §3.9 · LO 1.9

207 · Answer B — Unglazed or rough feldspathic porcelain.
Why: It is the most abrasive surface; polished monolithic zirconia is relatively kind. Always polish after adjustment.
Revise: Unit 3 §3.9 · LO 1.9

208 · Answer E — Low initial pH; incremental mixing on a cool slab.
Why: The acid can irritate the pulp; incremental mixing on a cool slab controls the exothermic reaction and prolongs working time.
Revise: Unit 3 §3.10 · LO 1.9

209 · Answer C — Hygroscopic expansion of RMGIC.
Why: RMGIC absorbs water and expands, which can crack weak glass ceramics; use resin cement for them.
Revise: Unit 3 §3.10 · LO 1.9

210 · Answer D — Fluoride, tooth-like CTE, early moisture sensitivity.
Why: GIC releases fluoride and expands like tooth, but must be protected from moisture while it sets.
Revise: Unit 3 §3.10 · LO 1.9

211 · Answer B — Resin cement.
Why: Thin, weak feldspathic or leucite veneers need an adhesive resin bond; light-cure resin gives colour stability.
Revise: Unit 3 §3.10 · LO 1.9

212 · Answer A — Eugenol inhibits resin polymerisation.
Why: Use eugenol-free temporary cement when definitive resin bonding is planned.
Revise: Unit 3 (exam traps) · LO 1.9

213 · Answer C — 25 µm.
Why: ISO sets 25 µm as the maximum film thickness for luting cements.
Revise: Unit 3 §3.10 · LO 1.9

214 · Answer D — 30–37% phosphoric acid for 15–30 s.
Why: This produces the type I–III etch patterns; enamel bonding is the most reliable bond in dentistry.
Revise: Unit 3 §3.11 · LO 1.9

215 · Answer E — Collagen collapse and a weaker bond.
Why: Etched dentine must stay moist so resin can infiltrate the collagen and form a hybrid layer.
Revise: Unit 3 §3.11 · LO 1.9

216 · Answer E — Roughen and selectively etch enamel.
Why: Sclerotic dentine bonds poorly; roughening removes the hypermineralised surface and enamel etching adds a reliable bond.
Revise: Unit 3 §3.11 · LO 1.9

217 · Answer D — Air abrasion and MDP resin.
Why: Modern RBB wings use air-abraded base metal bonded with MDP-containing cement (for example Panavia); Rochette, Maryland and Virginia are historical.
Revise: Unit 3 §3.11 · LO 1.9

218 · Answer C — Tin plating or a noble-metal primer.
Why: Noble alloys form little oxide, so they need an added layer to bond chemically.
Revise: Unit 3 §3.11 · LO 1.9

219 · Answer A — A 1.5–2 mm ferrule.
Why: The ferrule, not the post, protects the root. A post only retains the core; it does not reinforce the tooth.
Revise: Unit 3 §3.12 (exam traps) · LO 1.9, 2c.14

220 · Answer B — Most accurate over short spans.
Why: Errors accumulate as images are stitched across long spans, and mobile, featureless mucosa scans poorly.
Revise: Unit 3 §3.14, Unit 11 · LO 1.9, 2a.19

Under 70% on Block C? Re-read Unit 3 and Crash notebook Schematics 6–10, then re-sit in a fortnight.

Block D answers — Pharmacology, emergencies and the medically compromised patient

221 · Answer B — 6 cartridges.
Why: 4.4 mg/kg × 60 kg = 264 mg. One 2.2 mL cartridge of 2% lidocaine holds 44 mg, so 264 ÷ 44 = 6.
Revise: Unit 4 §4.1 · LO 1.7, 2c.7

222 · Answer E — Good bone diffusion; rapid metabolism.
Why: Articaine is an amide with an ester side chain. It diffuses well through bone, so buccal infiltration can reach mandibular molars, and plasma esterases give it a short half-life.
Revise: Unit 4 §4.1 · LO 1.7

223 · Answer A — Prilocaine with felypressin.
Why: Felypressin has oxytocic action, so avoid it in late pregnancy. Large doses of prilocaine also cause

methaemoglobinaemia.
Revise: Unit 4 §4.1 (exam traps) · LO 1.7

224 · Answer D — Early local anaesthetic toxicity.
Why: Toxicity affects the CNS first (perioral tingling, tinnitus, drowsiness, convulsions), then depresses the cardiovascular system. Stop, call for help and support the airway.
Revise: Unit 4 §4.1 · LO 1.7

225 · Answer C — Hypertension with reflex bradycardia.
Why: Non-selective beta-blockers leave adrenaline's alpha effects unopposed. Limit the dose.
Revise: Unit 4 §4.1 (exam traps) · LO 1.7

226 · Answer A — Low tissue pH.
Why: Inflamed, acidic tissue reduces the non-ionised (lipid-soluble) fraction that crosses the nerve membrane.
Not the others: Accessory supply comes from the mylohyoid nerve or cervical plexus, not the facial nerve.
Revise: Unit 4 §4.1 · LO 1.7

227 · Answer C — Akinosi (closed-mouth) block.
Why: It is given with the teeth together, suiting trismus. Gow-Gates is another alternative to the standard block.
Revise: Unit 4 §4.1 · LO 1.7

228 · Answer E — 30% oxygen.
Why: Machines must deliver at least 30% oxygen (maximum 70% nitrous oxide), with fail-safe and scavenging.
Revise: Unit 4 §4.2 · LO 2c.8

229 · Answer B — First-trimester pregnancy.
Why: It is a relative contraindication, with nasal obstruction, severe COPD, recent middle-ear or eye surgery with gas, and B12 deficiency.
Revise: Unit 4 §4.2 · LO 2c.8

230 · Answer D — Flumazenil.
Why: It is the benzodiazepine antagonist. Naloxone reverses opioids.
Revise: Unit 4 §4.2 · LO 2c.8

231 · Answer E — Escort; no driving or machinery until the next day.
Why: IACSD standards require an escort after IV sedation, because of residual sedation and amnesia.
Revise: Unit 4 §4.2 · LO 2c.8

232 · Answer B — ASA I–II.
Why: ASA III and above are generally treated in secondary care.
Revise: Unit 4 §4.2 · LO 2c.8

233 · Answer D — Phenoxymethylpenicillin 500 mg four times daily for 5 days.
Why: For spreading infection with systemic signs, SDCEP recommends this, or amoxicillin 500 mg three

times daily, after drainage.
Revise: Unit 4 §4.3 · LO 1.7, 2c.15

234 · Answer A — Clarithromycin (or metronidazole).
Why: These are the SDCEP alternatives for penicillin allergy.
Not the others: Co-amoxiclav and flucloxacillin are penicillins.
Revise: Unit 4 §4.3 · LO 1.7

235 · Answer C — Metronidazole 400 mg three times daily for 3 days.
Why: It is the SDCEP regimen for necrotising periodontal disease, alongside debridement and oral hygiene.
Revise: Unit 4 §4.3 · LO 1.7

236 · Answer D — Drainage and removal of the cause.
Why: Antibiotics are not a substitute for drainage and are not indicated without spreading infection or systemic involvement.
Revise: Unit 4 §4.3 · LO 1.7, 2c.15

237 · Answer A — Fluconazole (and miconazole).
Why: Azoles potentiate warfarin (raising the INR) and raise statin levels (myopathy). Use nystatin and denture measures.
Revise: Unit 4 §4.3 · LO 1.7

238 · Answer C — Not routine; consider with the cardiologist for special risk.
Why: NICE CG64 does not recommend routine prophylaxis. SDCEP advises considering it for special-risk patients (previous endocarditis, prosthetic valves, some congenital heart disease) with the cardiologist.
Revise: Unit 4 §4.3 (exam traps) · LO 1.7, 2b.12

239 · Answer E — SLS inactivates chlorhexidine.
Why: Wait about 30 minutes after brushing. Chlorhexidine also stains and alters taste.
Revise: Unit 4 §4.3 · LO 1.7

240 · Answer B — NSAID-sensitive asthma.
Why: NSAIDs can trigger bronchospasm. Also avoid or take care with peptic ulcer, renal impairment, anticoagulants and third-trimester pregnancy.
Revise: Unit 4 §4.4 · LO 1.7

241 · Answer C — Maximum 4 g in 24 hours, reduced in low weight or hepatic impairment.
Why: The dose is reduced in low body weight, hepatic impairment and alcohol dependence.
Revise: Unit 4 §4.4 · LO 1.7

242 · Answer D — Ibuprofen and paracetamol combined.
Why: The combination is more effective than either drug alone for dental pain.
Revise: Unit 4 §4.4 · LO 1.7

243 · Answer B — Proceed with local measures.
Why: SDCEP: check the INR ideally within 24 hours; if it is below 4, treat in primary care with haemostatic measures.
Revise: Unit 4 §4.5 · Schematic 23 · LO 2b.11

244 · Answer E — Defer and seek advice.
Why: An INR of 4 or above is outside the threshold for primary care treatment.
Revise: Unit 4 §4.5 · LO 2b.11

245 · Answer A — Miss the morning dose.
Why: For twice-daily apixaban or dabigatran and a higher-risk procedure, SDCEP advises missing the morning dose; low-risk procedures need no interruption.
Revise: Unit 4 §4.5 · LO 2b.11

246 · Answer B — Delay the morning dose until 4 hours after haemostasis.
Why: That is the SDCEP advice for once-daily rivaroxaban or edoxaban and a higher-risk procedure.
Revise: Unit 4 §4.5 · LO 2b.11

247 · Answer E — Do not stop antiplatelets.
Why: Stopping dual antiplatelet therapy after a stent risks thrombosis. Use local measures.
Revise: Unit 4, In one minute, §4.5 · LO 2b.11

248 · Answer A — Liaise with the haemophilia centre.
Why: An inferior alveolar block without factor cover risks a pterygomandibular haematoma and airway compromise.
Revise: Unit 4 §4.5 · LO 2b.11

249 · Answer C — No interruption.
Why: Crown preparation with cord, impressions and denture work are low bleeding-risk procedures.
Revise: Unit 4 §4.5 · LO 2b.11

250 · Answer D — More than 8 weeks.
Why: MRONJ is exposed (or probeable) bone for over 8 weeks, with antiresorptive or antiangiogenic history and no jaw radiotherapy or metastasis.
Revise: Unit 4 §4.6 · LO 2b.11

251 · Answer A — Denosumab wears off in months; bisphosphonates persist for years.
Why: Denosumab is a RANKL antibody; bisphosphonates bind to bone and persist.
Revise: Unit 4 §4.6 · LO 1.7

252 · Answer C — Cancer patient on high-dose IV zoledronic acid.
Why: High-dose IV bisphosphonate or denosumab for cancer carries the highest MRONJ risk.
Revise: Unit 4 §4.6 · Schematic 24 · LO 2b.11

253 · Answer E — Retain the root.
Why: In high-risk patients, root treatment and decoronation avoid an extraction and support the denture as an overdenture abutment.
Revise: Unit 4 §4.6 · LO 2b.11, 2c.9

254 · Answer B — No drug holidays for routine care.
Why: SDCEP does not recommend interrupting antiresorptives for routine dentistry.
Revise: Unit 4 §4.6 · LO 2b.11

255 · Answer D — Dental assessment first.
Why: Getting the mouth healthy before antiresorptive therapy, including well-fitting dentures, reduces MRONJ risk.
Revise: Unit 4 §4.6 · LO 2b.11, 2b.12

256 · Answer E — At least 10–14 days before radiotherapy.
Why: Teeth of poor prognosis in the field should be removed early enough to heal before irradiation, reducing later osteoradionecrosis risk.
Revise: Unit 4 §4.7 · LO 2b.11, 2b.12

257 · Answer D — 5000 ppm, fluoride trays, saliva substitutes, trismus exercises.
Why: Radiation caries, xerostomia and trismus are managed preventively; avoid extractions in the field where possible.
Revise: Unit 4 §4.7 · LO 2b.11

258 · Answer B — Lifelong, dose-dependent, mandible far more than maxilla.
Why: Osteoradionecrosis risk persists for life and rises with doses above about 50–60 Gy.
Revise: Unit 4 §4.7 · LO 2b.11

259 · Answer A — Level 3.
Why: Obturators and implants in irradiated bone are Level 3 (specialist) care under the NHS restorative standard.
Revise: Unit 4 §4.7, Unit 5 · LO 2b.12

260 · Answer C — 48 mmol/mol (6.5%).
Why: An HbA1c of 48 mmol/mol or above diagnoses diabetes.
Revise: Unit 4 §4.8 · LO 2a.13

261 · Answer D — Prediabetes.
Why: 42–47 mmol/mol is prediabetes. Refer to the GMP for confirmation and management, as the Level 2 curriculum expects.
Revise: Unit 4 §4.8 · LO 2a.13

262 · Answer A — Morning, after a normal meal, glucose available.
Why: This minimises the risk of hypoglycaemia during treatment.
Revise: Unit 4 §4.8 · LO 2a.13

263 · Answer C — Adrenaline 1:1000, 500 micrograms IM.
Why: Give it into the anterolateral thigh and repeat after 5 minutes if there is no improvement. IV adrenaline is for specialists only.
Revise: Unit 4 §4.9 · LO 2b.1

264 · Answer B — 300 micrograms.
Why: Children aged 6–12 receive 300 micrograms; under 6, 150 micrograms.
Revise: Unit 4 §4.9 · LO 2b.1

265 · Answer E — GTN 400 micrograms sublingually.
Why: GTN is the first drug for angina; it can be repeated.
Revise: Unit 4 §4.9 · LO 2b.1

266 · Answer C — 999, aspirin 300 mg chewed, oxygen if hypoxic.
Why: Persistent pain unrelieved by GTN suggests myocardial infarction.
Revise: Unit 4 §4.9 · LO 2b.1

267 · Answer B — Salbutamol via a spacer.
Why: Give 4–6 puffs, repeating up to 10. Call 999 if it is life-threatening or not improving.
Revise: Unit 4 §4.9 · LO 2b.1

268 · Answer D — Oral glucose.
Why: A conscious patient able to swallow is given 15–20 g of oral glucose.
Revise: Unit 4 (exam traps) · LO 2b.1

269 · Answer E — Glucagon 1 mg IM.
Why: An unconscious patient cannot swallow safely; give glucagon IM and call for help.
Revise: Unit 4 (exam traps) · LO 2b.1

270 · Answer A — Buccal midazolam 10 mg.
Why: For a seizure lasting 5 minutes or more, protect the patient from injury and give buccal midazolam.
Revise: Unit 4 §4.9 · LO 2b.1

271 · Answer E — Lie flat, raise legs, oxygen.
Why: Vasovagal syncope restores cerebral perfusion quickly when the patient is laid flat with the legs raised.
Revise: Unit 4 §4.9 · LO 2b.1

272 · Answer D — 999 and hydrocortisone if prescribed and trained.
Why: Adrenal crisis needs urgent steroid replacement and transfer.
Revise: Unit 4 §4.9 · LO 2b.1

273 · Answer A — 30:2, 100–120 a minute, 5–6 cm, AED early.
Why: These are the Resuscitation Council UK adult standards.
Revise: Unit 4 §4.9 · LO 2b.1

274 · Answer C — Back blows, then abdominal thrusts.
Why: For ineffective coughing in a conscious adult: up to 5 back blows, then up to 5 abdominal thrusts, repeated.
Revise: Unit 4 §4.9 · LO 2b.1

275 · Answer B — 15 L/min via a non-rebreathe mask.
Why: High-flow oxygen is given to the acutely ill patient.
Revise: Unit 4 §4.9 · LO 2b.1

276 · Answer A — Prefer a fixed option.
Why: In uncontrolled epilepsy, small removable prostheses can be inhaled or fracture during a seizure. Metal-based or reinforced dentures are used if removable. Check for phenytoin-induced gingival overgrowth.
Revise: Unit 4 §4.10 · LO 2c.15

277 · Answer E — Second trimester; elective work can wait.
Why: Necessary care is safest in the second trimester; take radiographs only if justified.
Revise: Unit 4 §4.10 · LO 2c.15

278 · Answer B — Non-dialysis day; avoid NSAIDs.
Why: Heparin is used during dialysis, and NSAIDs and nephrotoxic drugs harm residual renal function; bleeding and infection risk are raised.
Revise: Unit 4 §4.10 · LO 2c.15

279 · Answer D — Non-latex gloves and dam.
Why: Latex allergy requires a latex-free environment. Latex also inhibits addition silicone setting.
Revise: Unit 4 §4.10 · LO 2c.15

280 · Answer C — Fully cured heat-cured PMMA, polyamide or polycarbonate.
Why: True allergy is to residual monomer; fully cured or methacrylate-free bases avoid it.
Not the others: Self-cure and chairside resins contain the most residual monomer.
Revise: Unit 4 §4.10 · LO 2c.15

281 · Answer B — Electrosurgery.
Why: Electrosurgery, and some ultrasonic devices, may interfere with pacemakers and ICDs.
Revise: Unit 4 §4.10 · LO 2c.15

282 · Answer A — Continue; denture impressions are low risk.
Why: Do not tell patients to stop anticoagulants for low-risk procedures; follow procedure-specific guidance.
Revise: Unit 4 companion note (SBA traps) · LO 2b.11

283 · Answer D — Stabilise first.
Why: Uncontrolled periodontal disease is not a suitable foundation for definitive bridgework; reassess after therapy.
Revise: Unit 4 companion note 02 · LO 2b.3, 2c.1

284 · Answer E — All of the above.
Why: Shared decision-making covers benefit, alternatives (including no treatment), medical risks and maintenance, and is documented.
Revise: Unit 4 companion note 03, Unit 13 · LO 3b.1

285 · Answer C — Copy dentures and labelling.
Why: Neuromuscular disease limits adaptation; copying keeps learned control. Labelling prevents loss in care homes.
Revise: Unit 4 §4.10 · LO 2c.15

286 · Answer C — Candidosis and delayed healing.
Why: Adrenal suppression is rarely relevant for routine prosthodontics; infection and healing are.
Revise: Unit 4 §4.10 · LO 2c.15

287 · Answer B — Implant-retained overdenture after assessment.
Why: Without saliva, physical retention fails; implants provide mechanical retention.
Revise: Unit 4 §4.10, Unit 11 · LO 2c.15

288 · Answer E — Manage the cause first.
Why: Ongoing acid would undermine new restorations; stabilise with the GMP, then restore.
Revise: Unit 4 §4.10, Unit 10 · LO 2c.15

289 · Answer A — Stabilise, liaise, defer elective work.
Why: With unstable disease, stabilise oral disease and defer elective definitive care until it is medically safe.
Revise: Unit 4 companion note 02 · LO 2b.12

290 · Answer D — Refer to secondary care.
Why: ASA III patients, and severe COPD, are outside primary-care sedation standards.
Revise: Unit 4 §4.2 · LO 2c.8, 2b.12

Under 70% on Block D? Re-read Unit 4 and Crash notebook Schematics 23–24 and Number card 4, then re-sit in a fortnight.

Block E answers — Assessment, diagnosis, imaging, classifications and planning

291 · Answer C — Her complaint in her own words, and her expectations.
Why: The history starts with the presenting complaint and expectations; they shape the whole plan and its prospects of success.
Revise: Unit 5 §5.1, companion note 01 · LO 2a.1

292 · Answer A — Hysterical.
Why: House ranks attitudes philosophical (best), exacting, indifferent, hysterical (worst prognosis).
Revise: Unit 5 §5.1 · LO 2a.2

293 · Answer E — High aesthetic risk.
Why: A high smile line shows more than 3 mm of gingiva, so margins, gingival contour and pontic emergence are all visible.
Revise: Unit 5 §5.1 · LO 2a.4

294 · Answer B — Full six-point charting of the whole dentition.
Why: BSP guidance: code 4 or * in any sextant prompts full charting; code 3 prompts charting of that sextant after initial therapy.
Revise: Unit 5 §5.4 · LO 2a.9

295 · Answer D — All of the above.
Why: Existing dentures show what works and what failed: retention, stability, support, extension, OVD, occlusion, aesthetics, hygiene and wear.
Revise: Unit 5 §5.1 · LO 2a.3

296 · Answer B — Class I.
Why: Bilateral free-end spaces behind the remaining teeth; unreplaced missing third molars are ignored.
Revise: Unit 5 §5.2 · Schematic 12 · LO 2a.10

297 · Answer D — Class II modification 1.
Why: The right free end (46, 47) sets Class II; the 35 space is one modification.
Revise: Unit 5 §5.2 · LO 2a.10

298 · Answer A — Class III.
Why: A unilateral bounded space with teeth in front and behind.
Revise: Unit 5 §5.2 · LO 2a.10

299 · Answer C — Class IV.
Why: A single bilateral anterior space crossing the midline.
Revise: Unit 5 §5.2 · LO 2a.10

300 · Answer E — Class III.
Why: A third molar present and used as an abutment counts, so the space is bounded.
Revise: Unit 5 §5.2 (exam traps) · LO 2a.10

301 · Answer A — After the extraction (Class II).
Why: Applegate: classify after planned extractions. Once 47 is removed the space becomes free-ended.
Revise: Unit 5 §5.2 · LO 2a.10

302 · Answer C — Class IV has no modifications.
Why: Any space behind the anterior space would itself be the most posterior and set a different class.
Revise: Unit 5 §5.2 (exam traps) · LO 2a.10

303 · Answer D — Class III.
Why: A missing second molar that will not be replaced is ignored, leaving the bounded 24–25 space.
Revise: Unit 5 §5.2 · LO 2a.10

304 · Answer E — Craddock Class 3.
Why: Support comes from teeth (rests) and mucosa (the free-end saddle). Class 1 is tooth-borne; Class 2 mucosa-borne.
Revise: Unit 5 §5.2 · LO 2a.10

305 · Answer B — Class IV (knife-edge).
Why: Cawood and Howell IV: adequate height, inadequate width.
Revise: Unit 5 §5.3 · LO 2a.11

306 · Answer E — Class V.
Why: A flat ridge, inadequate in height and width.
Revise: Unit 5 §5.3 · LO 2a.11

307 · Answer B — Class VI.
Why: A depressed ridge with basal bone loss.
Revise: Unit 5 §5.3 · LO 2a.11

308 · Answer C — U-shaped.
Why: It gives the best retention and resistance to lateral displacement; V-shaped is unstable and flat resists lateral movement poorly.
Revise: Unit 5 §5.3 · LO 2a.11

309 · Answer A — House Class III.
Why: A steep soft palate gives a narrow post-dam area and a poorer posterior seal.
Revise: Unit 5 §5.3 · LO 2a.11

310 · Answer D — Stage III.
Why: Stage is set by the worst interproximal site: coronal third is Stage II, mid third Stage III, apical third Stage IV.
Revise: Unit 5 §5.4 · LO 2a.9

311 · Answer D — Grade B.
Why: Percentage bone loss ÷ age = 0.89, within 0.5–1.0.
Revise: Unit 5 §5.4 · LO 2a.9

312 · Answer E — Grade C.
Why: 60 ÷ 30 = 2.0, above 1.0, showing rapid progression.
Revise: Unit 5 §5.4 · LO 2a.9

313 · Answer B — Localised.
Why: 6 of 26 teeth is 23%, under the 30% threshold, so the extent is localised; 30% or more of teeth is generalised.
Revise: Unit 5 §5.4 · LO 2a.9

314 · Answer A — Currently stable.
Why: BOP under 10%, PPD 4 mm or less and no BOP at 4 mm sites.
Revise: Unit 5 §5.4 · LO 2a.9

315 · Answer C — Currently in remission.
Why: BOP 10% or more, but PPD 4 mm or less with no BOP at 4 mm sites.
Revise: Unit 5 §5.4 · LO 2a.9

316 · Answer C — Currently unstable.
Why: Any PPD of 5 mm or more, or a 4 mm site with bleeding, is unstable.
Revise: Unit 5 §5.4 · LO 2a.9

317 · Answer A — Stage IV.
Why: Stage IV involves tooth loss from periodontitis and a need for complex rehabilitation (fewer than 20 teeth, masticatory dysfunction, drifting).
Revise: Unit 5 §5.4 · LO 2a.9

318 · Answer E — Level 2.
Why: NHS standard: after steps 1–2 in primary care, Stage III Grade A is Level 2.
Revise: Unit 5 §5.4 · LO 2b.10

319 · Answer D — Level 3.
Why: Stage III Grade B or C, and all Stage IV, are Level 3.
Revise: Unit 5 §5.4 · LO 2b.10

320 · Answer B — Code 3.
Why: The band partly visible means a 3.5–5.5 mm pocket; code 4 is the band disappearing (over 5.5

mm).
Revise: Unit 5 §5.4 · LO 2a.9

321 · Answer B — Grade 3.
Why: Over 1 mm horizontal and/or vertical (depressible) mobility.
Revise: Unit 5 §5.4 · LO 2a.9

322 · Answer D — Grade III.
Why: Hamp III is through-and-through; I is under 3 mm horizontal loss; II is over 3 mm but not through.
Revise: Unit 5 §5.4 · LO 2a.9

323 · Answer A — BEWE 2.
Why: 0 none, 1 initial loss of texture, 2 distinct defect under 50%, 3 loss of 50% or more.
Revise: Unit 5 §5.5 · LO 2a.8

324 · Answer C — High risk.
Why: A cumulative BEWE of 14 or more is high risk: identify the cause, prevent, monitor and consider restoration.
Revise: Unit 5 §5.5 · Number card 4 · LO 2a.8

325 · Answer E — Class II division 2.
Why: Retroclined upper centrals with lower edges behind the cingulum plateau.
Revise: Unit 5 §5.6 · LO 2a.15

326 · Answer A — Class III.
Why: Lower incisal edges lie anterior to the upper cingulum plateau (BSI classification).
Revise: Unit 5 §5.6 · LO 2a.15

327 · Answer E — Class I.
Why: A normal ANB angle is about 2–4°.
Revise: Unit 5 §5.6 · LO 2a.15

328 · Answer D — Ideal 2:3, minimum 1:1.
Why: A crown longer than its root in bone gives an unfavourable lever for an abutment.
Revise: Unit 5 §5.6 · LO 2a.16

329 · Answer B — Periapical, paralleling technique.
Why: It shows bone level, crown–root ratio, root form, periapical status and root-filling quality with least distortion.
Revise: Unit 5 §5.7 · LO 2a.12

330 · Answer C — Vertical bitewings.
Why: Vertical bitewings show crestal bone in periodontitis; horizontal bitewings are for caries.
Revise: Unit 5 §5.7 · LO 2a.12

331 · Answer E — Ghost image: opposite side, higher, larger.
Why: Objects between the source and the rotation centre project as blurred ghosts on the opposite side, higher and magnified.
Revise: Unit 5 (exam traps) · LO 2a.12

332 · Answer C — CBCT only for a justified three-dimensional question.
Why: CBCT has the highest dental dose; it is never routine screening.
Revise: Unit 5 §5.7, companion note 02 · LO 2a.12, 2a.19

333 · Answer A — The practitioner.
Why: Under IR(ME)R the practitioner justifies, the operator exposes, the referrer requests and the employer sets procedures.
Revise: Unit 5 §5.7 · LO 2a.12, 3a.6

334 · Answer D — Justification, optimisation and a recorded evaluation.
Why: Every exposure must be justified, kept as low as reasonably practicable and clinically evaluated with a written report.
Revise: Unit 5 §5.7 · LO 2a.12

335 · Answer B — Intraoral, panoramic, CBCT.
Why: Roughly 0.001–0.008, 0.003–0.024 and 0.02–1 mSv respectively.
Revise: Unit 5 §5.7 · LO 2a.12

336 · Answer D — Cold on a pledget.
Why: Tetrafluoroethane or ethyl chloride cold tests are more reliable than heat.
Revise: Unit 5 §5.8 · LO 2a.14

337 · Answer B — EPT measures nerve response; monitor.
Why: After trauma, nerve conduction may be lost temporarily while the blood supply survives; a single negative result does not justify root canal treatment.
Revise: Unit 5 §5.8 (exam traps) · LO 2a.14

338 · Answer E — Laser Doppler or pulse oximetry.
Why: These assess pulpal blood flow, true vitality; thermal and electric tests assess sensibility only.
Revise: Unit 5 §5.8 · LO 2a.14

339 · Answer C — Cusp-by-cusp bite test.
Why: Pain on release when loading a single cusp localises a crack; transillumination also helps.
Revise: Unit 5 §5.8 · LO 2a.14

340 · Answer A — Interocclusal space, wax-up and surveying.
Why: Mounted casts in RCP show space, guidance and interferences, and let you survey the RPD path and design.
Revise: Unit 5 §5.8 · LO 2a.15, 2c.2

341 · Answer A — Diagnostic wax-up and mock-up.
Why: They test aesthetics and OVD reversibly and help communication with patient and laboratory.
Revise: Unit 5 §5.8 · LO 2a.17

342 · Answer B — Iron.
Why: A low MCV (microcytosis) suggests iron deficiency; confirm with ferritin.
Revise: Unit 5 §5.8 · LO 2a.7

343 · Answer C — B12 or folate deficiency.
Why: A high MCV (macrocytosis) suggests B12 or folate deficiency.
Revise: Unit 5 §5.8 · LO 2a.7

344 · Answer E — All of the above.
Why: The restorability checklist covers a feasible ferrule, no vertical root fracture, an acceptable crown–root ratio and strategic value.
Revise: Unit 5 §5.9 · LO 2a.16

345 · Answer D — Sound coronal tissue above the crest.
Why: The NHS entry criteria require about 2 mm high and 1 mm wide sound tissue, or over 30% of the original crown.
Revise: Unit 5 §5.9 · LO 2a.16, 2b.10

346 · Answer E — Vertical root fracture; hopeless.
Why: An isolated deep, narrow pocket and a J-shaped lesion on a post-crowned tooth suggest a vertical fracture, which is unrestorable.
Revise: Unit 5 §5.9 · LO 2a.16

347 · Answer C — Investigate only if it may change the plan.
Why: Choose investigations whose results could alter diagnosis, prognosis or treatment.
Revise: Unit 5 companion note · LO 2a.12

348 · Answer B — Refer urgently.
Why: A normal radiograph does not overrule a suspicious clinical lesion; an indurated persistent ulcer needs the suspected cancer pathway.
Revise: Unit 5 companion note (traps), Unit 2 · LO 2a.6

349 · Answer D — Urgent, stabilise, re-evaluate, definitive, maintain.
Why: This five-phase sequence ensures definitive work rests on controlled disease.
Revise: Unit 5 treatment strategies note 01 · LO 2b.3

350 · Answer A — All options, including leaving the space.
Why: Compare preparation, reversibility, aesthetics, cost, longevity, cleaning and failure for each, including no treatment.
Revise: Unit 5 treatment strategies note 02 · LO 2b.4, 2c.2

351 · Answer D — Conformative approach.
Why: A stable, comfortable ICP is kept when space and function allow.
Revise: Unit 5 treatment strategies note 03, Unit 9 · LO 2c.19

352 · Answer A — Complex reorganisation, often specialist (Level 3).
Why: OVD change and full-mouth rehabilitation need diagnostic testing and usually specialist involvement.
Revise: Unit 5 treatment strategies note 03 · LO 2b.10

353 · Answer B — Material risks and reasonable alternatives.
Why: Montgomery (2015) requires disclosure of what this patient would consider material, plus

alternatives including no treatment.
Revise: Unit 5 treatment strategies note 04, Unit 13 · LO 3b.1

354 · Answer C — Diagnosis, prognosis, phases, responsibilities and records.
Why: A good referral lets the receiving clinician act without repeating the assessment.
Revise: Unit 5 treatment strategies note 05 · LO 2b.10

355 · Answer E — Definitive work before disease control is unsound.
Why: Active periodontitis threatens margins, abutments and long-term outcome; stabilise first.
Revise: Unit 5 treatment strategies note 01 · LO 2b.3

356 · Answer C — Reduced OVD; compare RVD with OVD.
Why: Overclosure produces deep folds, a prominent chin and angular cheilitis. An excessive freeway space confirms it.
Revise: Unit 5 §5.1, Unit 7 · LO 2a.15

357 · Answer D — About 2–4 mm.
Why: It is usually about 3 mm. Less suggests excessive OVD; more suggests overclosure.
Revise: Unit 5 §5.1, Number card 2 · LO 2a.15

358 · Answer A — It affects the occlusal approach.
Why: A significant slide with symptoms or interferences may favour reorganisation; a stable one may be conformed to.
Revise: Unit 5 §5.1, Unit 9 · LO 2a.15

359 · Answer E — All of the above.
Why: Ridge form, flabby tissue, undercuts, tori and mucosa all shape impression technique and design.
Revise: Unit 5 §5.1 · LO 2a.5

360 · Answer B — Timed collection.
Why: Unstimulated flow is collected by spitting or drooling over a set time; under 0.1 mL/min is hyposalivation.
Revise: Unit 5 §5.1, Unit 2 · LO 2a.7

361 · Answer B — Edentulism by ridge, relationship, muscle attachments and complicating factors.
Why: The ACP PDI grades complete and partial edentulism into classes I–IV of difficulty.
Revise: Unit 5 §5.3 · LO 2a.11

362 · Answer E — Molar–incisor pattern.
Why: The 2017 classification describes extent as localised, generalised or molar–incisor pattern.
Revise: Unit 5 §5.4 · LO 2a.9

363 · Answer C — Level 3.
Why: Stage III Grade B is Level 3, whatever the extent.
Revise: Unit 5 §5.4 · LO 2b.10

364 · Answer A — Horizontal bitewings.
Why: They are the standard view for proximal caries and restoration margins.
Revise: Unit 5 §5.7 · LO 2a.12

365 · Answer D — Focal trough distortion and overlap.
Why: Panoramic images give an overview, not the fine detail needed for early caries.
Revise: Unit 5 §5.7 · LO 2a.12

366 · Answer A — Intraoral 0.001–0.008 mSv; CBCT 0.02–1 mSv.
Why: The CBCT dose depends heavily on the field of view.
Revise: Unit 5 §5.7 · LO 2a.12

367 · Answer C — Anxiety or liquefactive necrosis.
Why: Both can give a false-positive EPT response.
Revise: Unit 5 §5.8 · LO 2a.14

368 · Answer D — Calcification with age (false negative).
Why: Older and calcified teeth often fail to respond despite vital pulps.
Revise: Unit 5 §5.8 · LO 2a.14

369 · Answer B — Counted by number, not size.
Why: Applegate: the extent of a modification is not considered, only how many extra spaces there are.
Revise: Unit 5 §5.2 · LO 2a.10

370 · Answer E — All of the above.
Why: Prognosis integrates periodontal support, restorability, endodontic status, strategic value and patient risk factors.
Revise: Unit 5 §5.9 · LO 2a.16

Under 70% on Block E? Re-read Unit 5 and Crash notebook Schematic 12 and Number cards 2 and 4, then re-sit in a fortnight.

Block F answers — Removable partial dentures

371 · Answer D — The saddle outline.
Why: The design sequence is saddles, support, retention, bracing and reciprocation, connection, review. First decide which spaces need restoring.
Revise: Unit 6 §6.1 · Schematic 11 · LO 2c.3

372 · Answer B — Support.
Why: Support (tooth, mucosa or both) follows the saddles; clasps come later.
Revise: Unit 6 §6.1 · LO 2c.3

373 · Answer E — Bracing, reciprocal elements and guide planes.
Why: The three forces are towards the tissues (support), away from them (retention) and lateral (bracing).
Revise: Unit 6 §6.1 · LO 2c.3

374 · Answer A — Whether the spaces need restoring at all.
Why: A shortened dental arch may give adequate function; not every space needs filling.
Revise: Unit 6 §6.1, Unit 2 · LO 2c.2

375 · Answer C — 0.25 mm.
Why: Stiff Co-Cr clasps engage 0.25 mm; gauges are 0.25, 0.5 and 0.75 mm.
Revise: Unit 6 §6.2, §6.4 · LO 2c.3

376 · Answer C — Greatest convexity relative to the path.
Why: The undercut lies gingival to the survey line, which changes with the chosen tilt.
Revise: Unit 6 §6.2 · LO 2c.3

377 · Answer E — All of the above.
Why: Guide planes, retention, interference and aesthetics together determine the path.
Revise: Unit 6 §6.2 · LO 2c.3

378 · Answer B — Mesial undercuts increase.
Why: Lowering the front tips the crowns mesially relative to the rod. Lowering the heels increases distal undercuts.
Revise: Unit 6 §6.2 (exam traps) · LO 2c.3

379 · Answer D — Tripod the cast.
Why: Three marks at the same height let the technician reproduce the tilt exactly.
Revise: Unit 6 §6.2 · LO 2c.3

380 · Answer A — 2–4 mm, parallel to the path.
Why: Guide planes define a single path, add frictional retention, brace and reduce food trapping.
Revise: Unit 6 §6.2 · LO 2c.3

381 · Answer E — Recontour enamel.
Why: Enameloplasty lowers the height of contour so the reciprocal arm sits correctly.
Revise: Unit 6 §6.2 · LO 2c.3

382 · Answer A — Composite to create an undercut.
Why: Additive composite can create retentive contour or a rest seat without crowning.
Revise: Unit 6 §6.2 · LO 2c.3

383 · Answer C — Retention by engaging an undercut.
Why: That is a clasp's job. Rests support, keep the denture in position, protect the gingiva, keep clasps placed, act as indirect retainers and prevent over-eruption.
Revise: Unit 6 §6.3 · LO 2c.3

384 · Answer D — Spoon-shaped, 1–1.5 mm, one-third width, under 90°.
Why: This directs load along the long axis and gives enough metal thickness for Co-Cr.
Revise: Unit 6 §6.3 · Number card 2 · LO 2c.3

385 · Answer B — After guide planes, before the impression.
Why: Guide planes alter proximal contours; rest seats follow, then the master impression records both.
Revise: Unit 6 §6.3 · LO 2c.3

386 · Answer B — Cingulum rest.
Why: It is less visible, gives a shorter lever and directs load better than an incisal rest.
Revise: Unit 6 §6.3 · LO 2c.3

387 · Answer D — Mesially.
Why: A mesial rest tips the abutment towards the arch rather than distally when the saddle sinks (RPI principle).
Revise: Unit 6 §6.3 (exam traps) · LO 2c.3

388 · Answer E — Embrasure (double) rests.
Why: They support the clasp and stop it wedging the teeth apart.
Revise: Unit 6 §6.3, §6.4 · LO 2c.3

389 · Answer A — More than 180°.
Why: Encirclement stops the tooth escaping from the clasp. Other requirements are retention, support, reciprocation, passivity and bracing.
Revise: Unit 6 §6.4 · LO 2c.3

390 · Answer C — Ring clasp with a strut.
Why: It reaches the mesiolingual undercut of a tipped molar; its long, flexible arm needs a bracing strut.
Revise: Unit 6 §6.4 · Schematic 14 · LO 2c.3

391 · Answer A — Embrasure clasp.
Why: It gives cross-arch retention and bracing on the dentate side of a Kennedy II; prepare enough embrasure space.
Revise: Unit 6 §6.4 · LO 2c.3

392 · Answer C — Shallow sulcus, soft-tissue undercut or prominent frenum.
Why: A gingivally approaching I-bar needs a sulcus of about 4 mm or more and no tissue undercut.
Revise: Unit 6 §6.4 · LO 2c.3

393 · Answer B — Wrought wire.
Why: Wrought wire is flexible, adjustable and suits 0.5–0.75 mm undercuts on free-end abutments.
Revise: Unit 6 §6.4 · LO 2c.3

394 · Answer E — About 8 times more flexible.
Why: Flexibility rises with the cube of length.
Revise: Unit 6 §6.4 · LO 2c.3

395 · Answer D — On or above the survey line, contacting first.
Why: It must resist the retentive arm's push as that arm flexes over the height of contour.
Revise: Unit 6 §6.4 · LO 2c.3

396 · Answer D — Gingival third, passive when seated.
Why: An active clasp at rest moves the tooth orthodontically.
Revise: Unit 6 §6.4 · LO 2c.3

397 · Answer A — Differential support.
Why: Teeth move 0.02–0.1 mm; mucosa compresses 0.4–2 mm. The saddle rotates about the fulcrum line and torques the abutment.
Revise: Unit 6 §6.5 · LO 2c.3

398 · Answer C — Retromolar pad and buccal shelf.
Why: Maximum extension spreads load over the largest supporting area.
Revise: Unit 6 §6.5 · LO 2c.3

399 · Answer E — Altered cast technique.
Why: It records the saddle area under functional load on the framework, matching tooth and mucosal support.
Revise: Unit 6 §6.5 · LO 2c.3

400 · Answer B — Mesial rest, distal proximal plate, buccal I-bar.
Why: That is Kratochvil and Krol's RPI assembly.
Revise: Unit 6 §6.5 · LO 2c.3

401 · Answer B — They disengage gingivally and mesially.
Why: Releasing from the tooth under load minimises torque on the abutment.
Revise: Unit 6 §6.5 · LO 2c.3

402 · Answer E — RPA.
Why: Eliason's RPA replaces the I-bar with an Akers-type arm when a gingivally approaching clasp is contraindicated.
Revise: Unit 6 §6.5 · LO 2c.3

403 · Answer D — Through the rests nearest the free-end saddles.
Why: The saddle rotates about this line.
Revise: Unit 6 §6.6 · LO 2c.3

404 · Answer C — As far anterior to the fulcrum line as possible.
Why: The longer the lever arm, the better the indirect retainer resists lifting.
Revise: Unit 6 §6.6 · LO 2c.3

405 · Answer A — Support and preventing connector settling.
Why: Indirect retainers also support anterior extensions and stop the major connector sinking.
Revise: Unit 6 §6.6 · LO 2c.3

406 · Answer E — Narrow, reduced occlusal tables.
Why: Narrower tables reduce the load on the resorbing ridge and the torque on abutments.
Revise: Unit 6 §6.5 · LO 2c.3

407 · Answer C — Reline.
Why: Lifting rests on pressing the saddle show ridge resorption and rotation.
Revise: Unit 6 §6.9 · LO 2c.20

408 · Answer A — Full palatal plate.
Why: Few teeth, long saddles and a flat palate need maximum support and retention.
Revise: Unit 6 §6.7 · Schematic 13 · LO 2c.3

409 · Answer B — Mid-palatal strap, at least 8 mm wide.
Why: Short bounded saddles on both sides need a thin, rigid strap.
Revise: Unit 6 §6.7 · LO 2c.3

410 · Answer D — Anteroposterior strap.
Why: It is rigid with less coverage and avoids the torus.
Revise: Unit 6 §6.7 · LO 2c.3

411 · Answer C — Horseshoe.
Why: It is the least rigid maxillary connector, used only for anterior saddles or an inoperable torus.
Revise: Unit 6 §6.7 (exam traps) · LO 2c.3

412 · Answer D — Lingual bar.
Why: At least 7–8 mm of sulcus allows a half-pear bar about 4 mm high and 3 mm clear of the gingiva.
Revise: Unit 6 §6.7 · Number card 2 · LO 2c.3

413 · Answer A — Lingual plate.
Why: A shallow sulcus, mobile incisors needing splinting and likely additions favour a plate.
Revise: Unit 6 §6.7 · LO 2c.3

414 · Answer B — Sublingual bar.
Why: It sits deeper in the sulcus and is recorded in a functional impression.
Revise: Unit 6 §6.7 · LO 2c.3

415 · Answer E — Labial bar.
Why: It is used for lingually inclined teeth or large tori, despite its bulk.
Revise: Unit 6 §6.7 · LO 2c.3

416 · Answer A — It traps food.
Why: A continuous clasp gives indirect retention but lies on the cingula, where food collects.
Revise: Unit 6 §6.7 · LO 2c.3

417 · Answer B — 3 mm (mandible) to 6 mm (maxilla) clear, or fully covered with relief.
Why: Never end a connector on the gingival margin.
Revise: Unit 6 §6.7 · LO 2c.3

418 · Answer D — At right angles, narrow, in the embrasure.
Why: This minimises gingival coverage and is least noticed by the tongue.
Revise: Unit 6 §6.7 · LO 2c.3

419 · Answer C — Major connectors must be rigid.
Why: Rigidity distributes forces across the arch; a flexible bar concentrates load on single abutments.
Revise: Unit 6 §6.7 · LO 2c.3

420 · Answer E — Gingival stripping, periodontal disease and caries.
Why: Mucosa-borne acrylic RPDs covering the margins are known as "gum strippers".
Revise: Unit 6 §6.8 · LO 2c.3

421 · Answer E — Acrylic interim (transitional) denture.
Why: Acrylic is easy to add to as teeth are lost; a Co-Cr framework would be wasted on teeth of uncertain prognosis.
Revise: Unit 6 §6.8 · LO 2c.4

422 · Answer B — Every denture.
Why: Every (1949): point contacts, open embrasures, free gingival margins and distal stabilising wires maximise gingival health in an acrylic design.
Revise: Unit 6 §6.8 · LO 2c.4

423 · Answer A — Spoon denture.
Why: A small upper anterior mucosa-borne interim denture with no gingival contact.
Revise: Unit 6 §6.8 · LO 2c.4

424 · Answer D — Swing-lock denture.
Why: A hinged labial bar engages labial undercuts of many teeth, suiting compromised teeth where conventional clasps fail.
Revise: Unit 6 §6.8 · LO 2c.4

425 · Answer C — Hard to adjust, reline or add to.
Why: Nylon is aesthetic and clasp-free but mainly mucosa-borne and difficult to modify.
Revise: Unit 6 §6.8 · LO 2c.4

426 · Answer C — Crowned abutments; usually beyond Level 2.
Why: Attachments need crowns, laboratory precision and maintenance.
Revise: Unit 6 §6.8 · LO 2c.4

427 · Answer A — Pre-prosthetic optimisation.
Why: Correct over-eruption, tilts and the occlusal plane before design; this is within Level 2.
Revise: Unit 6 §6.9 · LO 2c.3

428 · Answer E — All of the above.
Why: At framework try-in check fit with disclosing medium, full seating of rests, passive clasps and undisturbed occlusion before setting teeth.
Revise: Unit 6 §6.9 · LO 2c.20

429 · Answer D — Along the path, using the saddles.
Why: Pulling on clasp tips distorts them and loosens retention.
Revise: Unit 6 §6.9 · LO 2c.20

430 · Answer B — About 1 week, then regular recall.
Why: Early review catches pressure points and high rests.
Revise: Unit 6 §6.9 · LO 2c.20

431 · Answer D — Monitoring abutments and relining saddles.
Why: Caries and periodontal disease at abutments, and ridge resorption under free-end saddles, are the main long-term risks.
Revise: Unit 6 §6.9 · LO 2c.20

432 · Answer C — Well-designed Co-Cr does little harm.
Why: Bergman showed tooth-supported Co-Cr RPDs, with good hygiene and recall, do little harm to the remaining teeth.
Revise: Unit 6, In one minute · LO 2c.3

433 · Answer E — The dentist.
Why: The dentist is responsible for the design; the prescription must specify every component.
Revise: Unit 6, In one minute, Unit 13 · LO 2c.3, 3a.3

434 · Answer B — Without reciprocation the tooth is pushed.
Why: A retentive clasp without reciprocation is not a complete design.
Revise: Unit 6 companion note (SBA traps) · LO 2c.3

435 · Answer A — A high rest.
Why: Adjust the rest or deepen the seat; pain after fitting is often from a high rest, over-extension or tissue pressure.
Revise: Unit 6 companion note · LO 2c.20

436 · Answer B — Tooth-borne with rests at both ends.
Why: A Kennedy III saddle is tooth-supported, so it does not rotate about a fulcrum; indirect retention is generally unnecessary.
Revise: Unit 6 §6.1, §6.6 · LO 2c.3

437 · Answer E — All of the above.
Why: A Kennedy IV saddle needs rests on the adjacent teeth, indirect retention posteriorly, and attention to appearance and lip support.
Revise: Unit 6 §6.6 · LO 2c.3

438 · Answer A — Mesial rests, RPI, canine cingulum rests, lingual bar, maximum extension.
Why: This classic free-end design controls torque and rotation.
Revise: Unit 6 §6.5–6.7 · Schematic 14 · LO 2c.3

439 · Answer C — Full palatal plate.
Why: With only four teeth and bilateral free ends, palatal coverage gives most support and retention.
Revise: Unit 6 §6.7 · LO 2c.3

440 · Answer D — Survey and plan, prepare, then the master impression.
Why: Preparing teeth to a surveyed design ensures guide planes and rests match the path.
Revise: Unit 6 companion note 02 · LO 2c.3

441 · Answer A — Short, about 2–3 mm.
Why: A short guide plane lets the proximal plate move gingivally and disengage.
Revise: Unit 6 §6.5 · LO 2c.3

442 · Answer E — Wrought wire.
Why: Wrought wire engages 0.5–0.75 mm; Co-Cr 0.25 mm; cast gold 0.5 mm.
Revise: Unit 6 §6.4 · LO 2c.3

443 · Answer B — Distal undercuts increase.
Why: Lowering the heels tips the crowns distally relative to the rod.
Revise: Unit 6 §6.2 · LO 2c.3

444 · Answer C — About one-eighth.
Why: Flexibility falls with the cube of thickness.
Revise: Unit 6 §6.4 · LO 2c.3

445 · Answer D — I-bar, or move retention to a premolar.
Why: A gingivally approaching bar is less visible; retention can also be placed on less visible teeth or gained from guide planes.
Revise: Unit 6 §6.4 · LO 2c.3

446 · Answer E — Excellent oral hygiene.
Why: A lingual plate covers the gingival margins, so plaque control is critical.
Revise: Unit 6 §6.7 · LO 2c.3

447 · Answer D — Minimal, open design.
Why: Reducing tooth and gingival coverage limits plaque stagnation in a high-risk mouth.
Revise: Unit 6 §6.8, Unit 2 · LO 2c.3

448 · Answer B — Alkaline peroxide.
Why: Hypochlorite corrodes Co-Cr.
Revise: Unit 6 (exam traps) · LO 2c.20

449 · Answer A — Remove it at night.
Why: Night removal rests the tissues and reduces candidosis and plaque stagnation.
Revise: Unit 6 §6.9 · LO 2c.20

450 · Answer C — Reline the saddle.
Why: Rocking and lifting indirect retainers mean ridge resorption; tightening clasps would add torque.
Revise: Unit 6 §6.9 · LO 2c.20

451 · Answer B — Preventing over-eruption.
Why: Occlusal rests maintain the abutment's vertical position as well as supporting the denture.
Revise: Unit 6 §6.3 · LO 2c.3

452 · Answer A — Guide planes parallel to the path.
Why: Preparing parallel guide planes on a tilted abutment removes the obstruction to a single path.
Revise: Unit 6 §6.2 · LO 2c.3

453 · Answer D — A proximal plate on a guide plane.
Why: Close contact between plate and guide plane reduces food trapping.
Revise: Unit 6 §6.2 · LO 2c.3

454 · Answer E — In harmony with the existing ICP.
Why: RPD occlusion conforms to the natural occlusion; balanced occlusion is for complete dentures.
Revise: Unit 6 §6.9 · LO 2c.19

455 · Answer C — Pre-prosthetic optimisation within Level 2.
Why: Adjusting over-eruption and tilts before design is part of Level 2 care.
Revise: Unit 6 §6.9 · LO 2c.3

456 · Answer D — Lingual plate.
Why: A lingual plate splints mobile incisors and provides indirect retention.
Revise: Unit 6 §6.7 · LO 2c.3

457 · Answer C — Inadequate rest support.
Why: Without seated rests the denture settles and the bar impinges on the gingiva.
Revise: Unit 6 §6.3, §6.7 · LO 2c.3

458 · Answer A — Cingulum rests, lingual plate or premolar rests anterior to the fulcrum.
Why: Any rigid component seated well forward of the fulcrum line can act as an indirect retainer.
Revise: Unit 6 §6.6 · LO 2c.3

459 · Answer E — It reverses the design logic.
Why: Support and the path of insertion come first; clasps are chosen to suit them.
Revise: Unit 6 companion note (SBA traps) · LO 2c.3

460 · Answer B — Far forward, such as a canine cingulum rest.
Why: An indirect retainer far anterior to the fulcrum resists the saddle lifting.
Revise: Unit 6 §6.6 · LO 2c.3

461 · Answer C — Teeth and mucosa together.
Why: A distal-extension RPD is tooth- and mucosa-borne (Craddock 3).
Revise: Unit 6 companion note 01 · LO 2c.3

462 · Answer D — Distal torque; use mesial rests with RPI or RPA.
Why: Rigid clasps in distal undercuts lever the abutments as the saddle sinks.
Revise: Unit 6 §6.5 · LO 2c.3

463 · Answer B — Swing-lock.
Why: It gains retention from labial undercuts across many teeth through a hinged labial bar.
Revise: Unit 6 §6.8 · LO 2c.4

464 · Answer A — Acrylic interim RPD.
Why: It suits uncertain prognosis and can be modified as treatment progresses.
Revise: Unit 6 §6.8 · LO 2c.4

465 · Answer E — Beyond Level 2.
Why: Precision attachments on crowned abutments are usually specialist care.
Revise: Unit 6 §6.8 · LO 2b.10

466 · Answer A — Adjust the marked area and re-check.
Why: Disclosing medium identifies the interference; relieve only where it marks. Remake only if no interference explains the misfit.
Revise: Unit 6 §6.9 · LO 2c.20

467 · Answer B — Adjust the high rest.
Why: A high rest overloads the abutment and causes tenderness.
Revise: Unit 6 companion note · LO 2c.20

468 · Answer C — A full drawn design on a surveyed, tripoded cast.
Why: The prescription must specify every component; the laboratory must not design it.
Revise: Unit 6, In one minute · LO 2c.3, 3a.3

469 · Answer D — Anterior strap thickness and position.
Why: Bulk over the anterior palate disturbs tongue contact for "s"; thin or reposition it.
Revise: Unit 6 §6.7 · LO 2c.20

470 · Answer E — All of the above.
Why: Tooth support, minimal gingival coverage, hygiene and recall together explain why Co-Cr RPDs do little harm.
Revise: Unit 6 §6.8 · LO 2c.3

Under 70% on Block F? Re-read Unit 6 and Crash notebook Schematics 11–14 and Number card 2, then re-sit in a fortnight.

Block G answers — Complete dentures

471 · Answer C — Stability.
Why: Stability resists horizontal and rotational forces; it depends on ridge form, flange extension, balanced occlusion and teeth in the neutral zone. Retention resists displacement away from the tissues.
Revise: Unit 7 §7.1 · LO 2c.9

472 · Answer E — All of the above.
Why: Retention has physical, muscular and mechanical components.
Revise: Unit 7 §7.1 · LO 2c.9

473 · Answer A — Support; maximum extension.
Why: Support resists movement towards the tissues and is improved by covering all stress-bearing areas.
Revise: Unit 7 §7.1 · LO 2c.9

474 · Answer D — Muscular factors.
Why: Polished surfaces shaped to the neutral zone let the tongue and cheeks hold the denture in function.
Revise: Unit 7 §7.1 · LO 2c.9

475 · Answer B — About 2 mm short.
Why: It leaves room for border moulding to record the functional sulcus.
Revise: Unit 7 §7.2 · LO 2c.9

476 · Answer B — Record the full sulcus depth.
Why: The primary cast must show the whole denture-bearing area so special trays can be made.
Revise: Unit 7 §7.2 · LO 2c.9

477 · Answer D — Greenstick compound.
Why: Its thermoplastic handling allows sectional border moulding.
Revise: Unit 7 §7.2 · LO 2c.9

478 · Answer E — Type III stone.
Why: Dental stone gives adequate strength for denture master casts; die stone is for crowns.
Revise: Unit 7 §7.2 · LO 2c.9

479 · Answer A — ZOE, light-bodied silicone or polyether.
Why: These record detail in a close-fitting, border-moulded special tray.
Revise: Unit 7 §7.2 · LO 2c.9

480 · Answer C — OVD, RCP, plane, lines, mould and shade.
Why: Everything the technician needs to set the teeth is recorded at this visit, with a facebow if a semi-adjustable articulator is used.
Revise: Unit 7 §7.2 · LO 2c.9

481 · Answer E — Patient approval and post-dam marking.
Why: Try-in is the last chance to change appearance; approval is recorded and the vibrating line marked for the post-dam.
Revise: Unit 7 §7.2 · LO 2c.9

482 · Answer A — Pressure-indicating paste.
Why: It shows exactly where the fitting surface presses on the mucosa.
Revise: Unit 7 §7.2 · LO 2c.9

483 · Answer B — About 1 week.
Why: Early review lets you adjust ulcers and occlusion.
Revise: Unit 7 §7.2 · LO 2c.27

484 · Answer C — 8–10 mm.
Why: This rule of thumb restores lip support.
Revise: Unit 7 §7.3 · LO 2c.9

485 · Answer D — About 1–2 mm.
Why: Incisal display reduces with age.
Revise: Unit 7 §7.3 · LO 2c.9

486 · Answer D — Interpupillary line anteriorly, Camper's plane posteriorly.
Why: These references orient the occlusal plane.
Revise: Unit 7 §7.3 (exam traps) · LO 2c.9

487 · Answer C — Two-thirds up the retromolar pad.
Why: At this height, level with the lateral border of the tongue, the lower plane is stable.
Revise: Unit 7 §7.3 · LO 2c.9

488 · Answer A — Philtrum, and the corners of the mouth at rest.
Why: They locate the midline and the distal of the canines.
Revise: Unit 7 §7.3 · LO 2c.9

489 · Answer E — About 67 mm.
Why: OVD = RVD minus a freeway space of 2–4 mm (average about 3 mm).
Revise: Unit 7 §7.4 · Number card 2 · LO 2c.9

490 · Answer B — Base of the nose to the underside of the chin.
Why: Willis noted this roughly equals the pupil-to-stomion distance.
Revise: Unit 7 §7.4 · LO 2c.9

491 · Answer A — About 1–2 mm.
Why: If the teeth touch on "s", the OVD is too high.
Revise: Unit 7 §7.4 · LO 2c.9

492 · Answer D — Rest position varies; measure several times.
Why: Head posture, fatigue and emotion all change it.
Revise: Unit 7 §7.4 · LO 2c.9

493 · Answer C — OVD too high.
Why: Generalised soreness, clicking and strained lips are classic signs of too little freeway space.
Revise: Unit 7 §7.4 (exam traps) · LO 2c.9, 2c.21

494 · Answer E — OVD too low.
Why: Overclosure causes commissural folds (angular cheilitis), a collapsed look and a prominent chin.
Revise: Unit 7 §7.4 · LO 2c.9

495 · Answer B — Muscle fatigue, soreness, faster resorption.
Why: Constant loading without rest accelerates resorption and fatigues the muscles.
Revise: Unit 7 §7.4 · LO 2c.9

496 · Answer E — RCP is reproducible.
Why: With no teeth to guide ICP, RCP is the repeatable reference.
Revise: Unit 7 §7.5 · LO 2c.19

497 · Answer B — Gentle guidance, patient relaxed, tongue curled back.
Why: Force or protrusion gives false records.
Revise: Unit 7 §7.5 · LO 2c.19

498 · Answer D — The apex of the arrowhead.
Why: It marks the most retruded, reproducible position.
Revise: Unit 7 §7.5 · LO 2c.19

499 · Answer C — New registration and re-articulate.
Why: Grinding the teeth to fit a wrong record makes things worse.
Revise: Unit 7 §7.5 · LO 2c.19

500 · Answer A — 30° condylar guidance; 10 cm Bonwill triangle.
Why: These fixed averages are adequate for most complete dentures.
Revise: Unit 7 §7.6 · Number card 2 · LO 2c.19

501 · Answer B — From a protrusive record.
Why: The protrusive record positions the condylar elements at the patient's guidance angle.
Revise: Unit 7 §7.6 · LO 2c.19

502 · Answer C — L = H/8 + 12.
Why: Hanau's formula estimates the Bennett angle (L) from the horizontal condylar guidance (H).
Revise: Unit 7 §7.6 · LO 2c.19

503 · Answer E — Condylar element on the lower member.
Why: Arcon designs (Whip Mix, Denar) mimic the anatomy; non-arcon (Dentatus, Hanau H2) have it on the upper member.
Revise: Unit 7 §7.6 · LO 2c.19

504 · Answer A — Maxilla to hinge axis.
Why: It makes the arcs of closure on the articulator match the patient's.
Revise: Unit 7 §7.6 · LO 2c.19

505 · Answer D — Christensen's phenomenon; compensating curve.
Why: The condyles move down the eminence in protrusion; a compensating curve keeps posterior contact.
Revise: Unit 7 §7.6 · LO 2c.19

506 · Answer C — Hanau's quint.
Why: Condylar guidance, incisal guidance, cusp angle, plane of orientation and compensating curve must be balanced together.
Revise: Unit 7 §7.6 · LO 2c.19

507 · Answer A — Simultaneous working and non-working contacts.
Why: They stop the dentures tipping in excursions.
Revise: Unit 7 §7.6 · LO 2c.19

508 · Answer B — Upper palatal cusps into lower central fossae.
Why: Lingualised occlusion gives good stability on resorbed ridges with acceptable appearance.
Revise: Unit 7 §7.6 · LO 2c.19

509 · Answer D — Monoplane teeth.
Why: Zero-degree teeth generate the least horizontal force, which suits severe resorption, Class III and poor control.
Revise: Unit 7 §7.6 · LO 2c.19

510 · Answer E — About one-sixteenth of bizygomatic width.
Why: Berry's proportion is one guide, alongside the canine lines and inter-alar width.
Revise: Unit 7 §7.7 · LO 2c.9

511 · Answer D — Weakly evidence-based.
Why: Williams' face form and Frush and Fisher's dentogenics can guide shape but lack strong evidence.
Revise: Unit 7 §7.7 · LO 2c.9

512 · Answer B — Involve her; favour an age-appropriate shade.
Why: Very light teeth create a "denture look"; shared decision-making avoids later disputes.
Revise: Unit 7 §7.7 · LO 2c.9, 3b.1

513 · Answer A — The denture is pushed forward.
Why: Occlusal force on an inclined surface drives the denture anteriorly; lower posteriors go over the crest, never up the ramus.
Revise: Unit 7 §7.7 · LO 2c.9

514 · Answer E — Crossbite.
Why: Centripetal maxillary resorption makes the upper arch narrower; setting in crossbite keeps teeth over the ridges.
Revise: Unit 7 §7.7, Unit 2 · LO 2c.9

515 · Answer C — Mouldable material shaped by function.
Why: Tissue conditioner, compound or soft wax on a lower base records where tongue and cheek forces balance.
Revise: Unit 7 §7.7 · LO 2c.9

516 · Answer A — Resorbed lower ridge with unstable dentures.
Why: That is where muscle control matters most.
Revise: Unit 7 §7.7 · LO 2c.9

517 · Answer E — Buccal overjet.
Why: Horizontal overlap keeps the cheek out from between the teeth.
Revise: Unit 7 §7.7 · LO 2c.9

518 · Answer C — 1–1.5 mm either side of the midline.
Why: It is deepest over the compressible glandular tissue and shallower over the thin raphe.
Revise: Unit 7 §7.8 · LO 2c.9

519 · Answer B — Compensates for polymerisation contraction.
Why: The post-dam also completes the posterior seal and reduces food entry and gagging.
Revise: Unit 7 §7.8 · LO 2c.9

520 · Answer D — "Ah" and the Valsalva manoeuvre.
Why: Saying "ah" shows the posterior vibrating line; the Valsalva manoeuvre shows the anterior one.
Revise: Unit 7 §7.8 · LO 2c.9

521 · Answer E — Upper incisal edges touching the wet–dry line.
Why: The f/v sounds test upper incisor length and labial position.
Revise: Unit 7 §7.8 · LO 2c.9

522 · Answer A — Narrow anterior palate or palatally set teeth.
Why: Whistling on "s" suggests too little anterior tongue space.
Revise: Unit 7 §7.8 · LO 2c.21

523 · Answer D — Excess palatal thickness at the rugae.
Why: The tongue contacts the rugae area for many sounds; bulk there causes a lisp.
Revise: Unit 7 §7.8 · LO 2c.21

524 · Answer B — Over-extended distobuccal flange.
Why: The coronoid process moves forward on opening and pushes the flange down.
Revise: Unit 7 §7.9 (exam traps) · Schematic 16 · LO 2c.21

525 · Answer C — Inadequate post-dam or seal.
Why: Looseness at rest means air enters at the border; under-extension or poor fit are other causes.
Revise: Unit 7 §7.9 · LO 2c.21

526 · Answer B — Over-extended lingual or retromolar border.
Why: Mylohyoid rises on swallowing; teeth outside the neutral zone also displace the denture.
Revise: Unit 7 §7.9 · LO 2c.21

527 · Answer C — OVD, RCP error, parafunction, support or systemic factors.
Why: Generalised soreness points to overload or systemic susceptibility rather than a local fault.
Revise: Unit 7 §7.9 · LO 2c.21

528 · Answer E — Over-extended or sharp flange.
Why: Sulcus ulcers match the flange edge; trim and polish.
Revise: Unit 7 §7.9 · LO 2c.21

529 · Answer A — Occlusal interference, pressure spot or sharp bone.
Why: Crestal ulcers usually come from occlusal or fitting-surface pressure.
Revise: Unit 7 §7.9 · LO 2c.21

530 · Answer D — Pressure on the mental nerve.
Why: In a resorbed mandible the foramen lies near the crest; relieve the fitting surface over it.
Revise: Unit 7 §7.9, Unit 1 · LO 2c.21

531 · Answer A — Lack of buccal overjet.
Why: Edge-to-edge posteriors and low OVD trap the cheek.
Revise: Unit 7 §7.9 · LO 2c.21

532 · Answer E — Excessive OVD, porcelain teeth or loose dentures.
Why: Clicking teeth with generalised soreness point to excessive OVD.
Revise: Unit 7 (exam traps) · LO 2c.21

533 · Answer C — Correct the border, then manage the gag reflex.
Why: Treat the technical cause first; training bases, desensitisation and palateless implant designs help persistent gaggers.
Revise: Unit 7 §7.9 · LO 2c.21

534 · Answer D — Poor seal or under-extension.
Why: Gaps at the border let food under the denture.
Revise: Unit 7 §7.9 · LO 2c.21

535 · Answer B — Stomatitis, xerostomia, deficiency, diabetes, rarely allergy.
Why: Investigate local and systemic causes before assuming allergy or a psychological cause.
Revise: Unit 7 §7.9, Unit 2 · LO 2c.21

536 · Answer D — Low OVD, Candida, haematinic deficiency.
Why: Angular cheilitis is multifactorial.
Revise: Unit 7 §7.9 · LO 2c.21

537 · Answer B — Fatigue over a midline fulcrum.
Why: Flexure over a torus or a poor fit, plus natural opposing teeth, causes midline fatigue fracture.
Revise: Unit 7 §7.9, Unit 3 · LO 2c.22

538 · Answer A — Clinical remount.
Why: It corrects processing errors more accurately than adjusting in the mouth.
Revise: Unit 7 §7.9 · LO 2c.21

539 · Answer E — Copy dentures.
Why: They keep the polished and occlusal surfaces the patient has learned, changing only what is needed.
Revise: Unit 7 §7.10 · LO 2c.9

540 · Answer C — Poor retention; cannot be relined labially.
Why: Socketed immediates look natural at first but lose the labial seal as the ridge resorbs.
Revise: Unit 7 §7.10 · LO 2c.4

541 · Answer C — 24 hours in, next-day review, reline at 3–6 months.
Why: Rapid early resorption loosens the denture.
Revise: Unit 7 §7.10 · LO 2c.4

542 · Answer D — Root-filled, 2–3 mm, domed.
Why: This reduces leverage while preserving bone and proprioception; canines are ideal abutments.
Revise: Unit 7 §7.10 · LO 2c.4

543 · Answer B — Fluoride gel in the denture fitting surface.
Why: Caries and periodontitis are the main risks to overdenture abutments.
Revise: Unit 7 §7.10 · LO 2c.4

544 · Answer A — Two-implant overdenture.
Why: This is the minimum standard of care for the edentulous mandible.
Revise: Unit 7 §7.10 (exam traps), Unit 11 · LO 2c.5

545 · Answer E — All of the above.
Why: Kelly's combination syndrome includes all these changes.
Revise: Unit 7 §7.10 · LO 2c.9

546 · Answer E — Restore posterior support and reduce anterior loading.
Why: Unsupported posterior lower segments throw load onto the premaxilla.
Revise: Unit 7 §7.10 · LO 2c.9

547 · Answer A — Windowed selective-pressure technique.
Why: It records the firm ridge under load and the flabby tissue undisplaced.
Revise: Unit 7 §7.10 · LO 2c.9

548 · Answer D — A rebase replaces the whole base.
Why: A reline adds a new fitting surface; a rebase replaces all the base material, keeping the teeth.
Revise: Unit 7 §7.10 · LO 2c.23

549 · Answer C — Biting too firmly increased OVD.
Why: Closed-mouth relines raise the OVD if the patient bites into the material.
Revise: Unit 7 §7.10 · LO 2c.23

550 · Answer B — Level 3.
Why: Tuberosity reduction, vestibuloplasty and torus removal are usually specialist care.
Revise: Unit 7 §7.10 · LO 2b.10

551 · Answer D — More porous material.
Why: Chairside reline is quick but more porous and less durable than a laboratory reline.
Revise: Unit 7 §7.10 · LO 2c.23

552 · Answer C — Correct the border.
Why: Adhesive or OVD changes rarely cure an over-extended flange.
Revise: Unit 7 companion note (SBA traps) · LO 2c.21

553 · Answer E — Discuss the option and refer.
Why: Implant-assisted dentures should be offered; the surgery itself is beyond Level 2.
Revise: Unit 7 companion note · LO 2c.5

554 · Answer B — RCP registration error.
Why: Uneven contacts with a slide mean the jaw record is wrong.
Revise: Unit 7 §7.5 · LO 2c.19

555 · Answer A — Cutting the post-dam.
Why: Without it, polymerisation contraction lifts the posterior border and breaks the seal.
Revise: Unit 7 §7.8 · LO 2c.9

556 · Answer A — The occlusal plane.
Why: Fox's guide checks the plane against the interpupillary and ala–tragal lines.
Revise: Unit 7 §7.3 · LO 2c.9

557 · Answer D — Reset in the neutral zone.
Why: Teeth encroaching on tongue space trap the tongue and lift the denture.
Revise: Unit 7 §7.7 · LO 2c.21

558 · Answer C — Copy dentures.
Why: Dementia limits adaptation; preserve the familiar shapes.
Revise: Unit 7 §7.10, Unit 4 · LO 2c.9

559 · Answer B — Metal or reinforced palate with balanced contacts.
Why: Natural opposing teeth overload a single denture and cause fatigue fracture.
Revise: Unit 7 §7.10 · LO 2c.9

560 · Answer E — Undercuts, implants and adhesives.
Why: These are mechanical, as opposed to physical or muscular, retention.
Revise: Unit 7 §7.1 · LO 2c.9

561 · Answer B — OVD too high.
Why: Lips that cannot meet at rest look strained.
Revise: Unit 7 §7.4 · LO 2c.21

562 · Answer E — New jaw record, then reset or remake.
Why: Excessive OVD can only be cured by re-establishing the correct vertical relation.
Revise: Unit 7 §7.4 · LO 2c.21

563 · Answer D — It acts as a splint and limits swelling.
Why: Once swelling develops, reinsertion may be impossible.
Revise: Unit 7 §7.10 · LO 2c.4

564 · Answer C — Two-implant overdenture assessment and referral.
Why: It is the minimum standard for the failing edentulous mandible.
Revise: Unit 7 §7.10 · LO 2c.5

565 · Answer A — Regular mucosal and denture review.
Why: Edentulous patients still need oral cancer screening, and their dentures loosen over time.
Revise: Unit 7, Unit 12 · LO 2c.27

566 · Answer C — Explain expectations and adaptation.
Why: Denture wearers chew far less efficiently than dentate people; with no fault found, counselling is appropriate.
Revise: Unit 7 companion note 04, Unit 2 · LO 2c.27

567 · Answer B — Canine lines and inter-alar width.
Why: They estimate the width of the six upper anterior teeth.
Revise: Unit 7 §7.7 · LO 2c.9

568 · Answer A — Average-value articulator.
Why: It is adequate for most complete dentures.
Revise: Unit 7 §7.6 · LO 2c.19

569 · Answer D — Average-value articulator.
Why: It uses a Bonwill triangle of about 10 cm and fixed condylar guidance of about 30°.
Revise: Unit 7 §7.6 · LO 2c.19

570 · Answer E — All of the above.
Why: The prescription specifies mould, shade, occlusal scheme, post-dam and relief.
Revise: Unit 7, Unit 13 · LO 2c.9, 3a.3

Under 70% on Block G? Re-read Unit 7 and Crash notebook Schematics 15–16, then re-sit in a fortnight.

Block H answers — Fixed prosthodontics

571 · Answer B — Preservation of tooth structure and the pulp.
Why: Excessive reduction on a young vital tooth risks pulpal damage. Shillingburg's other principles are retention and resistance, structural durability and preservation of the periodontium.
Revise: Unit 8 §8.1 · LO 2c.13

572 · Answer D — 10–20°.
Why: About 6° is the theoretical ideal; 10–20° is typical clinically. Retention falls sharply as convergence increases.
Revise: Unit 8 §8.1 · Number card 3 · LO 2c.13

573 · Answer A — 3 mm for incisors and premolars; 4 mm for molars.
Why: Shorter walls lack resistance to tipping forces.
Revise: Unit 8 §8.1 · LO 2c.13

574 · Answer E — Height-to-width of at least 0.4.
Why: Wide, short preparations rotate off easily; this ratio improves resistance.
Revise: Unit 8 §8.1 · LO 2c.13

575 · Answer C — Grooves or boxes, adhesive cement or crown lengthening.
Why: These add resistance and surface area; more cement or taper does not.
Revise: Unit 8 §8.1 · LO 2c.13

576 · Answer A — Sharp burs, water, light pressure, no desiccation, good temporaries.
Why: Heat, desiccation and leakage injure the pulp.
Revise: Unit 8 §8.1 · LO 2c.13

577 · Answer C — Lower buccal and upper palatal cusps.
Why: The functional (supporting) cusps need a bevel so the restoration has bulk where it is loaded.
Revise: Unit 8 §8.1 · LO 2c.13

578 · Answer E — 1.5 mm functional, 1 mm non-functional, light chamfer.
Why: Gold is strong in thin section, so less reduction is needed.
Revise: Unit 8 §8.1 · LO 2c.13

579 · Answer B — Labial shoulder or heavy chamfer; lingual chamfer.
Why: The labial margin must give space for metal, opaque and porcelain.
Revise: Unit 8 §8.1 · LO 2c.13

580 · Answer D — Rounded shoulder or deep chamfer of about 1 mm.
Why: Ceramics need even thickness and rounded internal angles to avoid stress concentration.
Revise: Unit 8 §8.1 · LO 2c.13

581 · Answer D — 0.3–0.7 mm within enamel.
Why: Bonding to enamel is the key to veneer survival.
Revise: Unit 8 §8.1, §8.11 · LO 2c.13

582 · Answer A — Supragingival.
Why: Supragingival margins are easier to clean, record, finish and inspect.
Revise: Unit 8 §8.2 · LO 2c.13

583 · Answer E — About 3 mm.
Why: The supracrestal attachment averages about 2 mm (Gargiulo), plus the sulcus.
Revise: Unit 8 §8.2 · Number card 3 · LO 2c.13

584 · Answer C — Invasion of the supracrestal attachment.
Why: It causes chronic inflammation, recession or bone loss.
Revise: Unit 8 §8.2 · LO 2c.13

585 · Answer B — Overcontour trapping plaque.
Why: Emergence profiles should be flat, especially over furcations.
Revise: Unit 8 §8.2 · LO 2c.13

586 · Answer E — Adrenaline-impregnated cord.
Why: Systemic absorption can raise blood pressure; use aluminium chloride, ferric sulphate or plain cord.
Revise: Unit 8 §8.2 · LO 2c.13

587 · Answer B — Pulp, position, gingiva, function and testing.
Why: Provisionals also let you trial the planned shape, OVD or occlusion.
Revise: Unit 8 §8.3 · LO 2c.13

588 · Answer D — Bis-acryl composite.
Why: It has a low exotherm and good strength; PMMA is stronger long-term but hotter and has more monomer.
Revise: Unit 8 §8.3 · LO 2c.13

589 · Answer A — Eugenol-free temporary cement.
Why: Eugenol inhibits polymerisation of the later resin cement.
Revise: Unit 8 §8.3 · LO 2c.13

590 · Answer C — Conformative.
Why: Single units and short bridges are restored to the existing ICP.
Revise: Unit 8 §8.4 · LO 2c.19

591 · Answer C — Reorganised, usually Level 3 when major.
Why: An unstable ICP, extensive work or an OVD change needs a reorganised approach in RCP.
Revise: Unit 8 §8.4 · LO 2c.19, 2b.10

592 · Answer E — Before preparation.
Why: Teeth dehydrate and lighten during treatment, so later shades are too light.
Revise: Unit 8 §8.4 (exam traps) · LO 2c.13

593 · Answer B — Value.
Why: Brightness errors are the most noticeable.
Revise: Unit 8 §8.4 · LO 2c.13

594 · Answer A — Single units with stable intercuspation only.
Why: Dual-arch trays flex and cannot record longer spans accurately.
Revise: Unit 8 §8.4 · LO 2c.13

595 · Answer D — 20–40 µm, short of the margin.
Why: It gives room for cement without opening the margin.
Revise: Unit 8 §8.4 · LO 2c.13

596 · Answer B — Cuspal coverage.
Why: Sorensen and Martinoff showed much better survival for root-filled posterior teeth with cusp coverage.
Revise: Unit 8 §8.5 · LO 2c.14

597 · Answer A — Composite.
Why: Anterior root-filled teeth with little tissue loss need no crown or post.
Revise: Unit 8 §8.5 · LO 2c.14

598 · Answer C — Nayyar core.
Why: The pulp chamber retains an amalgam or composite core without a post.
Revise: Unit 8 §8.5 · LO 2c.14

599 · Answer D — A prompt coronal seal.
Why: Coronal leakage contributes to root canal failure.
Revise: Unit 8 §8.5 · LO 2c.14

600 · Answer E — Crown lengthening or orthodontic extrusion.
Why: A ferrule needs sound tooth above the bone; posts cannot replace it.
Revise: Unit 8 §8.5, Unit 3 §3.13 · LO 2c.14

601 · Answer E — The restoration on the abutment.
Why: The abutment is the tooth, the pontic the replacement, the connector the joint.
Revise: Unit 8 §8.6 · LO 2c.11

602 · Answer C — Long and divergent multi-rooted.
Why: These give more support than conical or fused roots.
Revise: Unit 8 §8.6 · LO 2c.11

603 · Answer D — Abutment root area at least that of the replaced teeth.
Why: That is Ante's law.
Revise: Unit 8 §8.6 · LO 2c.11

604 · Answer B — Historical and weakly supported.
Why: Ante's law is still examined but lacks good evidence.
Revise: Unit 8 §8.6 · LO 2c.11

605 · Answer A — Canine.
Why: The upper canine has the largest root of the single-rooted teeth.
Revise: Unit 8 §8.6 · LO 2c.11

606 · Answer D — About eightfold.
Why: Deflection increases with the cube of span length.
Revise: Unit 8 §8.6 · Number card 3 · LO 2c.11

607 · Answer E — About one-eighth.
Why: Deflection decreases with the cube of connector thickness.
Revise: Unit 8 §8.6 · LO 2c.11

608 · Answer A — A lever arm.
Why: Pontics outside the line between abutments tip the bridge labially.
Revise: Unit 8 §8.6 · LO 2c.11

609 · Answer C — Most destruction, parallel preparations, unnoticed loose retainer.
Why: These are the drawbacks of a fixed-fixed design.
Revise: Unit 8 §8.7 · Schematic 18 · LO 2c.11

610 · Answer B — A cantilever from 13.
Why: A long-rooted canine supporting a lateral incisor pontic is a classic cantilever.
Revise: Unit 8 §8.7 · LO 2c.11

611 · Answer A — Pontic no larger than the abutment; avoid long posterior distal cantilevers.
Why: Cantilevers torque their abutment.
Revise: Unit 8 §8.7 · LO 2c.11

612 · Answer B — Fixed-movable bridge.
Why: A movable joint avoids preparing non-parallel abutments to a single path.
Revise: Unit 8 §8.7 · LO 2c.11

613 · Answer C — Distal of the mesial (minor) retainer.
Why: Mesial drift then seats the key.
Revise: Unit 8 §8.7 (exam traps) · LO 2c.11

614 · Answer D — Spring cantilever; unhygienic palatal bar.
Why: It is historic and rarely used now.
Revise: Unit 8 §8.7 · LO 2c.11

615 · Answer E — Single-wing resin-bonded cantilever.
Why: It is first choice for a single missing incisor or premolar with sound abutments.
Revise: Unit 8 §8.8 · LO 2c.11

616 · Answer C — One wing debonded unnoticed.
Why: Differential movement between two wings leads to caries; single-wing designs perform better.
Revise: Unit 8 §8.7 · LO 2c.11

617 · Answer D — Hybrid bridge, usually with a movable joint.
Why: It combines a conventional retainer at one end with a resin-bonded retainer at the other.
Revise: Unit 8 §8.7 · LO 2c.11

618 · Answer B — The pier acts as a fulcrum.
Why: Use a non-rigid connector at the pier, or separate bridges.
Revise: Unit 8 §8.7 (exam traps) · LO 2c.11

619 · Answer E — All of the above.
Why: RBBs are minimally invasive, reversible, quick, safe for the pulp and usually rebondable.
Revise: Unit 8 §8.8 · LO 2c.11

620 · Answer A — About 0.7 mm, by a Dahl approach.
Why: Placing the wing at a slightly raised occlusion lets the teeth re-establish contact, avoiding preparation.
Revise: Unit 8 §8.8 · LO 2c.11

621 · Answer D — Maximum enamel coverage, wraparound, a rest and minimal preparation.
Why: These maximise the bond area and give resistance and a defined path.
Revise: Unit 8 §8.8 · LO 2c.11

622 · Answer E — Air-abrade, etch enamel, MDP resin, rubber dam.
Why: MDP bonds chemically to the base-metal oxide; isolation protects the bond.
Revise: Unit 8 §8.8, Unit 3 · LO 2c.11

623 · Answer B — Bruxism and heavy occlusion.
Why: They are the main risk factors for debonding.
Revise: Unit 8 §8.7 · LO 2c.11

624 · Answer A — Clean and rebond.
Why: RBBs fail safely; an intact bridge on a sound abutment can be rebonded.
Revise: Unit 8 §8.7 · LO 2c.20

625 · Answer C — Sanitary pontic.
Why: A gap of at least 3 mm suits lower posterior, non-aesthetic sites.
Revise: Unit 8 §8.9 · Schematic 19 · LO 2c.11

626 · Answer E — Modified ridge lap.
Why: Labial contact only balances appearance and hygiene in upper anterior and premolar sites.
Revise: Unit 8 §8.9 · LO 2c.11

627 · Answer C — Ovate pontic.
Why: It gives the most natural emergence but needs a thick ridge or socket preservation.
Revise: Unit 8 §8.9 · LO 2c.11

628 · Answer A — Saddle (ridge lap) pontic; unhygienic.
Why: Its concave surface cannot be cleaned and causes inflammation.
Revise: Unit 8 §8.9 (exam traps) · LO 2c.11

629 · Answer B — Convex, smooth, glazed or polished; no pressure.
Why: Blanching pressure causes inflammation under fixed pontics.
Revise: Unit 8 §8.9 · LO 2c.11

630 · Answer D — Narrow the buccolingual width.
Why: A narrower table reduces load on the abutments.
Revise: Unit 8 §8.9 · LO 2c.11

631 · Answer B — The connector.
Why: Connectors are the weakest point of all-ceramic bridges.
Revise: Unit 8 §8.10 · LO 2c.6

632 · Answer A — About 16 mm² premolar; three units to the second premolar.
Why: Lithium disilicate bridges are limited in length and position; anterior connectors need about 12 mm².
Revise: Unit 8 §8.10 · Number card 3 · LO 2c.6

633 · Answer D — About 9 mm².
Why: That is for a three-unit posterior zirconia bridge; longer spans need more.
Revise: Unit 8 §8.10 · LO 2c.6

634 · Answer E — Height.
Why: Strength rises with height squared and deflection falls with height cubed.
Revise: Unit 8 §8.10 · LO 2c.6

635 · Answer C — Discolouration unresponsive to bleaching.
Why: That is a classic indication, with diastema closure, shape change and minor malalignment.
Revise: Unit 8 §8.11 · LO 2c.13

636 · Answer A — Bruxism with edge-to-edge incisors.
Why: Heavy occlusion and edge-to-edge or Class III relationships fracture or debond veneers.
Revise: Unit 8 §8.11 · LO 2c.13

637 · Answer E — Incisal overlap with a palatal finish line.
Why: It distributes stress and gives a positive seat.
Revise: Unit 8 §8.11 · LO 2c.13

638 · Answer D — HF, silane, light-cured resin on etched enamel.
Why: This is the standard protocol for silica-based veneers.
Revise: Unit 8 §8.11, Unit 3 · LO 2c.13

639 · Answer C — Cusp coverage with an onlay.
Why: A cusp narrower than half its height is weak and needs protection.
Revise: Unit 8 §8.11 · LO 2c.13

640 · Answer B — Diverging 6–10°, no undercuts.
Why: An intracoronal restoration needs a path of withdrawal.
Revise: Unit 8 §8.11 · LO 2c.13

641 · Answer E — All details of patient, clinician, materials and design.
Why: The prescription is a legal document.
Revise: Unit 8 §8.12 · LO 3a.3

642 · Answer C — Statement of manufacture from an MHRA-registered manufacturer.
Why: Custom-made devices fall under the UK Medical Devices Regulations; keep the records.
Revise: Unit 8 §8.12 · LO 3a.3

643 · Answer A — The GDC.
Why: Technicians and clinical dental technicians are GDC registrants with their own scope of practice.
Revise: Unit 8 §8.12, Unit 13 · LO 3a.3

644 · Answer B — From outside inward.
Why: Face and lips set the frame before teeth and gingiva are judged.
Revise: Unit 8 aesthetic note 01 · LO 2a.16

645 · Answer D — Not mandatory.
Why: The golden proportion is not a natural-smile rule.
Revise: Unit 8 aesthetic note 01 · LO 2a.16

646 · Answer B — Investigate, then non-vital bleaching.
Why: Least invasive first: whitening before any preparation.
Revise: Unit 8 aesthetic note 02 · LO 2b.6

647 · Answer D — Resin infiltration or microabrasion.
Why: These treat white spots without preparation.
Revise: Unit 8 aesthetic note 02 · LO 2b.6

648 · Answer C — Additive composite.
Why: It is the most conservative option for peg laterals and spacing.
Revise: Unit 8 aesthetic note 02 · LO 2b.6

649 · Answer E — Orthodontics first.
Why: Malalignment is corrected before any elective tooth preparation.
Revise: Unit 8 aesthetic note 02 · LO 2b.6

650 · Answer A — Wax-up, mock-up, assessment and consent.
Why: Veneers are irreversible and will need replacing; test the result and discuss risks first.
Revise: Unit 8 aesthetic note 03 · LO 2b.6, 3b.1

651 · Answer C — Less conservative than the alternatives.
Why: Crowning sound teeth for a minor concern is a classic trap.
Revise: Unit 8 aesthetic note 05 · LO 2b.6

652 · Answer B — Survived but not a success.
Why: Success includes biological health, function and satisfaction, not just presence.
Revise: Unit 8 companion note 05 · LO 2c.20

653 · Answer E — Monolithic materials and protection.
Why: Layered ceramics chip under parafunction.
Revise: Unit 8 companion note 05, Unit 3 · LO 2c.6

654 · Answer D — Level 3.
Why: Full-mouth reconstruction with OVD change is specialist care.
Revise: Unit 8 companion note · LO 2b.10

655 · Answer A — Adrenaline-impregnated cord.
Why: Systemic absorption can raise blood pressure.
Revise: Unit 8 §8.2 · LO 2c.13

656 · Answer D — Colour-corrected daylight, quickly, with photographs.
Why: Eye fatigue and poor lighting distort shade; photos with tabs help the laboratory.
Revise: Unit 8 §8.4 · LO 2c.13

657 · Answer A — Short spans.
Why: Stitching errors accumulate across full arches.
Revise: Unit 8 §8.4, Unit 11 · LO 2a.19

658 · Answer E — Caries under the loose retainer.
Why: A partly debonded fixed-fixed bridge stays in place while leakage causes caries.
Revise: Unit 8 §8.7 · LO 2c.20

659 · Answer B — Cuspal coverage.
Why: Root-filled posterior teeth with MOD cavities need cusp protection.
Revise: Unit 8 §8.5 · LO 2c.14

660 · Answer C — An adhesive onlay preserves more tooth.
Why: It covers cusps without full-coverage reduction.
Revise: Unit 8 §8.11 · LO 2c.13

661 · Answer A — Lower central incisor.
Why: In the mandible, root surface area runs 6 > 7 > 3 > 5 ≈ 4 > 2 > 1.
Revise: Unit 8 §8.6 · LO 2c.11

662 · Answer D — Fixed-movable bridge or uprighting.
Why: A tilted abutment cannot share a single path with an upright one.
Revise: Unit 8 §8.7 · LO 2c.11

663 · Answer C — Avoid a posterior distal cantilever.
Why: Under heavy load it torques the single abutment.
Revise: Unit 8 §8.7 · LO 2c.11

664 · Answer E — Pier fulcrum; non-rigid connector.
Why: A rigid bridge rocks over the pier and debonds the terminal retainers.
Revise: Unit 8 §8.7 · LO 2c.11

665 · Answer B — Heavy occlusion contraindicates veneers.
Why: Discuss alternatives rather than proceeding.
Revise: Unit 8 §8.11 · LO 2c.13

666 · Answer C — Direct composite.
Why: It is reversible and suits a young patient who is still growing.
Revise: Unit 8 aesthetic note 02 · LO 2b.6

667 · Answer B — Shade change and reduced bond strength.
Why: Residual oxygen after bleaching inhibits resin polymerisation for a period, and the shade settles.
Revise: Unit 8 aesthetic note 03 · LO 2b.6

668 · Answer A — Fluid control plus displacement.
Why: The deep cord stays in; the upper cord is removed just before the impression.
Revise: Unit 8 §8.2 · LO 2c.13

669 · Answer D — Recontour or remake with a flat emergence.
Why: The cause is the crown contour; mouthwash and antibiotics only mask it.
Revise: Unit 8 §8.2 · LO 2c.20

670 · Answer E — All of the above.
Why: Recall covers biological, technical and patient-reported outcomes.
Revise: Unit 8 companion note 05 · LO 2c.20

Under 70% on Block H? Re-read Unit 8 and Crash notebook Schematics 17–19 and Number card 3, then re-sit in a fortnight.

Block I answers — Occlusion and TMD

671 · Answer D — Maximum intercuspation, independent of condylar position.
Why: ICP is defined by the teeth, not the joints.
Revise: Unit 9 §9.1 · LO 2a.15

672 · Answer A — RCP.
Why: RCP is the first tooth contact on the retruded (terminal hinge) arc.
Revise: Unit 9 §9.1 · LO 2a.15

673 · Answer C — A jaw-to-jaw relation with condyles superior-anterior.
Why: Centric relation is independent of the teeth.
Revise: Unit 9 §9.1 · LO 2a.15

674 · Answer E — About 1 mm forwards in about 90%.
Why: A slide is normal and needs no treatment by itself.
Revise: Unit 9 §9.1 · LO 2a.15

675 · Answer B — About 2–4 mm.
Why: The freeway space is the gap between rest position and occlusion.
Revise: Unit 9 §9.1 · LO 2a.15

676 · Answer E — Canine guidance.
Why: Only the canines contact on the working side and disclude all other teeth.
Revise: Unit 9 §9.1 · Schematic 20 · LO 2a.15

677 · Answer B — Group function.
Why: Several working-side teeth share the load.
Revise: Unit 9 §9.1 · LO 2a.15

678 · Answer D — Mutually protected occlusion.
Why: Anteriors protect posteriors in excursions; posteriors protect anteriors in ICP.
Revise: Unit 9 §9.1 · LO 2a.15

679 · Answer A — Non-working interference.
Why: It is most damaging in natural dentitions; desirable only in complete dentures (balanced occlusion).
Revise: Unit 9 §9.1 · LO 2a.15

680 · Answer C — Working-side interference.
Why: A posterior contact prevents canine or group guidance on the working side.
Revise: Unit 9 §9.1 · LO 2a.15

681 · Answer A — Deflective contact.
Why: It diverts the mandible from its normal path of closure.
Revise: Unit 9 §9.1 · LO 2a.15

682 · Answer C — Bennett angle, about 7–15°.
Why: It is the angle of the non-working condyle path to the sagittal plane.
Revise: Unit 9 §9.1 · LO 2a.15

683 · Answer E — Curve of Wilson.
Why: Wilson is mediolateral; Spee is anteroposterior.
Revise: Unit 9 §9.1 · LO 2a.15

684 · Answer B — Restricted.
Why: Normal maximum opening is about 40–55 mm; under 40 mm is restricted.
Revise: Unit 9 §9.1 · LO 2a.20

685 · Answer D — About 8–10 mm each.
Why: Those are typical lateral excursion and protrusion ranges.
Revise: Unit 9 §9.1 · LO 2a.15

686 · Answer B — 20–25 mm.
Why: Further opening needs condylar translation.
Revise: Unit 9 §9.1 · LO 2a.15

687 · Answer E — All of the above.
Why: These are the features of a functional ("ideal") occlusion.
Revise: Unit 9 §9.1 · LO 2a.15

688 · Answer A — Myalgia.
Why: Familiar pain on palpating the masticatory muscles; it is the commonest TMD.
Revise: Unit 9 §9.2 · Schematic 21 · LO 2a.20

689 · Answer C — Arthralgia.
Why: Pain from the joint itself on palpation or movement.
Revise: Unit 9 §9.2 · LO 2a.20

690 · Answer D — Disc displacement with reduction.
Why: A reciprocal click shows the disc is recaptured on opening.
Revise: Unit 9 §9.2 · LO 2a.20

691 · Answer C — DDwR with intermittent locking.
Why: Clicking plus episodes of limited opening.
Revise: Unit 9 §9.2 · LO 2a.20

692 · Answer D — DDwoR with limited opening (right closed lock).
Why: Opening under 40 mm, deflection to the affected side and restricted contralateral excursion.
Revise: Unit 9 §9.2 · LO 2a.20

693 · Answer B — DDwoR without limited opening.
Why: A history of lock, now opening 40 mm or more.
Revise: Unit 9 §9.2 · LO 2a.20

694 · Answer A — Degenerative joint disease.
Why: Crepitus with radiographic change.
Revise: Unit 9 §9.2 · LO 2a.20

695 · Answer E — Dislocation (open lock).
Why: The condyle is trapped anterior to the eminence and the patient cannot close.
Revise: Unit 9 §9.2 · LO 2a.20

696 · Answer A — Headache attributed to TMD.
Why: Temporal headache modified by jaw function.
Revise: Unit 9 §9.2 · LO 2a.20

697 · Answer C — Multifactorial and biopsychosocial.
Why: Parafunction, trauma, stress, sleep and widespread pain contribute; occlusion has only a weak association.
Revise: Unit 9 §9.2 · LO 2a.20

698 · Answer E — Women aged about 20–40.
Why: That is the commonest demographic.
Revise: Unit 9 §9.2 · LO 2a.20

699 · Answer D — Reassurance only.
Why: A click without pain or locking needs no treatment.
Revise: Unit 9 §9.3 · LO 2a.20

700 · Answer B — Soft diet, jaw rest, warm compresses, habit control.
Why: Self-care is first-line and most TMD is self-limiting.
Revise: Unit 9 §9.3 · LO 2a.20

701 · Answer E — Full-coverage hard acrylic with even RCP contacts and canine guidance.
Why: That is a stabilisation (Michigan-type) splint.
Revise: Unit 9 §9.3 · LO 2a.20

702 · Answer B — Short-term; may increase clenching.
Why: Soft splints are not for long-term myalgia management.
Revise: Unit 9 §9.3 · LO 2a.20

703 · Answer D — Posterior open bite.
Why: Long-term anterior repositioning changes the occlusion irreversibly.
Revise: Unit 9 §9.3 (exam traps) · LO 2a.20

704 · Answer C — Over-eruption of the uncovered posteriors.
Why: Partial-coverage appliances worn continuously let uncovered teeth erupt.
Revise: Unit 9 §9.3 · LO 2a.20

705 · Answer A — Manipulation, then conservative care.
Why: A recent closed lock may be reduced; otherwise reassure, splint and exercise.
Revise: Unit 9 §9.3 · LO 2a.20

706 · Answer D — Down and back.
Why: This lowers the condyle below the eminence and guides it back into the fossa; then advise limited opening.
Revise: Unit 9 §9.3 (exam traps) · LO 2a.20

707 · Answer A — Same-day referral for giant cell arteritis.
Why: Jaw claudication over age 50 risks sight; ESR and CRP are urgent.
Revise: Unit 9 §9.4 · LO 2a.20

708 · Answer B — Malignancy.
Why: Progressive trismus, numbness, weight loss and referred otalgia are red flags.
Revise: Unit 9 §9.4 · LO 2a.20

709 · Answer E — Cardiac pain.
Why: Exertional jaw pain can be angina.
Revise: Unit 9 §9.4 · LO 2a.20

710 · Answer C — Trigeminal neuralgia; carbamazepine.
Why: Paroxysmal electric-shock pain triggered by light touch is characteristic.
Revise: Unit 9 §9.4 · LO 2a.20

711 · Answer B — Irreversible and not evidence-based.
Why: Selective grinding is not a TMD treatment.
Revise: Unit 9 (exam traps) · LO 2a.20

712 · Answer E — All of the above.
Why: Secondary care offers tricyclics, botulinum toxin, arthrocentesis, arthroscopy and surgery.
Revise: Unit 9 §9.3 · LO 2a.20

713 · Answer A — GMP or specialist.
Why: Low-dose tricyclics are prescribed medically, not by the dentist routinely.
Revise: Unit 9 §9.3 · LO 2a.20

714 · Answer D — Attrition, fractures, masseteric hypertrophy, scalloped tongue, linea alba.
Why: These are the classic signs of bruxism.
Revise: Unit 9 §9.5 · LO 2c.19

715 · Answer C — A hard night splint with metal or monolithic zirconia.
Why: Layered porcelain chips under parafunction.
Revise: Unit 9 §9.5 · LO 2c.19

716 · Answer C — Mounted-cast examination.
Why: It identifies interferences, the slide and guidance before multiple restorations.
Revise: Unit 9 §9.5 · LO 2c.19

717 · Answer D — Reorganise to RCP.
Why: An unstable ICP or OVD change needs a reproducible reference.
Revise: Unit 9 §9.5 · LO 2c.19

718 · Answer A — Only for specific restorative interferences.
Why: Equilibration is irreversible.
Revise: Unit 9 §9.5 · LO 2c.19

719 · Answer B — No routine imaging.
Why: Imaging is targeted to a specific clinical question.
Revise: Unit 9 companion note · LO 2a.20

720 · Answer E — Referral.
Why: A closed lock not responding to conservative care is a referral scenario.
Revise: Unit 9 companion note · LO 2a.20

721 · Answer A — Posterior disclusion.
Why: Canine guidance takes lateral load off the posterior restorations.
Revise: Unit 9, In one minute · LO 2c.19

722 · Answer E — All of the above.
Why: Physiotherapy, exercises, CBT and stress management address the biopsychosocial components.
Revise: Unit 9 §9.3 · LO 2a.20

723 · Answer C — Simple analgesia.
Why: Short-term NSAIDs suit acute arthralgia.
Revise: Unit 9 §9.3 · LO 2a.20

724 · Answer B — Towards the affected side.
Why: The locked condyle cannot translate, so the jaw swings towards it.
Revise: Unit 9, In one minute · LO 2a.20

725 · Answer D — Posselt's envelope.
Why: It describes the border movements of the incisal point.
Revise: Unit 9 §9.1 · LO 2a.15

726 · Answer B — Full-coverage hard maxillary stabilisation splint.
Why: That is a Michigan splint.
Revise: Unit 9 §9.3 · LO 2a.20

727 · Answer D — Location, limitation, noises, locking, impact.
Why: These support a DC/TMD diagnosis.
Revise: Unit 9 companion note 01 · LO 2a.20

728 · Answer C — Group function.
Why: It shares lateral load when a canine cannot take it alone.
Revise: Unit 9, In one minute · LO 2c.19

729 · Answer E — Conform.
Why: A stable, comfortable ICP is kept for single restorations.
Revise: Unit 9 §9.5 · LO 2c.19

730 · Answer A — Clenching or bruxism.
Why: Linea alba and a scalloped tongue are soft-tissue signs of clenching.
Revise: Unit 9 §9.5 · LO 2c.19

Under 70% on Block I? Re-read Unit 9 and Crash notebook Schematics 20–21, then re-sit in a fortnight.

Block J answers — Tooth wear, the failing dentition, interfaces and trauma

731 · B — A stable oral environment. Good hygiene, managed caries risk, treated caries and stable periodontal disease come before NHS prosthodontics. *Unit 10, intro · LO 2c.1*

732 · E — Urgent, stabilisation, reassessment, definitive, maintenance. *Unit 10 §10.1 · Schematic 22 · LO 2c.1*

733 · A — Hygiene, caries control, periodontal steps 1–2, hopeless extractions, interim prostheses. Smoking cessation and medical risk control belong here too. *Unit 10 §10.1 · LO 2c.1*

734 · D — An engaging patient. About a 50% improvement in plaque and bleeding scores shows motivation. *Unit 10 §10.1 · LO 2c.1*

735 · C — Endodontics, periodontal surgery, then prosthodontics. *Unit 10 §10.1 · LO 2c.1*

736 · C — Risk-matched recall. Include supportive periodontal care and prosthesis review. *Unit 10 §10.1 · LO 2c.20*

737 · A — Identify and remove the cause. Use a diet diary and history; refer intrinsic causes to the GMP. *Unit 10 §10.2 · LO 2c.10*

738 · E — Acids at mealtimes, straw, delay brushing, high fluoride. *Unit 10 §10.2 · LO 2c.10*

739 · B — Photographs, casts, silicone indices, BEWE or Smith and Knight. *Unit 10 §10.2 · LO 2c.10*

740 · D — Symptoms, concern, structural or pulpal risk, or progression. Wear alone is not an indication. *Unit 10 §10.2 · LO 2c.10*

741 · D — Direct composite. It is additive, adhesive, repairable and reversible. *Unit 10 §10.2 · LO 2c.11*

742 · B — Last. Crowns remove the most tooth. *Unit 10 §10.2 · LO 2c.11*

743 · C — A removable Co-Cr anterior bite platform. Today the principle uses direct composite at an increased OVD. *Unit 10 §10.3 · LO 2c.11*

744 · A — Anterior intrusion and posterior over-eruption. There is also some mandibular repositioning. *Unit 10 §10.3 · LO 2c.11*

745 · E — About 4–6 months, occasionally up to 18. *Unit 10 §10.3 · LO 2c.11*

746 · A — Dahl. It suits localised anterior wear where dentoalveolar compensation has removed the space. *Unit 10 §10.3 · LO 2c.11*

747 · D — Implants cannot erupt. Implants or ankylosed posterior teeth stop contacts re-establishing. *Unit 10 (exam traps) · LO 2c.11*

748 · C — Bisphosphonates. Drugs affecting bone turnover are a relative contraindication, with active periodontal disease, TMD, an anterior open bite tendency and poor compliance. *Unit 10 §10.3 · LO 2c.11*

749 · E — Temporary speech change, tenderness, eating difficulty and months to contact. *Unit 10 §10.3 · LO 2c.11, 3b.1*

750 · B — Wax-up and a provisional trial at the new OVD. Extensive reorganisation is generally Level 3. *Unit 10 §10.3 · LO 2c.11*

751 · E — Elective root canal treatment. It sacrifices pulp vitality just to gain space. *Unit 10 §10.4 · LO 2c.11*

752 · C — Level 3, with risks. Crown lengthening can expose roots and worsen the crown–root ratio. *Unit 10 §10.4 · LO 2c.11*

753 · B — Compensation often maintains OVD. Do not infer reduced OVD from worn teeth alone. *Unit 10 companion note (traps) · LO 2a.15*

754 · D — Stabilise the cause first. Then review the remaining tissue. *Unit 10 companion note (traps) · LO 2c.10*

755 · A — Mixed mechanisms are common. Address every relevant risk factor. *Unit 10 companion note 01 · LO 1.4*

756 · B — Abutment potential, occlusal stops and canine guidance. *Unit 10 §10.5 · LO 2c.1*

757 · E — Shortened dental arch. Avoid heroic molar treatment when function is adequate. *Unit 10 §10.5 · LO 2c.1*

758 · A — Transitional acrylic RPDs and overdenture abutments. Avoid sudden total clearance. *Unit 10 §10.5 · LO 2c.1*

759 · C — Root-filled or asymptomatic, and caries-free. Retained roots preserve bone and proprioception. *Unit 10 §10.5 · LO 2c.1*

760 · D — Doubtful teeth are not bridge abutments. Design RPDs so teeth can be added instead. *Unit 10 §10.5 · LO 2c.1*

761 · C — PPD 4 mm or less with no bleeding. *Unit 10 §10.6 · LO 2c.16*

762 · A — Plaque under 20%, bleeding under 30%. Level 2–3 settings use stricter targets (bleeding under 10%). *Unit 10 §10.6 · LO 2c.16*

763 · E — Supragingival margins, correct contour, cleansable design. *Unit 10 §10.6 · LO 2c.16*

764 · B — Tooth-supported Co-Cr, open design, minimal torque. *Unit 10 §10.6 · LO 2c.16*

765 · D — When mobility affects function or comfort; adhesive splints. *Unit 10 §10.6 · LO 2c.16*

766 · E — Poor abutments; root resection is specialist. *Unit 10 §10.6 · LO 2c.16*

767 · C — Mucositis Level 2; peri-implantitis Level 3. *Unit 10 (exam traps) · LO 2c.16*

768 · D — Restorability. Decide on sound tissue, ferrule potential and periodontal support before root canal treatment. *Unit 10 §10.7 · LO 2c.17*

769 · A — About 10–15%. This is higher than for single crowns. *Unit 10 §10.7 (exam traps) · LO 2c.17*

770 · B — Level 3. Combined endodontic–periodontal–prosthodontic problems on strategic teeth are specialist care. *Unit 10 §10.7 · LO 2c.18*

771 · A — Appearance, drift and over-eruption, often minimal. *Unit 10 §10.8 · LO 2c.2*

772 · D — Surgery, cost, time, bone, peri-implant disease, NHS access. *Unit 10 §10.8 · LO 2c.5*

773 · B — Level 1. *Unit 10 §10.9 · LO 2b.10*

774 · E — Replacing and temporising multiple fixed units. That is a moderately difficult technical need (Level 2). *Unit 10 §10.9 · LO 2b.10*

775 · C — Level 3. It covers developmental conditions and head and neck cancer rehabilitation. *Unit 10 §10.9 · LO 2b.10*

776 · D — Level 3. Severe gagging is Level 3; minor gagging is Level 2. *Unit 10 §10.9 · LO 2b.10*

777 · B — Level 3. Undiagnosed pain is complex care. *Unit 10 §10.9 · LO 2b.10*

778 · A — Replant promptly if appropriate. Follow current IADT guidance and arrange urgent care. *Unit 10 trauma note 01 · LO 2a.18*

779 · E — Never replant a primary tooth. It risks damage to the permanent successor. *Unit 10 trauma note 01 · LO 2a.18*

780 · C — Appropriate storage medium and urgent care. *Unit 10 trauma note 01 · LO 2a.18*

781 · B — Necrosis, resorption, ankylosis, colour change, fracture healing. *Unit 10 trauma note 02 · LO 2a.18*

782 · E — Defer until growth is complete. Implants do not erupt with a growing jaw. *Unit 10 trauma note 03 · LO 2a.18*

783 · D — RBB, RPD or orthodontic space management. These keep future options open. *Unit 10 trauma note 03 · LO 2a.18*

784 · C — Ankylosis. An ankylosed tooth cannot erupt with growth and becomes infra-occluded. *Unit 10 trauma note 02 · LO 2a.18*

785 · A — Fracture position, root length, support and ferrule. *Unit 10 trauma note 04 · LO 2a.18*

786 · A — Multidisciplinary advice. *Unit 10 trauma note 04 · LO 2a.18*

787 · C — Liaise, prevent, delay definitive work. *Unit 10 §10.2, Unit 2 · LO 2c.10*

788 · B — A night-time hard splint. *Unit 10 §10.2 · LO 2c.10*

789 · D — 5000 ppm fluoride toothpaste. *Unit 10 §10.2, Unit 2 · LO 2c.10*

790 · E — Indirect composite, gold or ceramic onlays and palatal veneers. These come before conventional crowns. *Unit 10 §10.2 · LO 2c.11*

791 · E — Prevention and monitoring. Mild, stable, symptomless wear needs no restoration. *Unit 10 §10.2 · LO 2c.10*

792 · A — Posterior teeth out of contact. That is the intended starting point. *Unit 10 §10.3 · LO 2c.11*

793 · C — Assess and record pulp status. Prepared abutments carry a long-term necrosis risk. *Unit 10 §10.7 · LO 2c.17*

794 · B — Canines. Long roots and a strategic position make them ideal overdenture abutments. *Unit 10 §10.5, Unit 7 · LO 2c.1*

795 · D — A planned, staged transition. *Unit 10 §10.5 · LO 2c.1*

796 · C — Conservative composite. Anterior root-filled teeth with little tissue loss need no crown. *Unit 10 §10.7 · LO 2c.17*

797 · E — Only if hygiene allows. A plate covers the gingival margins. *Unit 10 §10.6 · LO 2c.16*

798 · D — Disease control first. Her scores are far above the targets for definitive work. *Unit 10 §10.6 · LO 2c.16*

799 · A — Supportive periodontal care with prosthesis review. *Unit 10 §10.1 · LO 2c.20*

800 · B — Time, appliances and Level 3 planning. *Unit 10 §10.8 · LO 2c.2*

Under 70% on Block J? Re-read Unit 10 and Crash notebook Schematic 22, then re-sit in a fortnight.

Block K answers — Implants and digital prosthodontics

801 · C — Simple implant prostheses and diagnosing peri-implant disease. Assessment, planning and placement are Level 3. *Unit 11, In one minute · LO 2c.5*

802 · E — Direct bone–implant connection with no soft tissue. That is Brånemark's definition. *Unit 11 §11.1 · LO 2c.5*

803 · A — Primary mechanical; secondary biological. *Unit 11 §11.1 · LO 2c.5*

804 · B — About 2–4 weeks. Primary stability falls before secondary stability builds. *Unit 11 (exam traps) · LO 2c.5*

805 · D — 1.5 mm. *Unit 11 §11.1 · Number card 3 · LO 2a.19*

806 · A — 3 mm. *Unit 11 §11.1 · LO 2a.19*

807 · D — 2 mm. *Unit 11 §11.1 · LO 2a.19*

808 · E — At least 1.5–2 mm. Thin buccal bone risks recession and resorption. *Unit 11 §11.1 · LO 2a.19*

809 · C — Immediate (within 1 week). *Unit 11 §11.1 · LO 2c.5*

810 · B — Early (1 week to 2 months). Conventional loading is after 2 months. *Unit 11 §11.1 · LO 2c.5*

811 · B — External hex. This historic design has more screw loosening. *Unit 11 §11.1 · LO 2c.5*

812 · D — Conical (Morse taper). It gives a better seal and stability. *Unit 11 §11.1 · LO 2c.5*

813 · C — Platform switching. It may reduce crestal bone loss. *Unit 11 §11.1 · LO 2c.5*

814 · A — History of periodontitis with smoking. Other risks: poorly controlled diabetes, poor plaque control, no maintenance, radiotherapy, IV antiresorptives and bruxism. *Unit 11 §11.1 · LO 2c.5*

815 · E — Two-implant overdenture. *Unit 11 §11.2 · LO 2c.5*

816 · A — Easier hygiene, less space; inserts wear. *Unit 11 §11.2 · LO 2c.5*

817 · A — A bar. It splints divergent implants but needs more vertical space and hygiene effort. *Unit 11 §11.2 · LO 2c.5*

818 · B — Usually 4 or more; palateless possible. *Unit 11 §11.2 · LO 2c.5*

819 · D — Screw-retained. It is retrievable with no subgingival cement. *Unit 11 §11.2 · LO 2c.5*

820 · C — Residual cement. Excess subgingival cement is a recognised cause of peri-implant disease. *Unit 11 §11.2 · LO 2c.5*

821 · D — Level 3; passive fit; short cantilevers. *Unit 11 §11.2 · LO 2c.5*

822 · A — Insert wear. Loss of retention is the commonest overdenture complication. *Unit 11 §11.2 · LO 2c.21*

823 · E — Screw loosening and ceramic chipping. *Unit 11 §11.2 · LO 2c.21*

824 · C — A calibrated torque driver. Tighten to the manufacturer's value. *Unit 11 §11.2 · LO 2c.5*

825 · B — No periodontal ligament. Differential mobility makes rigid tooth–implant connection a problem. *Unit 11 (exam traps) · LO 2c.5*

826 · C — No erythema, bleeding or suppuration; no bone loss. *Unit 11 §11.3 · LO 2c.21*

827 · B — Mucositis; Level 2. It is reversible with debridement, OHI and a design check. *Unit 11 §11.3 · LO 2c.21*

828 · D — Bone 3 mm or more apical to the platform, PPD 6 mm or more with bleeding. That is the 2017 World Workshop definition without baseline records. *Unit 11 §11.3 · LO 2c.21*

829 · A — Baseline radiograph and probing depths. *Unit 11 §11.3 · LO 2c.21*

830 · E — Interdental brushes, floss under the bar, non-scratch debridement, annual review. *Unit 11 §11.3 · LO 2c.21*

831 · A — Instruments that will not scratch titanium. *Unit 11 (exam traps) · LO 2c.21*

832 · C — Osseoperception only. There is no periodontal ligament or physiological movement. *Unit 11 §11.2, Unit 2 · LO 2c.5*

833 · B — Survived, not a success. *Unit 11 companion note 04 · LO 2c.21*

834 · E — Prosthesis-first assessment. *Unit 11 companion note 01 · LO 2a.19*

835 · D — Refer. Sinus grafting and augmentation are not Level 2. *Unit 11 companion note 05 · LO 2b.10*

836 · B — Short spans best; stitching errors over full arches. *Unit 11 §11.4 · LO 2a.19*

837 · E — All of the above. *Unit 11 §11.4 · LO 2a.19*

838 · C — Surface triangles, open format, no colour. *Unit 11 §11.4 · LO 2a.19*

839 · D — PLY or OBJ. These carry colour and texture. *Unit 11 §11.4 · LO 2a.19*

840 · A — DICOM merged with STL. *Unit 11 §11.4 · LO 2a.19*

841 · E — Trueness is closeness to the reference; precision is repeatability. *Unit 11 digital note 03 · LO 2a.19*

842 · B — Dense and homogeneous; limited by bur size. *Unit 11 §11.4 · LO 2a.19*

843 · D — SLA, DLP and selective laser melting. *Unit 11 §11.4 · LO 2a.19*

844 · A — Models, guides, trays, splints, temporaries. *Unit 11 §11.4 · LO 2a.19*

845 · C — Fewer visits, stored records, low-monomer bases; limited functional borders. *Unit 11 §11.4 · LO 2c.4*

846 · D — A physical mock-up. It remains useful for approval and consent. *Unit 11 §11.4 · LO 2a.19, 3b.1*

847 · C — Jaw motion tracking. *Unit 11 §11.4 · LO 2a.19*

848 · A — Secure storage and documented revisions. *Unit 11 digital note 05 · LO 3a.5*

849 · E — Post-curing and material indication. *Unit 11 digital note 04 · LO 2a.19*

850 · B — Validated technique with verification. Passive fit on multiple implants is a known digital limitation. *Unit 11 digital note 02 · LO 2a.19*

851 · C — Torque, occlusal load and connection. *Unit 11 §11.2 · LO 2c.21*

852 · A — Fracture of the upper denture. It is a recognised complication opposite implants. *Unit 11 §11.2 · LO 2c.21*

853 · E — A protective night guard. Bruxism raises complication risk. *Unit 11 §11.1 · LO 2c.5*

854 · D — Higher failure; Level 3. *Unit 11 §11.1, Unit 4 · LO 2c.5*

855 · B — Careful risk assessment; generally avoided. MRONJ risk is high with IV antiresorptives. *Unit 11 §11.1, Unit 4 · LO 2c.5*

856 · E — Optimise glycaemic control first. *Unit 11 §11.1, Unit 4 · LO 2c.5*

857 · D — Debridement, OHI and a cleansability check. *Unit 11 §11.3 · LO 2c.21*

858 · B — Refer as peri-implantitis. *Unit 11 §11.3, Unit 10 · LO 2c.21*

859 · A — Replace the matrix. *Unit 11 §11.2 · LO 2c.21*

860 · C — Facial references in the plan. They support, not replace, examination and consent. *Unit 11 §11.4 · LO 2a.19*

Under 70% on Block K? Re-read Unit 11 and Number card 3, then re-sit in a fortnight.

Block L answers — Complications, failures and maintenance

861 · C — Find the cause first. Repeated decementation points to resistance form; an identical replacement will fail the same way. *Unit 12 §12.1 · Schematic 27 · LO 2c.21*

862 · A — Remove, treat caries, redesign. Differential movement loosened one retainer while the other held. *Unit 12 §12.1 · LO 2c.22*

863 · E — Polish or repair with composite. Use HF and silane on ceramic, or MDP on any exposed metal. *Unit 12 §12.1 · LO 2c.22*

864 · B — Replace and redesign. Unsupported porcelain over 2 mm will chip again; give metal support or a metal or zirconia occlusal surface. *Unit 12 §12.1 · LO 2c.22*

865 · D — Check occlusion, desensitise, monitor. Most early sensitivity settles. *Unit 12 §12.1 · LO 2c.21*

866 · E — Root canal treatment through the crown. Seal the access, or replace the crown if needed. *Unit 12 §12.1 · LO 2c.21*

867 · B — Usually extraction. Prevention depends on ferrule and post design. *Unit 12 §12.1 · LO 2c.22*

868 · D — Residual cement. Suspect it until proven otherwise. *Unit 12 (exam traps) · LO 2c.21*

869 · A — An open or weak contact. Adjust or replace the restoration. *Unit 12 §12.1 · LO 2c.21*

870 · C — Clean, sandblast and rebond with MDP. An intact debonded RBB with no caries is rebonded, not remade. *Unit 12 (exam traps) · LO 2c.22*

871 · A — Caries and loss of retention. *Unit 12 §12.1 · LO 2c.22*

872 · D — Later need for endodontics. It is a common biological complication of crowned vital teeth. *Unit 12 §12.1 · LO 2c.21, 3b.1*

873 · C — Distorted impression or cast. Widespread misfit means re-impression and remake. *Unit 12 §12.2 · LO 2c.22*

874 · E — Distortion by removal on the clasp. Work hardening, a wrong undercut and wear are other causes. *Unit 12 §12.2 · LO 2c.22*

875 · B — Fatigue. Repeated adjustment of low-ductility Co-Cr, or casting porosity, makes it worse. *Unit 12 §12.2 · LO 2c.22*

876 · D — Overload, unseated rest, saddle rotation or active clasp. *Unit 12 §12.2 · LO 2c.21*

877 · E — Reline the saddle. *Unit 12 (exam traps) · LO 2c.23*

878 · A — Plaque at clasps and minor connectors, and sugar. *Unit 12 §12.2 · LO 2c.21*

879 · C — At least annually. Check the mucosa, fit and occlusal wear. *Unit 12 §12.3 · LO 2c.27*

880 · B — Reline or rebase. Remake only when OVD, appearance or occlusion are also wrong. *Unit 12 §12.3 · LO 2c.23*

881 · B — Accurate reassembly; investigate repeated fractures. Self-cure repairs are weaker. *Unit 12 §12.3 · LO 2c.22*

882 · C — Soft brush and non-abrasive cleanser over water, overnight soak, night removal. *Unit 12 §12.3 · LO 2c.27*

883 · D — Raised OVD from flash. It is a processing error. *Unit 12 §12.4 · LO 2c.22*

884 · A — Clear prescriptions, quality control, photographs, agreed times. *Unit 12 §12.4 · LO 3a.3*

885 · E — Survival is in place; success is complication-free. Ask which one a paper reports. *Unit 12 §12.5 · LO 2c.21*

886 · C — About 89% at 10 years. *Unit 12 §12.5 · LO 2c.21*

887 · A — About 95% at 5 years. *Unit 12 §12.5 · LO 2c.21*

888 · B — About 80–90% at 5–10 years. Cantilever designs perform well. *Unit 12 §12.5 · LO 2c.21*

889 · E — Implant overdenture attachments. *Unit 12 §12.5 · LO 2c.27*

890 · D — Diagnose the cause and change the design. Treat failure as feedback to the plan. *Unit 12 companion note 03, 05 · LO 2c.22*

Under 70% on Block L? Re-read Unit 12 and Crash notebook Schematic 27, then re-sit in a fortnight.

Block M answers — Governance, law and communication

891 · B — Informed, voluntary, capacitous, specific, documented, withdrawable. Consent is a process, not a signature. *Unit 13 §13.1 · LO 3b.1*

892 · E — Montgomery (2015). It requires disclosure of material risks and reasonable alternatives, and replaced Bolam for disclosure. *Unit 13 §13.1 · Schematic 25 · LO 3b.1*

893 · A — Bolam qualified by Bolitho. They still apply to diagnosis and treatment. *Unit 13 §13.1 · LO 3a.1*

894 · C — Duty, breach, causation, harm. *Unit 13 §13.1 · LO 3a.1*

895 · D — Vicarious liability. *Unit 13 §13.1 · LO 3a.1*

896 · D — An unwise decision is not incapacity. *Unit 13 §13.1 · LO 3b.2*

897 · A — Impairment, then understand, retain, use or weigh, and communicate. *Unit 13 §13.1 · LO 3b.2*

898 · E — Decision-specific and time-specific. Reassess on the day. *Unit 13 §13.1 · LO 3b.2*

899 · B — Presumed capacity at 16–17. *Unit 13 (exam traps) · LO 3b.2*

900 · C — Gillick competence. *Unit 13 §13.1 · LO 3b.2*

901 · E — Adults with Incapacity (Scotland) Act 2000. *Unit 13 §13.1 · LO 3b.2*

902 · C — BRAN. Discuss benefits, risks, alternatives and doing nothing. *Unit 13 §13.1 · LO 3b.1*

903 · D — Maximising income. It is not a GDC principle. *Unit 13 §13.2 · LO 3a.2*

904 · A — 100 hours in 5 years, at least 10 in every 2, with a PDP. *Unit 13 §13.2 · LO 3a.2*

905 · B — Indemnity is mandatory. *Unit 13 §13.2 · LO 3a.2*

906 · A — All registrants. *Unit 13 §13.3 · LO 3a.7*

907 · E — An apology is not an admission of liability. *Unit 13 (exam traps) · LO 3a.7*

908 · C — The statutory duty of candour. It applies to CQC-registered providers for notifiable incidents. *Unit 13 §13.3 · LO 3a.7*

909 · D — MHRA Yellow Card. *Unit 13 §13.3 · LO 3a.7*

910 · B — MHRA. *Unit 13 §13.3 · LO 3a.7*

911 · C — RIDDOR to the HSE. *Unit 13 §13.3 · LO 3a.7*

912 · A — A never event. *Unit 13 (exam traps) · LO 3a.7*

913 · B — Acknowledge within 3 working days; 12 months. *Unit 13 §13.3 · LO 3a.8*

914 · E — PHSO (NHS) or Dental Complaints Service (private). *Unit 13 §13.3 · LO 3a.8*

915 · D — Public Interest Disclosure Act 1998. *Unit 13 §13.3 · LO 3a.9*

916 · B — A dated, signed correction. Never alter the original. *Unit 13 §13.4 · LO 3a.5*

917 · D — Adults 11 years; children to 25 or 11 years, whichever is longer. *Unit 13 §13.4 · LO 3a.5*

918 · A — One month, normally free. *Unit 13 §13.4 · LO 3a.5*

919 · C — Consent, law, or public interest. Share the minimum necessary. *Unit 13 §13.4 · LO 3a.5*

920 · E — Caldicott principles. *Unit 13 §13.4 · LO 3a.5*

921 · D — HSE registration, RPA, RPS, local rules, controlled area. *Unit 13 §13.5 · LO 3a.6*

922 · B — About 1.5 m, and the primary beam. *Unit 13 §13.5 · LO 3a.6*

923 · E — Employer, referrer, practitioner, operator. *Unit 13 §13.5, Unit 5 · LO 3a.6*

924 · C — Rectangular collimation, holders, fast receptor, QA audit. *Unit 13 §13.5 · LO 3a.6*

925 · A — 134–137°C for at least 3 minutes. *Unit 13 §13.6 · Schematic 26 · LO 3a.4*

926 · E — A type B vacuum steriliser. It is needed for wrapped and hollow instruments. *Unit 13 §13.6 · LO 3a.4*

927 · C — A washer-disinfector. *Unit 13 §13.6 · LO 3a.4*

928 · B — Automatic control test, plus steam penetration for vacuum. *Unit 13 §13.6 · LO 3a.4*

929 · A — Disinfect both ways and label. *Unit 13 §13.6 · LO 3a.4*

930 · D — 2 minutes at the start, 20–30 seconds between patients. *Unit 13 §13.6 · LO 3a.4*

931 · A — Bleed, wash, cover, report, occupational health urgently. HIV PEP ideally within 1 hour, up to 72 hours. *Unit 13 §13.6 · LO 3a.4*

932 · D — Hepatitis B vaccination with documented response. *Unit 13 §13.6 · LO 3a.4*

933 · C — Recognise, record, refer. Do not investigate yourself. *Unit 13 §13.7 · LO 3a.10*

934 · B — Report to the police. It is mandatory for known FGM in under-18s in England and Wales. *Unit 13 §13.7 · LO 3a.10*

935 · E — Reasonable adjustments. Disability is one of nine protected characteristics. *Unit 13 §13.7 · LO 3a.11*

936 · C — The seven pillars. *Unit 13 §13.8 · LO 3a.12*

937 · B — Audit. Audit measures against a standard; research creates new knowledge. *Unit 13 (exam traps) · LO 3a.12*

938 · D — Re-audit. *Unit 13 §13.8 · LO 3a.12*

939 · E — PDSA cycles. *Unit 13 §13.8 · LO 3a.12*

940 · A — A systematic review and meta-analysis of RCTs. *Unit 13 §13.8 · LO 3a.12*

941 · E — GRADE. *Unit 13 §13.8 · LO 3a.12*

942 · C — Sensitivity. *Unit 13 §13.8 · LO 3a.12*

943 · A — PPV falls with low prevalence. *Unit 13 §13.8 · LO 3a.12*

944 · D — 10. The absolute risk reduction is 10%, so NNT = 1 ÷ 0.10. *Unit 13 §13.8 · LO 3a.12*

945 · B — Not significant. The interval crosses 1. *Unit 13 §13.8 · LO 3a.12*

946 · A — CQC, HIS, HIW, RQIA. The GDC is the professional regulator. *Unit 13 §13.8 · LO 3a.2*

947 · B — Network link, caseload, PROMs and PREMs, audit, complication reporting. *Unit 13 §13.8 · LO 3a.12*

948 · C — SPIKES. *Unit 13 §13.9 · LO 3b.3*

949 · E — A professional interpreter. Never use family members or children. *Unit 13 §13.9 · LO 3b.4*

950 · D — A written plan and estimate (FP17DC). *Unit 13 §13.9 · LO 3b.1*

Under 70% on Block M? Re-read Unit 13 and Crash notebook Schematics 25–26, then re-sit in a fortnight.

Block N answers — Integrated cases

951 · C — Retain the root as an overdenture abutment. Over 5 years of oral bisphosphonate raises MRONJ risk, so root retention beats extraction; apixaban is not stopped. *Units 4, 6, 7 · LO 2b.11, 2c.9*

952 · A — Manage the cause and monitor. Stabilise reflux erosion before any irreversible veneers. *Units 2, 8, 10 · LO 2c.10*

953 · E — Capacity, best interests, simple dentures, labelling. Copying is impossible because the dentures are lost. *Units 4, 7, 13 · LO 3b.2, 2c.9*

954 · B — Remove the allergen; adrenaline 500 micrograms IM. Anaphylaxis needs IM adrenaline first. *Units 4, 13 · LO 2b.1*

955 · D — Mucositis from residual cement; Level 2. Remove the cement and debride; there is no bone loss. *Unit 11 · LO 2c.21*

956 · B — Specialist referral. Osteoradionecrosis risk is lifelong; avoid extraction in the field where possible. *Units 4, 6 · LO 2b.11*

957 · E — Empathy, consented liaison, prevention first. She is an adult, so her confidentiality is respected. *Units 2, 10, 13 · LO 2c.10, 3a.5*

958 · A — Hygiene, night removal, fit; nystatin if needed. Azoles interact with warfarin and simvastatin. *Units 2, 4 · LO 1.8, 1.7*

959 · D — Review the occlusion and design; discuss alternatives. Repeated debonding in a bruxist with heavy contact means the design must change. *Units 8, 12 · LO 2c.22*

960 · C — Wax-up, provisional trial, Level 3. Generalised wear with an OVD increase is reorganised rehabilitation. *Units 9, 10 · LO 2c.11*

961 · D — Tooth-supported Co-Cr with RPI, lingual bar and open design. A 9 mm sulcus allows a bar; high caries risk favours minimal coverage. *Units 6, 10 · LO 2c.3, 2c.16*

962 · C — Trim the distobuccal flange and thin the posterior border. Keep the post-dam seal. *Unit 7 · LO 2c.21*

963 · A — Monitor with multidisciplinary advice; defer implants. Ankylosed teeth cannot be extruded orthodontically. *Units 10, 11 · LO 2a.18*

964 · E — A well-fitting conventional denture. High-dose IV antiresorptives carry high MRONJ risk, so implants are generally avoided. *Units 4, 7, 11 · LO 2b.11*

965 · B — Periodontal therapy and diabetic control first. Definitive work waits for stability. *Units 2, 4, 5, 10 · LO 2b.3, 2a.13*

966 · A — Prompt cuspal coverage, monolithic material, night guard. *Units 3, 8, 9 · LO 2c.14*

967 · D — Candour, apology, remedy, never-event report, documentation. *Unit 13 · LO 3a.7*

968 · C — Acknowledge within 3 working days. *Unit 13 · LO 3a.8*

969 · E — Cord retraction, margin about 3 mm from the crest. Avoid electrosurgery with an ICD; respect the supracrestal attachment. *Units 4, 8 · LO 2c.13*

970 · B — Anteroposterior strap. *Unit 6 · LO 2c.3*

971 · E — Record and refer via the safeguarding lead. Do not investigate yourself. *Unit 13 · LO 3a.10*

972 · B — Interim prostheses and Level 3 referral. Hypodontia is complex multidisciplinary care. *Units 10, 11 · LO 2b.10*

973 · D — Encourage coughing; image if not retrieved. Back blows are only for ineffective coughing. *Units 4, 12 · LO 2b.1*

974 · C — Peri-implantitis: refer. *Units 10, 11 · LO 2c.21*

975 · A — Reversible care first; no equilibration for TMD. *Unit 9 · LO 2a.20*

976 · C — Mental nerve pressure. Symptoms that resolve when the denture is removed point to pressure; relieve the fitting surface. *Units 1, 7 · LO 2c.21*

977 · A — Fluoride gel in the denture, hygiene, diet. *Units 2, 7 · LO 2c.9*

978 · B — Selective pressure, balanced occlusion, light anterior contact. This limits combination-syndrome-type loading. *Units 7, 11 · LO 2c.9*

979 · D — High fluoride, neutral substitute, open design, short recall. *Units 2, 6 · LO 2c.3*

980 · E — 10. The absolute risk reduction is 10%, so NNT = 1 ÷ 0.10. *Unit 13 · LO 3a.12*

981 · A — Individual justification. *Units 5, 11, 13 · LO 2a.12*

982 · D — Retain the records; offer copies. *Unit 13 · LO 3a.5*

983 · B — Presumed capacity at 17. *Unit 13 · LO 3b.2*

984 · C — Best interests, least restrictive option. *Units 4, 13 · LO 3b.2*

985 · E — Statement of manufacture and MHRA registration. *Units 8, 13 · LO 3a.3*

986 · E — Standard precautions for every patient. *Units 3, 13 · LO 3a.4*

987 · C — Remove, treat, redesign at the pier. *Units 8, 12 · LO 2c.22*

988 · D — Dehydration and poor lighting. *Unit 8 · LO 2c.13*

989 · B — Haemophilia centre liaison. *Units 4, 7 · LO 2b.11*

990 · A — Standard care, no drug holiday. Low-risk MRONJ profile. *Unit 4 · LO 2b.11*

991 · B — Implant-retained palateless prosthesis. *Units 7, 11 · LO 2c.5*

992 · E — Survived, not a success. *Unit 12 · LO 2c.21, 3b.1*

993 · C — Single-wing resin-bonded cantilever. *Unit 8 · LO 2c.11*

994 · A — Posterior support and selective pressure. That manages combination syndrome. *Units 6, 7 · LO 2c.9*

995 · D — Same-day referral for suspected GCA. *Unit 9 · LO 2a.20*

996 · D — Liaise and defer elective care. *Unit 4 · LO 2b.12*

997 · B — Continue antiplatelets. *Unit 4 · LO 2b.11*

998 · E — Urgent suspected cancer referral. *Units 2, 7 · LO 1.11*

999 · A — Re-audit. *Unit 13 · LO 3a.12*

1000 · C — Risk-based recall with maintenance and PROMs. *Units 10, 11, 12 · LO 2c.27*

Under 70% on Block N? Find which units you dropped marks in and re-read those first.

Mark sheet

Score each block as a percentage. Under 70%: re-read the unit and re-sit after a fortnight. Aim for 80% or more on every block by the final attempt; the exam itself has no negative marking, so answer every question.

Block	Topic	Questions	Attempt 1 (%)	Attempt 2 (%)	Attempt 3 (%)
A	Applied anatomy	1–70 (70)			
B	Physiology and pathology	71–130 (60)			
C	Dental materials	131–220 (90)			
D	Pharmacology and emergencies	221–290 (70)			
E	Assessment, diagnosis and imaging	291–370 (80)			
F	Removable partial dentures	371–470 (100)			
G	Complete dentures	471–570 (100)			
H	Fixed prosthodontics	571–670 (100)			
I	Occlusion and TMD	671–730 (60)			
J	Tooth wear, failing dentition, trauma	731–800 (70)			
K	Implants and digital	801–860 (60)			
L	Complications and maintenance	861–890 (30)			
M	Governance, law and communication	891–950 (60)			
N	Integrated cases	951–1000 (50)			
	Total	**1,000**			

Mock papers. To mirror the real exam (two papers of 90 SBAs, 2 hours each), pick 90 questions spread across the blocks in blueprint proportions for each paper and time them strictly.

Mock	Date	Score (/90)	%	Weakest blocks
Paper 1				
Paper 2				